

IBM Z NetView  
Version 6 Release 3

*User's Guide: Automated Operations  
Network*



**Note**

Before using this information and the product it supports, read the information in [“Notices” on page 391.](#)

This edition applies to version 6, release 3 of IBM Z NetView (product number 5697-NV6 ) and to all subsequent versions, releases, and modifications until otherwise indicated in new editions.

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## About this publication

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The IBM Z® NetView® product provides advanced capabilities that you can use to maintain the highest degree of availability of your complex, multi-platform, multi-vendor networks and systems from a single point of control. This publication, the *IBM Z NetView User's Guide: Automated Operations Network*, describes how to use NetView Automated Operations Network (AON) functions to improve system and network efficiency and to eliminate or simplify much of the routine work that operators perform. It also describes customization and programming activities used to tailor and extend the automated operations capabilities of AON, which provides event-driven network automation implemented from a NetView base.

## Intended audience

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This publication is for network system programmers and network operators who use Automated Operations Network (AON) to perform network automation. The publication is for network operators who need to control and to manage the NetView network automation policy and functions. It is also for network system programmers who are responsible for customizing AON and who should have experience using the NetView program and should understand the requirements of their network.

## Publications

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This section lists publications in the IBM Z NetView library and related documents. It also describes how to access NetView publications online and how to order NetView publications.

### IBM Z NetView library

The following documents are available in the IBM Z NetView library:

- *Administration Reference*, SC27-2869, describes the NetView program definition statements required for system administration.
- *Application Programmer's Guide*, SC27-2870, describes the NetView program-to-program interface (PPI) and how to use the NetView application programming interfaces (APIs).
- *Automation Guide*, SC27-2846, describes how to use automated operations to improve system and network efficiency and operator productivity.
- *Command Reference Volume 1 (A-N)*, SC27-2847, and *Command Reference Volume 2 (O-Z)*, SC27-2848, describe the NetView commands, which can be used for network and system operation and in command lists and command procedures.
- *Installation: Configuring Additional Components*, GC27-2851, describes how to configure NetView functions beyond the base functions.
- *Installation: Configuring the NetView Enterprise Management Agent*, GC27-2853, describes how to install and configure the IBM Z NetView Enterprise Management Agent.
- *Installation: Getting Started*, GI11-9443, describes how to install and configure the base NetView program.
- *Installation: Migration Guide*, GC27-2854, describes the new functions that are provided by the current release of the NetView product and the migration of the base functions from a previous release.
- *IP Management*, SC27-2855, describes how to use the NetView product to manage IP networks.
- *Messages and Codes Volume 1 (AAU-DSI)*, GC27-2856, and *Messages and Codes Volume 2 (DUI-IHS)*, GC27-2857, describe the messages for the NetView product, the NetView abend codes, the sense codes that are included in NetView messages, and generic alert code points.
- *Programming: Pipes*, SC27-2859, describes how to use the NetView pipelines to customize a NetView installation.

- *Programming: REXX and the NetView Command List Language*, SC27-2861, describes how to write command lists for the NetView product using the Restructured Extended Executor language (REXX) or the NetView command list language.
- *Security Reference*, SC27-2863, describes how to implement authorization checking for the NetView environment.
- *Troubleshooting Guide*, GC27-2865, provides information about documenting, diagnosing, and solving problems that occur in the NetView product.
- *Tuning Guide*, SC27-2874, provides tuning information to help achieve certain performance goals for the NetView product and the network environment.
- *User's Guide: Automated Operations Network*, SC27-2866, describes how to use the NetView Automated Operations Network (AON) component, which provides event-driven network automation, to improve system and network efficiency. It also describes how to tailor and extend the automated operations capabilities of the AON component.
- *User's Guide: NetView*, SC27-2867, describes how to use the NetView product to manage complex, multivendor networks and systems from a single point.
- *User's Guide: NetView Enterprise Management Agent*, SC27-2876, describes how to use the NetView Enterprise Management Agent.
- *Using Tivoli System Automation for GDPS/PPRC HyperSwap Manager with NetView*, GI11-4704, provides information about the Tivoli® System Automation for GDPS®/PPRC HyperSwap® Manager with NetView feature, which supports the GDPS and Peer-to-Peer Remote Copy (PPRC) HyperSwap Manager services offering.
- *Licensed Program Specifications*, GC31-8848, provides the license information for the NetView product.
- *Program Directory for IBM Z NetView US English*, GI11-9444, contains information about the material and procedures that are associated with installing the NetView product.
- *Program Directory for IBM Z NetView Japanese*, GI11-9445, contains information about the material and procedures that are associated with installing the NetView product.
- *Program Directory for IBM Z NetView Enterprise Management Agent*, GI11-9446, contains information about the material and procedures that are associated with installing the IBM Z NetView Enterprise Management Agent.

The following books are archived:

- *Customization Guide*, SC27-2849, describes how to customize the NetView product and points to sources of related information.
- *Data Model Reference*, SC27-2850, provides information about the Graphic Monitor Facility host subsystem (GMFHS), SNA topology manager, and MultiSystem Manager data models.
- *Installation: Configuring Graphical Components*, GC27-2852, describes how to install and configure the NetView graphics components.
- *Programming: Assembler*, SC27-2858, describes how to write exit routines, command processors, and subtasks for the NetView product using assembler language.
- *Programming: PL/I and C*, SC27-2860, describes how to write command processors and installation exit routines for the NetView product using PL/I or C.
- *Resource Object Data Manager and GMFHS Programmer's Guide*, SC27-2862, describes the NetView Resource Object Data Manager (RODM), including how to define your non-SNA network to RODM and use RODM for network automation and for application programming.
- *SNA Topology Manager Implementation Guide*, SC27-2864, describes planning for and implementing the NetView SNA topology manager, which can be used to manage subarea, Advanced Peer-to-Peer Networking, and TN3270 resources.
- *User's Guide: NetView Management Console*, SC27-2868, provides information about the NetView management console interface of the NetView product.

## Related publications

You can find additional product information on the IBM Z NetView web site at <https://www.ibm.com/us-en/marketplace/ibm-tivoli-netview-for-zos>.

For information about the NetView Bridge function, see *Tivoli NetView for OS/390® Bridge Implementation*, SC31-8238-03 (available only in the V1R4 library).

## Terminology in this Library

The following terms are used in this library:

### **CNMCMC**

For the CNMCMC member and the members that are included in it using the %INCLUDE statement

### **CNMSTYLE**

For the CNMSTYLE member and the members that are included in it using the %INCLUDE statement

### **DSIOPF**

For the DSIOPF member and the members that are included in it using the %INCLUDE statement

### **IBM® Tivoli Netcool®/OMNIbus**

For either of these products:

- IBM Tivoli Netcool/OMNIbus
- IBM Tivoli OMNIbus and Network Manager

### **MVS™**

For z/OS® operating systems

### **MVS element**

For the base control program (BCP) element of the z/OS operating system

### **NetView**

For the following products:

- IBM Z NetView version 6 release 3
- IBM Tivoli NetView for z/OS version 6 release 2 modification 1
- NetView releases that are no longer supported

### **PARMLIB**

For SYS1.PARMLIB and other data sets in the concatenation sequence

### **VTAM®**

For Communications Server - SNA Services

Unless otherwise indicated, topics to programs indicate the latest version and release of the programs. If only a version is indicated, the topic is to all releases within that version.

When a topic is made about using a personal computer or workstation, any programmable workstation can be used.

## Using IBM Z NetView online help

The following types of IBM Z NetView mainframe online help are available, depending on your installation and configuration:

- General help and component information
- Command help
- Message help
- Sense code information
- Recommended actions

## Accessing publications online

IBM posts publications for this and all other products, as they become available and whenever they are updated, to the IBM Knowledge Center at <https://www.ibm.com/support/knowledgecenter>. You can find IBM Z NetView documentation on [IBM Z NetView Knowledge Center](#).

**Note:** If you print PDF documents on other than letter-sized paper, set the option in the **Print** window that enables Adobe Reader to print letter-sized pages on your local paper.

## Ordering publications

You can order many Tivoli publications online at <http://www.ibm.com/e-business/linkweb/publications/servlet/pbi.wss>

You can also order by telephone by calling one of these numbers:

- In the United States: 800-426-4968
- In Canada: 800-879-2755

In other countries, contact your software account representative to order Tivoli publications. To locate the telephone number of your local representative, perform the following steps:

1. Go to <http://www.ibm.com/e-business/linkweb/publications/servlet/pbi.wss>.
2. Select your country from the list and click the grey arrow button beside the list.
3. Click **About this site** to see an information page that includes the telephone number of your local representative.

## Accessibility

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Accessibility features help users with a physical disability, such as restricted mobility or limited vision, to use software products successfully. Standard shortcut and accelerator keys are used by the product and are documented by the operating system. Refer to the documentation provided by your operating system for more information.

For additional information, see the Accessibility appendix in the *User's Guide: NetView*.

## Tivoli user groups

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Tivoli user groups are independent, user-run membership organizations that provide Tivoli users with information to assist them in the implementation of Tivoli Software solutions. Through these groups, members can share information and learn from the knowledge and experience of other Tivoli users.

## Support information

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If you have a problem with your IBM software, you want to resolve it quickly. IBM provides the following ways for you to obtain the support you need:

### Online

Please follow the instructions located in the support guide entry: <https://www.ibm.com/support/home/pages/support-guide/?product=4429363>.

### Troubleshooting information

For more information about resolving problems with the IBM Z NetView product, see the *IBM Z NetView Troubleshooting Guide*. You can also discuss technical issues about the IBM Z NetView product through the NetView user group located at <https://groups.io/g/NetView>. This user group is for IBM Z NetView customers only, and registration is required. This forum is also monitored by interested parties within IBM who answer questions and provide guidance about the NetView



product. When a problem with the code is found, you are asked to open an official case to obtain resolution.

## Conventions used in this publication

---

This section describes the conventions that are used in this publication.

### Typeface conventions

This publication uses the following typeface conventions:

#### **Bold**

- Lowercase commands and mixed case commands that are otherwise difficult to distinguish from surrounding text
- Interface controls (check boxes, push buttons, radio buttons, spin buttons, fields, folders, icons, list boxes, items inside list boxes, multicolumn lists, containers, menu choices, menu names, tabs, property sheets), labels (such as **Tip:**, and **Operating system considerations:**)
- Keywords and parameters in text

#### ***Italic***

- Citations (examples: titles of publications, diskettes, and CDs)
- Words defined in text (example: a nonswitched line is called a *point-to-point line*)
- Emphasis of words and letters (words as words example: "Use the word *that* to introduce a restrictive clause."; letters as letters example: "The LUN address must start with the letter *L*.")
- New terms in text (except in a definition list): a *view* is a frame in a workspace that contains data.
- Variables and values you must provide: ... where *myname* represents...

#### **Monospace**

- Examples and code examples
- File names, programming keywords, and other elements that are difficult to distinguish from surrounding text
- Message text and prompts addressed to the user
- Text that the user must type
- Values for arguments or command options

### Operating system-dependent variables and paths

For workstation components, this publication uses the UNIX convention for specifying environment variables and for directory notation.

When using the Windows command line, replace **\$variable** with **%variable%** for environment variables and replace each forward slash (/) with a backslash (\) in directory paths. The names of environment variables are not always the same in the Windows and UNIX environments. For example, %TEMP% in Windows environments is equivalent to \$TMPDIR in UNIX environments.

**Note:** If you are using the bash shell on a Windows system, you can use the UNIX conventions.

### Syntax diagrams

The following syntax elements are shown in syntax diagrams. Read syntax diagrams from left-to-right, top-to-bottom, following the horizontal line (the main path).

- [“Symbols” on page xxvi](#)
- [“Parameters” on page xxvi](#)
- [“Punctuation and parentheses” on page xxvi](#)

- [“Abbreviations” on page xxvii](#)

For examples of syntax, see [“Syntax examples” on page xxvii](#).

## Symbols

The following symbols are used in syntax diagrams:



Marks the beginning of the command syntax.



Marks the end of the command syntax.



Indicates that the command syntax is continued on the next line.



Indicates that a statement is continued from the previous line.



Marks the beginning and end of a fragment or part of the command syntax.

## Parameters

The following types of parameters are used in syntax diagrams:

### Required

Required parameters are shown on the main path.

### Optional

Optional parameters are shown below the main path.

### Default

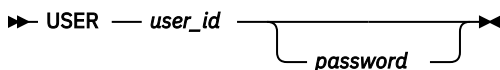
Default parameters are shown above the main path. In parameter descriptions, default parameters are underlined.

Syntax diagrams do not rely on highlighting, brackets, or braces. In syntax diagrams, the position of the elements relative to the main syntax line indicates whether an element is required, optional, or the default value.

When you issue a command, spaces are required between the parameters unless a different separator, such as a comma, is specified in the syntax.

Parameters are classified as keywords or variables. Keywords are shown in uppercase letters. Variables, which represent names or values that you supply, are shown in lowercase letters and are either italicized or, in NetView help, displayed in a differentiating color.

In the following example, the `USER` command is a keyword, the `user_id` parameter is a required variable, and the `password` parameter is an optional variable.



## Punctuation and parentheses

You must include all punctuation that is shown in the syntax diagram, such as colons, semicolons, commas, minus signs, and both single and double quotation marks.

When an operand can have more than one value, the values are typically enclosed in parentheses and separated by commas. For a single value, the parentheses typically can be omitted. For more information, see [“Multiple operands or values” on page xxviii](#).

If a command requires positional commas to separate keywords and variables, the commas are shown before the keywords or variables.

When examples of commands are shown, commas are also used to indicate the absence of a positional operand. For example, the second comma indicates that an optional operand is not being used:

```
COMMAND_NAME opt_variable_1,opt_variable_3
```

You do not need to specify the trailing positional commas. Trailing positional and non-positional commas either are ignored or cause a command to be rejected. Restrictions for each command state whether trailing commas cause the command to be rejected.

## Abbreviations

Command and keyword abbreviations are listed in synonym tables after each command description.

## Syntax examples

The following examples show the different uses of syntax elements:

- [“Required syntax elements” on page xxvii](#)
- [“Optional syntax elements” on page xxvii](#)
- [“Default keywords and values” on page xxvii](#)
- [“Multiple operands or values” on page xxviii](#)
- [“Syntax that is longer than one line” on page xxviii](#)
- [“Syntax fragments” on page xxviii](#)

### Required syntax elements

Required keywords and variables are shown on the main syntax line. You must code required keywords and variables.

➤ REQUIRED\_KEYWORD — *required\_variable* ➤

A required choice (two or more items) is shown in a vertical stack on the main path. The items are shown in alphanumeric order.

➤  
└─ REQUIRED\_OPERAND\_OR\_VALUE\_1  
└─ REQUIRED\_OPERAND\_OR\_VALUE\_2  
└─

### Optional syntax elements

Optional keywords and variables are shown below the main syntax line. You can choose not to code optional keywords and variables.

➤  
└─ OPTIONAL\_OPERAND  
└─

A required choice (two or more items) is shown in a vertical stack below the main path. The items are shown in alphanumeric order.

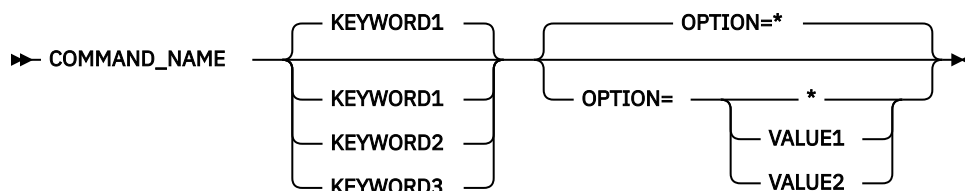
➤  
└─ OPTIONAL\_OPERAND\_OR\_VALUE\_1  
└─ OPTIONAL\_OPERAND\_OR\_VALUE\_2  
└─

### Default keywords and values

Default keywords and values are shown above the main syntax line in one of the following ways:

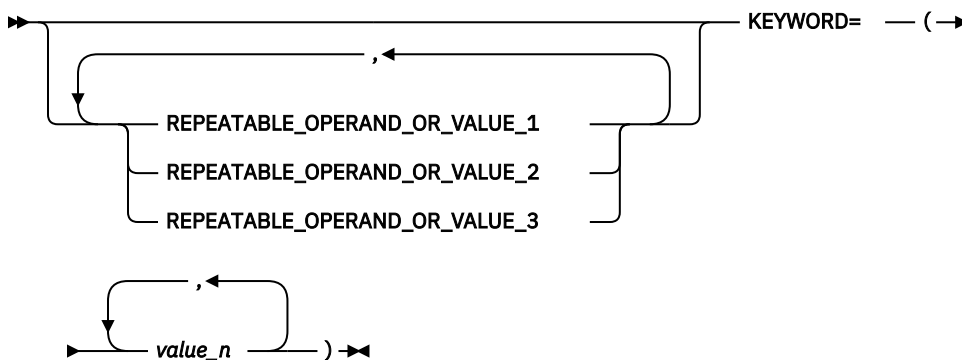
- A default keyword is shown only above the main syntax line. You can specify this keyword or allow it to default. The following syntax example shows the default keyword KEYWORD1 above the main syntax line and the rest of the optional keywords below the main syntax line.

- If an operand has a default value, the operand is shown both above and below the main syntax line. A value below the main syntax line indicates that if you specify the operand, you must also specify either the default value or another value shown. If you do not specify the operand, the default value above the the main syntax line is used. The following syntax example shows the default values for operand OPTION=\* above and below the main syntax line.



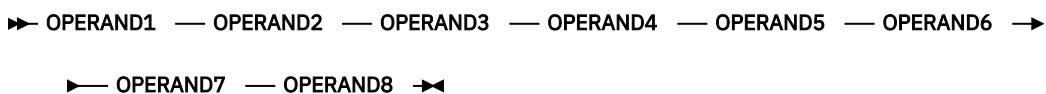
### Multiple operands or values

An arrow returning to the left above a group of operands or values indicates that more than one can be selected or that a single one can be repeated.



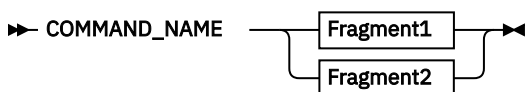
### Syntax that is longer than one line

If a diagram is longer than one line, each line that is to be continued ends with a single arrowhead and the following line begins with a single arrowhead.



### Syntax fragments

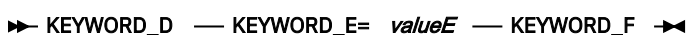
Some syntax diagrams contain syntax fragments, which are used for lengthy, complex, or repeated sections of syntax. Syntax fragments follow the main diagram. Each syntax fragment name is mixed case and is shown in the main diagram and in the heading of the fragment. The following syntax example shows a syntax diagram with two fragments that are identified as **Fragment1** and **Fragment2**.



#### Fragment1



#### Fragment2



---

## **Part 1. Introducing Automated Operations Network**



# Chapter 1. Introducing Automated Operation Network (AON)

This chapter describes how to use the Automated Operations Network (AON) component of NetView by showing you how to display and use its panels. Many of the AON functions can be accessed without displaying a panel if you provide all the parameters with the command and function. For a list of these commands, refer to the *IBM Z NetView Command Reference Volume 2 (O-Z)*.

The AON operator interface enables you to use operator functions to view color-coded status displays, change automation settings, receive messages, issue commands, and perform many other functions that control automation and resources.

To use the operator interface, log on to the NetView program. If you log on as a NetView-NetView task (NNT), the operator interface is bypassed; however, you can use AON by issuing commands from the command line.

## Displaying the AON: Operator Commands Main Menu

You can display the operator interface from any command line within the NetView program. The main panel of the AON operator interface is the AON: Operator Commands Main Menu panel.

To display the AON: Operator Commands Main Menu panel:

1. Type **AON** on the command line.
2. Press Enter.

The AON: Operator Commands Main Menu panel shown in [Figure 1 on page 3](#) is displayed.

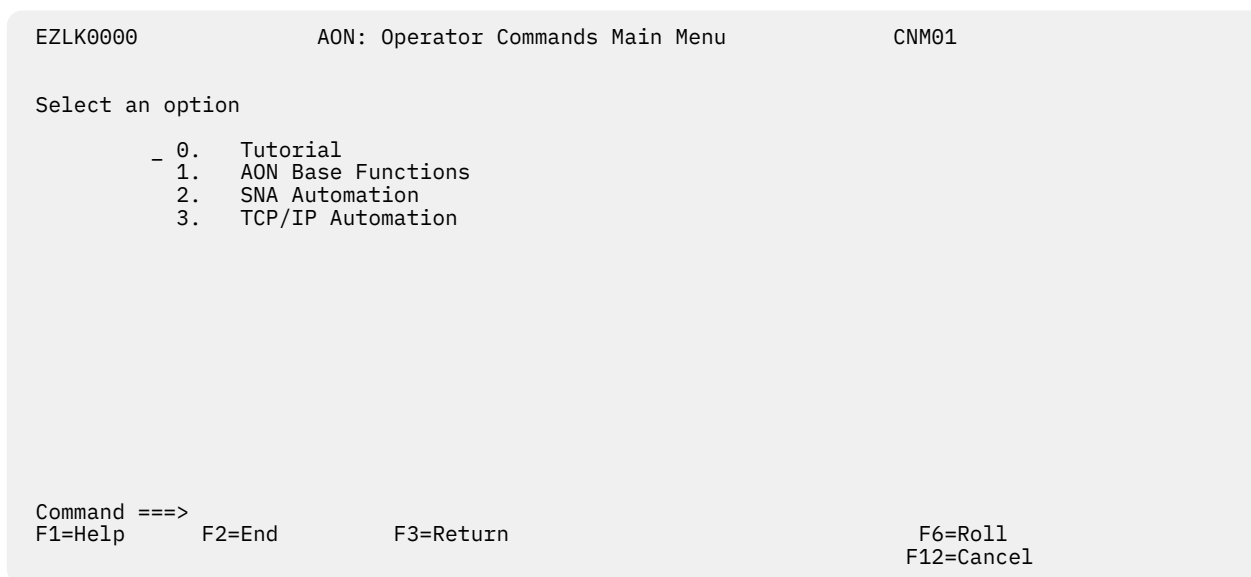


Figure 1. AON: Operator Commands Main Menu Panel

If an automation component is not enabled, the menu option for that component is displayed in a different, dimmer color from that of the installed options.

If you use the AON functions to disable all or part of AON automation, the option on the main menu panel for the disabled component is displayed in the dimmer color. The main menu panel enables you to access base AON, SNA, and TCP/IP automation functions.

## Displaying the AON Base Functions Panel

To access AON functions, use the AON Base Functions panel:

1. Display the **AON: Operator Commands Main Menu** panel.

**Note:** To display the AON: Operator Commands Main Menu panel, see [“Displaying the AON: Operator Commands Main Menu”](#) on page 3.

2. On the AON: Operator Commands Main Menu panel, type **1** in the entry field.
3. Press Enter.

The AON: Base Functions panel shown in [Figure 2 on page 4](#) is displayed.

**Note:** You can also display the AON: Base Functions panel by typing **AON 1** on any command line. The following chapters explain how to use several of the options on this panel.

```
EZLK0100          AON: Base Functions          CNM01

Select an option

- 0. Tutorial
  1. Help Desk
  2. AutoView
  3. DDF
  4. Automation Settings
  5. Cross Domain Functions
  6. Timer
  7. Task and Log Maintenance
  8. Support Functions
  9. Display the Inform Log

Command ===>
F1=Help      F2=Main Menu  F3=Return

F6=Roll
F12=Cancel
```

*Figure 2. AON: Base Functions Panel*

## Understanding how a panel is organized

This section describes using the AON operator interface panels. The panels have a similar structure. Each panel has a heading that displays basic information about the panel. Under the heading, many panels display lists or rows of data, as well as interactive entry fields, such as menus and data entry fields.

Some panels display pop-up command windows, providing additional entry fields for issuing commands. Some panels support selection lists from which you can select previously defined entries. Some panels support the use of wildcard characters.

Finally, each panel has an area near the bottom where messages are displayed in response to actions you issue on the panel. At the bottom of each panel, a set of function keys is available for navigating through the interface.

Each of these panel parts is described in detail in the following subsections.



# Headings

The panel heading, as shown in [Figure 3 on page 5](#), provides information about the panel.

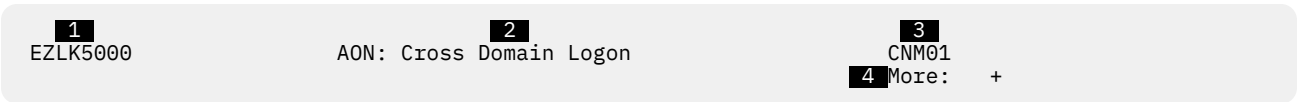


Figure 3. Example of a Panel Heading

- 1** The panel ID.
- 2** The name of the panel.
- 3** The domain name. (The name is not displayed on all panels.)
- 4** If there is more information for this panel, you can display the information by scrolling the panel. (This field is not displayed on all panels.)

**Note:** When applicable, the time of day is also displayed on panels.

# Entry fields

Most panels have menu choices or data entry fields, as shown in [Figure 4 on page 5](#), in which you can type data to be processed. Menu choices and data entry fields differ from panel to panel.

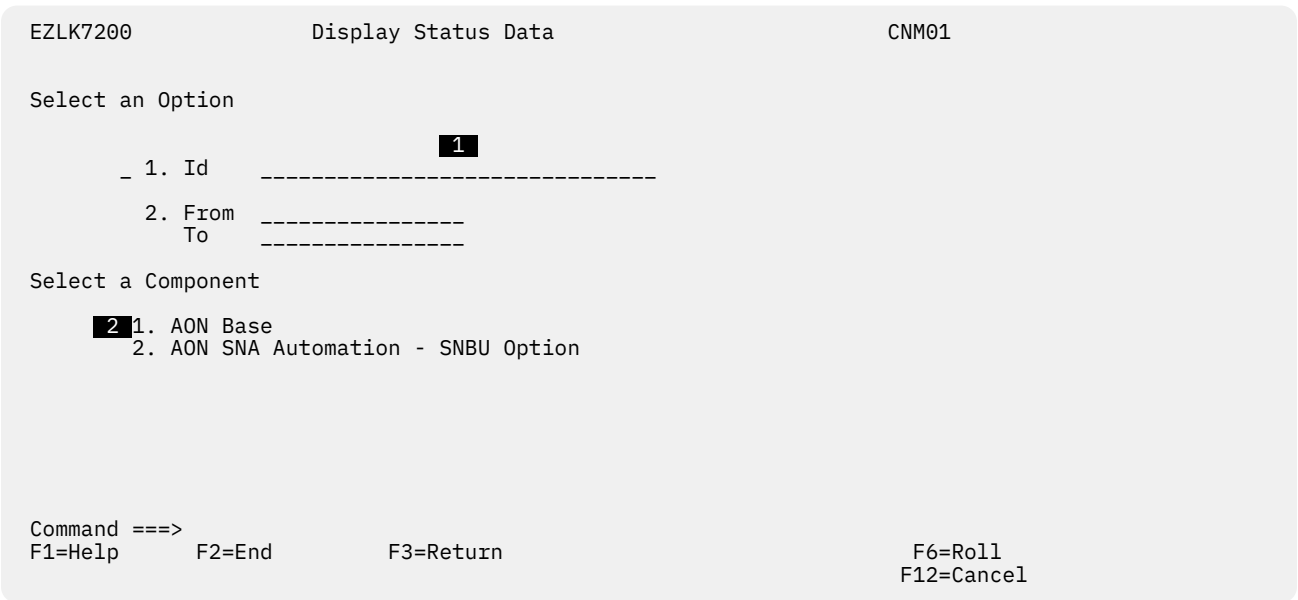


Figure 4. Example of a Panel with Entry Fields and Menu Options

- 1** An example of a data entry field
- 2** An example of a menu choice entry field

# Data panels

Some panels display data arranged in columns or rows for informational purposes. The Domain, Status, Operator, Type, Init, and Description columns shown in [Figure 5 on page 6](#) are examples. The data that is displayed on a particular panel differs from panel to panel.

EZLK5000

AON: Cross Domain Logon

CNM01

1

1=Start

2=Stop

3=Send

Origin

Operator

OPER1

2

-

-

-

-

Domain

CNM01

CNM02

CNM10

Status

Inactive

Inactive

Inactive

Operator

Type

NNT

NNT

NNT

Init

Description

Figure 5. Example of Information Displays

- 1
- Action codes
- 2
- Entry fields

Pop-up command windows

On some panels, the F4 key is activated. When you press the **F4** key, a pop-up window is displayed that lists commands that you can issue against a resource. The commands that are displayed in the pop-up window depend on the type of resource that has been selected. [Figure 6 on page 6](#) shows an example of a pop-up command window.

EZLK5100

AON: Cross Domain Logon

CNM01

1=Start

2=Stop

3=Send

Origin

Operator

OPER1

Domain

Status

Operator

Type

AON01

Inactive

NNT

AON02

Inactive

NNT

AOC10

Inactive

NNT

CNM10

Inactive

NNT

AOF10

Inactive

NNT

AOC06

Inactive

NNT

CNM06

Inactive

NNT

AOC05

Inactive

NNT

CNM05

Inactive

NNT

CNM01

Inactive

RMTCMD

.....

Select an Action

.....

To act on a single session:

1. Start CNM01 Session

2. Stop CNM01 Session

3. Send to CNM01

4. Start New RMTCMD Session

5. View Start Error Message

To act on all defined sessions:

6. Start All Sessions

7. Stop All Sessions

.....

F1=Help

F12=Cancel

.....

To issue commands, tab to the Domain and press F4.

Command ==>

F1=Help

F2=Main Menu

F3=Return

F4=Commands

F5=Refresh

F6=Roll

F7=Backward

F8=Forward

F12=Cancel

Figure 6. Pop-up Command Window

From this pop-up command window, you can choose any of the seven actions.

Message display area

AON frequently displays messages on the panels of the operator interface in response to actions. These messages are displayed in the lower portion of a panel, where the message EZL910I ENTER A SELECTION is displayed in [Figure 7 on page 6](#).

|                                  |  |              |  |           |  |            |
|----------------------------------|--|--------------|--|-----------|--|------------|
| <b>EZL910I ENTER A SELECTION</b> |  |              |  |           |  |            |
| Command ==>                      |  |              |  |           |  |            |
| F1=Help                          |  | F2=Main Menu |  | F3=Return |  | F5=Refresh |
| F7=Backward                      |  | F8=Forward   |  | F6=Roll   |  | F12=Cancel |

Figure 7. Message Displayed in Message Area

These messages can be of various types. Some tell you whether a function completed successfully, and others tell you what further actions you must take, why a function failed, and other useful information.

**Note:** Some functions use the NetView command facility to display messages. For example, if you are defined as a notification operator for a resource, AON optionally sends messages to you when there are problems with that resource. Those messages are displayed on your NetView command facility and stay there until you clear them. You can use the DM command to clear them.

Online help for AON messages is available. To view message help:

1. Type **HELP** *messagenumber*.
2. Press Enter.

For more information about AON messages, see *IBM Z NetView Messages and Codes Volume 2 (DUI-IHS)*.

## Function keys

The function keys are displayed at the bottom of each panel (except tutorials), as shown in [Figure 8](#) on [page 7](#). Use the function keys to perform functions on the interface.

|            |              |           |             |            |            |
|------------|--------------|-----------|-------------|------------|------------|
| F1=Help    | F2=Main Menu | F3=Return | F4=Commands | F5=Refresh | F6=Roll    |
| F7=Forward | F8=Backward  |           | F10=Left    | F11=Right  | F12=Cancel |

*Figure 8. Function Keys*

The keys used on each panel vary according to the functions required. The following keys usually have these functions:

### **F1**

Displays contextual help. Detailed help for the entire AON product, the AON base, or the specialized automation components is included in the tutorials. The Help key shows you only the instructions you need for using the panel currently displayed.

### **F2**

Displays the AON: Operator Commands Main Menu panel.

### **F3**

Displays the previous panel.

### **F4**

Displays a pop-up command window.

### **F5**

Updates the information on the panel.

### **F6**

Rolls the display among the various active functions.

### **F7**

Moves the interface backward one panel if MORE is indicated.

### **F8**

Moves the interface forward one panel if MORE is indicated.

### **F10**

Scrolls the panel to the left.

### **F11**

Scrolls the panel to the right.

### **F12**

Cancels the current function.

## Selection lists

Some of the panels of the AON interface display the following prompt next to an entry field:

```
(Type ? for a selection list)
```

Use selection lists to see a complete list of the available responses.

## Using the wildcard function

Some of the fields of the AON interface require that you fill in information and specify search parameters. When you want to locate all instances of a certain resource you can define your search parameters with a *wildcard* character. AON enables you to use two different wildcard characters:

**\***

Multiple character wildcard

**%**

Single character wildcard

For example, both PU0\* and PU%% are matches for PU01. ENTRY=ENVI\* searches for all entries starting with ENVI.

## Navigating through AON panels

---

You can navigate through the panels of the AON operator interface by using several methods:

- Selecting menu choices
- Using fast path commands
- Using AON command synonyms

Selecting menu choices enables new AON operator interface users to navigate through the panels. Users experienced with the arrangement of the panels throughout the interface can use the fast path method or command synonyms to navigate through the panels more quickly. Each of these methods of navigation is described in the following subsections.

### Selecting menu options

One way to navigate the AON operator interface is by selecting menu choices. The following example shows how you use menu choices to display the comprehensive AON tutorial:

1. Display the AON: Operator Commands Main Menu panel shown in [Figure 1 on page 3](#).

**Note:** To display the AON: Operator Commands Main Menu panel, see [“Displaying the AON: Operator Commands Main Menu” on page 3](#).

2. Type 0 for Tutorial in the Select an option entry field.
3. Press Enter.

The AON tutorial shown in [Figure 9 on page 9](#) is displayed:

The following tutorial defines automated operations and how they are implemented on a Tivoli system using the NetView product.

Before discussing automated operations, it is helpful to know what is meant by :

- o Availability
- o System operations
- o Network operations
- o The operator

Command ==>

F2=End  
F8=Forward

F3=Return

F4=Top

F5=Bottom  
F11=Entry Point

F6=Roll

*Figure 9. AON Tutorial*

## Using fast path commands

Fast path commands enable you to display a panel directly from a panel that is not adjacent without displaying intervening panels. You can use the fast path from the NetView command facility or from any command line in the AON operator interface.

For example, to use the fast path method to display the tutorial shown in [Figure 9 on page 9](#):

1. Type **AON 0** on any command line.
2. Press Enter.

You can use the fast path method to reach a panel that is up to three levels away from the AON: Operator Commands Main Menu.

For example, the Reinitialize Automation panel is three levels away from the AON: Operator Commands Main Menu panel. To display that panel by using the fast path method:

1. Type **AON 1.8.2** on any command line.
2. Press Enter.

The AON: Reinitialize Automation panel is displayed:

```
Select Confirmation Option      . . _ 1. Confirm
                                2. Cancel
```

```
Reload with Automation Table    . . DSITBL01
Generate Listing File           . . LISTNAME
```

```
Reload with Control File       . . NVPOLICY
```

```
Trace Setting                   . . OFF
```

Command ==>

F1=Help F2=Main Menu F3=Return

F6=Roll  
F12=Cancel

Figure 10. AON: Reinitialize Automation Panel

## Using AON command synonyms

You can also display specific panels by using AON command synonyms. AON command synonyms are commands you type on the command line to display another panel.

**Note:** If you issue a command from a panel and you select an option or fill in an input field at the same time, the command takes precedence over the panel entries. AON does not process any of the panel entries.

As an example, use the command synonym AONINFO to display the AON tutorial in [Figure 9 on page 9](#).

1. Type **AONINFO** on any NetView or AON command line.
2. Press Enter.

The AON tutorial shown in [Figure 9 on page 9](#) is displayed.

You can use command synonyms to reach a panel that is up to three levels away from the AON: Operator Commands Main Menu.

For example, the Reinitialize Automation panel is three levels away from the AON: Operator Commands Main Menu panel. To display that panel by using the command synonym:

1. Type **AONINIT** on any command line.
2. Press Enter.

The AON: Reinitialize Automation panel shown in [Figure 10 on page 10](#) is displayed.

## Using AON commands

You can issue AON commands from any NetView command line.

For most AON functions, you can bypass the operator interface completely by using these commands. This enables you to use the automation functions from within your own user-written programs. To issue commands, type the command and its parameters. If you issue the command with all of the required parameters entered correctly, AON processes the command without going through the operator interface. If you enter the command name without any parameters or with the parameters entered incorrectly, AON displays the appropriate operator interface panel.

**Note:** If you issue a command from a panel and you select an option or fill in an input field at the same time, the command takes precedence over the panel entries. AON does not process any of the panel entries.

## Getting help

To access help for AON, use the extensive online help facilities that come with the program. AON offers online tutorials, which answer basic questions about how AON works, and contextual help, which answers questions about the fields of a particular panel.

### Displaying the AON tutorial

The AON comprehensive tutorial is one of several help facilities that come with the AON program. This tutorial explains how the AON program works by defining automated operations and how they are implemented on a Tivoli system using AON.

To display the main AON tutorial:

1. Display the AON: Operator Commands Main Menu panel.

**Note:** To display the AON: Operator Commands Main Menu panel, see [“Displaying the AON: Operator Commands Main Menu”](#) on page 3.

2. On the AON: Operator Commands Main Menu panel, type **O** in the entry field.
3. Press Enter. The following panel is displayed:

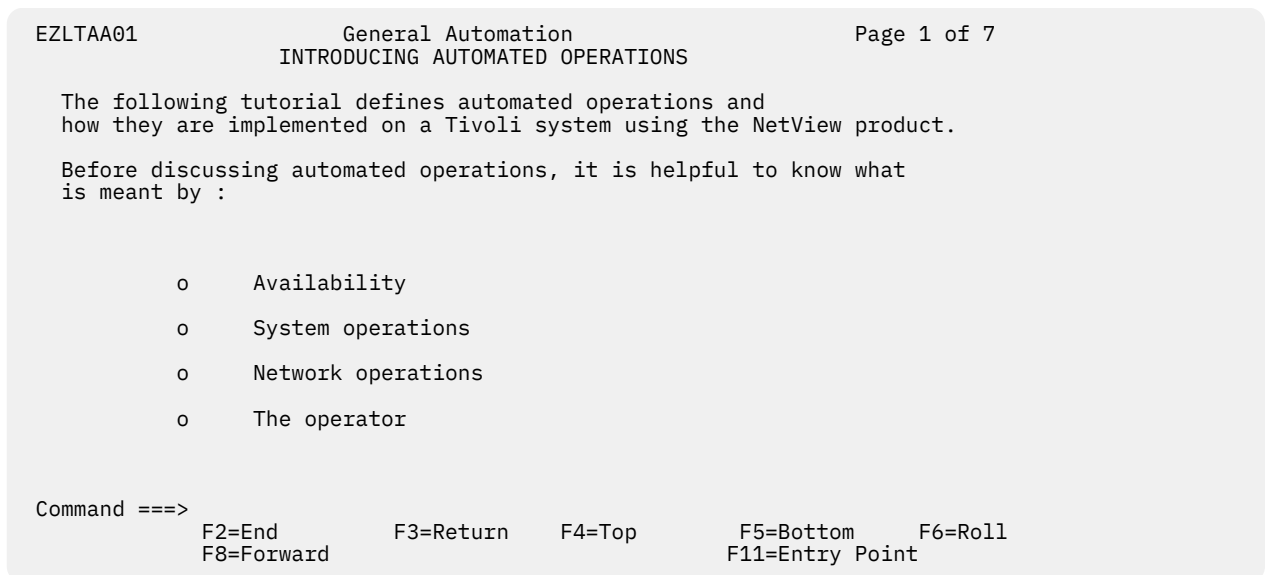


Figure 11. AON Tutorial

As you can see in the top right corner of the panel, the first AON tutorial consists of seven panels. Use the function keys to scroll through the panels.

### Using context-sensitive help

In addition to extensive online tutorials, AON provides context-sensitive help for each of the panels of the operator interface. Press **F1** to display help for a panel and the entry fields on it.

### Using message help

You can get online help for any of the messages you receive. To see the message help:

1. Type **HELP** message number.
2. Press Enter.

A help panel is displayed that explains the message.

For more information about AON messages, see *IBM Z NetView Messages and Codes Volume 2 (DUI-IHS)*.

## Using command help

You can get online help for most of the AON commands. If the command takes you to a panel, you can display online help for that panel. If the command performs other functions, you can display a help panel that explains what you can do with the command.

To see the online help for a command:

1. Type **HELP** *command name* on the command line of any panel.
2. Press Enter.

A help panel is displayed for the command.



---

## Chapter 2. Solving network problems with Help Desks

You can solve problems on the network by using automated help desks. You use the AON: Help Desk panel to display all of the help desks that are available to you. The help desks for all of the components function similarly, but the displays and options vary according to the needs of the type of network.

Any automation component that has a defined help desk is displayed as an option on the AON: Help Desk panel. You can use help desks to display color-coded pictures of how resources are connected to networks, stop and start resources, perform problem determination, and issue commands to solve network problems.

---

### Displaying the AON Help Desk

To display the AON Help Desk panel:

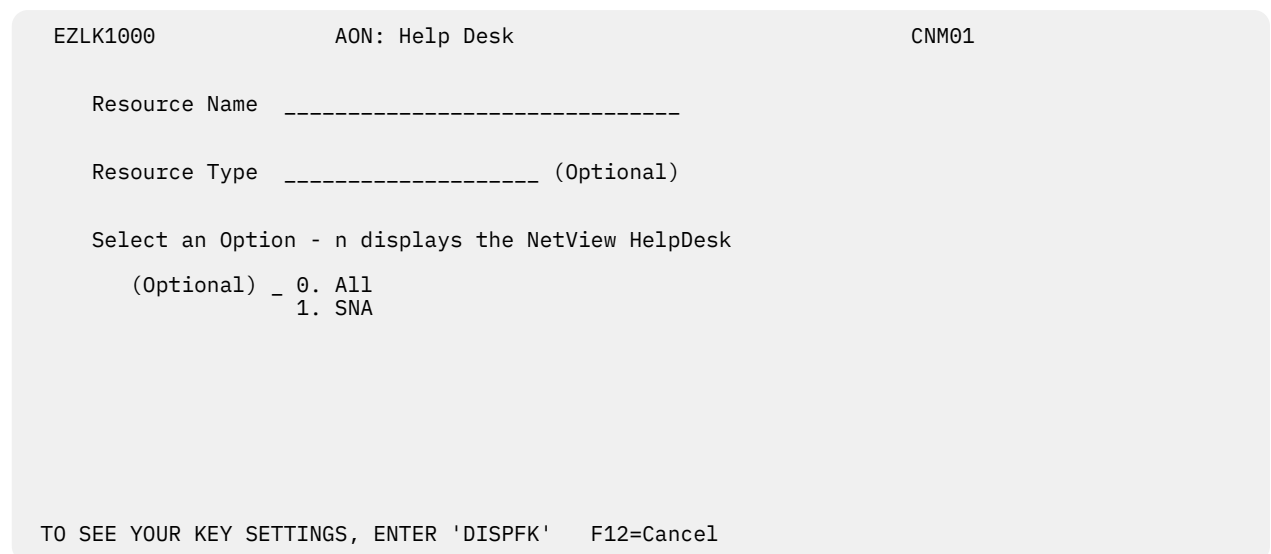
1. Display the Base Functions panel.

**Note:** To display the Base Functions panel, see [“Displaying the AON Base Functions Panel”](#) on page 4.

2. Type **1** in the entry field.
3. Press **Enter**.

The AON: Help Desk panel shown in [Figure 12](#) on page 13 is displayed.

**Note:** You can also display the AON: Help Desk panel by entering **AON 1.1** or **AONHD** on any command line.



```
EZLK1000          AON: Help Desk          CNM01

Resource Name  _____

Resource Type  _____ (Optional)

Select an Option - n displays the NetView HelpDesk
(Optional) _ 0. All
              1. SNA

TO SEE YOUR KEY SETTINGS, ENTER 'DISPFK'   F12=Cancel
```

*Figure 12. AON: Help Desk Panel*

The AON: Help Desk panel displays the following entry fields and options:

**Resource Name**

Enables you to specify the resource that is having a problem. The resource name is required.

### Resource Type

Enables you to specify the type of resource. For example, resource types in the SNA environment include physical unit (PU), logical unit (LU), Network Control Program (NCP), and application (APPL). The resource type is optional.

### Select an Option

Enables you to specify the help desk for the kind of network on which the specified resource is located. You can select one of the following options:

#### All

Searches all of the automation components listed on the panel for the resource and displays the resource on its help desk.

#### SNA

Searches the SNA automation component for the resource.

If you leave this field blank, AON searches all of the automation components for the resource.

The following sections explain how to use each option.

## Searching AON components for a resource

To search all AON components listed on the AON: Help Desk panel for a resource and display the resource in each of the help desks:

1. Display the AON: Help Desk panel.

**Note:** To display the AON: Help Desk panel, see [“Displaying the AON Help Desk” on page 13](#).

2. Type the name of the resource in the Resource Name field on the AON Help Desk panel.
3. Optionally, type the resource type in the Resource Type field.
4. Leave the Select an Option entry field blank.
5. Press **Enter**.

If the resource is defined to more than one automation component, the AON: Component Selection panel shown in [Figure 13 on page 14](#) lists each occurrence of the resource.

```
EZLKIDNT          AON: Component Selection          CNM01

The resource chosen has valid definitions in the following installed
options.  Select the applicable option.

Resource Chosen    test

  Select an Option
                - 0. All
                  1. SNA
                  2. SNBU
                  3. TCPIP

Command ===>
F1=Help      F2=Main Menu  F3=Return
F6=Roll
F12=Cancel
```

*Figure 13. AON: Component Selection Panel*

6. Type the number of the option you want in the Select an Option entry field. You can select any of the options displayed on the panel. In the following example, AON found the resource in AON/SNA.
7. Press **Enter**.

AON displays the Help Desk panel that you selected. See the following sections for more information about the AON/SNA Help Desk.

## Using the SNA Help Desk

You can use the SNA Help Desk when a user terminal is experiencing a problem. If the problem is with a SNA resource on a remote NetView domain, you must have a NetView-NetView Task (NNT) or a remote commands session to that domain to use the SNA Help Desk.

To use the SNA Help Desk:

1. Display the AON: Help Desk panel.

**Note:** To display the AON: Help Desk panel, see “Displaying the AON Help Desk” on page 13.

2. Type the name of the resource in the Resource Name field on the AON: Help Desk panel. This example uses resource TA1PT209.
3. Type the resource type in the Resource Type field. AON/SNA resource types are generic resource groups such as PU, LU, NCP, and APPL.
4. Type **2** in the Select an Option entry field.
5. Press **Enter**.

The SNA Automation: Help Desk panel shown in [Figure 14](#) on page 15 is displayed.

```
FKVK1000                      SNA Automation: Help Desk

**** *  **  **          ** **          ***          *****          **
**  * **  **  ****          ** **          **          ** **          **
**    *** ** ** **          ** **  ****          **  *****          ***** ** **
**  ***** ****          ***** ** **          ** ** ** ** ** ** ** ** ** ** **
** ** ** ** ** ** **          ** ** *****          ** ** ** ** ** ** ** ** ** **
*  ** ** ** ** ** **          ** ** **          ** ** ** ** ** ** ** ** ** ** **
**** **  * ** **          ** **  ****          ***** *****          ***** ** *
```

Enter name...: TA1PT209

Select option: - 1. Recycle resource  
2. Problem Determination  
3. NetView Access Services User ID  
n. NetView Help Desk

EZL910I ENTER A SELECTION  
Command ==> F1=Help F2=Main Menu F3=Return F12=Cancel F6=Roll

Figure 14. SNA Automation: Help Desk Panel

6. To search for a different resource than is displayed in the Enter Name field, type over the existing resource name.
7. Type the option you want in the Select Option entry field. You can select one of the following options:

### Recycle resource

Forces the resource inactive then activates it again. If AON/SNA cannot not recycle the resource, it displays additional panels with further options.

### Problem Determination

Displays the current status of the resource. You can perform additional queries and tests on the resource.

### NetView Access Services User ID

Works with user IDs that are logged on NetView Access Services and their applications on the same VTAM where AON/SNA resides.

**NetView Help Desk**

This option takes you to the main NetView Help Desk facility.

8. Press **Enter**.

# Chapter 3. Displaying resource information with AutoView

The AutoView function works with more than one setting for a single resource. The AutoView function displays all known data for the resource and tries to determine which automation components are interested in the resource.

## Displaying the AON: AutoView panel

To display the AON: AutoView panel:

- 1. Display the Base Functions panel.
- 2. Type **2** in the entry field.
- 3. Press **Enter**.

The AON: AutoView panel shown in [Figure 15 on page 17](#) is displayed.

**Note:** You can also display the AON: AutoView panel by entering **AON 1.2** or **AUTOVIEW** on any command line.

EZLK2000

AON: AutoView

NTVE1

Resource Name

\_\_\_\_\_

Resource Type

\_\_\_\_\_ (Optional)

Select an Option

(Optional) \_

1. TCPIP

2. IP390

3. SNA

4. APPN

5. SNBU

Command ===>

F1=Help

F2=Main Menu

F3=Return

F6=Roll

F12=Cancel

Figure 15. AON: AutoView Panel

The AON: AutoView panel displays the following entry fields and options:

**Resource Name**

Enables you to specify a resource for which you want to display current information or change settings.

**Resource Type**

Enables you to specify the resource type.

**Select an Option**

Specifies the component to which the resource is defined. Each component has predefined information that it displays about the resource. Each component has commands that you can use to get further information about the resource or set and change automation settings. The list of options varies, depending on which components have the resource defined. The following components are provided:

**APPN**

Looks for the resource as an Advanced Peer-to-Peer Networking (APPN) network resource.

**IP390**

Looks for the resource as an IP resource.

**SNA**

Looks for the resource as an SNA network resource.

**SNBU**

Looks for the resource as a switched network backup (SNBU) network resource.

**TCPIP**

Looks for the resource as a z/OS Communication Server IP resource.

**Note:** These options might be displayed in a different order on your panels, because the panel changes depend on the configuration of your site.

The following sections provide an example of how AutoView works for an SNA resource.

**USAGE :**

- The AutoView function utilizes UNIX System Services to monitor and display IP resource status through TCP autotasks. These autotasks are defined in the control file and must be authorized to access UNIX System Services.
- The AutoView function queries SNMP MIB variables to determine the status of IP resources. The SNMP Community name defined in CNMPOLCY for each TCP390 stack must match the one configured by the z/OS communication server for each stack. The Community name can be dynamically updated by using the IP Resource Manager function.

For more information about UNIX System Services authorization and the SNMP Community name, refer to the *IBM Z NetView Security Reference*. The Community name definition is described in the TCP390 definition in the *IBM Z NetView Administration Reference*.

## Displaying resource information (an SNA example)

---

To display information about an SNA resource:

1. Display the AON: AutoView panel.

**Note:** To display the AON: AutoView panel, see [“Displaying the AON: AutoView panel” on page 17](#).

2. Type the name of an SNA resource in the **Resource Name** field. This example uses resource TA1P523A.
3. Optionally, type the resource type in the **Resource Type** field. Resource types are LU, PU, NCP, or APPL.
4. Optionally, to select SNA, type the number for SNA in the **Select an Option** entry field.
5. Press **Enter**.

If you selected SNA, or left the option blank, and the resource is defined to an SNA network, the panel in [Figure 17 on page 19](#) is displayed. Proceed to Step [“6” on page 20](#).

If you left the option blank and the resource is defined to more than one type of network, the AON: Component Selection panel shown in [Figure 16 on page 19](#) is displayed. Proceed to Step [“5.a” on page 19](#).

|                                |                           |                       |
|--------------------------------|---------------------------|-----------------------|
| EZLK2000                       | AON: AutoView             | CNM01                 |
| Resource Name _____            |                           |                       |
| Resource Type _____ (Optional) |                           |                       |
| Select an Option               |                           |                       |
| (Optional) _ 1. TCPIP          |                           |                       |
| 2. IP390                       |                           |                       |
| 3. SNA                         |                           |                       |
| 4. APPN                        |                           |                       |
| 5. SNBU                        |                           |                       |
| Command ==>                    |                           |                       |
| F1=Help                        | F2=Main Menu    F3=Return | F6=Roll<br>F12=Cancel |

*Figure 16. AON: Component Selection Panel*

On the AON: Component Selection panel, follow these steps:

- a. Type **1** in the select an option field.
- b. Press **Enter**.

The following panel is displayed:

|                                                                                                                                                                                  |                           |                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----------------------|
| EZLKVIEW                                                                                                                                                                         | AON: AutoView             | CNM01                 |
| Resname . . . . . TA1P523A                                                                                                                                                       |                           |                       |
| Restype . . . . . PU                                                                                                                                                             |                           |                       |
| Option . . . . . SNA                                                                                                                                                             |                           |                       |
| Status . . . . . PCTD2 <span style="background-color: black; color: white; padding: 0 5px;">1</span>                                                                             |                           |                       |
| Automation Status . . . TREAT                                                                                                                                                    |                           |                       |
| Automation Flag . . . . ON                                                                                                                                                       |                           |                       |
| DDF message . . . . . NONE                                                                                                                                                       |                           |                       |
|                                                                                                                                                                                  |                           |                       |
| Resource Definitions <span style="background-color: black; color: white; padding: 0 5px;">2</span> <span style="background-color: black; color: white; padding: 0 5px;">3</span> |                           |                       |
| _ 1. Automation (RECOVERY DEFAULTS)                                                                                                                                              |                           |                       |
| 2. Thresholds (THRESHOLDS DEFAULTS)                                                                                                                                              |                           |                       |
| 3. Active Monitoring (ACTMON DEFAULTS)                                                                                                                                           |                           |                       |
| 4. Monitor Intervals (MONIT DEFAULTS)                                                                                                                                            |                           |                       |
| 5. Timer (NONE)                                                                                                                                                                  |                           |                       |
| 6. Display Network LOG Information for TA1P523A                                                                                                                                  |                           |                       |
| Command ==>                                                                                                                                                                      |                           |                       |
| F1=Help                                                                                                                                                                          | F2=Main Menu    F3=Return | F6=Roll<br>F12=Cancel |

*Figure 17. AON: AutoView Panel*

Although the appearance of the AON: AutoView panel differs among automation components, the panel in [Figure 17 on page 19](#) displays the following information and options:

- 1 Summarizes information about the resource including the resource name, resource type, status, automation settings for the resource, DDF message for the resource (if any) and other information, depending on the automation component.
- 2 Displays a list you can use for the resource. The following actions are supported:

### Automation

Sets the RECOVERY flags for the resource. If none are found, DEFAULTS is displayed. For detailed information about Automation settings, see [“Displaying the recovery settings panel” on page 24.](#)

### Thresholds

Sets the THRESHOLDS processing for errors received on this resource. The default is DEFAULTS if none is defined. For detailed information about setting thresholds, see [“Displaying thresholds” on page 30.](#)

### Active Monitoring

Sets the ACTMON entry for this resource. If none are found, the DEFAULTS value is used. For detailed information about setting active monitoring, see [“Displaying the Active Monitor settings panel” on page 34.](#)

### Monitor Intervals

Sets the MONIT intervals for the resource, defaulting to the type of resource if intervals are defined for that type. Otherwise, DEFAULTS are used if no intervals are found. For detailed information about setting monitor intervals, see [“Displaying the Monitor Intervals panel” on page 32.](#)

### Timer

Sets a timer command for the resource. For detailed information about setting timers, refer to the information about timer commands in the *IBM Z NetView User's Guide: NetView*.

### Display Network LOG Information

Displays only the Netlog information for the resource using the NetView BLOG command.

**3**

Displays which control file entry is currently defining the settings for the commands listed in the second group. For example, in Figure 17 on page 19, the settings for option 1 (Automation) are defined in the RECOVERY DEFAULTS control file entry.

For commands that do not have a corresponding control file entry, such as option 5 (Timer), the information in parentheses tells you if any settings exist. In [Figure 17 on page 19](#), no timers are set for the resource TA1P523A.

6. To continue with the example, type **6** for Display Network LOG Information in the **Resource Definitions** field.

7. Press **Enter**.

The NetView Log shown in [Figure 18 on page 20](#) is displayed with the resource you specified.

```
STATMON.BROWSE      ACTP  NETWORK LOG FOR 04/29/19 (11119) COLS 017 096 15:20
HOST: HOST1          *1*   *2*   *3*   *4*                               SCROLL: CSR
---2-----3-----+---4-----5-----+---6-----7-----+---8-----9-----
CNM01 P% 11:49:06    IST259I  INOP RECEIVED FOR TA1P523A CODE = 01
CNM01 P% 11:49:06    IST619I  ID = TA1P523A FAILED - RECOVERY IN PROGRESS
CNM01 % 11:49:08     CNM094I  STATUS UPDATE FOR RESOURCE = TA1P523A IN NETWORK =
CNM01 % 11:49:20     CNM094I  STATUS UPDATE FOR RESOURCE = TA1P523A IN NETWORK =
CNM01 P% 11:49:20    IST621I  RECOVERY SUCCESSFUL      FOR NETWORK NODE TA1P523A
CNM01 % 11:49:20     CNM094I  STATUS UPDATE FOR RESOURCE = TA1P523A IN NETWORK =
CNM01      14:42:25   EZL112I  ACTIVE TYPE= TA1P523A
```

```
CMD==>
1=HLP 2=END 3=RET 4=TOP 5=BOT 6=ROL 7=BCK 8=FWD 9=RPF 10=LFT 11=RG 12=ALL
```

Figure 18. NetView Log Panel Displaying One Resource



8. Browse the log as needed.

**Note:** This option uses the NetView BLOG sample. If there are no entries in the log for the resource you are using, no action is taken and no message is displayed.



## Chapter 4. Specifying automation policy settings

Specify automation policy settings using the AON: Automation Settings panel and the subordinate panels. Use these panels to add, display, change, and delete automation policy settings that control various parts of the network.

When NetView initializes, it loads a copy of the automation policy into storage. When you use the AON: Automation Settings panels to change automation settings, you change the in-storage copy of the policy without altering the original. When the policy is reloaded into storage, the changes you made to the automation settings are reset.

If you need to make permanent changes to the policy, your system programmer can edit and reload it so that the changes are available the next time AON initializes.

**Note:** The AON: Automation Settings panel lists settings for all network resources. However, to view and change numerous settings for a single resource, use the AutoView panel. The AutoView panel consolidates commands for automation settings onto a single menu. The AutoView function displays settings panels that apply only to a single resource.

### Displaying the AON: Automation Settings Panel

To display the AON: Automation Settings panel:

1. Display the Base Functions panel.

**Note:** To display the Base Functions panel, see [“Displaying the AON Base Functions Panel”](#) on page 4.

2. On the Base Functions panel, type **4** in the entry field.
3. Press **Enter**.

The AON: Automation Settings panel shown in [Figure 19 on page 23](#) is displayed.

**Note:** You can also display the AON: Automation Settings panel by entering **AON 1.4** on any command line.

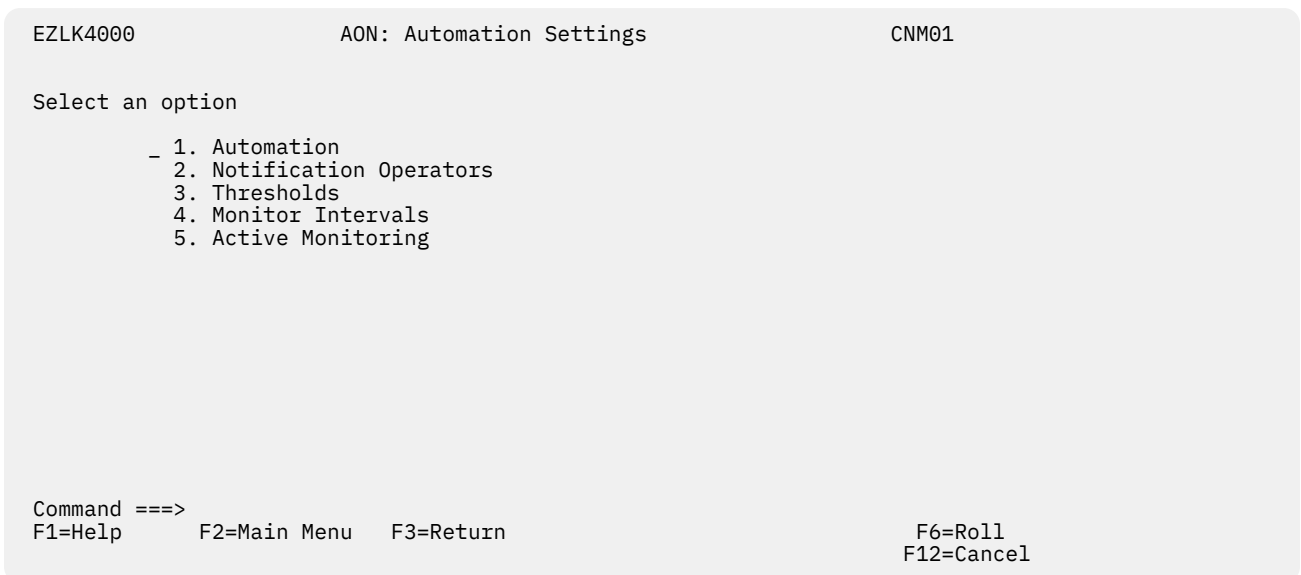


Figure 19. Automation Settings Panel

The AON: Automation Settings panel displays the following options:

**Automation**

Defines the times when AON automation recovery actions occur on resources. These values are retrieved from the RECOVERY control file entry.

**Notification Operators**

Specifies which operators receive AON messages when automation events occur. These values are retrieved from the NTFYOP control file entry.

**Thresholds**

Specifies the number of times an event must occur within a given time span before AON sends notifications to the selected operators. These values are retrieved from the THRESHOLDS control file entry.

**Monitor Intervals**

Controls reactivation intervals for failing resources. These intervals determine how often AON attempts to recover the failed resources and send messages to the notification operators. These values are retrieved from the MONIT control file entry.

**Active Monitoring**

Controls active monitoring intervals for resources. AON regularly checks the status of important resources that must be available to the network. The active monitoring intervals determine how often AON monitors those resources. These values are retrieved from the ACTMON control file entry.

The following sections describe how to use these options.

## Specifying recovery settings

---

Use the Recovery Settings panel to add, display or change, and delete automation settings. Recovery settings set automation on or off for one resource or group of resources. You can also specify times when automation does not operate.

**Note:** A RECOVERY DEFAULTS policy definition is required. The DEFAULTS entry is used during recovery processing and a specific resource name or type is not coded. The DEFAULTS entry cannot be deleted.

### Displaying the recovery settings panel

To display the Recovery Settings panel:

1. Display the AON: Automation Settings panel.

**Note:** To display the AON: Automation Settings panel, see [“Displaying the AON: Automation Settings Panel”](#) on page 23.

2. Type **1** in the entry field on the AON: Automation Settings panel.
3. Press **Enter**.

The Recovery Settings panel shown in [Figure 20 on page 25](#) is displayed.

**Note:** You can also display the Recovery Settings panel by entering **AON 1.4.1, SETAUTO, DISAUTO,** or **DELAUTO** on any command line.

| EZLK4100                                                                          | Recovery Settings | CNM01                  |
|-----------------------------------------------------------------------------------|-------------------|------------------------|
| Select one of the following. Then press enter.<br>1=Add 2=Display/Change 3=Delete |                   |                        |
| Type                                                                              | Automation        | Scheduled Recovery     |
| - AOF01*                                                                          | OFF               |                        |
| - APPL                                                                            | OFF               |                        |
| - CNM01*                                                                          | OFF               |                        |
| - DEFAULTS                                                                        | ON                |                        |
| - LINKSTA                                                                         | ON                |                        |
| - TAF*                                                                            | OFF               |                        |
| - TS0*                                                                            | OFF               |                        |
| Command ==>                                                                       |                   |                        |
| F1=Help                                                                           | F2=Main Menu      | F3=Return              |
| F7=Backward                                                                       | F8=Forward        | F5=Refresh             |
|                                                                                   |                   | F6==Roll<br>F12==Cacel |

Figure 20. Recovery Settings Panel

The Recovery Settings panel displays three columns of data:

#### Type

Lists all the resources that have recovery settings defined.

#### Automation

Indicates whether automation is set on or off for the resources by looking at the AUTO= parameter in the control file setting.

#### Scheduled Recovery

Looks at the NOAUTO= parameter. For resources that have intervals scheduled when automation does *not* occur, this field has one of the following values:

##### ON

When the current time is outside the times defined with the NOAUTO= parameter.

##### OFF

When the current time is within one of the times specified with the NOAUTO= parameter.

If you leave the field blank, scheduled recovery is not in place for the resource.

Scheduled recovery intervals are optional. If no such intervals are coded, automation is either on all the time or off all the time depending on the setting in the Automation column. The Scheduled Recovery column is blank if no scheduled intervals are specified for the resource.

**Note:** Scheduled Recovery settings, if any, and automation must be set to ON for automation to occur on the resource.

Use the Recovery Settings panel to add, display or change, and delete recovery settings. The following sections explain how to perform these actions.

## Displaying or changing recovery settings

To display or change an existing recovery setting:

1. Display the Recovery Settings panel.

**Note:** To display the Recovery Settings panel, see [“Displaying the recovery settings panel”](#) on page 24.

2. Type **2** in the entry field next to the resource, the settings for which you want to change, on the Recovery Settings panel.
3. Press **Enter**.

The Display/Change Recovery Setting panel shown in [Figure 21 on page 26](#) is displayed.



- 6 SATurday
- 7 SUNday
- 8 WEEKDAY
- 9 WEEKEND
- \* everyday
- delete a NOAUTO window

**Note:** The example in [Figure 21 on page 26](#) shows an asterisk (\*) to represent that everyday is selected. You can also specify the valid days by typing the capital letters shown in the Valid Days pop-up window, in the second column text field under Day.

#### Start Time

Sets automation off starting at this time. Specify the time in the *hh:mm* format, where *hh* is a number in the range of 00–23 and *mm* is a number in the 00–59 range.

#### End Time

Determines the end of the interval when automation is not active for the resource. Specify the time in the *hh:mm* format, where *hh* is a number in the range of 00–23 and *mm* is a number in the 00–59 range.

**Note:** The ending time must be later in the day than the starting time *except* when you specify \* (every day).

## Specifying notification operators

---

Use the Notification Operators panel to:

- add new notification operators
- display or change the settings for existing operators
- delete settings for existing operators

Notification operators are operators who receive messages from AON and its components when automation events occur. These values are retrieved from the NTFYOP control file entry.

### Displaying the Notification Operators panel

To display the Notification Operators panel:

1. Display the AON: Automation Settings panel.

**Note:** To display the AON: Automation Settings panel, see [“Displaying the AON: Automation Settings Panel” on page 23](#).

2. Type **2** in the entry field.
3. Press **Enter**.

The Notification Operators panel is displayed in [Figure 22 on page 28](#).

**Note:** You can also display the Notification Operators panel by entering **AON 1.4.2, SETNTFY, DISNTFY, or DELNTFY** on any command line.

|                                                |                        |             |            |                       |
|------------------------------------------------|------------------------|-------------|------------|-----------------------|
| EZLK4200                                       | Notification Operators |             |            | CNM01                 |
| Select one of the following. Then press Enter. |                        |             |            | More: +               |
| 1=Add 2=Display/Change 3=Delete                |                        |             |            |                       |
|                                                | Operator               | Notify Flag | Logged On  | Description           |
| -                                              | OPER1                  | Y           | N          | 'Operator One'        |
| -                                              | OPER2                  | Y           | N          | 'Operator Two'        |
| -                                              | OPER3                  | Y           | Y          | 'Operator Three'      |
| -                                              | OPER4                  | Y           | Y          | 'Operator Four'       |
| -                                              | OPER5                  | Y           | N          | 'Operator Five'       |
| -                                              | OPER6                  | Y           | N          | 'Operator Six'        |
| -                                              | OPER7                  | Y           | N          | 'Operator Seven'      |
| -                                              | NISTMP1                | Y           | N          | 'Test Operator One'   |
| -                                              | NISTMP2                | Y           | N          | 'Test Operator Two'   |
| -                                              | NISTMP3                | Y           | N          | 'Test Operator Three' |
| -                                              | NISTMP4                | Y           | Y          | 'Test Operator Four'  |
| -                                              | NISTMP5                | Y           | N          | 'Test Operator Five'  |
| -                                              | NISTMP6                | Y           | N          | 'Test Operator Six'   |
| -                                              | NISTMP7                | Y           | N          | 'Test Operator Seven' |
| Command ==>                                    |                        |             |            |                       |
| F1=Help                                        | F2=Main Menu           | F3=Return   | F5=Refresh | F6=Roll               |
| F7=Backward                                    | F8=Forward             |             |            | F12=Cancel            |

Figure 22. Notification Operators Panel

The Notification Operators panel lists the following columns of data:

#### Operator

The operator ID of the notification operator.

#### Notify Flag

Indicates if messages are sent to the notification operator. Use this flag to temporarily stop the notification messages without removing the operator from the list of valid notification operators. The data is taken from the NOTIFY= parameter in the NTFYOP control file entry.

#### Logged On

Indicates if this notification operator is currently logged on to the system.

If the operator is logged on and the Notify Flag is set to Yes (Y), then the operator is receiving messages as defined by the notification settings.

#### Description

Describes the notification operator, usually, the name of the operator (optional).

## Adding a Notification Operator

To add an operator ID to the list of valid notification operators:

1. Display the Notification Operators panel.

**Note:** To display the Notification Operators panel, see [“Displaying the Notification Operators panel” on page 27](#).

2. Type **1** in the field next to an existing notification operator.
3. Press **Enter**. A pop-up window shown in [Figure 23 on page 29](#) is displayed.



```

EZLK4210          Notification Operators          CNM01

Select one of the following. Then press Enter.          More:  +
1=Add 2=Display/Change 3=Delete

  Operator      Notify      Logged      Description
  1 OPER1       Y          N          'Operator One'
- OPER2
- OPER3       : Operator      -----
- OPER4       : Description   -----
- OPER5       : Classes      -----
- OPER6       : Messages     - Information
- OPER7       :              - Warning
- NISTMP1     :              - Error
- NISTMP2     :              - Action
- NISTMP3     : Notify       -          (1=Yes 2=No)
- NISTMP4     :
- NISTMP5     : F1=Help      F12=Cancel
- NISTMP6     : -----
- NISTMP7     :              Y          N          'Test Operator Seven'

Command ==>
F1=Help      F2=Main Menu  F3=Return      F5=Refresh  F6=Roll
F7=Backward  F8=Forward

```

Figure 23. Notification Operators Panel

4. Type the operator ID in the **Operator** field. This is the only required field.
5. Type a brief description for the operator in the **Description** field.
6. Type the numbers for the message classes in the **Classes** field. The default message class is 10. The data is taken from the CLASS= parameter in the NTFYOP control file entry.

**Note:** See *IBM Z NetView Messages and Codes Volume 2 (DUI-IHS)* for more information about message classes.

7. In the **Messages** fields, type any character in the field for each message type that is held on the command facility until the notification operator clears it. The data is taken from the HELDMSG= parameter in the NTFYOP control file entry. The message types are:

**I or INFO**

Informational messages

**W or WARN**

Warning messages

**E or ERROR**

Error messages

**A or ACTION**

Action messages

8. Type **1** to set notifications on for the operator or **2** to suppress the notification messages in the **Notify** field. You can also leave this field blank. By default, the notification operator receives the messages in the message classes assigned.
9. Press **Enter**.

If the information is entered correctly, AON displays this confirmation message on the panel:

```
EZL919I ALL ACTIONS SUCCESSFULLY COMPLETED
```

10. Press F3 to return to the Notification Operators panel.

The new notification operator is displayed in the alphabetized list on the updated Notification Operators panel.

## Setting automation thresholds

To add, change, or delete threshold settings use the Thresholds panel. Threshold settings trigger notifications to operators when the thresholds are reached. An event is defined as an alert generated by

Define the threshold settings so that each threshold type is unique for the resource being monitored. Define critical thresholds to have the smallest interval or highest occurrence frequency, followed by frequent, then infrequent.

**Note:** You can also display the Thresholds panel by entering **AON 1.4.3, SETTHRES, DISTHRES, or DELTHRES** on any command line.

**Note:** A THRESHOLD DEFAULTS policy definition is required. The DEFAULTS entry is used when there is threshold processing and a specific resource name or type is not specified. You cannot delete the DEFAULTS entry.

Each threshold setting follows the same syntax. Two settings are entered to set the threshold:

**number**

The number of events that cause the threshold to be exceeded. The valid range is 0–10.

**interval**

The time span, in hours and minutes, in which the number of events must occur for the threshold to be exceeded. The valid range is 00:00–99:59.

You can use the Thresholds panel to add, change, and delete threshold settings. The following sections explain how to perform these actions in detail.

## Adding a threshold

To add a threshold:

1. Display the **Thresholds** panel.

**Note:** To display the Thresholds panel, see [“Displaying thresholds” on page 30](#).

2. Type **1** in entry field on the Thresholds panel.
3. Press **Enter**.

A pop-up window shown in [Figure 25 on page 31](#) is displayed.

EZLK4310
Thresholds
CNM01

Select one of the following. Then press Enter.

1=Add 2=Change 3=Delete

Type

- DEFAULTS

|               |           |    |           |    |       |    |         |
|---------------|-----------|----|-----------|----|-------|----|---------|
| In            | Frequent: | 05 | Errors in | 24 | Hours | 00 | Minutes |
| Fr            | :         |    |           |    |       |    | Minutes |
| Cr            | :         |    |           |    |       |    | Minutes |
| : Type: ----- |           |    |           |    |       |    |         |

1 HOST :

|    |    |           |    |           |    |      |         |
|----|----|-----------|----|-----------|----|------|---------|
| In | In | Frequent: | 00 | Errors in | 00 | : 00 | Minutes |
| Fr | :  | Frequent: | 00 | Errors in | 00 | : 00 | Minutes |
| Cr | :  | Critical: | 00 | Errors in | 00 | : 00 | Minutes |
|    |    | Notify:   | 1  |           |    |      |         |

- INFC : F1=Help F12=Cancel :

|           |           |    |           |    |       |    |         |
|-----------|-----------|----|-----------|----|-------|----|---------|
| In        | Frequent: | 00 | Errors in | 00 | Hours | 00 | Minutes |
| Fr        | :         | 00 | Errors in | 00 | Hours | 00 | Minutes |
| Critical: | :         | 10 | Errors in | 00 | Hours | 10 | Minutes |

Command ==>

|             |              |           |            |         |
|-------------|--------------|-----------|------------|---------|
| F1=Help     | F2=Main Menu | F3=Return | F5=Refresh | F6=Roll |
| F7=Backward | F8=Forward   |           | F12=Cancel |         |

*Figure 25. Thresholds Panel*

4. In the pop-up window, type in the values you want.
5. Press **Enter**.

When AON adds the thresholds, the following message is displayed in the message area near the end of the panel:

EZL919I ALL ACTIONS SUCCESSFULLY COMPLETED

## Changing Monitor Intervals

Use the Monitor Intervals panel to add, display and change, and delete monitor intervals. Monitor intervals determine how often AON tries to reactivate failed resources and if AON sends messages to the notification operators during these attempts. These values are retrieved from the MONIT control file entry.

**Note:** A DEFAULTS entry in the control file is required. The DEFAULTS entry is used when the parameters do not exist on the specific resource definition.

### Displaying the Monitor Intervals panel

To display the Monitor Intervals panel:

1. Display the **AON: Automation Settings** panel.

**Note:** To display the AON: Automation Settings panel, see [“Displaying the AON: Automation Settings Panel”](#) on page 23.

2. Type **4** in the entry field on the AON: Automation Settings panel.
3. Press **Enter**.

The Monitor Intervals panel shown in [Figure 26 on page 32](#) is displayed.

**Note:** You can also display the Monitor Intervals panel by entering **AON 1.4.4, SETMONIT**, or **DELMONIT** on any command line.

```
EZLK4400          Monitor Intervals          CNM01

Select one of the following. Then press enter.
1=Add 2=Display/Change 3=Delete

    Type
-   CDRM
-   CP
-   CPCPSESS
-   DEFAULTS
-   HOST
-   INFC
-   IPHOST
-   IPINFC
-   IPNAMESERV
-   IPPORT
-   IPTELNET
-   NAMESERV
-   NCP
-   SP

Command ===>
F1=Help      F2=Main Menu    F3=Return
F7=Backward  F8=Forward
F5=Refresh   F6==Roll
F12==Cancel
```

*Figure 26. Monitor Intervals Panel*

The types of monitor intervals are displayed in the Type column. You can use the Monitor Intervals panel to add, display, change, or delete monitor intervals. The following sections explain how to perform these actions.

### Displaying or changing a current monitor interval

To display or change recovery monitoring intervals:

1. Display the **Monitor Intervals** panel.

**Note:** To display the Monitor Intervals panel, see [“Displaying the Monitor Intervals panel”](#) on page 32.

2. Type **2** in the entry field next to the resource type whose monitor interval you want to display or change.

3. Press **Enter**.

A pop-up window shown in [Figure 27](#) on [page 33](#) is displayed.

```

EZLK4410                               Monitor Intervals                               CNM01

Select one of the  : .....
1=Add  2=Display/  : Type: ----- :
                   :      Hour Minute :
   Type            :   Interval: 00 : 00  Notify: -- :
1  CDRM            :   Interval: 00 : 00  Notify: -- :
-  CP              :   Interval: 00 : 00  Notify: -- :
-  CPCPSESS        :   Interval: 00 : 00  Notify: -- :
-  DEFAULTS        :   Interval: 00 : 00  Notify: -- :
-  HOST            :   Interval: 00 : 00  Notify: -- :
-  INFC            :   Interval: 00 : 00  Notify: -- :
-  IPHOST          :   Interval: 00 : 00  Notify: -- :
-  IPINFC          :   Interval: 00 : 00  Notify: -- :
-  IPNAMESERV      :   Interval: 00 : 00  Notify: -- :
-  IPPORT          :   Interval: 00 : 00  Notify: -- :
-  IPTELNET        :   Interval: 00 : 00  Notify: -- :
-  NAMESERV        :   Interval: 00 : 00  Notify: -- :
-  NCP             :   F1=Help      F12=Cancel :
-  SP              : ..... :

Command ==>
F1=Help      F2=Main Menu  F3=Return      F5=Refresh      F6==Roll
F7=Backward  F8=Forward                               F12==Cancel

```

Figure 27. Monitor Intervals Panel

4. To change a monitoring interval, type over the settings you want to change. The variables on this setting are:

## hh:mm

The length of the interval between reactivation attempts expressed as hours (*hh*) and minutes (*mm*). The range is 00:00–24:00.

## Notify

The setting that determines whether messages are sent to the notification operators when AON attempts to reactivate the resource. The settings can be:

**Y**

Consult the notification policy to determine which notifications to issue.

**N**

Do not send notifications.

**YF**

Consult the notification policy to determine which notifications to issue. Repeat recovery monitoring at the last interval specified.

**NF**

Do not send notifications. Repeat recovery monitoring at the last interval specified.

5. Press **Enter**.

If the settings are entered correctly, AON displays this message:

EZL919I ALL ACTIONS SUCCESSFULLY COMPLETED

## Changing Active Monitor settings

Use the Active Monitor Setting panel to set intervals for active monitoring and to verify that resources are active. If the resource is not active, recovery monitoring starts for the resource and a timer is set to check the resource again for the next interval. These values are retrieved from the ACTMON control file entry.

Active monitoring expects specific entries only. Wildcard entries are not allowed. Example SNA entries and types are:

- Specific resource name (PU01, LINE23)
- General resource type (NCP, LU, PU)

**Note:** A DEFAULTS entry in the control file is required. The DEFAULTS entry is used when the parameters do not exist on the specific resource definition.

## Displaying the Active Monitor settings panel

To display the Active Monitor Settings panel:

1. Display the **AON: Automation Settings** panel.

**Note:** To display the AON: Automation Settings panel, see “Displaying the AON: Automation Settings Panel” on page 23.

2. Type **5** in the entry field on the AON: Automation Settings panel.
3. Press **Enter**.

The Active Monitor Settings panel is displayed, as shown in [Figure 28 on page 34](#).

**Note:** You can also display the Active Monitor Settings panel by entering **AON 1.4.5** or **ACTMON** on any command line.

|                                                                                   |                         |            |
|-----------------------------------------------------------------------------------|-------------------------|------------|
| EZLK4500                                                                          | Active Monitor Settings | CNM01      |
| Select one of the following. Then press Enter.<br>1=Add 2=Display/Change 3=Delete |                         | More: - +  |
| Name                                                                              | Resource Type           | Interval   |
| - SULU                                                                            | CPCPSESS                | Timer ID   |
| - TA1L5023                                                                        | LINE                    | FKV00011   |
| - TA1N500                                                                         | NCP                     | FKV00009   |
| - TA1T1046                                                                        | LU                      | FKV00010   |
| - TA1T1047                                                                        | LU                      |            |
| - TA1T1048                                                                        | LU                      |            |
| - TEST1                                                                           | SESSION                 |            |
| - TEST2                                                                           | SESSION                 |            |
| - TEST3                                                                           | SESSION                 |            |
| - USIBMTA.TA1CPBAD                                                                | CP                      | 01:00      |
| - USIBMTA.TA1CP207                                                                | CP                      | 01:00      |
| - USIBMTA.TA1CP208                                                                | CP                      | 01:00      |
| - 9.67.6.2                                                                        | HOST                    | 00:25      |
| - 9.67.6.3                                                                        | HOST                    |            |
| Command ==>                                                                       |                         |            |
| F1=Help                                                                           | F2=Main Menu            | F3=Return  |
| F7=Backward                                                                       | F8=Forward              | F5=Refresh |
|                                                                                   |                         | F6=Roll    |
|                                                                                   |                         | F12=Cancel |

*Figure 28. Active Monitor Settings Panel*

The Active Monitor Settings panel displays these columns of data:

### Name

The name of a resource.

### Resource Type

The type of resource. This data is taken from the RESTYPE= keyword of the ACTMON control file entry.

### Interval

The interval setting. The interval can be in minutes or in both hours and minutes. This data is taken from the INTVL= keyword of the ACTMON control file entry.

### Timer ID

The ID of the timer. The timer issues the ACTMON routine for this resource to see if the resource is active.

Use the Active Monitoring Settings panel to add, display or change, and delete active monitoring settings. The following sections explain how to perform these actions.

# Displaying or changing Active Monitor settings

To display or change active monitoring settings:

1. Display the **Active Monitor Settings** panel.

**Note:** To display the Active Monitor Settings panel, see [“Displaying the Active Monitor settings panel”](#) on page 34.

2. Move the cursor to the field next to the resource you want to display or change and type **2**.
3. Press **Enter**.

A pop-up window shown in [Figure 29 on page 35](#) is displayed.

EZLK4520

Active Monitor Settings

CNM01

Select one of the following. Then press Enter.  
1=Add 2=Display/Change 3=Delete

More: - +

| Name        | Resource Type                        | Interval   | Timer ID |
|-------------|--------------------------------------|------------|----------|
| SULU        | CPCPSESS                             |            | FKV00011 |
| 2 TA1L5023  | .....                                | .....      | FKV00009 |
| - TA1N500   | :                                    | :          | FKV00010 |
| - TA1T1046  | : Name: TA1L5023                     | :          | :        |
| - TA1T1047  | :                                    | :          | :        |
| - TA1T1048  | : Type: LINE                         | :          | :        |
| - TEST1     | :                                    | :          | :        |
| - TEST2     | : Interval: 00 : 13                  | :          | :        |
| - TEST3     | :                                    | :          | :        |
| - USIBMTA.T | : Timer Id: FKV00009                 | :          | :        |
| - USIBMTA.T | : Timer Date/Time: 04/15/19 13:15:59 | :          | FKV00012 |
| - USIBMTA.T | :                                    | :          | :        |
| - 9.67.6.2  | : F1=Help                            | F12=Cancel | :        |
| - 9.67.6.3  | .....                                | .....      | :        |

Command ==>  
F1=Help F2=Main Menu F3=Return F5=Refresh F6=Roll  
F7=Backward F8=Forward F12=Cancel

Figure 29. Active Monitoring Settings Panel

4. Type the resource type in the **Type** field.
5. Type the interval values you want in the **Interval** field.

An interval is defined by *hh:mm* with valid values that are in the range of 00:00–24:00. The intervals specified are cumulative time delays from the previous activation setting. You cannot change any other fields on this panel.

6. Press **Enter**.

If the data is entered correctly, the following message is displayed:

EZL919I ALL ACTIONS SUCCESSFULLY COMPLETED





## Chapter 5. Using Cross-Domain functions

This chapter documents general-use programming interface and associated guidance information.

To establish communication with the NetView program and other applications use the AON: Cross-Domain Functions panel and its subordinate panels. You can use these panels to manage cross-domain sessions for your operators, for AON Gateway operators, and for terminal access facility (TAF) full-screen sessions.

## Displaying the Cross-Domain Functions panel

To display the Cross-Domain Functions panel:

1. Display the Base Functions panel.

**Note:** To display the **Base Functions** panel, see “Displaying the AON Base Functions Panel” on page 4.

2. Type **5** in the entry field.

3. Press **Enter**.

The AON: Cross-Domain Functions panel shown in Figure 30 on page 37 is displayed.

**Note:** You can also display the AON: Cross-Domain Functions panel by entering **AON 1.5** on any command line.

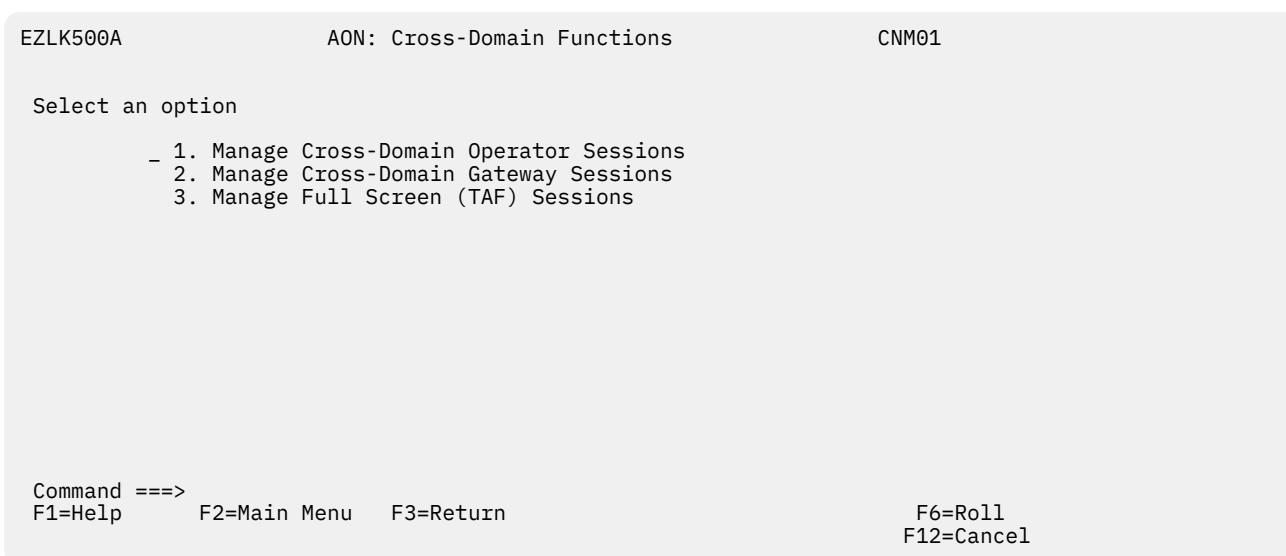


Figure 30. AON: Cross-Domain Functions Panel

The AON: Cross-Domain Functions panel displays the following options:

## Manage Cross-Domain Operator Sessions

Enables you to establish sessions with other NetView domains using your NetView-NetView Task (NNT) sessions or through remote command (RTMCD) sessions. After you log on to the sessions, you can send commands across those sessions. These values are retrieved from the CDLOG control file entry.

## Manage Cross-Domain Gateway Sessions

Enables you to start and stop the AON gateway sessions to other NetView domains and send commands across the gateway sessions. When you use the gateways, an automation operator logs on to the other domains and handles communications, so you do not need to establish your own sessions

with the domains with which you want to communicate. These values are retrieved from the GATEWAY control file entry.

### **Manage Full-Screen (TAF) Sessions**

Enables you to establish a full-screen session with another application using the NetView Terminal Access Facility (TAF). After you log on to another application using TAF, the display on your domain looks like the full-screen display for the application that has the established session. These values are retrieved from the FULLSESS control file entry.

The following sections explain how to use the options.

## **Managing Cross-Domain Operator Sessions**

---

Use the AON: Cross-Domain Functions panel to communicate with other NetView domains by establishing personal operator sessions with those domains. These values are retrieved from the CDLOG control file entry.

You can use the AON: Cross-Domain Logon panel to:

- Start an operator session to another domain
- Stop an active session
- Send a command to an active session

You can establish NNT or RMTCMD sessions.

**Note:** To use the help desk to recover SNA resources on other NetView domains, you must have NNT sessions established to those domains to reactivate the failed SNA resources. Use the Manage Cross-Domain Operator Sessions option to establish those sessions.

### **Displaying the AON: Cross-Domain Logon Panel**

To display the AON: Cross-Domain Logon panel:

1. Display the AON: Cross-Domain Functions panel.

**Note:** To display the AON: Cross-Domain Functions panel, see [“Displaying the Cross-Domain Functions panel” on page 37](#).

2. Type **1** in the entry field on the **AON: Cross-Domain Functions** panel.
3. Press **Enter**.

The AON: Cross-Domain Logon panel, shown in [Figure 31 on page 39](#), is displayed.

**Note:** You can also display the AON: Cross-Domain Logon panel by entering **AON 1.5.1** or **CDLOG** on any command line.

EZLK5000

AON: Cross-Domain Logon

CNM01

| 1=Start<br>Domain | 2=Stop<br>Status | 3=Send<br>Operator | Origin<br>Operator | OPER2<br>Type | Init | Description |
|-------------------|------------------|--------------------|--------------------|---------------|------|-------------|
| - CNM01           | Inactive         |                    |                    | RMTCMD        |      |             |
| - CNM01           | Inactive         |                    |                    | NNT           |      |             |
| - CNM02           | Inactive         |                    |                    | NNT           |      |             |
| - AOC10           | Inactive         |                    |                    | NNT           |      |             |
| - CNM10           | Inactive         |                    |                    | NNT           |      |             |
| - AOF10           | Inactive         |                    |                    | NNT           |      |             |
| - AOC06           | Inactive         |                    |                    | NNT           |      |             |
| - CNM06           | Inactive         |                    |                    | NNT           |      |             |
| - AOC05           | Inactive         |                    |                    | NNT           |      |             |
| - CNM05           | Inactive         |                    |                    | NNT           |      |             |

To issue commands, tab to the Domain and press F4.

Command ==>

F1=Help

F2=Main Menu

F3=Return

F4=Commands

F5=Refresh

F6=Roll

F7=Backward

F8=Forward

F12=Cancel

Figure 31. AON: Cross-Domain Logon Panel

The AON: Cross-Domain Logon panel displays the following data:

#### Domain

Target NetView domain.

#### Status

Status of the operator session, which is either active or inactive.

#### Operator

Target operator ID for session.

#### Type

Type of operator session, either NetView-NetView Task (NNT) or remote command (RMTCMD). This data is taken from the SESSTYPE= keyword of the CDLOG control file entry.

#### Init

Specifies whether this operator session starts automatically when you log on to your home NetView domain. This data is taken from the INIT= keyword of the CDLOG control file entry.

#### Description

A brief description for the session.

You can use the AON: Cross Domain Logon panel to start and stop cross-domain operator sessions and to send a command from your NetView domain to another domain, using an active NNT or remote command session. The following sections explain how to perform these actions.

## Starting a Cross-Domain operator session

To start a cross-domain operator session:

1. Display the **AON: Cross-Domain Logon** panel.

**Note:** To display the AON: Cross-Domain Logon panel, see [“Displaying the AON: Cross-Domain Logon Panel” on page 38](#).

2. Type **1** in the field next to the domain name in which you want to start a session on the Cross-Domain Logon panel.
3. Press **Enter**.

When all the information needed to start the session is available from the control file, the session starts immediately. However, usually, you must provide further information to start the session. If further information is needed, AON displays the pop-up window shown in [Figure 32 on page 40](#).



```

EZLK5102                AON: Cross-Domain Logon                CNM01

1=Start    2=Stop    Origin Operator OPER2
Domain Status Operator Type : Select an Action:
- CNM01 Inactive : .....
- CNM01 Inactive : .....
- CNM02 Inactive : .....
- AOC10 Inactive : .....
- CNM10 Inactive : .....
- AOF10 Inactive : .....
- AOC06 Inactive : .....
- CNM06 Inactive : .....
- AOC05 Inactive : .....
2 CNM05 Active OPER2 : .....
: .....
: F1=Help F12=Cancel :
To issue commands, tab to the Dom .....

Command ==>
F1=Help F2=Main Menu F3=Return F4=Commands F5=Refresh F6=Roll
F7=Backward F8=Forward F12=Cancel

```

Figure 33. AON: Cross-Domain Logon Panel with Stop Session Confirmation Pop-up Window

4. Press **Enter** to confirm that you want to stop the session or press **F12** to cancel the session.

When the session ends, the status of your domain is shown as inactive, and the following message is displayed in the panel:

```
EZL930I SESSION(S) ENDED
```

## Sending a Command to another domain

To send a command from your NetView domain to another domain using an active NNT or remote command session:

1. Display the **AON: Cross-Domain Logon** panel.

**Note:** To display the Cross-Domain Logon panel, see [“Displaying the AON: Cross-Domain Logon Panel” on page 38](#).

2. On the Cross-Domain Logon panel, type **3** in the field next to the name of the domain to which you want to send the command.
3. Press **Enter**.

If you try to send a command across an inactive session, you see the following message:

```
EZL881I NNT SESSION TO CNM02 IS NOT ACTIVE
```

If the operator session is active, the pop-up window shown in [Figure 34 on page 42](#) is displayed.

```

EZLK5103          AON: Cross-Domain Logon          CNM01

          Origin Operator OPER2
1=Start      2=Stop      3=Send
Domain Status Operator Type : Select an Action:
- CNM01 Inacti .....
- CNM02 Inacti : Send Command To CNM05
- AOC10 Inacti : Enter Command Below: (50 character limit)
- CNM10 Inacti : ==>
- AOF10 Inacti :
- AOC06 Inacti :
- CNM06 Inacti : Note: Output is displayed in NCCF, ROLL to return.
- AOC05 Inacti : F1=Help F12=Cancel
3 CNM05 Active .....
- CNM01 Inactive. :
          . F1=Help F12=Cancel :
          .....

To issue commands, tab to the Domain and press F4.

Command ==>
F1=Help      F2=Main Menu  F3=Return  F4=Commands  F5=Refresh  F6=Roll
F7=Backward  F8=Forward      F12=Cancel

```

Figure 34. AON: Cross-Domain Logon Panel with Send Command to Remote Domain Pop-up Window

4. Type the command you want to enter in the **Enter Command Below** field.
5. Press **Enter**.

AON displays the command facility (NCCF). AON displays the results from issuing the command on the command facility, along with the following message:

```

EZL938I ISSUING COMMAND = "command". PRESS PF6 TO ROLL BACK
TO CDLOG

```

6. To return to the operator interface, press the Roll key, **F6**.

To display a pop-up command window from which you can issue session commands, press the Commands key, **F4**. For more information about using these commands, see [“Pop-up command windows”](#) on page 6.

## Managing cross-domain gateway sessions

Use the AON: Cross-Domain Gateway Display panel to display the active and inactive sessions available for your domain. The list that is displayed is generated from the configuration file entries for your domain. You can use the AON: Cross-Domain Gateway Display panel to start and stop a cross-domain gateway session and to send a command to a cross-domain gateway session. To send a command to a cross-domain gateway session, the session selected must be active. These values are retrieved from the GATEWAY control file entry.

**Note:** Gateway sessions must be NNT sessions. For more information, see the *IBM Z NetView Administration Reference*.

### Displaying the AON: Cross-Domain Gateway panel

To display the AON: Cross-Domain Gateway Display panel:

1. Display the **AON: Cross-Domain Functions** panel.

**Note:** To display the AON: Cross-Domain Functions panel, see [“Displaying the Cross-Domain Functions panel”](#) on page 37.

2. Type **2** in the entry field on the AON: Cross-Domain Functions panel.
3. Press **Enter**.

The first half of the AON: Cross-Domain Gateway Display panel shown in [Figure 35 on page 43](#) is displayed.

**Note:** You can also display the AON: Cross-Domain Gateway Display panel by entering **AON 1.5.2** or **AONGW** on any command line.

```
EZLK5200          AON: Cross-Domain Gateway Display          CNM01
                  FOCAL POINT = CNM01
                  OPERATOR ID = OPER2                          More: >

Select one of the following options, then press ENTER.
1=Start    2=Stop    3=Send

      DESCRIPTION          DOMAIN      STATUS      VIA
      -----          -
-      TO AOC06            AOC06      INACTIVE    AOC06
-      TO CNM06            CNM06      ACTIVE      CNM06
-      TO CNM10            CNM10      ACTIVE      CNM10

Command==>
F1=Help    F2=Main Menu    F3=Return          F5=Refresh    F6=Roll
F7=Backward F8=Forward      F11=Right     F12=Cancel
```

Figure 35. Cross-Domain Gateway Display Panel

The Cross-Domain Gateway Display panel displays the domains with which you are communicating through gateway operators and gives information to you about the gateway operators that are controlling the communications.

The columns on the first half of the panel (Figure 35 on page 43) provide the following information:

#### DESCRIPTION

The **Description** field shows a user-written description that describes the gateway to the remote NetView domain. For example, the first description says that this line of the status display states TO AOC06.

#### DOMAIN

The **Domain** field shows the name of the remote NetView domain as it is defined to the NetView program.

#### STATUS

The **Status** field indicates if communications between your home NetView domain and this remote NetView domain are ACTIVE or INACTIVE. In Figure 35 on page 43, communications between the remote NetView domain, CNM06, and NetView domain, CNM01, are active.

#### VIA

The **Via** field shows the NetView domain that is physically between this remote NetView domain and your NetView domain that is used for routing communications between them. In Figure 35 on page 43, the communications between your NetView domain (CNM01) and CNM06 travel through CNM06, which means that your NetView domain and CNM06 are physically connected.

**Note:** Only one NetView domain can be physically located between your NetView domain and the remote NetView domain with which you establish gateway sessions.

4. To display the second half of the Cross-Domain Gateway Display panel, press F11. Figure 36 on page 44 shows the adjoining gateway display panel for the sample display shown in Figure 35 on page 43.

```

EZLK5201          AON: Cross-Domain Gateway Display          CNM01
                  FOCAL POINT = CNM01
                  OPERATOR ID = OPER2                          More: <

Select one of the following options, then press ENTER.
1=Start   2=Stop   3=Send
  DIRECT INB/OUTB STAT  SYSNAME  PRODUCT  ADJ DOM  ALT DOM  USER ID
  -----
- INACTIVE  INV OPID   NA        NA V.R
- ACTIVE    ACTIVE    CNM06   TIV-AON 6.2
- ACTIVE    ACTIVE    CNM10   TIV-AON 6.2
                                     GATCNM06
                                     GATCNM10

Command==>
F1=Help      F2=Main Menu  F3=Return      F5=Refresh    F6=Roll
F7=Backward  F8=Forward   F10=Left      F12=Cancel

```

Figure 36. Cross-Domain Gateway Display Panel: Scrolled-right View

The column headings in [Figure 36 on page 44](#) provide the following information about gateways between your NetView domain and this remote NetView domain:

#### DIRECT INB

The **Direct Inbound** field indicates if the remote domain has logged on to your NetView domain. Thus, for direct communications between your NetView domain and a physically adjacent NetView domain, the direct inbound status can be either ACTIVE or INACTIVE. If your NetView domain is not physically next to the remote NetView domain, communications must be routed across an intermediate NetView domain that is physically next to both your NetView domain and the remote NetView domain. The status is NO DIRECT, indicating that there are no direct communication gateways between your NetView domain and the remote NetView domain.

If an error occurred during a logon attempt by a gateway operator, the inbound status displays in yellow.

#### OUTB STAT

The **Outbound Status** field indicates if the gateway operator handling communications for your NetView domain is successfully logged on to this remote NetView domain and sending communications. A status of ACTIVE means that communications from your NetView domain (outbound) to this remote NetView domain are going to it successfully. If outbound communications are not active, the status displayed indicates the cause of the problem, if it is known.

If an error occurred during a logon attempt by a gateway operator, the inbound status displays in yellow.

#### SYSNAME

The **System Name** field shows you the name by which the Dynamic Display Facility (DDF) knows this remote NetView domain. This name might be the same name used for the NetView domain under the DOMAIN heading on the scrolled-left view of the gateway panel (see [Figure 35 on page 43](#)), but it is not required. If the connection has not been established, the panel displays NA to indicate that the name is not available. Your system programmer defines the DDF names for the NetView domains using the SYSNAME parameter on the ENVIRON SETUP control file entry.

#### PRODUCT

The **Product** field provides the name of the product being run on this target NetView domain, for example, TIV-AON 6.2. If the product is back-level, the entry says OLD ? . ?, which indicates that communications have been established, but the product name is not supplied to this panel by the back-level product.



If communications with the domain have not been established since the last NetView startup, the name of the product cannot be determined, so you see NA V.R, which is an abbreviation for Not Available Version.Release.

#### **ADJ DOM**

The **Adjacent Domain** field indicates a NetView domain that is a path to the remote domain. The adjacent domain must be physically adjacent. If a NetView domain that is physically next to both domains is defined as an adjacent NetView domain, you can use gateways to communicate with a NetView domain that is not physically next to your own. Your system programmer defines adjacent NetView domains in the control file with the ADJNETV control file entry.

#### **ALT DOM**

The **Alternate Domain** field displays the name of the physically adjacent NetView domain that can function as a backup to the adjacent domain. If you are using an adjacent NetView domain as an intermediate routing domain, it is useful to have another pathway in case the preferred adjacent NetView domain goes down. Your system programmer defines backup NetView domains in the control file using the ADJNETV entry.

#### **USER ID**

The **User ID** field shows the user ID for the gateway operator logged on to your NetView domain from another NetView domain. The gateway operator is a program, but it logs on to the other NetView domains much like a human operator does. In Figure 36 on page 44, GATCNM06 is the user ID for the gateway operator on NetView domain CNM06 that logs on to your NetView domain. That is, GATCNM06 handles outbound communications for CNM06, but it brings inbound communications to your NetView domain.

**Note:** Each NetView domain uses one gateway operator for outbound communications. That gateway operator logs on to all the other NetView domains it communicates with. Your NetView domain can have many gateway operators logged on to it.

### **Sending a command to a Cross-Domain Gateway session**

To send a command to a cross-domain gateway session, use the following steps. You can also use the SENDCMD command from the NetView command facility. See the NetView online help or the *IBM Z NetView Command Reference Volume 2 (O-Z)* for more information.

1. Display the Cross-Domain Gateway Display panel.

**Note:** To display the AON: Cross-Domain Display panel, see [“Displaying the AON: Cross-Domain Gateway panel”](#) on page 42.

2. Type **3** in the field next to the domain name for which you want to send a command.
3. Press **Enter**.

The AON: Command Routing panel shown in [Figure 37 on page 46](#) is displayed.

EZLK5300

AON: Command Routing

CNM01

RESP

=>

\_

1=Yes, 2=Ack, 3=No

OPERATOR ID

=>

\_\_\_\_\_

Can be left blank

To DOMAIN

=>

CNM06

Command

=>

\_\_\_\_\_

Command==>

F1=Help

F2=Main Menu

F3=Return

F6=Roll

F12=Cancel

Figure 37. AON: Command Routing Panel

4. Type the number of the entry you want in the **RESP** field. The choices are:
  - 1=Yes**  
Displays the result of the command you issued on the command facility (NCCF).
  - 2=Ack**  
Displays acknowledgment on the command facility indicating whether the command was issued.
  - 3=No**  
Displays neither the result of whether the command was issued nor the acknowledgment.
5. Optionally, specify the operator ID where the command is to be issued in the **OPERATOR ID** field. If you do not specify an operator ID, the GATOPER defined in the control file is used as the operator ID.
6. Type the name of the NetView domain to which you are sending the command in the **To DOMAIN** field.
7. Type the command to issue on the other domain in the **Command** field.
8. Press **Enter**.  
The command you specified runs on the remote domain.

## Managing full-screen TAF sessions

You can log on to selected applications in the system by using the AON: Terminal Access Facility Menu panel. AON enables automated setup of TAF sessions.

The AON: Terminal Access Facility Menu panel displays the status of all TAF full-screen applications that are defined in the control file for operators. You can start and stop TAF full-screen sessions from this panel. When you establish a TAF session, your current domain looks like the full screen for the application with which you are in session. These values are retrieved from the FULLSESS control file entry.

**Note:** It is not required that you establish sessions with applications using TAF.

### Displaying the AON: TAF Menu panel

To display the AON: Terminal Access Facility Menu panel:

1. Display the **Cross-Domain Gateway Display** panel.
  - Note:** To display the AON: Cross-Domain Display panel, see [“Displaying the AON: Cross-Domain Gateway panel”](#) on page 42.
2. Type **3** in the entry field on the AON: Cross-Domain Display panel.

3. Press **Enter**.

The AON: Terminal Access Facility Menu panel is displayed in [Figure 38](#) on page 47.

**Note:** You can also display the AON: Terminal Access Facility Menu panel by entering **AON 1.5.3** or **AONTAF** on any command line.

```
EZLK5400          AON: Terminal Access Facility Menu          CNM01

      SRCLU = TAFANF00  OPERATOR ID = OPER1

Select one of the following. Then press Enter.
1=Start/Return  2=Stop
  Description      Applid      System      Status
  -----
  _  TSO Application  TSO        CNM01      ACTIVE

Command ==>
F1=Help      F2=Main Menu  F3=Return          F5=Refresh      F6=Roll
F7=Backward  F8=Forward
```

*Figure 38. AON: Terminal Access Facility Menu Panel*

The **AON: Terminal Access Facility Menu** panel displays the following data:

**Description**

A short description that identifies the application to be started.

**Applid**

The short identifier for the application.

**System**

The name of the system on which the application runs.

**Status**

Specifies whether the application is active or inactive.

You can use the AON: Terminal Facility Menu panel to start, return to, and stop a TAF full-screen session. The following sections explain how to perform these actions.

## Starting an application session

To start a TAF session:

1. Display the AON: Terminal Access Facility Menu panel.

**Note:** To display the AON: Terminal Access Facility Menu panel, see [“Displaying the AON: TAF Menu panel”](#) on page 46.

2. Type **1** next to the session you want to start.
3. Press **Enter**.

The application you selected is displayed. The next step you take depends on the application. For example, if the application is Time Sharing Option (TSO), the logon panel for that TSO session is displayed.

## Stopping an application session

To stop a TAF session:

1. Display the **AON: Terminal Access Facility Menu** panel.

**Note:** To display the AON: Terminal Access Facility Menu panel, see [“Displaying the AON: TAF Menu panel”](#) on page 46.

2. Type **2** next to the session you want to stop.
3. Press **Enter**.

The TAF session deactivates.

4. Press **F5** to refresh the status.

## Chapter 6. Performing task and log maintenance

To perform task and log maintenance, use the AON: Task and Log Maintenance panel and the subordinate panels. Use these panels to:

- Display data in the configuration and status files
- Browse the NetView and automation logs
- Start and stop automation components
- Work with automation tasks

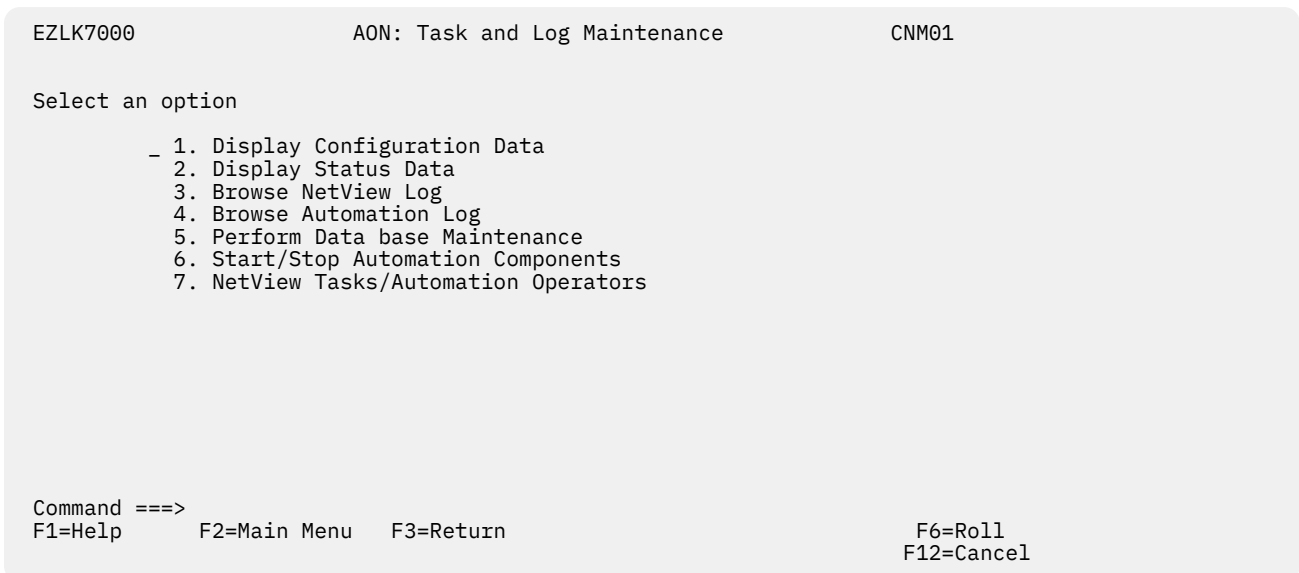
### Displaying the AON: Task and Log Maintenance panel

To display the Task and Log Maintenance panel:

1. Display the Base Functions panel.
2. Type **7** in the entry field on the AON: Base Functions panel.
3. Press **Enter**.

The AON: Task and Log Maintenance panel shown in [Figure 39 on page 49](#) is displayed.

**Note:** You can also display the AON: Task and Log Maintenance panel by typing **AON 1.7** or **AONMAINT** on any command line.



*Figure 39. AON: Task and Log Maintenance Panel*

The AON: Task and Log Maintenance panel displays the following options:

#### **Display Configuration Data**

Displays current configuration data from the control file for an entry and type. The control file is checked and results displayed. You can add, change, and delete the configuration data.

#### **Display Status Data**

Displays the status for specific IDs or a range of IDs. Deletions from the status file are enabled.

#### **Browse NetView Log**

The command BR NETLOGA enables you to browse the NetView log.

### Browse Automation Log

The NLOG command enables you to browse the AON log.

### Perform Data Base Maintenance

Enables selective purging of outdated VSAM database records. The databases that can be maintained are the NetView Hardware Monitor database, the NetView Session Monitor database, the AON status file, and other databases as installed by the AON components.

### Start/Stop Automation Components

Enables you to start, stop, or reload the automation components. The components are DDF, LOG file, status file, and control file. In addition, you can change which control file member is loaded again or you can load another control file.

### Automation Tasks/Operators

Finds all tasks on the domain and displays the information in a list. From this list, you can stop, start, force the task off the system, display detailed information about the task, or show the utilization of the task.

The following sections explain how to use these options.

## Displaying configuration data

---

Use the AON: Configuration Data Display panel to display data in the control file.

### Displaying the AON: Configuration Data Display panel

To display the AON: Configuration Data Display panel:

1. Display the **AON: Task Log and Maintenance** panel.

**Note:** To display the AON: Task Log and Maintenance panel, see [“Displaying the AON: Task and Log Maintenance panel”](#) on page 49.

2. Type **1** in the entry field.
3. Press **Enter**.

The AON: Configuration Data Display panel, shown in [Figure 40 on page 50](#) is displayed.

**Note:** You can also display the AON: Configuration Data Display panel by entering **AON 1.7.1** or **DSPCFG** on any command line.

```
EZLK7100          AON: Configuration Data Display          CNM01

Entry: _____
Type:  _____

Command ==>
F1=Help      F2=Main Menu  F3=Return      F6=Roll
F12=Cancel
```

*Figure 40. AON: Configuration Data Display Panel*

4. Type the name of the entry in the control file that you want to display in the **Entry** field. You can enter from 1 to 15 characters without embedded blanks, commas, or quotation marks.

**Note:** The following wildcards are enabled for this field:

**\***

Multiple character wildcard

**%**

Single character wildcard

You can display the entire control file by entering an asterisk (\*) in this field.

5. Type any data associated with the entry in the **Type** field. You can enter from 1 to 32 characters without embedded blanks, commas, or quotation marks.

**Note:** The following wildcards are enabled for this field:

**\***

Multiple character wildcard

**%**

Single character wildcard

You can display the entire control file by entering an asterisk (\*) in this field.

For example, to display all settings for notification operators, type **NTFYOP** in the entry field.

6. Press **Enter**.

The AON: Configuration Data Display panel, shown in [Figure 41 on page 51](#), displays the control file entries for notification operators.

|                                                                           |                            |            |
|---------------------------------------------------------------------------|----------------------------|------------|
| EZLK7110                                                                  | Display Configuration Data | CNM01      |
| Select one of the following. Then press Enter.<br>1=Add 2=Change 3=Delete |                            | More: +    |
| - OPER1                                                                   | NTFYOP                     |            |
| - OPER                                                                    | 'OPERATOR ONE'             |            |
| - CLASS                                                                   | (60,65,66)                 |            |
| - HELDMSG                                                                 | (I,W,E,A)                  |            |
| - OPER2                                                                   | NTFYOP                     |            |
| - OPER                                                                    | 'OPERATOR TWO'             |            |
| - CLASS                                                                   | (10,20,30,50)              |            |
| - HELDMSG                                                                 | (I,W,E,A)                  |            |
| - OPER3                                                                   | NTFYOP                     |            |
| - OPER                                                                    | 'OPERATOR THREE'           |            |
| - CLASS                                                                   | (10,30)                    |            |
| - OPER4                                                                   | NTFYOP                     |            |
| - OPER                                                                    | 'OPERATOR FOUR'            |            |
| - CLASS                                                                   | 10                         |            |
| - OPER5                                                                   | NTFYOP                     |            |
| Command ==>                                                               |                            |            |
| F1=Help                                                                   | F2=Main Menu               | F3=Return  |
| F7=Backward                                                               | F8=Forward                 | F5=Refresh |
|                                                                           |                            | F6=Roll    |
|                                                                           |                            | F12=Cancel |

*Figure 41. Display Configuration Data Panel*

Use the AON: Configuration Data Display panel to add, change, and delete notification operators. The following sections explain how to perform these actions. The changes are made to the in-storage copy of your policy and overwritten the next time the policy is reloaded.

## Adding data to the control file

To add an entry to the control file:

1. Display the control file through the **AON: Configuration Data Display** panel.

**Note:** To display the control file with the AON: Configuration Data Display panel, see [“Displaying the AON: Configuration Data Display panel” on page 50](#).

2. Type **1** next to an existing control file entry on the AON: Configuration Data Display panel.

**Note:** To add only a keyword, type **1** in a field for a keyword.

3. Press **Enter**.

The Configuration Data Display, shown in [Figure 42 on page 52](#), is displayed with the ADD ENTRY pop-up window.

```

EZLK7111          Configuration Data Display          CNM01
Select one of the following. Then press Enter.          More:  +
1=Add 2=Change 3=Delete

1  OPER1          : ADD ENTRY
-   OPER          : Entry . . -----
-   CLASS         :
-   HELDMSG       : Type . . -----
-   OPER2         : F1=Help                      F12=Cancel
-   CLASS         :
-   HELDMSG       : (I,W,E,A)
-   OPER3         : NTFYOP
-   OPER          : 'OPERATOR THREE'
-   CLASS         : (10,30)
-   OPER4         : NTFYOP
-   OPER          : 'OPERATOR FOUR'
-   CLASS         : 10
-   OPER5         : NTFYOP

Command ==>
F1=Help      F2=Main Menu  F3=Return          F5=Refresh  F6=Roll
F7=Backward  F8=Forward   F12=Cancel

```

*Figure 42. Configuration Data Display Panel with Add Entry Pop-up Window*

4. Type the name of the entry you want to add in the **Entry** field in the pop-up window.
5. Type the type of the entry you want to add in the **Type** field in the pop-up window.
6. Press **Enter**.

The Define Data pop-up window shown in [Figure 43 on page 52](#) is displayed.

```

EZLK7112          Configuration Data Display          CNM01
Select one of the following. Then press Enter.          More:  +
1=Add 2=Change 3=Delete

1  OPER1          : ADD ENTRY
-   OPER          : Entry . . ntfyop-----
-   CLASS         :
-   HELDMSG       : Type . . tstoper-----
-   OPER2         :
-   OPER          : DEFINE DATA
-   CLASS         : KEYWORD . . .
-   HELDMSG       :
-   OPER3         : KEYWORD VALUE .
-   OPER          :
-   CLASS         :
-   OPER4         : F1=Help                      F12=Cancel
-   OPER          : F10=Save
-   CLASS         :
-   OPER5         : NTFYOP

Command ==>
F1=Help      F2=Main Menu  F3=Return          F5=Refresh  F6=Roll
F7=Backward  F8=Forward   F12=Cancel

```

*Figure 43. DEFINE DATA Pop-up Window*

7. Type a keyword in the keyword field of the **DEFINE DATA** pop-up window. The keyword can be any name from 1 to 31 characters in length. You cannot use embedded blanks, commas, or quotation marks.



8. Type a value in the **KEYWORD VALUE** field. This is the value for the keyword entered. You can use as many as 62 characters for the value. You can use embedded blanks and commas only if the keyword value is enclosed in quotation marks.
9. To save the settings press **F10**.

The following message is displayed:

```
EZL001I REQUEST EZLSCMOD WAS SUCCESSFUL FOR ntfyop.
```

## Displaying status data

---

Use the Display Status Data panel and the subordinate panels to display data in the status file.

The Display Status Data panel uses a set of command processors to write and update records for the AON status files. These records are written to a VSAM data set. The file contains the following type of information for each resource:

- Automation Status
- The operator ID that last changed the **Automation Status** field
- The last exceeded threshold
- The time stamps for errors that resulted in a lost session
- The operator ID that last changed a record in the Automation Status file.
- Additional information depending on the component.

### Displaying the Status Data panel

To display the Display Status Data panel:

1. Display the **AON: Task Log and Maintenance** panel.

**Note:** To display the AON: Task Log and Maintenance panel, see [“Displaying the AON: Task and Log Maintenance panel”](#) on page 49.

2. Type **2** in the entry field.
3. Press **Enter**.

The Display Status Data panel shown in [Figure 44](#) on page 54 is displayed.

**Note:** You can also display the AON: Configuration Data Display panel by typing **AON 1.7.2** or **DSPSTS** on any command line.

|                                     |                     |                       |
|-------------------------------------|---------------------|-----------------------|
| EZLK7200                            | Display Status Data | CNM01                 |
| Select an Option                    |                     |                       |
| _ 1. Id                             |                     |                       |
| 2. From                             |                     |                       |
| To                                  |                     |                       |
| Select a Component                  |                     |                       |
| - 1. AON Base                       |                     |                       |
| 2. AON SNA Automation - SNBU Option |                     |                       |
| Command ==>                         |                     |                       |
| F1=Help                             | F2=End              | F3=Return             |
|                                     |                     | F6=Roll<br>F12=Cancel |

Figure 44. Display Status Data Panel

The Display Status Data panel displays the following options:

#### Select an Option

Searches the status file by using one of two methods that you specify:

##### Id

Searches for a specific resource by searching on the ID of the resource. The ID is the 16-character resource name for the resource records. To use this option, see the [“Example” on page 54](#) section.

##### From/To

Searches an alphabetic range within which the ID of the resource you want to view falls. Both the **From** and **To** fields are required for this option. These entries are also 16-character keys to the resource records.

#### Select a Component

Specifies an AON component associated with the resource. If your component is not listed on the panel, select **1** for AON base.

The following example explains how to display the status of a resource. If you need status file records for SNBU automation, you must select those components. Status data for a resource managed by those components use different record formats and the data is accessible only by selecting that component.

#### Example

When you know the ID of a resource and want to display status data for that resource, do the following:

1. Type **1** in the **Select an Option** entry field on the Display Status Data panel.
2. Type the name of the resource in the **Id** entry field. This example uses resource **TA1N400**.
3. Type the number corresponding to the component type of the resource in the **Select a Component** entry field. This example uses **1** for AON Base.
4. Press **Enter**.

The Display Status Data panel, shown in [Figure 45 on page 55](#), is displayed with the data for the resource you specified.

```

EZLK7210          Display Status Data          CNM01

Select one of the following.  Then press Enter.
1=Delete

- ID= TA1N400          , TYPE= UNKNOWN          , STATUS= ACTIV
  LAST UPDATE BY OPERATOR AUTMSG
  LAST THRESHOLD EXCEEDED -
  OPERATOR NOTIFIED: Y , TIMER SET:
  LAST STATUS CHANGE DATE= 04/18/19 , TIME= 19:27 , OPID= AUTMSG
  LAST MONITORED DATE= 04/18/19 , TIME= 19:27
  NO ERROR DATA AVAILABLE

Command ==>
F1=Help      F2=Main Menu   F3=Return          F5=Refresh    F6=Roll
F7=Backward  F8=Forward                F12=Cancel

```

Figure 45. Display Status Data Panel

5. Optionally, type **1** to delete the record from the status file.
6. Press **Enter**.

## Maintaining databases

Use the AON: Database Maintenance panel to maintain databases. The panel is built dynamically, based on the components installed. However, if a component does have a status file, it is defined in the component option definition table using the STSCMD= keyword. Select the database with which you want to work, and specify the Purge Limit and Purge Time.

**Note:** The DBMAINT command reproduces, without records marked for deletion, the VSAM database that is currently active. When the DBMAINT command finishes, the database that was active when the DBMAINT command started is activated again. The DBMAINT command works for an active primary or secondary VSAM database for the NetView hardware monitor (NPDA) or the NetView session monitor (NLDM). The AON status file (EZLSTS) has only a primary database.

### Displaying the AON: Database Maintenance panel

To display the Database Maintenance panel:

1. Display the **AON: Task Log and Maintenance** panel.

**Note:** To display the AON: Task Log and Maintenance panel, see [“Displaying the AON: Task and Log Maintenance panel”](#) on page 49.

2. Type **5** in the entry field.
3. Press **Enter**.

The AON: Database Maintenance panel, shown in [Figure 46](#) on page 56, is displayed.

**Note:** You can also display the AON: Database Maintenance panel by typing **AON 1.7.5** or **DBMAINT** on any command line.

```
EZLK7500                      AON: Database Maintenance          CNM01

Select one of the following
- 1. Session Monitor
  2. Hardware Monitor
  3. AON Base
  4. AON SNA Automation - SNBU Option

Compress      : 1                      (1=Yes  2=No)
Purge Limit   : 007 Days
Purge Time    : 045 Minutes

Command ==>
F1=Help      F2=Main Menu  F3=Return          F6==Roll
F7=Backward  F8=Forward    F12==Cancel
```

*Figure 46. AON: Database Maintenance Panel*

The AON: Database Maintenance panel displays the following fields and options:

**Select one of the following**

Enables you to purge records and compress the databases for hardware and session monitors and for AON status files.

**Compress**

Specifies whether to compress a database after deleting records from it.

**Purge Limit**

Specifies the number of days to leave in the file. If the number is 7, then all records older than 7 days are purged.

**Purge Time**

Specifies the amount of time to wait for the purge of the records to complete.



Turns off automation for AON and its components, or turns off automation for the components and subcomponents only.

Lists all common global variables used in NetView program. Use this option to add, change, or delete common global variables.

Starts the AUTOMAN function to assist you in managing your automation tables.

Lists the tables used to load the common global variables. Use this option to browse and reload tables.

## Setting traces

## Displaying the Set Trace panel

**Note:** The AONTRACE and RXTRACE commands do not require AON. These functions are documented in this chapter as reference items.

- Note:** To display the Support Functions panel, see [“Displaying the Support Functions panel”](#) on page 57.

3. Press **Enter**.

**Note:** You can also display the Set Trace panel by entering **AON 1.8.1** or **AONTRACE** or **RXTRACE** on any command line.

Figure 48. Set Trace panel

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### Entry/Exit tracing

Traces all the entry and exit parameters of the AON code. This includes command lists, REXX programs, and some AON command processors. You can trace the programs running on an operator ID, a NetView domain, or both.

### Program tracing

Traces the command lists and interpreted REXX programs you specify. Also, specify a trace option that limits the trace. Use the Program tracing option for the operator ID or the domain.

### Administrative Functions

Authorized operators can enable Entry/Exit tracing and Program tracing. By default, tracing is disabled and can be enabled only through this option.

The following sections explain how to use these options.

## Setting Entry/Exit traces

To trace all the programs that AON uses:

1. Display the **Set Trace** panel.

**Note:** To display the Set Trace panel, see [“Setting traces” on page 58](#).

2. Type **1** in the entry field on the Set Trace panel.
3. Press **Enter**.

The Set Entry/Exit Tracing panel is shown in [Figure 49 on page 59](#).

|                                                                    |                        |                       |
|--------------------------------------------------------------------|------------------------|-----------------------|
| EZLK8110                                                           | Set Entry/Exit Tracing | NTVE1                 |
| Operator id OPER1 __ (? for list)                                  |                        |                       |
| Select trace option . . .                                          |                        |                       |
| - 1. ON Turn on entry and exit tracing for operator                |                        |                       |
| 2. OFF Suppress entry and exit tracing for operator                |                        |                       |
| 3. <b>DEFAULT</b> Use trace option for domain (domainwide default) |                        |                       |
| Domain id NTVE1                                                    |                        |                       |
| Select trace option . . .                                          |                        |                       |
| - 1. ON Turn on entry and exit tracing for domain                  |                        |                       |
| 2. <b>OFF</b> Suppress entry and exit tracing for domain           |                        |                       |
| Command ==>                                                        |                        |                       |
| F1=Help                                                            | F2=Main Menu F3=Return | F6=Roll<br>F12=Cancel |

*Figure 49. Set Entry/Exit Tracing panel*

4. Select Entry/Exit tracing for an operator ID, a domain, or both on the **Set Entry/Exit Tracing** panel. Domain tracing occurs only on the current domain. Current® settings are highlighted.

To select Entry/Exit tracing for a domain only, use the following steps:

- a. Type **1** for ON or **2** for OFF in the **Domain ID Select trace option** entry field.
- b. Press **Enter**.

The following message is displayed:

```
EZL908I SETTINGS REPLACED
```

## Setting program traces

To set a trace for a particular program:

1. Display the **Set Trace** panel.

**Note:** To display the Set Trace panel, see [“Setting traces”](#) on page 58.

2. Type **2** in the entry field on the Set Trace panel.
3. Press **Enter**. The Set Program Tracing panel is shown in [Figure 50](#) on page 60.

```

EZLK8120                      Set Program Tracing                      NTVE1

Enter module name and trace option.                                     More:  +
R=Result I=Intermediate C=Command E=Error F=Failure L=Label O=Off

.. Settings for Operator OPER1___ .. .. Settings for Domain NTVE1 .....
:  Module      Option      :  :  Module      Option      :
:  -----      -          :  :  -----      -          :
:  -----      -          :  :  -----      -          :
:  -----      -          :  :  -----      -          :
:  -----      -          :  :  -----      -          :
:  -----      -          :  :  -----      -          :
:  -----      -          :  :  -----      -          :
:  -----      -          :  :  -----      -          :
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:  -----      -          :  :  -----      -          :
:  -----      -          :  :  -----      -          :
:  -----      -          :  :  -----      -          :
:  -----      -          :  :  -----      -          :
:  -----      -          :  :  -----      -          :

Command ==>
F1=Help      F2=Main Menu  F3=Return
F7=Backward  F8=Forward
F6=Roll
F12=Cancel

```

*Figure 50. Set Program Tracing panel*

You can trace the programs by operator ID or domain ID. If you trace by domain ID, NetView traces the programs in the domain the operator is logged on to. You can also select a tracing option on this panel that enables you to limit the trace.

4. Type the name of the program or programs you want to trace in the **Module** column for either the operator ID or domain ID.
5. Type the letter for the trace option you want to use in the **Option** column. You can select one of the following trace options:

#### **R (Result)**

Use this option for general debugging. This traces all the clauses before running them and traces the final results of evaluating expressions.

#### **I (Intermediate)**

Use this option to trace all clauses before they are run and trace any intermediate results during expression evaluation and substitution.

#### **C (Command)**

Use this option to trace all commands before running them and display any error return codes from the commands.

#### **E (Error)**

Use this option to trace any command that has an error or fails after it is run. This option also displays the return codes.

#### **F (Failure)**

Use this option to trace any command that fails after it is issued. This option is the same as the Trace Normal command.

#### **L (Label)**

Use this option to trace all labels passed. Make a note of all subroutine calls and signals when you use this option.

#### **O (Off)**

Use this option to turn off all traces and reset any previous trace settings.



**Note:** If the program being traced is a NetView command list, the C, E, and O options are valid and all other selections result in a trace ALL.

6. Press **Enter**.

The following message is displayed:

```
EZL908I SETTINGS REPLACED
```

## Administrative functions

By default, entry/exit and program tracing is disabled for performance purposes. This is defined on your environment setup policy definition statement. To enable tracing:

1. Display the **Set Trace** panel.

**Note:** To display the Set Trace panel, see [“Setting traces” on page 58](#).

2. Type **3** in the entry field on the Set Trace panel.

3. Press **Enter**.

The Trace Administrative Functions panel is shown in [Figure 51 on page 61](#).

```
EZLK8130                      Trace Administrative Functions                      NTVE1

Current Trace Setting is . . . NONE
To modify the trace setting select an option:

      Trace Setting      Entry/Exit Trace      Program Trace
      - 1. ON            Yes                    Yes
        2. OFF            No                     Yes
        3. NONE           Disabled                Disabled

Selecting NONE will disable both Entry/Exit and Program tracing.
You should select NONE for best overall performance and only
select ON or OFF when you need to debug a possible problem.

Command ==>
      F2=Main Menu   F3=Return
                                F6=Roll
                                F12=Cancel
```

*Figure 51. Trace Administrative Functions panel*

[Figure 51 on page 61](#) displays that tracing is currently set to NONE, which means that all tracing is disabled for this domain. To enable tracing, select Options 1 or 2. Option 1 enables both entry/exit and program tracing. Option 2 enables only Program tracing.

When you select options 1 or 2 you can turn on program tracing for any AON program by going back to the Trace Menu panel and selecting the **Program Trace** option.

## Reinitializing automation

Use the AON: Reinitialize Automation panel to reinitialize the AON program. You can use the automation table and control file currently being used. To display the AON: Reinitialize Automation panel:

1. Display the **Support Functions** panel.

**Note:** To display the Support Functions panel, see [“Displaying the Support Functions panel” on page 57](#).

2. Type **2** in the entry field on the Support Functions panel.

3. Press **Enter**.

The Reinitialize Automation panel is shown in [Figure 52](#) on page 62.

**Note:** You can also display the AON: Reinitialize Automation panel by typing **AON 1.8.2** or **AONINIT** on any command line.

|                              |                              |            |
|------------------------------|------------------------------|------------|
| EZLK8200                     | AON: Reinitialize Automation | CNM01      |
|                              |                              |            |
| Select Confirmation Option   | . . _ 1. Confirm             |            |
|                              | 2. Cancel                    |            |
|                              |                              |            |
| Reload with Automation Table | . . DSITBL01                 |            |
| Generate Listing File        | . . LISTNAME                 |            |
|                              |                              |            |
| Reload with Control File     | . . NVPOLICY                 |            |
|                              |                              |            |
| Trace Setting                | . . OFF                      |            |
|                              |                              |            |
| Command ===>                 |                              |            |
| F1=Help                      | F2=Main Menu                 | F3=Return  |
|                              |                              | F6=Roll    |
|                              |                              | F12=Cancel |

*Figure 52. Reinitialize Automation panel*

4. Verify that the name of the automation table is correct. The default name for the automation table is DSITBL01. The policy file is defined in the CNMSTYLE member. The policy file name is fixed and cannot be changed.
5. Type **1** in the first entry field to confirm that you want to reinitialize AON or type **2** to cancel the reinitialization. For information about how AON loads the automation table, refer to the *IBM Z NetView Installation: Getting Started*.
6. Press **Enter**.  
AON confirms or cancels your request.

## Enabling and disabling automation

You can use the AON: Enable/Disable Automation panel to turn off message processing for components. The panel supports multiple selections. If you enable or disable message processing for the AON base, you also enable or disable automation for all of the components. Therefore, select AON with caution.

When you select options 1–4 on the listing panel, special confirmation panels enable you to confirm your action on the Enable/Disable panel. When you select a component that has subcomponents, the confirmation panel that is displayed lists all the subcomponents that the selection affects.

**Note:** Using this panel to enable or disable automation does not permanently change the settings. When you recycle AON or select the default option, AON resets the settings.

### Displaying the AON: Enable/Disable Automation panel

To display the AON: Enable/Disable automation panel:

1. Display the **AON: Support Functions** panel.

To display the AON: Support Functions panel, see [“Displaying the Support Functions panel”](#) on page 57.

2. Type **3** in the entry field on the Support Functions panel.

### 3. Press **Enter**.

The AON: Enable/Disable Automation panel is shown in [Figure 53](#) on page 63.

**Note:** You can also display the Enable/Disable Automation panel by entering **AON 1.8.3** or **AONENABL** on any command line.

```
EZLK8300          AON: Enable/Disable Automation          CNM01

Select one of the following. Then press enter.
1=Enable 2=Disable 3=Default 4=Initialize 5=Product Information

  Component Name                               Initialized Enabled
- AON Base                                     Y           Y
- AON TCP/IP Automation                       Y           Y
- AON TCP/IP Automation - TCP/IP 390 Option   Y           Y
- AON SNA Automation                         Y           Y
- AON SNA Automation - SNBU Option           Y           N
- AON SNA Automation - SubArea Option        Y           Y
- AON SNA Automation - APPN Option           Y           N

Command ==>
F1=Help      F2=Main Menu  F3=Return          F5=Refresh    F6==Roll
F7=Backward  F8=Forward
```

*Figure 53. Enable/Disable Automation panel*

The AON: Enable/Disable panel displays the following information:

#### **Component name**

Lists the components.

#### **Initialized**

States whether the component has been initialized when AON is initialized or when you specified for it to be initialized.

#### **Enabled**

States the current ENABLE/DISABLE setting of the component. A component can be listed as initialized but not enabled, but when a component is not initialized, it cannot be enabled.

You can use the Enable/Disable Automation panel to enable and disable automation, to specify default automation for a component, to initialize a component, and to display information about a product.

The following sections explain how to perform these actions.

## Using the Common Global Editor

You can use the Common Global Editor panel to view all of the current common global variables being used in the NetView program. You can use the Common Global Editor panel to add, change, or delete common global variables (CGLOBALS).

**Note:** The Common Global Editor does not require AON. These functions are documented in this chapter as reference items.

Use this editor with caution because any changes you make can affect AON and NetView processing. Each common global variable is restricted to 255 characters. Mixed case is supported for common global variable values and is set exactly as you type.

Any changes that you make by using the editor are for the current AON session only. When AON recycles, the common global variables are reset to values at initialization time.

## Displaying the Common Global Editor panel

To display the Common Global Editor panel:

1. Display the **Support Functions** panel.

**Note:** To display the Support Functions panel, see [“Displaying the Support Functions panel”](#) on page 57.

2. Type **4** in the entry field on the Support Functions panel.
3. Press **Enter**.

The Common Global Editor panel is shown in [Figure 54](#) on page 64.

**Note:** You can also display the Common Global Editor panel by typing **AON 1.8.4** or **CGED** on any command line.

EZLK8400 Common Global Editor NTVE1

Select an option

- 1. EZLRT
- 2. EZLOPT
- 3. Generic

Search Criteria

-----

Command ==>

F1=Help F2=Main Menu F3=Return F6=Roll F12=Cancel

*Figure 54. Common Global Editor panel*

The Common Global Editor panel displays the following options:

### Select an option.

Specifies a common global. The options are:

#### EZLRT

Shows the resource type common global variables set from the option definition tables at AON startup. Do not delete these common global variables.

#### EZLOPT

Shows the option definitions set from the option definition tables at AON startup. Do not delete these common global variables.

#### Generic

Shows the common global variables specified by the operator. If you select this option and do not fill in the **Search Criteria** field, AON displays all of the common global variables. If you type a value in the **Search Criteria** field, AON appends an asterisk (\*) to the global name. You can also use a wildcard.

### Search Criteria

Specifies the search criteria that you want to use.

The following sections explain how to use these options.

## Changing Common Global Variables

To change a common global variable:

1. Display the Common Global Editor panel.

**Note:** To display the Common Global Editor panel, see [“Displaying the Common Global Editor panel”](#) on page 64.

2. Select an option and the search criteria you want on the Common Global Editor panel. For example, type **1** for EZLRT common global variables.

**Note:** If you select EZLRT or EZLOPT, the editor checks for an entry in the **Search Criteria** field. The entry in this field is then appended to EZLRT or EZLOPT. The search criteria must not exceed 31 characters.

For example, if you type **1** in the **Select Option** field and specify **AON** in the **Search Criteria** field, the editor searches for all common global variables that are EZLRT.AON.\*

Or, if you type **2** in the **Select Option** entry field and enter **APPN.ACT\*** in the **Search Criteria** field, the editor looks for all common global variables that are EZLOPT.APPN.ACT\*

3. Press **Enter**.

The CGLOBAL EDITOR panel is shown in [Figure 55 on page 65](#).

```

EZLK8410          Operator Command : CGLOBAL EDITOR          NTVE1

Select one of the following. Then press enter.
1=Add 2=Change 3=Delete

    Name                                0.....1.....2.....3.....4.
- EZLOPT.AON.AONALRT                    00010204
- EZLOPT.AON.AONNAME                    AON Base
- EZLOPT.AON.AONPROD                    5697ENV00
- EZLOPT.AON.AONREL                     050400
- EZLOPT.AON.ENABLE                     Y
- EZLOPT.AON.MAINPANELPOS               1
- EZLOPT.AON.MAXFEATURES                3
- EZLOPT.AON.MSGCLASS                   90
- EZLOPT.AON.OCMDCMD                   EZLE0100
- EZLOPT.AON.OCMDDESC                   AON Base Functions
- EZLOPT.AON.OPERLIST                   ALRTOPER MSGOPER MSGOPER2 BASEOPER
- EZLOPT.AON.STSCMD                     EZLSTS
- EZLOPT.AON.STSPIPE                     EZLSPIPS

EZL919I ALL ACTIONS SUCCESSFULLY COMPLETED
Command ==>
F1=Help      F2=Main Menu   F3=Return          F5=Refresh   F6=Roll
F7=Backward  F8=Forward      F11=Right    F12=Cancel

```

*Figure 55. Operator Command: CGLOBAL Editor panel*

4. Type **2** in the entry field next to the CGLOBAL you want to change.
5. Type over the existing value of the CGLOBAL to the right of the entry field with the new value.
6. Press **Enter**.

The following message is displayed:

```
EZL919I ALL ACTIONS SUCCESSFULLY COMPLETED
```

AON changes the CGLOBAL.



---

## Chapter 8. Managing multiple automation tables

This chapter documents general-use programming interface and associated guidance information.

The AUTOTBL command enables you to load multiple automation tables. An automation table, typically, is made up of many included members. The automation table management (AUTOMAN) command enables you to change selected tables or changes that have an effect on all automation tables. To help you work with automation tables, AUTOMAN provides a full-screen panel interface.

AUTOMAN and the full-screen panel interface enable you to do the following tasks:

- View and manage single or multiple automation tables
- Enable or disable individual automation tables or statements
- View existing tables and their status

**Note:** AUTOMAN does not require AON.

---

### Getting started

AUTOMAN provides individual table commands and global commands. The individual table commands apply to one or more selected tables, and global commands apply to all automation tables. See the following features and options of each type of command:

- With individual table commands, you can enable or disable automation tables. You can also enable or disable automation table statements, based on the following:
  - Sequence number
  - Label
  - Endlabel
  - Block
  - Group
  - Include

With individual table commands, you can also issue requests for the following:

- Display disabled statements
- Display labels, blocks, and groups
- Load or unload tables
- Test tables
- Display the %INCLUDE structure
- Display synonyms
- With global commands, you can enable, disable, or unload automation tables. You can enable disabled statements or enable and disable blocks, groups, and labels. Global commands affect all automation tables.

Automation statements can be enabled or disabled across all tables based on the following:

- Label
- Block
- Group

With global commands, you can also issue requests for the following:

- Locate disabled statements

- Display labels, blocks, and groups
- Display the %INCLUDE structure

## Using Automation Table Management

From the command line, enter AUTOMAN. The panel in [Figure 56 on page 68](#) is displayed. This panel enables you to see your automation table structure and do what is necessary.

```

EZLK8500                      Automation Table Management

AUTOMATION TABLE             Enter any character in the selection fields
SEL      POS      NAME        STATUS    MARKERS    TASK      DATE      TIME
-        1        DISTABLE    ENABLED          NETOP2    03/18/19  13:15:24
-        2        DSITBL01    ENABLED    (AON)      NETOP2    03/18/19  13:11:09

Command ==>
F1=Help      F2=Main Menu  F3=Return    F4=Commands  F5=Refresh  F6=Roll
F7=Backward  F8=Forward   F9=Responses F10=Global Commands F12=Cancel

```

Figure 56. Automation Table Structure

## Using commands for selected tables

The Commands pop-up in [Figure 57 on page 68](#) provides options to help you work with one or more selected automation tables. In the following figure, options 1–7 apply to one or more selected tables, in contrast to global commands in *IBM Z NetView Automation Guide*, which apply to all tables. Options 8–9 apply to only one table.

In [Figure 56 on page 68](#), pressing **F4**, for Commands, displays [Figure 57 on page 68](#) where DSITBL01 is selected to be disabled.

Selecting option 2 causes a pop-up to be displayed to confirm that you want to disable the selected table. When DSITBL01 is disabled, a message indicates whether the command was successful or whether failures were detected. Press **F9** in [Figure 56 on page 68](#) to view the results of your command.

```

EZLKATBC                      AUTOMATION TABLE MANAGEMENT

AUTOMATION TABLE             Enter any character in the selection fields
SEL  POS  NAME                  : .....:
/    1    DISTABL : COMMANDS (Choose a highlighted command option) :
-    2    DSITBL0 : .....:
:                  : 1 -ENABLE the selected tables :
:                  : 2 -DISABLE the selected tables :
:                  : 3 -RELOAD the selected tables :
:                  : 4 -RELOAD and REINSTATE disabled elements :
:                  : 5 -TEST the selected tables :
:                  : 6 -ENABLE/DISABLE parts of the selected :
:                  : tables :
:                  : 7 -UNLOAD selected tables :
:                  : 8 -INSERT a table :
:                  : 9 -DISPLAY options :
:                  : .....:
:                  : Enter=Execute Command      F3 or F12=Cancel :
:                  : .....:

Command ==>
F1=Help      F2=Main Menu                      F6=Roll

```

Figure 57. Automation Table Management Commands Pop-up panel



# Chapter 9. Using the Inform Log utility

This chapter documents general-use programming interface and associated guidance information.

Operators can use the inform log utility to display inform log entries. The entries are records of the automated notification actions that have taken place, such as a pager call or email. An operator can view the log, acknowledge receipt of an inform notification, reinform a specific contact, or delete an entry. All automated actions are logged, by default, when inform logging is enabled. The logging of operator calls with INFORM/EZLECALL can be enabled in the inform policy. Refer to the SETUP policy member in the *IBM Z NetView Administration Reference* for more details.

**Note:** It is not the purpose of the inform log utility to keep a log that tracks all inform actions. Rather, for tracking purposes, messages are generated and can be found in the NetView log. The purpose of the ILOG® data set is to help you dynamically track, respond to, and delete inform log entries. Because of the I/O required and data set constraints, the ILOG function might not be practical in all environments.

To start the inform log utility, you can enter one of the following commands:

```
ILOG
AON 1.9
```

If you enter ILOG, the following full-screen panel containing the contents of your inform log is displayed:

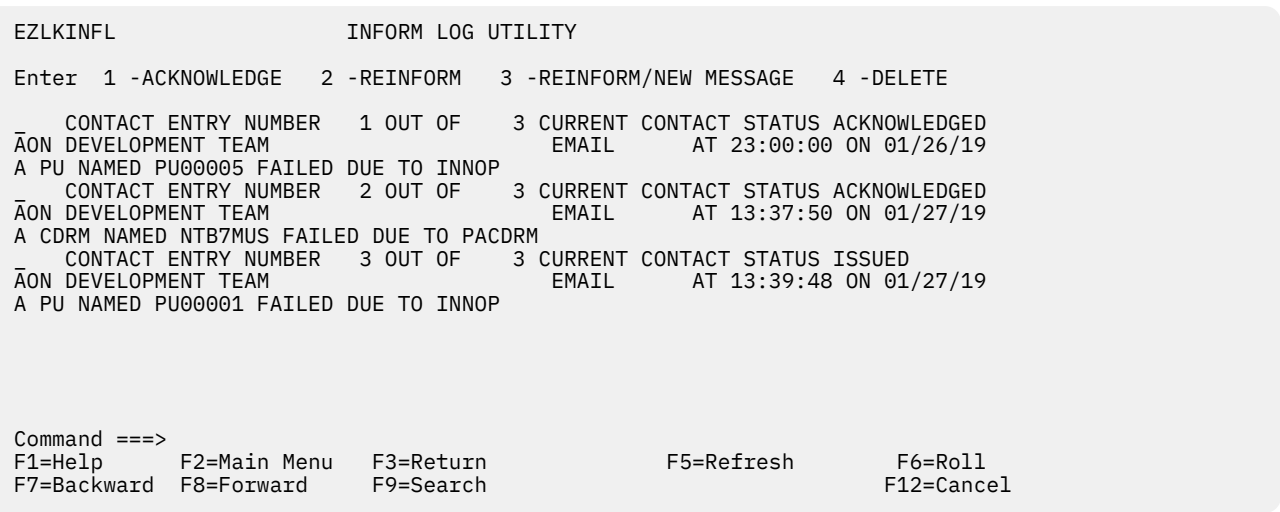


Figure 58. Inform Log Utility panel

You can choose any of the following functions, which are shown in [Figure 58 on page 69](#), for a single inform log entry:

### 1 -ACKNOWLEDGE

Changes the displayed status to ACKNOWLEDGED.

### 2 -REINFORM

Reissues the message against the active inform policy member using the original policy name.

### 3 -REINFORM/NEW MESSAGE

Reissues the message against the active inform policy member using the original policy name and provides a pop-up window with the original message text. The message text can be edited or replaced prior to confirming the REINFORM.

### 4 -DELETE

Removes the entry from the inform log.

The following function keys provide functions that are independent of an individual log entry:

**F5**

Refreshes and displays a current version of the inform log.

**F9**

Searches for specified log entries. From a pop-up window, you can search for a name, partial name, message text, and date and time. The default is to search through all log entries from the oldest to the most current. For more information, see [“Searching log entries” on page 72](#).

The lines of a log entry in [Figure 58 on page 69](#) displays the following options:

**First line**

Contains the current entry number, the total number of entries, and the status of the entry.

```
CONTACT ENTRY NUMBER 1 OUT OF 3 CURRENT CONTACT STATUS ISSUED
```

**Second line**

Contains the contact name field, the connection type used, and the time and date of the notification.

```
AON DEVELOPMENT TEAM          EMAIL      AT 23:00:00 ON 01/26/19
```

**Third line**

Contains the message issued by the notification.

```
A PU NAMED PU00005 FAILED DUE TO INNOP
```

You can also access the inform log utility by entering the AON 1.9 command synonym. When the AON Base Functions panel, shown in [Figure 59 on page 70](#), is displayed select option 9:

|                  |                             |            |
|------------------|-----------------------------|------------|
| EZLK0100         | AON: Base Functions         | NTVGD      |
| Select an option |                             |            |
| -                | 0. Tutorial                 |            |
|                  | 1. Help Desk                |            |
|                  | 2. AutoView                 |            |
|                  | 3. DDF                      |            |
|                  | 4. Automation Settings      |            |
|                  | 5. Cross Domain Functions   |            |
|                  | 6. Timer                    |            |
|                  | 7. Task and Log Maintenance |            |
|                  | 8. Support Functions        |            |
|                  | 9. Display the Inform Log   |            |
| Command ==>      |                             |            |
| F1=Help          | F2=Main Menu                | F3=Return  |
|                  |                             | F6=Roll    |
|                  |                             | F12=Cancel |

*Figure 59. AON Base Functions panel*

## Checking notification status

The status conditions, which are located at the end of the first line in each log entry, are described in the following list. See [Figure 58 on page 69](#) where the status is ISSUED.

**ISSUED**

An inform request was sent to a service point.

**ACKNOWLEDGED**

An operator acknowledged the notification.

**DELETED**

The request was deleted from the inform log database.

**ROUTED**

The request was routed to another NetView domain that owns the service point.

**REINFORMED**

The request was reissued after the current inform policy was applied.

**REINFORMED/NEW**

The request was reissued after the current inform policy was applied and a new message has been sent.

**FAILED**

The request failed. The probable cause is a communication error with the service point.

**NOTROUTED**

The request could not be routed to the target NetView domain.

**UNKNOWN ENTRY**

The inform log contains a corruption error.

## Reissuing notifications

To display the following pop-up window where you can reissue notifications, enter 3 -REINFORM/NEW MESSAGE on the panel shown in [Figure 58 on page 69](#):

```

EZLKINFR                                INFORM LOG UTILITY
Enter  1 -ACKNOWLEDGE   2 -REINFORM   3 -REINFORM/NEW MESSAGE   4 -DELETE

CONTACT ENTRY NUMBER   1 OUT OF   3 CURRENT CONTACT STATUS ISSUED
AON .....
A P : REISSUE an INFORM action via policy PAUL .....
3 : .....
AON : Update or Replace original message (60 character limit) .....
A C : => A CDRM NAMED NTB7MVS FAILED DUE TO PACDRM .....
: .....
AON : Note: Numeric message is required for Numeric Pagers .....
A P : F1=Help      Press ENTER to REINFORM      F12=Cancel .....
.....

Command ==>
F1=Help      F2=Main Menu   F3=Return      F5=Refresh      F6=Roll
F7=Backward  F8=Forward     F9=Search      F12=Cancel

```

*Figure 60. Inform Log Utility with Reissue Request Pop-up panel*

Depending on how your policy is defined, the REISSUE request might change. For example, at 16:00, the policy might indicate a notification is to be sent as an e-mail, but at 19:00 the same policy might have no active contacts or might issue a notification through a pager.

**Usage :**

- An active inform policy member is required to view the inform log. Only automated notifications generated by EZLENFRM (the notification policy) are logged in the inform log by default. For more information about the notification policy, refer to the *IBM Z NetView Administration Reference*
- Inform entries are logged in the DSILIST DD data set.
- The ILOG command uses the INFORM command to perform REINFORM actions. For information about the ILOG, INFORM, and INFORMTB commands, refer to the *IBM Z NetView Command Reference Volume 1 (A-N)*.

# Searching log entries

The inform log search panel, shown in [Figure 61 on page 72](#), enables you to search the inform log entries.

EZLKINFS

INFORM LOG UTILITY

Enter 1 -ACKNOWLEDGE 2 -REINFORM 3 -REINFORM/NEW MESSAGE 4 -DELETE

CONTACT ENTRY NUMBER1 OUT OF3 CURRENT CONTACT STATUS ISSUED

.....

AON: ENTER THE SEARCH CRITERIA:

: specify Name of Message field search:

AON: Name Field: \*:

AC: Message Field: \*:

::

AON: Start Date: 01/26/19 23:00:00 -Oldest Log Entry Date/Time:

AP: Stop Date: 02/02/19 09:20:05 -Current Date/Time:

::

: F1=Help Press ENTER to REFRESH F12=Cancel:

: with Search Criteria:

.....

Command ==>

F1=Help F2=Main Menu F3=Return F5=Refresh F6=Roll

F7=Backward F8=Forward F9=Search F12=Cancel

Figure 61. Inform Log Utility Search panel

By default, the search proceeds through all entries beginning with the oldest date and time and ending with the current date and time. You can limit your search by any, a combination of, or all of, the following ways:

- The name or message fields
- Start and stop date and time

In a partial search, you can specify part of a name followed by and asterisk in the name field and the inform log utility finds every occurrence of the partial name. For example, you can search for XYZ or XYZ\* and the inform log utility finds all entries containing XYZ. The asterisk (\*) is optional. If you narrow the search still further using specific start and stop dates and times, the search is limited to the specified criteria beginning and ending on the specified dates and times.

If search criteria is not updated, press F5. An error message is displayed.

---

## Part 2. Using AON/SNA



## Chapter 10. Getting started with AON/SNA

This topic explains navigation through the AON/SNA operator interface. The AON/SNA operator interface is a series of full-screen, menu-driven panels that provide all the operator functions that are available in AON/SNA.

You can identify an AON/SNA panel by the prefix of FKV in the panel name, located in the upper left of the panel. Use the operator interface to look at color-coded displays of your network, resolve network problems, receive messages, issue commands, and perform many other functions that control automation and resource availability.

To perform a task in AON/SNA, use the operator interface panels. For each task, there is a primary panel, which might have one or more subordinate panels.

In addition to AON/SNA panels, some AON/SNA tasks use AON panels, which are identified by an EZL prefix in the upper left corner of the panel, and some AON/SNA tasks use NetView panels.

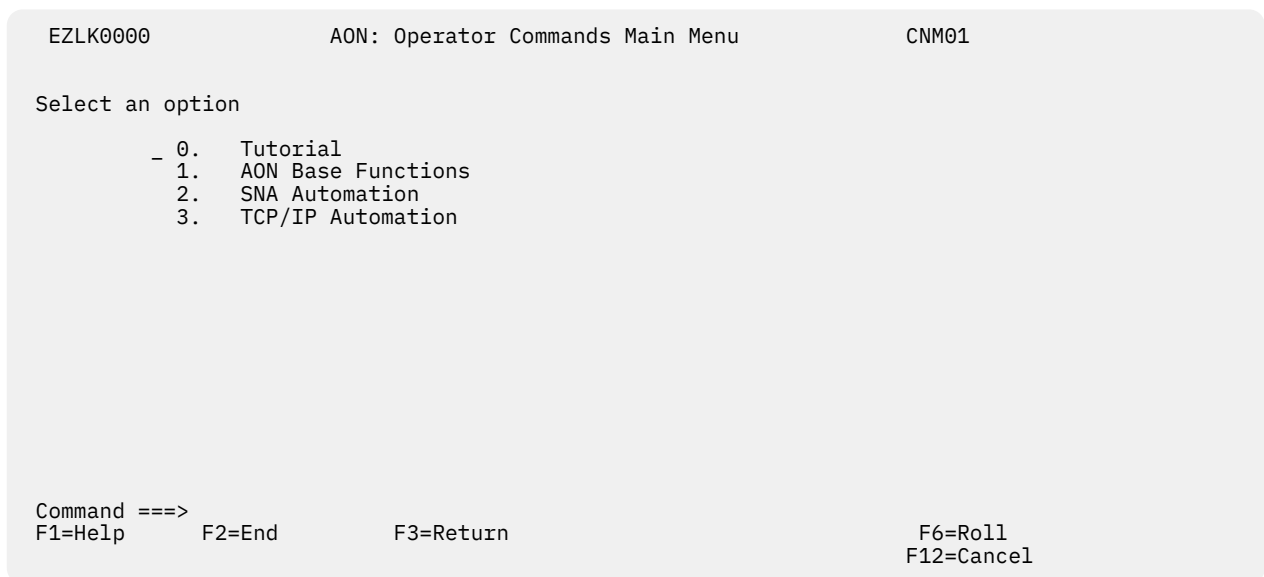
### Accessing the Operator Interface

You can access the AON/SNA operator interface from anywhere within NetView or the AON panels. You can also reach the AON/SNA panels from the other AON component (AON/TCP) if this component is installed and initialized by your organization.

To access the AON: Operator Commands Main Menu panel:

1. Type **AON** from the NetView command line.
2. Press **Enter**.

The NetView program displays the AON: Operator Commands Main Menu panel shown in [Figure 62 on page 75](#). This panel displays all of the available components of AON. If a component (like AON/SNA) is not available, the name of the component on the panel is not available.



*Figure 62. AON: Operator Commands Main Menu panel*

3. Type **2** in the entry field, which is located next to the first option.
4. Press Enter.

AON displays the SNA Automation: Menu panel shown in [Figure 63 on page 76](#). Use this panel to access all the functions of AON/SNA.

**Note:** You can also display the AON/SNA operator interface by entering **AONSNA** on any command line within the NetView program, AON, or from the other AON components, provided these components are installed and initialized on your system.

```
FKVK0000      SNA Automation: Menu      CNM01

Select an option

  _ 0. Tutorial
    1. Help Desk
    2. SNAMAP
    3. VTAM Options Management
    4. NetStat
    5. VTAM Commands
    6. APPN menu
    7. Switched Network Backup menu
    8. X.25 menu
    9. NCP Recovery Definitions (NCP name=_____)

Command ===>
F1=Help      F2=Main Menu  F3=Return
F6=Roll      F12=Cancel
```

*Figure 63. SNA Automation: Menu panel*

## Using the SNA Help Desk

The SNA Help Desk is an application program that enables NetView operators to solve network problems and to support users. The SNA Help Desk systematically finds the cause of the network problems and helps you perform recovery actions through a series of panels. The SNA Help Desk resolves network problems for AON/SNA by primarily attending to SNA resources with support for remote systems and NetView Access Services IDs (NVAS).

The SNA Help Desk enables you to select a resource and recycle it. The SNA Help Desk also enables you to use the problem determination panels to solve problems with that resource. Because AON/SNA performs problem determination through the operator interface panels, you receive immediate feedback on the status of a resource with any possible problems highlighted.

The SNA Help Desk enables you to view a resource and its higher connected nodes. This contrasts with SNAMAP that enables you to zoom to lower connected nodes.

The SNA Help Desk enables inexperienced help desk operators to solve network problems. The resource name is the only required field, so all you need to know is the terminal ID of the user. After you enter the terminal ID into the system, the SNA Help Desk displays a pictorial representation of how the terminal of the user is attached to the system.

If your enterprise installed NetView Access Services (NVAS), you can select the NetView Access Services IDs (NVAS) option. This option determines the location of network problems using the user ID of the caller, thus eliminating the need to know even the terminal ID.

To increase your productivity, the SNA Help Desk:

- Reduces the amount of input you enter
- Automates problem determination
- Enables you to be productive immediately, even if you do not know the network configuration



- Teaches problem determination skills while you resolve network failures

To recover resources in another NetView domain, AON/SNA needs an active NNT or remote command (RMTCMD) NetView session with domains other than the one on which AON/SNA is located. This connection is necessary for the SNA Help Desk to find resources that are not on the operator's domain. If you need to establish these sessions, use the AON CDLOG command to go directly to the panel for establishing those sessions.

## Accessing the SNA Help Desk

To display the SNA Help Desk, select the Help Desk option on the SNA Automation: Menu panel, or use the fast path command or a panel synonym.

```
FKVK0000      SNA Automation: Menu      CNM01

Select an option

  1 0. Tutorial
    1. Help Desk
    2. SNAMAP
    3. VTAM Options Management
    4. NetStat
    5. VTAM Commands
    6. APPN menu
    7. Switched Network Backup menu
    8. X.25 menu
    9. NCP Recovery Definitions (NCP name=_____)

Command ===>
F1=Help      F2=Main Menu  F3=Return

                        F6=Roll
                        F12=Cancel
```

*Figure 64. Selecting the SNA Automation Help Desk*

To display the SNA Help Desk from the SNA Automation: Menu panel:

1. Type **1** in the entry field.
2. Press **Enter**.

AON/SNA displays the SNA Automation: Help Desk panel shown in [Figure 65 on page 78](#).

**Note:** You can also display the SNA Automation: Help Desk panel by typing **AON 2.1** or **SNAHD** on any command line and pressing **Enter**.

```

FKVK1000                                SNA Automation: Help Desk

**** *  **  **          ** **          ***          *****          **
** * ** ** ** ***** ** **          **          ** **          **
**   **** ** ** ** **   ** **          **          **** **          **** **
**   **** ** **** **   **** ** ** ** **   ** ** ** **   ** ** **   ****
**   **** ** **** **   **** ** ** **   ** ** ** **   ** ** **   ****
*   ** ** ** ** ** **   ** ** **          **          ** ** **   ** **** **   ****
**** ** * ** **   ** **          ****          ****          **** ** **
                                     **
                                     **

Enter name...: -----

Select option: - 1. Recycle resource
                  2. Problem Determination
                  3. NetView Access Services User ID
                  n. NetView Help Desk

Command ==>
F1=Help      F2=Main Menu  F3=Return

                                     F6=Roll
                                     F12=Cancel

```

Figure 65. SNA Automation: Help Desk Panel

3. Type the terminal ID of the user in the **Enter name** field.
4. Type the number of the option you want in the **Select option** entry field. You can select one of the following options:

#### Recycle resource

Attempts to activate the resource and its hierarchy.

#### Problem Determination

Displays the status of the resource and its higher node to help determine the problem.

#### NetView Access Services User ID

Enables you to use a NetView Access Services user ID to start problem determination.

#### NetView Help Desk

This option takes you to the main NetView Help Desk facility.

5. Press **Enter**.

AON/SNA displays the panel for the option you selected. See the following sections for more information about those options.

## Recycling resources

When you recycle a resource, AON/SNA attempts to deactivate and then activate the resource. Before attempting to activate the resource you select, AON/SNA checks all of the higher nodes to determine the highest inactive node and tries to recycle those nodes first.

For example, to recycle a resource from the SNA Automation: Help Desk panel that is shown in [Figure 66](#) on [page 79](#), follow these steps:

1. Type the terminal ID in the **Enter name** field, as shown in [Figure 66](#) on [page 79](#). This example uses the TA1PT209 terminal ID.
2. Type **1** in the entry field.
3. Press **Enter**.

```
FKVK1000                      SNA Automation: Help Desk

**** *  **  **          ** **          ***          *****          **
** * ** ** ****          ** **          **          **          **
** **** ** ** **          ** **          **          **          **
** **** ** ** **          **** ** ** ** **          ** ** ** **          **
** **** ** ** **          **** ** ** **          ** ** ** **          **
** ** ** ** **          ** ** ** **          ** ** ** **          **
*  ** ** ** **          ** ** **          ** ** ** **          ** ** **
**** **  * ** **          ** **          ****          ****          ** **
                                     **
                                     **

Enter name...:  TA1PT209

Select option:      1 1. Recycle resource
                   2 2. Problem Determination
                   3 3. NetView Access Services User ID
                   n. NetView Help Desk

EZL333I PU TA1PT209 ON CNM01 HAS BEEN RECYCLED
Command ==>
F1=Help      F2=Main Menu  F3=Return
                                     F6=Roll
                                     F12=Cancel
```

Figure 66. Message Indicating Recycled Resources — SNA Automation: Help Desk

If AON/SNA cannot activate the resource, it displays a panel that displays a message, explaining the current status of the resource. [Figure 66 on page 79](#) shows a message that indicates AON/SNA successfully recycled the resource.

When AON/SNA cannot recycle the resource, it displays an Operator Command Interface: SNA Help Desk panel shown in [Figure 67 on page 79](#).

```
FKVK1RC1                      Operator Command Interface: SNA Help Desk                      CNM01

Current Status of  PU : TA1P519A is NEVAC
Resource Domain is: CNM01

Select one of the following or Press F12 to Cancel:

_ 1. View Status Information for  NEVAC
  2. Force activate resource
  3. View Hardware Monitor for LINE : TA1L5019
  4. View terminal Error Code explanation
  5. View X.25 Error Code explanation
  6. View Sense Code Information for  08220000

EZL910I ENTER A SELECTION
Command ==>
F1=Help      F2=Main Menu  F3=Return
                                     F6=Roll
                                     F12=Cancel
```

Figure 67. Operator Command Interface: SNA Help Desk Panel

[Figure 67 on page 79](#) shows the domain that you are on and the status of the resource. To use this panel, do the following:

- 1. Type the number of the command you want to process in the entry field. You can select one of the following options:

**View Status information**

Displays the status of the resource. Provides an explanation of the status as supplied by the NetView STATUS command.

**Force activate resource**

Tries to force the resource active.

### View Hardware Monitor

Displays the most recent events for the highest inactive node of the resource.

### View terminal Error Code

Displays a pop-up window, prompting you for a three-letter error code that you get from the person who called. You see an explanation of the terminal error code.

### View X.25 Error Code explanation

Provides an explanation of the five-character error code provided by the person who called.

### View Sense Code Information

Displays any sense code data and provides an explanation of it. This field might show sense code data when a VARY active occurs. You can look at this information using the NetView SENSE command. If there is no sense data, AON/SNA sets the sense field to 00000000.

2. Press **Enter**.

AON/SNA processes the command you selected.

## Determining Problem Cause

AON/SNA displays a panel that shows the pictorial representation of your network and displays a message that explains the status of the resource. For example, if the resource is active, AON/SNA displays a message similar to the following:

```
EZL043I  TA1PT209 IS ACTIVE
```

You can use the SNA Help Desk to determine the cause of network problems. For example, suppose a user, whose terminal ID is RL523A1, detects problems. The user calls the help desk operator, who enters the terminal ID on the AON Help Desk panel. The AON/SNA Help Desk determines that the terminal is a SNA resource, so it displays the SNA Automation: Help Desk panel shown in [Figure 68 on page 80](#).

```
FKVK1000                      SNA Automation: Help Desk

**** *  **  **                ** **                ***                *****                **
**  *  **  **  ****          ** **                **                ** **                **
**    **** ** ** **          ** **  ****          **  ***** ** **  ****  ***** ** **
**    ***** *****          ***** ** **          ** ** ** ** ** ** ** ** ** ** ** **
** ** ** ** ** ** **          ** ** *****          ** ** ** ** ** ** ** ** ** ** ** **
*  ** ** ** ** ** **          ** ** **          ** ** ** ** ** ** **          ** ** **
**** **  *  **  **          ** **  ****          ***** *****          ***** ** **

                                **
                                **

Enter name...:  RL523A1

Select option:  2  1. Recycle resource
                  2. Problem Determination
                  3. NetView Access Services User ID
                  n. NetView Help Desk

Command ==>
F1=Help      F2=Main Menu  F3=Return

                                F6=Roll
                                F12=Cancel
```

Figure 68. SNA Automation: Help Desk panel – Selecting Problem Determination

To use the Problem Determination option on the SNA Automation: Help Desk panel:

1. Type the terminal ID in the **Enter name** field. This example uses the RL523A1 terminal ID.
2. Type **2** in the entry field.
3. Press **Enter**.

AON/SNA displays the Operator Command Interface: SNA Help Desk panel shown in [Figure 69 on page 81](#).



FKVK1111

SNA Help Desk: LU : RL523A1

CNM01

Problem Determination:

Resource RL523A1 on Domain CNM01 is not active and  
not in CONCT state.

Choose one of the following options or Press F12 to Return:

- 1. Activate the Hierarchy for the resource
- 2. Force Recycle the resource
- 3. SNBU Connect PU TA1P523A

Command ==>

F1=Help

F2=Main Menu

F3=Return

F6=Roll

F12=Cancel

Figure 70. SNA Help Desk Problem Determination panel

Figure 70 on page 82 shows the error message at the top of the panel. The panel lists actions to take to attempt to solve the problem. The Problem Determination panel provides the following actions:

**Activate the Hierarchy for the resource**

Attempts to activate the hierarchy of the resource starting from the highest inactive resource to the resource you entered.

**Force Recycle the resource**

Attempts to deactivate, then activate the resource you entered.

**SNBU Connect PU**

Takes you to SNBU so you can initiate the dial backup connection.

See the following sections for more information about these options.

**Activating the hierarchy**

You can use the Problem Determination panel to activate the LU and the other resources in its hierarchy (the PU and the line). To do this, follow these steps:

1. Type **1** in the entry field shown in [Figure 71 on page 83](#).
2. Press **Enter**.

AON/SNA attempts to activate all of the resources in the hierarchy, starting from the highest inactive resource continuing down to the resource you entered on the panel.

```

FKVK1111          SNA Help Desk: LU   :  RL523A1          CNM01

                Problem Determination:

Resource RL523A1  on Domain CNM01  is not active and
                  not in CONCT state.

Choose one of the following options or Press F12 to Return:

    1 1.  Activate the Hierarchy for the resource
    2 2.  Force Recycle the resource
    3 3.  SNBU Connect PU

Command ==>
F1=Help      F2=Main Menu   F3=Return

                                F6=Roll
                                F12=Cancel

```

*Figure 71. Activating the Hierarchy for the Resource*

AON/SNA cannot always resolve the problem by activating the hierarchy for the resource. [Figure 72 on page 83](#) shows a message indicating that AON/SNA did not activate the hierarchy for RL523A1.

```

FKVK1111          SNA Help Desk: LU   :  RL523A1          CNM01

                Problem Determination:

Resource RL523A1  on Domain CNM01  is not active and
                  not in CONCT state.

Choose one of the following options or Press F12 to Return:

    _ 1.  Activate the Hierarchy for the resource
    2 2.  Force Recycle the resource
    3 3.  SNBU Connect PU

FKV339I UNABLE TO ACTIVATE THE HIERARCHY FOR RL523A1. SENSECODE=08220000
Command ==>
F1=Help      F2=Main Menu   F3=Return

                                F6=Roll
                                F12=Cancel

```

*Figure 72. Unable to Activate Message on the Problem Determination Panel*

If AON/SNA did not activate the resource, try the **Force Recycle the resource** option (option 2) on the Problem Determination panel. To force the recycle of a resource on the Problem Determination panel as shown in the panel in [Figure 71 on page 83](#):

1. Type **2** in the entry field.
2. Press **Enter**.

If AON/SNA successfully activates the resource, it returns a message similar to the one that is shown in boldface in [Figure 66 on page 79](#). If AON/SNA cannot recycle the resource, it displays a message similar to the following:

```
EZL334I  UNABLE TO VARY restype resname ON domain ACTIVE
```

The Operator Command Interface: SNA Help Desk panel enables you to issue commands for a resource. To issue commands, press the F4 function key. AON/SNA displays a pop-up command window that lists the commands you can issue for the resource you select. To issue a command for a resource:

- AON/SNA displays a pop-up command window with the commands you can issue for that resource. Figure 73 on page 84 shows the command window for the TA1PT209 resource.

Figure 73. Operator Command Interface: SNA Help Desk panel with Pop-up Command Window

- ## Recycle *resname*

## AutoView

## SNAMAP

## DDF Details

## Automation commands

## NetView commands

4. Press **Enter**.

AON/SNA issues the command that you selected.



## Using the Automation Commands

You can use the Automation Commands panel to issue commands that control automation. You can access this panel by selecting Automation Commands from a pop-up command window shown in [Figure 73 on page 84](#). To issue automation commands:

1. Type **6** in the entry field on the pop-up command window.
2. Press **Enter**.

AON/SNA displays the SNA Help Desk Automation Commands panel shown in [Figure 74 on page 85](#).

```
FKVK1ACM          SNA Help Desk          CNM01
                   Automation Commands

Current Resource Name: TA1PT209

Select a command:

- 1. Automation Flag
  2. Thresholds
  3. Monitor Intervals
  4. Active Monitoring
  5. SNBU Connect
  6. Change SNBU Speed
  7. Automation Definition
  8. Status History
  9. LUDRPOOL

Command ===>
F1=Help      F2=Main Menu  F3=Return
                                     F6=Roll
                                     F12=Cancel
```

*Figure 74. Using the SNA Help Desk Automation Commands panel*

3. Type the number of the command you want in the entry field. You can select one of the following commands:

### **Automation Flag**

Enables you to set, change, or delete the automation recovery settings in the control file.

### **Thresholds**

Enables you to set, change, or delete the threshold settings in the control file.

### **Monitor Intervals**

Enables you to set or change the monitoring intervals.

### **Active Monitoring**

Enables you to set or change active monitoring.

### **SNBU Connect**

Enables you to change the SNBU connection, if SNBU is one of your resources.

### **Change SNBU Speed**

Enables you to change the modem speed, if SNBU is one of your resources.

### **Automation Definition**

Displays the configuration data for this resource.

### **Status History**

Displays all the status information for this resource.

### **LUDRPOOL**

Displays X.25 results from the LUDRPOOL command.

4. Press **Enter**.

AON/SNA issues the command that you selected.

## Issuing NetView Commands

You can issue six different NetView commands from the NetView Commands panel. You can access this panel by selecting NetView Commands from a pop-up command window shown in [Figure 73 on page 84](#). To issue NetView commands:

1. Type **7** in the entry field on the pop-up commands window.
2. Press **Enter**.

AON/SNA displays the SNA Help Desk NetView Commands panel shown in [Figure 75 on page 86](#).

```
FKVK1NCM          SNA Help Desk          CNM01
                   NetView Commands

Current Resource Name: TA1PT209
Resource Domain      : CNM01

Select a command:

  _ 1. Hardware Monitor
    2. Session Monitor
    3. Timer
    4. Activate
    5. Inactivate
    6. Help Status  ACTIV--L--

Command ===>
F1=Help      F2=Main Menu  F3=Return
                                     F6=Roll
                                     F12=Cancel
```

*Figure 75. Using the SNA Help Desk NetView Commands panel*

3. Type the number of the command you want in the entry field. You can select one of the following commands:

### Hardware Monitor

Displays the most recent events for the resource.

### Session Monitor

Displays the summary information for the latest response time.

### Timer

Enables you to set or change the timers for the resource.

### Activate

Attempts to activate the resource.

### Inactivate

Attempts to deactivate the resource.

### Help Status

Shows the status of the resource that is displayed in the **Current Resource Name** field on the NetView Commands panel.

4. Press **Enter**.

AON/SNA issues the command that you selected.

## Using NetView Access Services (NVAS)

NetView Access Services (NVAS) provides simultaneous access to one or several applications from a single terminal, using one user ID and password. The applications include any of the following:

- System (for example, CICS®)
- Subsystem (for example, TSO/E)

- Application
- Transaction within a system

AON/SNA supports only the relay mode sessions for NetView Access Services (NVAS).

To select NetView Access Services from the SNA Automation: Help Desk panel shown in [Figure 76 on page 87](#):

1. Type your NVAS user ID in the **Enter name** field.
2. Type **3** in the entry field.
3. Press **Enter**.

```
FKVK1000                                SNA Automation: Help Desk

**** * ** ** ** ** ** ** ** ** ** ** ** ** ** ** ** ** ** **   ** **          ***          *****          **
** * ** ** ** ***** ** **          **          **          **          **
**   *** ** ** **   ** ** ** ***** **          ***** **          ***** ** **
**   ***** ***** **          ***** ** ** ** ** ** ** ** ** ** ** ** ** ** ** ** ** ** **   ***
** ** ** ** ** ** ** **   **   ** ** ***** **          ** ** ** ** ** ** **   ** ***** **   ****
*   ** ** ** ** **   ** ** **   **   ** ** **          **   ** ** ** **   ** ***** **   ****
**** **   * ** **   **   ** ** ***** **          ***** ***** ***** **   **

Enter name...: OPER1__

Select option:   3 1. Recycle resource
                  2. Problem Determination
                  3. NetView Access Services User ID
                  n. NetView Help Desk

Command ==>
F1=Help      F2=Main Menu  F3=Return

F6=Roll
F12=Cancel
```

Figure 76. Selecting the Option for NetView Access Services (NVAS) Problems

AON/SNA displays the SNA Help Desk panel for NetView Access Services shown in [Figure 77 on page 87](#).

```
FKVKCHP1                                Operator Command Interface: SNA Help Desk                                CNM01
                                         NetView Access Services: EMS01

USERID= OPER1                                GROUP= PROD
Enter a Non-blank character next to selection to Cancel. Then press enter.

Terminal Address                                Relay LUs      Application LUs  ACB Type
_ TA1TT170                                - EMS01F01     AON01026        I
                                           - EMS01F03     AON06012        U
                                           - EMS01F01     AON05012        I
                                           - EMS01F02     AON04008        U

Command ==>
F1=Help      F2=Main Menu  F3=Return      F5=Refresh      F6=Roll
F7=Backward  F8=Forward                                F12=Cancel
```

Figure 77. SNA Help Desk panel for NetView Access Services

The SNA Help Desk panel for NetView Access Services lists all available applications that are active for your user ID. The SNA Help Desk panel for NetView Access Services can display up to 99 applications for

a single NetView Access Services user ID. You can use the F7 and F8 function keys to look at all of the applications.

The SNA Help Desk panel for NetView Access Services displays information about the ID and what LUs the ID is logged on to. From here, you can cancel the NetView Access Services user ID. This action cancels one or more of the applications under the NetView Access Services user ID.

The SNA Help Desk panel for NetView Access Services provides the following information for the user ID:

**Terminal Address**

The address on which the NetView Access Services user ID is logged on. If the address shows as DISCONNECTED, the ID is disconnected. However, AON/SNA might still run applications that you originally logged on to by the user ID.

**Relay LU**

The pseudo terminal name given when an application is selected by the NetView Access Services user ID. If the name shows as NO SESSIONS, the NetView Access Services user ID is logged on and is currently on the selection list.

**Application LU**

The primary logical unit for the application. The naming conventions are important to help identify the application.

**ACB Type**

Explains the type of session the ID is logged on to. The settings are:

**U**

Unique

**I**

Individual

**S**

Shared

A shared session is a special case because it uses the same relay LU as the other NetView Access Services user ID. To cancel a relay LU that is shared, AON/SNA cancels the terminal address instead. If AON/SNA cancels a shared user ID, the user of that ID can log on to NetView Access Services with that ID and issue the LF command against the application that is stopped.

**Canceling a NetView Access Services user ID**

To cancel a NetView Access Services user ID and all NetView Access Services applications:

1. Type a non-blank character in the entry field next to the terminal address.
2. Press **Enter**.

AON/SNA cancels the NetView Access Services user ID and all the NetView Access Services applications.

To cancel one or more applications running under the NetView Access Services user ID:

1. Press the **Tab** key to move the cursor to the **Relay LUs** column.
2. Type a non-blank character in the entry field next to the relay LUs and applications you want to cancel.
3. Press **Enter**.

AON/SNA cancels the applications you selected.

To cancel any other type of ACB session:

1. Type a non-blank character in the entry field next to the terminal address.
2. Press **Enter**.

AON/SNA cancels the ACB session you selected.

**Attention:** AON/SNA does not display a confirmation panel before canceling the NetView Access Services user ID. Use this function carefully.

## Chapter 11. Using SNAMAP

SNAMAP provides a tool to view a list of all the resources on a domain. You create the list by selecting one of the following resource types:

- Major nodes
- Applications
- Cross-domain resource managers
- Cross-domain resources
- Link stations
- Cluster controllers
- Terminals
- User entered resource name

SNAMAP displays lower connected nodes. This contrasts with the SNA Help Desk which provides a view of a resource and its connected higher nodes.

```
FKVK0000          SNA Automation: Menu          CNM01

Select an option

  2  0.  Tutorial
    1.  Help Desk
    2.  SNAMAP
    3.  VTAM Options Management
    4.  NetStat
    5.  VTAM Commands
    6.  APPN menu
    7.  Switched Network Backup menu
    8.  X.25 menu
    9.  NCP Recovery Definitions (NCP name=_____)

Command ===>
F1=Help      F2=Main Menu  F3=Return

                                F6=Roll
                                F12=Cancel
```

Figure 78. Selecting SNAMAP on the SNA Automation: Menu panel

To use SNAMAP:

1. Type **2** in the entry field on the **SNA Automation: Menu** panel shown in [Figure 78 on page 89](#).
2. Press **Enter**.

AON/SNA displays the SNA Automation: SNAMAP panel shown in [Figure 79 on page 90](#).

**Note:** You can also get to the SNA Automation: SNAMAP panel by typing **AON 2.2** or **SNAMAP** on any command line and pressing **Enter**.

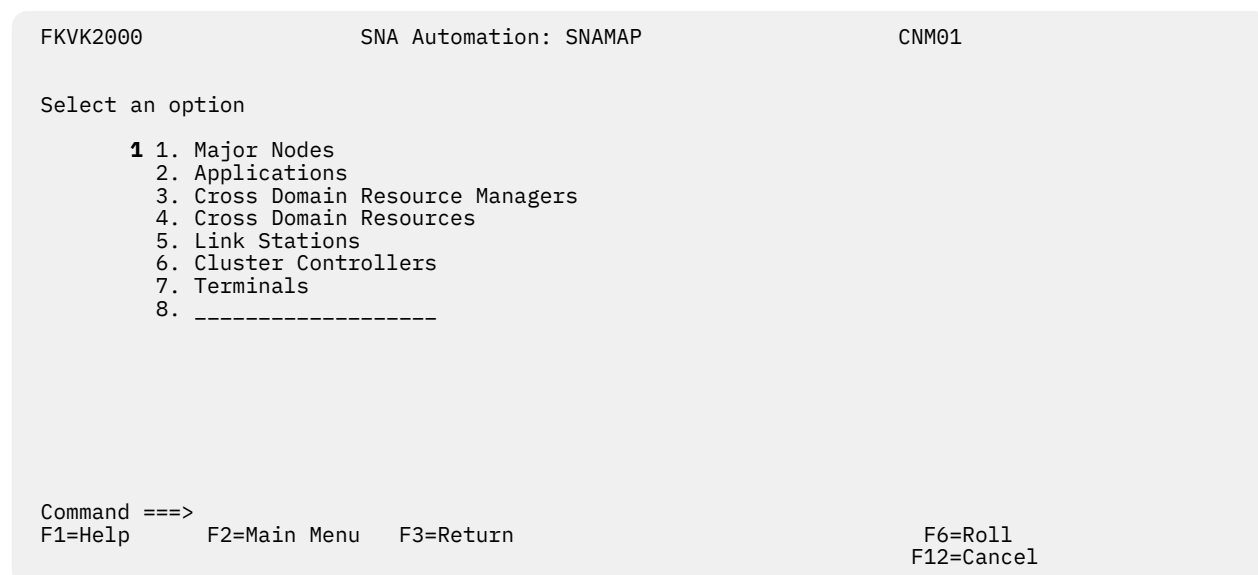


Figure 79. SNA Automation: SNAMAP panel

3. Type the number of the option you want in the entry field. This example selects option 1 (Major Nodes) shown in [Figure 79 on page 90](#).
4. Press **Enter**.

AON/SNA displays the Operator Command Interface: SNAMAP panel shown in [Figure 80 on page 90](#).

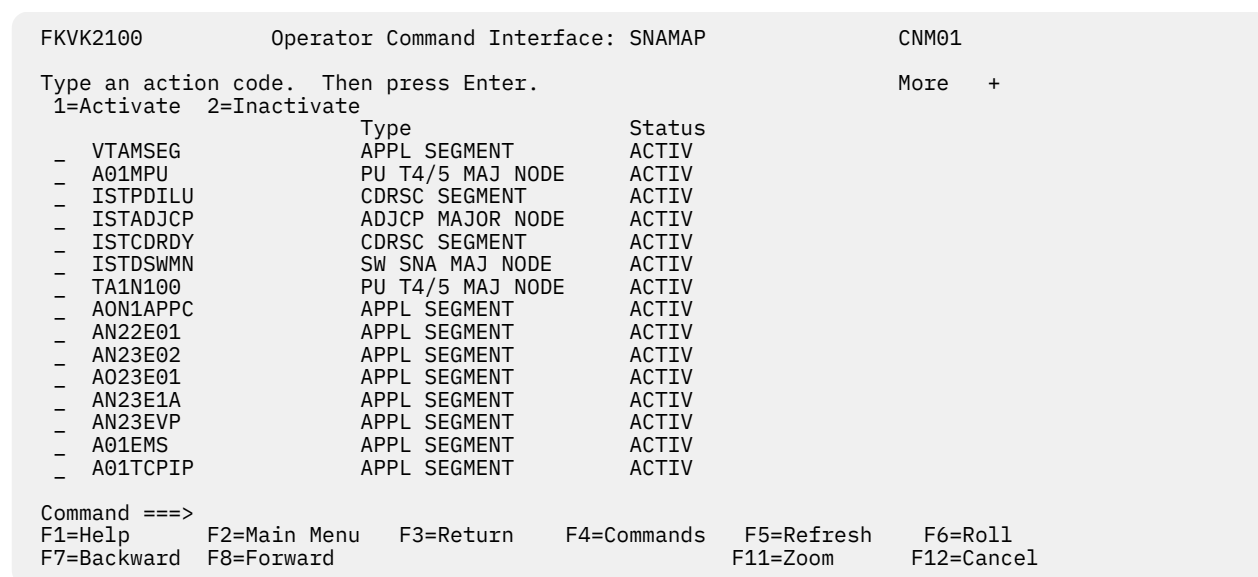


Figure 80. Operator Command Interface: SNAMAP panel

The Operator Command Interface: SNAMAP panel lists the available resources and shows the type and status of each resource. Press the F7 and F8 function keys to scroll through the list of resources if they are displayed on more than one panel.

You can use one of the following action codes on a specified resource:

**1=Activate**

Activates the resource you specify

**2=Inactivate**

Deactivates the resource you specify

See the following sections for more information about activating and deactivating a resource.

## Activating resources

You can activate a specific resource from the Operator Command Interface: SNAMAP panel. For example, to activate the AN23E02 resource:

1. Press the **Tab** key to move the cursor to the AN23E02 resource, as shown on the Operator Command Interface: SNAMAP panel in [Figure 80 on page 90](#).
2. Type **1** in the entry field next to the AN23E02 resource.
3. Press **Enter**.

AON/SNA displays a message indicating that the resource is now active, as shown in [Figure 81 on page 91](#).

```
FKVK2100          Operator Command Interface: SNAMAP          CNM01
Type an action code. Then press Enter.          More:  +
  1=Activate  2=Inactivate
Type                                     Type                                     Status
- VTAMSEG                                     APPL SEGMENT                                     ACTIV
- A01MPU                                     PU T4/5 MAJ NODE                                 ACTIV
- ISTPDILU                                    CDRSC SEGMENT                                    ACTIV
- ISTADJCP                                    ADJCP MAJOR NODE                                ACTIV
- ISTCDRDY                                    CDRSC SEGMENT                                    ACTIV
- TA1N100                                    PU T4/5 MAJ NODE                                ACTIV
- AON1APPC                                    APPL SEGMENT                                    ACTIV
- AN22E01                                    APPL SEGMENT                                    ACTIV
* AN23E02                                    APPL SEGMENT                                    ACTIV
- A023E01                                    APPL SEGMENT                                    ACTIV
- AN23E1A                                    APPL SEGMENT                                    ACTIV
- A01EMS                                     APPL SEGMENT                                    ACTIV
- A01TCPIP                                    APPL SEGMENT                                    ACTIV
- A01TSO                                     APPL SEGMENT                                    ACTIV
- CDRM01                                    CDRM SEGMENT                                    ACTIV
IST1132I AN23E02          IS ACTIVE, TYPE = APPL SEGMENT
Command ==>
F1=Help      F2=Main Menu  F3=Return   F4=Commands  F5=Refresh  F6=Roll
F7=Backward  F8=Forward   F11=Zoom   F12=Cancel
```

Figure 81. Activate Message

## Deactivating resources

You can deactivate a specific resource from the Operator Command Interface: SNAMAP panel. For example, to deactivate the AON1APPC resource:

1. Press the **Tab** key to move the cursor to the AON1APPC resource, as shown on the Operator Command Interface: SNAMAP panel in [Figure 80 on page 90](#).
2. Type **2** in the entry field next to the AON1APPC resource.
3. Press **Enter**.

AON/SNA displays a message indicating that the resource is now inactive, as shown in [Figure 82 on page 92](#).

```

FKVK2100          Operator Command Interface: SNAMAP          CNM01

Type an action code. Then press Enter.          More:  +
1=Activate 2=Inactivate

- VTAMSEG          APPL SEGMENT          ACTIV
- A01MPU           PU T4/5 MAJ NODE      ACTIV
- ISTDILU          CDRSC SEGMENT         ACTIV
- ISTDJCP          ADJCP MAJOR NODE      ACTIV
- ISTDY           CDRSC SEGMENT         ACTIV
- ISTDW           SW SNA MAJ NODE        ACTIV
- TA1N100          PU T4/5 MAJ NODE      ACTIV
* AON1APPC         APPL SEGMENT          ACTIV
- AN22E01          APPL SEGMENT          ACTIV
- AN23E02          APPL SEGMENT          ACTIV
- A023E01          APPL SEGMENT          ACTIV
- AN23E1A          APPL SEGMENT          ACTIV
- A01EMS           APPL SEGMENT          ACTIV
- A01TCPIP         APPL SEGMENT          ACTIV
- A01TSO           APPL SEGMENT          ACTIV
IST1133I AON1APPC      IS NOW INACTIVE, TYPE = APPL SEGMENT
Command ==>
F1=Help      F2=Main Menu  F3=Return  F4=Commands  F5=Refresh  F6=Roll
F7=Backward  F8=Forward   F11=Zoom   F12=Cancel

```

Figure 82. Deactivate Message

4. Press **F5** to refresh the panel.

You can also press the **F4** function key to display the pop-up command window, or the **F11** function key to display the next level of information for a resource. Both of these function keys are cursor sensitive and display the information for the resource at the position of the cursor. The following sections describe the use of these keys.

## Viewing lower connected nodes

You can use the F11 function key to view the connected lower nodes for a specific resource. AON/SNA labels this function key as a zoom key. To see the connected lower nodes:

1. On the **Operator Command Interface: SNAMAP** panel, press the **Tab** key to move the cursor to the resource you want. This example shows the TA1N100 resource.
2. Press **F11**.

AON/SNA displays the lower connected nodes for the TA1N100 resource in a pop-up window on the right side of the panel shown in Figure 83 on page 92.

```

FKVK2110          Operator Command Interface: SNAMAP          CNM01

Type an action code. Then press Enter.          More  +
1=Activate 2=Inactivate

- VTAMSEG          : Type an action code. Then press Enter.      More:  +
- A01MPU           : 1=Activate 2=Inactivate                      :
- ISTDILU          :                                          Type      Status
- ISTDJCP          : - TA07LNPA          LINES          ACTIV----T
- ISTDY           : - TA07L000          LINES          NEVAC
- ISTDW           : - TA07L004          LINES          NEVAC
- TA1N100          : - L07CA000          LINES          NEVAC
- AON1APPC         : - L07CA001          LINES          NEVAC
- AN22E01          : - L07CA002          LINES          NEVAC
- AN23E02          : - L07CA003          LINES          NEVAC
- A023E01          : - L07CA004          LINES          NEVAC
- AN23E1A          :
- AN23EVP          : F1=Help      F3=Return  F4=Commands  F5=Refresh  F6=Roll
- A01EMS           : F7=Backward F8=Forward          F11=Zoom   F12=Cancel
- A01TCPIP         :
Command ==>
F1=Help      F2=Main Menu  F3=Return  F4=Commands  F5=Refresh  F6=Roll
F7=Backward  F8=Forward   F11=Zoom   F12=Cancel

```

Figure 83. Viewing lower connected nodes



You can activate or deactivate the resources listed in the pop-up window. For more information about activating and deactivating the resources, see [“Activating resources” on page 91](#) and [“Deactivating resources” on page 91](#).

## Using SNAMAP pop-up commands

You can use the F4 function key to display a list of pop-up commands that you can issue for a specified resource. For example, you might want to display the status history of a resource before you activate it.

To display the command list and issue a command:

1. On the **Operator Command Interface: SNAMAP** panel, press the **Tab** key to move the cursor to the resource you want. This example shows the TA07LNPA resource.
2. Press **F4**.

AON/SNA displays the commands you can use in a pop-up window on the panel as shown in [Figure 84 on page 93](#).

```

FKVK2111          Operator Command Interface: SNAMAP          CNM01
Type an action code. Then press Enter.                      More +
1=Activate 2=Inactivate                                     : ID: TA07LNPA :
VTAMSEG : Type an action code. Then pre : More: + :
A01MPU : 1=Activate 2=Inactivate : :
ISTPDILU : : Type : 5 1. Activate :
ISTADJCP : TA07LNPA : LINES : 2. AutoView :
ISTCDRDY : TA07L000 : LINES : 3. Display :
ISTDSWMN : TA07L004 : LINES : 4. Display Config :
TA1N100 : L07CA000 : LINES : 5. Display History :
A0N1APPC : L07CA001 : LINES : 6. Help Desk :
AN22E01 : L07CA002 : LINES : 7. Inactivate :
AN23E02 : L07CA003 : LINES : :
A023E01 : L07CA004 : LINES : F1=Help F12=Cancel :
AN23E1A : : : :
AN23EVP : F1=Help F3=Return F4=Commands F5=Refresh F6=Roll :
A01EMS : F7=Backward F8=Forward : F11=Zoom F12=Cancel :
A01TCPIP : : : :
Command ===>
F1=Help F2=Main Menu F3=Return F4=Commands F5=Refresh F6=Roll
F7=Backward F8=Forward F11=Zoom F12=Cancel

```

Figure 84. Displaying the Command List Pop-up panel

3. Type the number of the command you want to issue in the entry field in the pop-up command window. In this example, you can issue one of the following commands:

### Activate

Activates the resource.

### AutoView

Shows all the automation settings for the resource and enables you to change the settings.

### Display

Displays the details about this resource.

### Display Config

Displays the control file for the resource.

### Display History

Displays the status file history for the resource.

### Help Desk

Displays the problem determination information from the SNA Help Desk.

### Inactivate

Deactivates the resource.

**Monitor**

Displays the monitoring intervals for the resource.

**Recovery**

Displays the recovery settings for the resource.

**Thresholds**

Displays the error threshold settings for the resource.

**Timer**

Displays the timer settings for the resource.

Press the **F7** and **F8** function keys to scroll through the list of commands.

4. Press **Enter**.

AON/SNA processes the command you selected.

## Chapter 12. Displaying network status

To display the status of specified resource groups, use the NetStat option. To check the status of your network, specify the type of resource and which resources of that type to display.

```
FKVK0000          SNA Automation: Menu          CNM01

Select an option

  4 0. Tutorial
    1. Help Desk
    2. SNAMAP
    3. VTAM Options Management
    4. NetStat
    5. VTAM Commands
    6. APPN menu
    7. Switched Network Backup menu
    8. X.25 menu
    9. NCP Recovery Definitions (NCP name=_____)

Command ==>
F1=Help      F2=Main Menu  F3=Return

F6=Roll
F12=Cancel
```

Figure 85. SNA Automation: Menu Panel – Selecting the NetStat option

To check the status of your network from the **SNA Automation: Menu** panel:

1. Type **4** in the entry field.
2. Press **Enter**.

AON/SNA displays the SNA Automation: NetStat panel shown in [Figure 86 on page 95](#).

**Note:** You can also display the NetStat panel by typing **AON 2.4** or **NETSTAT** on any command line and pressing **Enter**.

```
FKVK4000          SNA Automation: NetStat          CNM01

Select a Type      1 1. All
                   2. Physical resources
                   3. Applications
                   4. Cross Domain Resource Managers
                   5. Cross Domain Resources

Select Scope       2 1. All
                   2. Not active (EXCEPT)
                   3. _____

Check Automation: 1          (1=Yes 2=No)

Send to DDF.....: 2          (1=Yes 2=No)

Command ==>
F1=Help      F2=Main Menu  F3=Return

F6=Roll
F12=Cancel
```

Figure 86. SNA Automation: NetStat panel

3. Type the number of the resource type you want in the **Select a Type** entry field. You can select one of the following options:

**All**

Displays all known resources. The default is All.

**Physical resources**

Displays only the physical resources. For example, this option displays NCPs, lines, PUs, and LUs.

**Applications**

Displays all of the applications.

**Cross Domain Resource Managers**

Displays all of the cross-domain resource managers.

**Cross Domain Resources**

Displays all the cross-domain resources.

4. Type the number of the scope you want in the **Select Scope** entry field. You can select one of the following options:

**All**

Displays all resources regardless of their status.

**Not active (EXCEPT)**

Displays all resources currently not in an active state. The default is EXCEPT.

**User-defined field**

Displays the scope of resources you specify. If you do not want to display resources with a certain status, type an ~ before the status.

5. Specify whether you want to check automation. You can specify one of the following values:

**1=Yes**

Checks the recovery setting for the resource before it displays the resource. If you specify Yes and recovery is turned off, the resource is not displayed. This is the default.

**2=No**

Does not check automation.

6. Specify if you want to send the resource to DDF. You can specify one of the following values:

**1=Yes**

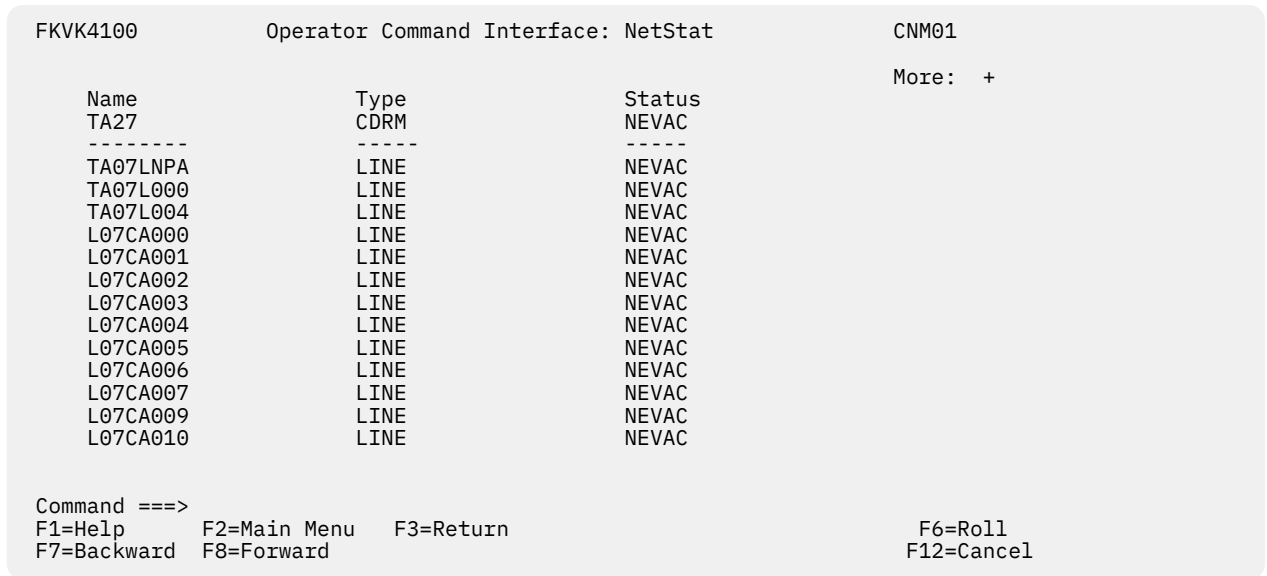
Updates DDF with the status of the resource you display. Specify Yes to reset the status of the resource or reload DDF.

**2=No**

Does not update DDF with the status of the resource. The default is No.

7. Press **Enter**.

AON/SNA displays the status of the resources you specified on the Operator Command Interface: NetStat panel shown in [Figure 87 on page 97](#).



*Figure 87. Operator Command Interface: NetStat Panel*

**Note:** For information about the programmatic interface to the AON NETSTAT function, see the SNA resource automation information, see [“SNA Resource Automation \(FKVESYNC\)”](#) on page 322.



## Chapter 13. Issuing VTAM commands

To issue VTAM commands and see the results of the commands on a panel, use the VTAM commands option on the SNA Automation: Menu panel or the VTAMCMD command. The VTAM commands option saves commands across user task sessions. If any command is left on the panel when you exit the panel, it is displayed in the same place when you return to the panel. This is helpful if you have a small set of frequently used commands, and you do not want to look up the syntax of a command each time you use it.

The VTAM command option is cursor sensitive. If you have several commands on the panel, AON/SNA issues the command at the position of the cursor. You can issue a command by typing a new command and pressing **Enter**, or use the **Tab** key to move the cursor to the command you want to select and press **Enter**.

```
FKVK0000          SNA Automation: Menu          CNM01

Select an option

  5 0.  Tutorial
    1.  Help Desk
    2.  SNAMAP
    3.  VTAM Options Management
    4.  NetStat
    5.  VTAM Commands
    6.  APPN menu
    7.  Switched Network Backup menu
    8.  X.25 menu
    9.  NCP Recovery Definitions (NCP name=_____)

Command ==>
F1=Help      F2=Main Menu  F3=Return

                                F6=Roll
                                F12=Cancel
```

Figure 88. Selecting the VTAM Commands option

To issue VTAM commands from the SNA Automation: Menu panel:

1. Type **5** in the entry field.
2. Press **Enter**.

AON/SNA displays the SNA Automation: VTAM Commands panel shown in [Figure 89 on page 100](#).

**Note:** You can also get to the VTAM Commands panel by typing **AON 2.5** or **VTAMCMD** on any command line and pressing **Enter**.

Type in a VTAM Command or Tab to a Command. Then press Enter.

```
F NET,CHKPT,TYPE=ALL_____
ID=TA1PT209_____
TOPO_____
D NET,TOPO,LIST=SUMMARY_____
D NET,DIRECTRY,ID=USIBMTA.TA01_____
D NET,TOPO,LIST=ICN_____
D NET,ADJCP,ID=*______
ID=TA1T1048,E_____
ID=CICS01A,E_____
```

NOTE: If the word NET does not occur in the command,  
D NET is assumed.

Command ==>

F1=Help F2=Main Menu F3=Return

F6=Roll  
F12=Cancel

Figure 89. SNA Automation: VTAM Commands panel

When you select this panel for the first time, the command input fields are blank. However, because AON/SNA saves the commands across user sessions, including NetView sessions, the panel fills up with the commands you use. This saves time if you use a few commands frequently.

You can shorten the commands on this panel. For example, you can enter the D NET,TOPO,LIST=ICN command as TOPO,LIST=ICN because AON/SNA assumes that you want to use D NET if you do not specify the word NET.

To change a command, use the Tab key to move the cursor to the command and type over it.

To add a VTAM command to this panel and issue it:

1. Type the command in the entry field. You can type only one command on each line.
2. Press **Enter**.

AON/SNA issues the command and displays the Operator Command Interface: VTAM Commands panel shown in Figure 90 on page 100. In this example, the DISPLAY NET, ID=TA1T1048 command response is displayed.

Output of: D NET,ID=TA1T1048,E

More: +

```
IST097I  DISPLAY  ACCEPTED
IST075I  NAME = USIBMTA.TA1T1048 , TYPE = CDRSC
IST486I  STATUS= ACT/S---Y, DESIRED STATE= ACTIV
IST977I  MDLTAB=***NA*** ASLTAB=***NA***
IST861I  MODETAB=***NA*** USSTAB=***NA*** LOGTAB=***NA***
IST934I  DLOGMOD=***NA*** USS LANGTAB=***NA***
IST597I  CAPABILITY-PLU ENABLED ,SLU ENABLED ,SESSION LIMIT NONE
IST231I  CDRSC MAJOR NODE = ISTCDRDY
IST479I  CDRM NAME = TA06 , VERIFY OWNER = NO
IST1131I DEVICE = CDRSC
IST654I  I/O TRACE = OFF, BUFFER TRACE = OFF
IST171I  ACTIVE SESSIONS = 0000000001, SESSION REQUESTS = 0000000000
IST206I  SESSIONS:
IST634I  NAME STATUS SID SEND RECV VR TP NETID
IST635I  CNM01024 ACTIV-P F9DB431A7D93BAD8 000D 0024 0 1 USIBMTA
```

Command ==>

F1=Help F2=Main Menu F3=Return  
F7=Backward F8=Forward

F6=Roll  
F12=Cancel

Figure 90. Operator Command Interface: VTAM Commands panel



The output from the command is displayed on this panel. If the output is more than one panel long, you can scroll through the panels to see all of the information.

To issue previously saved commands:

1. Press the **Tab** key to move the cursor to the command you want to issue.
2. Press **Enter**.

AON/SNA issues the command and displays the output on the Operator Command Interface: VTAM Commands panel.



## Chapter 14. Using Advanced Peer-to-Peer Networking (APPN)

AON/SNA Advanced Peer-to-Peer Networking (APPN) is a powerful, flexible, easy-to-use networking solution for client-server and distributed applications.

In an AON/SNA APPN environment, AON provides menu-driven commands to simplify VTAM topology and directory database management. This environment accepts operator commands for common AON/SNA APPN VTAM functions. It also provides active monitoring of control points and control point sessions.

You can use the AON/SNA: APPN Command Menu panel to perform the following APPN functions:

- Issue checkpoint commands
- Display control points
- Display directory
- Display transmission group profiles

```
FKVK0000          SNA Automation: Menu          CNM01

Select an option

  6 0.  Tutorial
    1.  Help Desk
    2.  SNAMAP
    3.  VTAM Options Management
    4.  NetStat
    5.  VTAM Commands
    6.  APPN menu
    7.  Switched Network Backup menu
    8.  X.25 menu
    9.  NCP Recovery Definitions (NCP name=_____)

Command ==>
F1=Help      F2=Main Menu  F3=Return

                                F6=Roll
                                F12=Cancel
```

*Figure 91. SNA Automation: Menu Panel – Selecting the APPN Menu Option*

To use the APPN functions from the **SNA Automation: Menu** panel:

1. Type **6** in the entry field.
2. Press **Enter**.

AON/SNA displays the SNA Automation: APPN Commands Menu panel, which is shown in [Figure 92 on page 104](#).

**Note:** You can also get to the APPN Commands Menu panel by typing **AON 2.6** or **APPN** on any command line and pressing **Enter**.

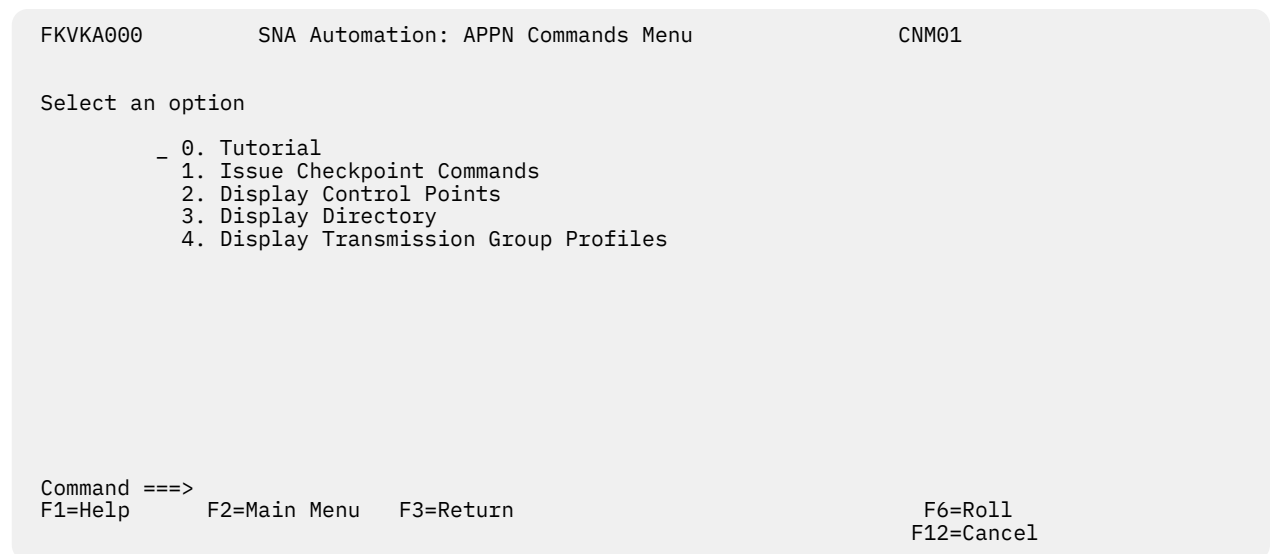


Figure 92. SNA Automation: APPN Commands Menu Panel

## Displaying Control Points

You can select the display control points command from the SNA Automation: APPN Commands Menu panel. After you make this selection, you can select from the following actions for the resource:

- Details
- Delete topology
- Delete directory
- Active monitoring
- Timers
- AutoView

You can use the SNA Automation: APPN Control Points Display panel to identify control points to specify in your statements in the CPCPSESS control file entry.

You can use the Display Control Points option on the SNA Automation: APPN Commands Menu panel to work with the control points on your APPN network. To do this:

1. Type **2** in the entry field on the **SNA Automation: APPN Commands Menu** panel. [Figure 92 on page 104](#) shows this panel.
2. Press **Enter**.

AON/SNA displays the APPN CP Display panel shown in [Figure 93 on page 105](#).

**Note:** You can also display the SNA Automation: APPN CP Display panel, type **AON 2.6.2** on any command line and pressing **Enter**.

```

FKVKA200          SNA Automation: APPN CP Display          CNM01

Type an action code. Then press Enter.          More: +
1=Details 2=Delete Topology 3=Delete Directory 4=Active Monitoring
5=Timers 6=AutoView
Control Point      Node Type
1 ISTADJCP        ADJCP MAJOR NODE
- USIBMTA.TA1PT106  EN
- TA1CP213          *NA*
- TA1CP214          *NA*
- USIBMTA1.OPER1    EN
- USIBMTA.NTC0PUN6  *NA*
- USIBMTA.TA1CP210  EN
- APPN.TA1PT209     EN
- USIBMXXX.YYY00000 EN
- USIBMTA.TA1PT107  EN
- USIBMTA.TA1PT220  EN
- USIBMTA.TA1CP207  NN
- USIBMTA.TA1PT203  EN
- TA1CP208          *NA*

Command ==>
F1=Help      F2=Main Menu  F3=Return          F5=Refresh  F6=Roll
F7=Backward  F8=Forward    F12=Cancel

```

Figure 93. SNA Automation: APPN CP Display Panel

3. Type an action code in the entry field next to the resource you want. This example shows the Details action code next to the ISTADJCP control point. You can select one of the following actions:

#### 1=Details

Displays detailed VTAM information for the control point on the Operator Command Interface: APPN CP Detail panel. This is obtained from the combined output of the VTAM D NET,E commands and its link station. The output includes both session and LU information, which is helpful in solving problems. The status of the link station is often a clue to physical network problems, while the control point status information shows configuration or application program problems.

#### 2=Delete topology

Deletes the control point from the topology database. Displays the messages about the deletion on the Operator Command Interface: VTAM commands panel.

#### 3=Delete directory

Deletes the control point from the directory database. Displays the messages about the deletion on the Operator Command Interface: VTAM commands panel.

#### 4=Active monitoring

Starts or stops active monitoring of the control point. AON/SNA displays the Active Monitoring Settings panel and enables you to change the settings.

#### 5=Timers

Displays the AON Automation: Timer Set panel where you can add, display, change, and delete the timers for the control point you select.

#### 6=AutoView

Displays automation information for the control point. Displays the AON: AutoView panel.

4. Press **Enter**.

AON/SNA displays the Operator Command Interface: APPN CP Detail panel shown in [Figure 94 on page 106](#).

```

FKVKA210          Operator Command Interface: APPN CP Detail          CNM01

                  Control Point ISTDJCP                                More:  +

IST097I  DISPLAY  ACCEPTED
IST075I  NAME = ISTDJCP          , TYPE = ADJCP MAJOR NODE
IST486I  STATUS= ACTIV          , DESIRED STATE= ACTIV
IST1100I  ADJACENT CONTROL POINTS FROM MAJOR NODE ISTDJCP
IST1102I  NODENAME          NODETYPE CONNECTIONS CP CONNECTIONS
IST1103I  USIBMTA.TA1PT106    EN          1          1
IST1103I  TA1CP213            *NA*        0          0
IST1103I  TA1CP214            *NA*        0          0
IST1103I  USIBMTA1.DOWNING    EN          1          1
IST1103I  USIBMTA.NTC0PUN6    *NA*        0          0
IST1103I  USIBMTA.TA1CP210    EN          1          1
IST1103I  APPN.TA1PT209       EN          1          1
IST1103I  USIBMXXX.YYY00000    EN          1          1
IST1103I  USIBMTA.TA1PT107    EN          1          1
IST1103I  USIBMTA.TA1PT220    EN          1          1
IST1103I  USIBMTA.TA1CP207    NN          1          1

Command ==>
F1=Help      F2=Main Menu   F3=Return          F6=Roll
F7=Backward  F8=Forward     F12=Cancel

```

Figure 94. Operator Command Interface: APPN CP Detail panel

## Chapter 15. NCP recovery definitions

To display the NCPRECOV Control File definitions for a particular NCP, or for all NCPs you have defined, use the NCP Recovery Definitions menu option.

To display all defined NCPs choose option 9. To display a particular NCP, choose option 9 and enter the name of the NCP in the **NCP name=** field on the panel.

Selecting the NCP Recovery Definition menu option causes the DSPCFG command to be issued based upon your NCP selection.

[Figure 95 on page 107](#) displays the SNA Automation: Menu panel.

```
FKVK0000          SNA Automation: Menu          CNM01

Select an option

  9 0.  Tutorial
    1.  Help Desk
    2.  SNAMAP
    3.  VTAM Options Management
    4.  NetStat
    5.  VTAM Commands
    6.  APPN menu
    7.  Switched Network Backup menu
    8.  X.25 menu
    9.  NCP Recovery Definitions (NCP name= _____ )

Command ==>
F1=Help      F2=Main Menu  F3=Return          F6=Roll
F12=Cancel
```

*Figure 95. Selecting NCP Recovery Definitions*

To access NCP Recovery Definitions from the SNA Automation: Menu panel:

1. Type 9 in the entry field.
2. Press **Enter**.

AON/SNA issues the DSPCFG command for all of your defined NCPs, which is shown in [Figure 96 on page 108](#).

```
EZLK7110          Display Configuration Data          CNM01

Select one of the following.  Then press Enter.
1=Add 2=Change 3=Delete

-  NCP01          NCPRECOV
-  HOST           CNM01
-  DUMP           (Y,N)
-  RELOAD         (Y,N)
-  LINKSTA        001-S
-  DUMPSTA        001-S
-  LOADTIME       00:05
-  DUMPTIME       00:10
-  EXIT01         FKVEX01
-  EXIT02         FKVEX02
-  EXIT03         FKVEX03
-  EXIT04         FKVEX04
-  LOADMOD        NCP01A

Command ==>
F1=Help      F2=Main Menu  F3=Return          F5=Refresh    F6=Roll
F7=Backward  F8=Forward          F12=Cancel
```

Figure 96. Selecting NCP Recovery Definitions

This is the DSPCFG panel for the NCP01 NCPRECOV Control File statement. In this example, this is the only NCP being automated on this system.



## Chapter 16. Displaying SNA resource information with AutoView

The AutoView function displays a summary of information for a single resource and provides a list of commands you can issue for more information about the resource and changing its automation settings. To use the AutoView panel, you must know the name of the resource with which you want to work.

To display the AutoView selection panel:

1. Type **SNAVIEW** on the command line of any panel.
2. Press **Enter**. A panel similar to the one shown in Figure 97 on page 109 is displayed. This panel is dynamically ordered according to your installation, so the order of the options on your panel might be different.

```
EZLK2000          AON: AutoView          NTVE1

Resource Name  _____

Resource Type  _____ (Optional)

Select an Option
(Optional) _  1. TCPIP
              2. IP390
              3. SNA
              4. APPN
              5. SNBU

Command ==>
F1=Help      F2=Main Menu  F3=Return

F6=Roll
F12=Cancel
```

Figure 97. AON: AutoView panel

3. Type the resource name in the **Resource Name** field.
4. Optionally, tab to the **Resource Type** field and type the resource type (such as LU, PU, NCP, or APPL).
5. Type **5** for SNA in the entry field. If you do not select an option, AON searches all of the automation components to find the resource.
6. Press **Enter**.

For example, to use AutoView for the TA1P523A resource, type **TA1P523A** in the **Resource Name** field, which is shown in Figure 98 on page 110.

|                                                                   |               |                       |
|-------------------------------------------------------------------|---------------|-----------------------|
| EZLK2000                                                          | AON: AutoView | CNM01                 |
| Resource Name <b>TA1P523A</b> _____                               |               |                       |
| Resource Type <b>PU</b> _____ (Optional)                          |               |                       |
| Select an Option                                                  |               |                       |
| (Optional) _ 1. TCPIP<br>2. IP390<br>3. SNA<br>4. APPN<br>5. SNBU |               |                       |
| Command ==>                                                       |               |                       |
| F1=Help                                                           | F2=Main Menu  | F6=Roll<br>F12=Cancel |
| F3=Return                                                         |               |                       |

Figure 98. AON: AutoView Panel Sample Entry

## Finding defined resources

Each automation component displays a different set of predefined information and a different list of commands. Because no particular automation component is selected in the example shown in [Figure 98 on page 110](#), AON searches all the automation components for the resource, TA1P523A. When the resource is defined to more than one automation component, AON lists all the places the resource is defined.

In [Figure 99 on page 110](#), AON found the resource, TA1P523A, defined as both a SNA and a SNBU resource.

|                                                                                                             |                          |                       |
|-------------------------------------------------------------------------------------------------------------|--------------------------|-----------------------|
| EZLKIDNT                                                                                                    | AON: Component Selection | CNM01                 |
| The resource chosen has valid definitions in the following installed options. Select the applicable option. |                          |                       |
| Resource Chosen TA1P523A                                                                                    |                          |                       |
| Select an Option                                                                                            |                          |                       |
| <b>1</b> 0. All<br>1. SNA<br>2. SNBU                                                                        |                          |                       |
| Command ==>                                                                                                 |                          |                       |
| F1=Help                                                                                                     | F2=Main Menu             | F6=Roll<br>F12=Cancel |
| F3=Return                                                                                                   |                          |                       |

Figure 99. AON: Component Selection panel

To select the SNA AutoView display:

1. Type **1** in the entry field of the **AON: Component Selection** panel.
2. Press **Enter**.

[Figure 100 on page 111](#) shows the SNA AutoView display for the resource, TA1P523A.

```

EZLKVIEW                      AON: AutoView                      CNM01

Resname . . . . . TA1P523A
Restype . . . . . PU
Option . . . . . SNA
Status . . . . . PCTD2
Automation Status . . . TREAT
Automation Flag . . . . ON
DDF message . . . . . NONE

Resource Definitions
- 1. Automation (RECOVERY DEFAULTS)
  2. Thresholds (THRESHOLDS DEFAULTS)
  3. Active Monitoring (ACTMON DEFAULTS)
  4. Monitor Intervals (MONIT DEFAULTS)
  5. Timer (NONE)
  6. Display Network LOG Information for TA1P523A

Command ==>
F1=Help      F2=Main Menu  F3=Return
F6=Roll
F12=Cancel

```

Figure 100. AON: AutoView panel

## Viewing resource information

The following resource information is available:

- 1** The first group of information provides a summary of information about the resource including the resource name, resource type, status, automation settings for the resource, DDF message for the resource, if any, and other information depending on the automation component that is used..
  - 2** The second group is a list of commands you can issue for the resource to display more information, change automation settings, or perform other functions depending on the automation component used.
  - 3** The third group of information (shown in parentheses) tells you which control file entry is currently defining the settings for the commands shown in the second group. For example, in [Figure 100 on page 111](#), the settings for option 1 (Automation) are defined by the RECOVERY DEFAULTS control file entry.
- For commands that do not have a corresponding control file entry, such as option 5 (Timer) the information in the parentheses indicates whether a setting exists. In [Figure 100 on page 111](#), there are no timers set for TA1P523A.

You can select one of the following resource definitions:

### Automation

Turns automation on and off.

### Thresholds

Adds, changes, or deletes the threshold settings. The settings are infrequent, frequent, and critical threshold.

### Active Monitoring

Indicates if the resource is active.

### Monitor Intervals

Sets the intervals for trying to reactivate a resource.

### Timer

Adds, changes, or deletes a timer.

**Display Network LOG Information**

Displays only the Netlog information for the resource.

---

## Part 3. Using AON with TCP/IP



---

## Chapter 17. Using the AON/TCP Operator Interface

Full-screen, menu-driven panels make up the operator interface for AON/TCP.

As an operator, you can do most of your work from within the AON/TCP operator interface. This topic introduces the operator interface and describes how to access it.

---

### Introducing the AON/TCP Operator Interface

Automated Operations Network (AON) provides a rich operator interface for managing TCP/IP resources through a z/OS (IP390 option) system.

You can manage TCP/IP resources through z/OS Communications Server IP. For example, you can use z/OS Communications Server IP to issue PING and TRACERTE commands. Also, you can manage Telnet and FTP sessions connecting to your z/OS system. Use the connection management function to display connection information, to determine connection status, and, if necessary, to take corrective actions. You can issue SNMP requests (for example, GET), enable TCP/IP tracing, monitor resources for specific performance thresholds, and automate and monitor critical sockets.

**Note:** Some of these z/OS functions no longer require AON. For more information, see the topics that describe these functions.

---

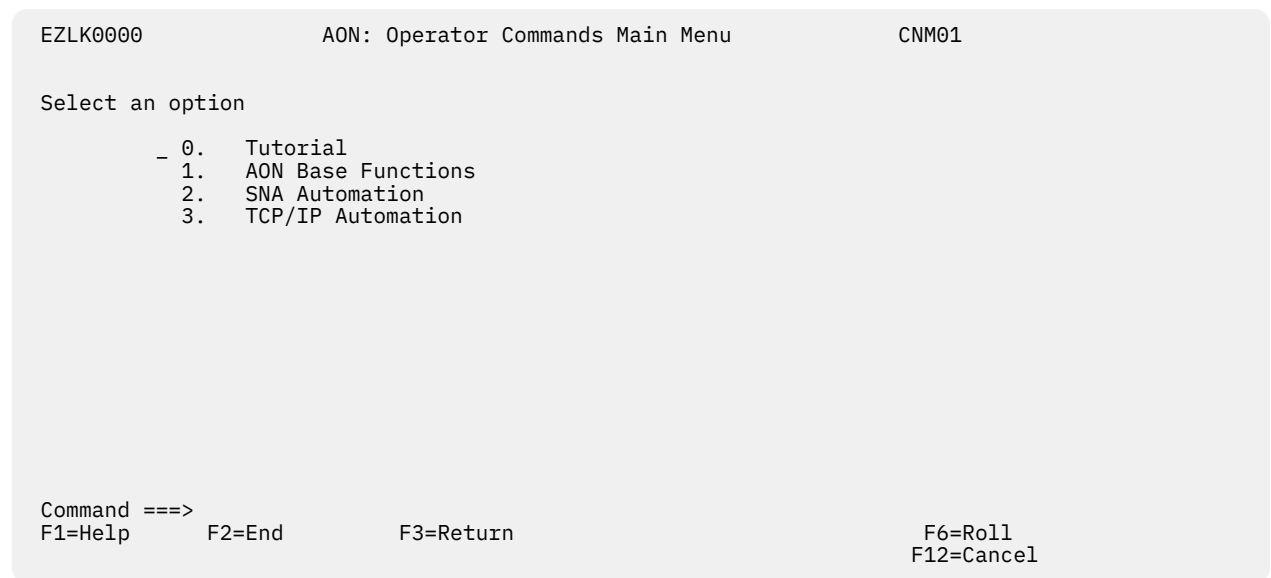
### Accessing the Operator Interface

You can perform any of the operator functions from the operator interface. With the operator interface, you can look at color-coded status displays of the resources on your network, change automation settings, receive messages, issue commands, and perform many other functions that control automation and resource availability. New AON/TCP users might find the panels an easier way to reach the different functions.

You can access the AON/TCP operator interface from anywhere within the NetView program or from the AON panels. To access AON/TCP from the NetView program, perform the following steps:

1. Type AON on the command line.
2. Press Enter.

The AON: Operators Commands Main Menu panel, which is displayed in [Figure 101 on page 116](#), is displayed:



*Figure 101. AON: Operator Commands Main Menu Panel*

The AON: Operator Commands Main Menu panel is the main panel for AON. This panel displays all available components. AON checks the control file entry for each AON component to determine if the component is installed and initialized. If the component is not available, the name of the component on the panel is not available.

3. To select the TCP/IP Menu option from the AON: Commands Main Menu panel, type 3 in the entry field, and press Enter.

The MVS TCP/IP: Commands Menu panel is displayed. See [Chapter 18, “TCP/IP for z/OS,” on page 117](#).

**Note:** You can also reach the AON/TCP interface by entering AONTCP on any NetView command line.



## Chapter 18. TCP/IP for z/OS

This section explains the function that uses z/OS Communications Server IP.

Selecting option **3** on the panel in [Figure 101 on page 116](#) (AON: Operator Commands Main Menu) displays the MVS TCP/IP commands menu, which is shown in [Figure 102 on page 117](#).

```
FKXK2000          MVS TCP/IP Commands Menu          NTV9D

Select an option

-  1. Issue Ping
   2. MVS IP Session Status
   3. Issue Tracerte Command
   4. Issue Commands
   5. SNMP MENU
   6. IP Server Management
   7. IP Resource Manager
   8. SNMP View
   9. CISCOWorks Blue Inter-network Status Monitor
  10. TCP/IP Trace Control Center. SP: -----

Command ==>
F1=Help      F2=Main Menu  F3=Return

                                F6=Roll
                                F12=Cancel
```

*Figure 102. MVS TCP/IP Automation: Commands Menu Panel*

The options on this panel provide the following functions:

**Option 1**

Ping a resource through z/OS Communications Server IP.

**Option 2**

Solve problems and manage IP connections for your z/OS system, such as TN3270 and FTP sessions.

**Option 3**

Issue a TCP/IP TraceRte command against a resource.

**Option 4**

View a full-screen panel that contains TSO or UNIX line mode commands. Responses are correlated and displayed in a full-screen panel.

**Option 5**

Link to the SNMP menu where you can issue SNMP commands.

**Option 6**

Monitor and control TSO and UNIX command servers.

**Option 7**

Provides management functions for TCP/IP resources, such as managing policy definitions and proactive monitoring.

**Option 8**

Provides generic interface resource views through SNMPView.

**Option 9**

Provides Cisco router management functions through Cisco Works Blue.

**Note:** AON/TCP provides a link to Cisco Works Blue from this panel, but before attempting to select this option, perform all of the installation or customization steps required to run Cisco Works Blue, for example, authorizing all NetView operators to use Cisco Works Blue.

### Option 10

Perform diagnostic traces to resolve TCP/IP problems.

**Usage :** For options **5**, **8**, and **10**, the resources must support SNMP requests. Some functions of options 2 and 7 might also require SNMP support.

## Using PING

**Note:** This function no longer requires AON. The information about this function is included in both *IBM Z NetView IP Management* and the *IBM Z NetView User's Guide: Automated Operations Network*.

Pinging is used to test connectivity to an IP host and can often be useful in determining if a resource can be reached. To ping a resource, select option **1** from the panel that is shown in [Figure 102 on page 117](#) or enter PING, MVSPING, or AON 3.2.1 from any command line. These commands provide the same function.

**Note:** PING is also available from the menus on the NetView management console.

When you issue a PING or MVSPING command without any parameters, the panel that is shown in [Figure 103 on page 118](#) is displayed.

```
FKXK2100      MVS TCP/IP Automation: Ping from a Service Point      CNM01
Host Name or IP Address
-----

Service Point Name _____ (? for Selection list)

Ping Count      4__
Ping Timeout    3___
Ping Length     16___

Command ==>
F1=Help      F2=Main Menu   F3=Return

F6=Roll
F12=Cancel
```

Figure 103. TCP/IP Automation: Ping Panel

The resource name can be an IP host name or an IP address. If no service point name is specified, the TCP/IP policy definitions for the service point associated with the resource are searched. Optionally, you can change the ping count, the ping timeout, and the ping length.

If you issue a PING or MVSPING command for a particular resource from an NCCF command line, the results are correlated and displayed from the command line rather than in panels.

For the syntax of and detailed information about the PING and MVSPING commands, see the online help.

## Checking Connection Status (IPSTAT)

**Note:** This function no longer requires AON. The information about this function is included in both *IBM Z NetView IP Management* and *IBM Z NetView User's Guide: Automated Operations Network*.

Users who connect to your z/OS environment can encounter various problems. You can diagnose these problems by using the capabilities of the connection management function:

- Determine connection endpoints
- Identify the type of connection (such as TN3270, FTP, or SMTP)

- Determine if a connection is stopped
- Drop a connection
- Run TCP/IP problem determination commands, such as PING and TRACERTE, to determine the cause of session problems
- Query sessions on multiple z/OS hosts and multiple TCP/IP stacks
- Access the SNMP command menu
- Display detailed information about specified Telnet connections
- Cause specified ports not to accept any new Telnet connections
- Cause previously quiesced ports to begin accepting Telnet connections

To access the connection management function, select option **2** displayed in [Figure 102 on page 117](#), enter AON 3.2.2, or issue the IPSTAT command. A panel similar to the one in [Figure 104 on page 119](#) is displayed.

```

FKXK2200          TCP/IP for z/OS Connection Status          NTV70

Enter TCP/IP address or HOSTNAME:
-----
Service Point  System  IP Address  Host Name
- NMPIPL10     TCPIP    9.42.45.10  nmpipl10.tivlab.raleigh.ibm.com
- NMPIPL10_B   TCPIP    9.42.45.101 nmpipl10-b.tivlab.raleigh.ibm.co
- NMP101       TCPIP    9.42.45.101 nmp101.tivlab.raleigh.ibm.com
- NMP190       TCPIP    9.42.45.190 NMP190.TIVLAB.RALEIGH.IBM.COM

Command ==>
F1=Help      F2=Main Menu  F3=Return    F6=Roll
F7=Backward  F8=Forward   F9=Filters   F10=Details  F12=Cancel

```

*Figure 104. Connection Status: Main Panel*

You can set filters for viewing connections by pressing F9 (Filters). For information about setting filters, see [“Setting Connection Management Filters” on page 122](#).

To see the details about a stack, select a stack by placing the cursor next to the stack and press F10 (Details). After you view the details, return to the main connection status panel by pressing F3 (Return).

## Displaying the Connections for a Stack

From the main connection status panel (FKXK2200), display one or more stacks and the number of connections for each by following these steps:

1. Type a TCP/IP address or host name (or a wildcard, for example, an asterisk (\*)). Use of wild cards is the same as for the TCPCONN command; for more information, see the online help for the TCPCONN command.
2. Select one or more stacks by placing any non-blank character next to each stack.
3. Press Enter.

If, for example, you type an asterisk (\*) for the TCP/IP address and select the first stack (NMPIPL10), a panel similar to the one in [Figure 105 on page 120](#) is displayed.

### Notes:

1. Depending on how you have customized your IPPORT policy definitions, this panel might not show all active sessions. For additional information, see IPPORT in the *IBM Z NetView Administration Reference*.

2. The IBM and Cisco TN3270 servers do not support the use of a wildcard (\*) in any part of the **IP Address** field. If you try to use a wildcard with a TN3270 server, the FKX962I message is displayed.

```
FKXK2210          TCP/IP for z/OS Connection Management

CLIENT

*

Service           Active           IP
Point             Hostname        Connections  Address
NMPIPL10 nmpipl10.tivlab.r 7      9.42.45.10

Command ==>
F1=Help      F2=Main Menu  F3=Return    F4=Commands  F5=Refresh    F6=Roll
F7=Backward  F8=Forward    F9=Filters    F11=Zoom     F12=Cancel
```

*Figure 105. Connection Management: Selected Stacks*

To display a list of connections for a specific stack, select the stack by placing the cursor next to it and press F11 (Zoom). For example, if you select NMPIPL10 on the panel shown in [Figure 105 on page 120](#), a panel similar to the one in [Figure 106 on page 120](#) is displayed. If you press F4 (Commands) next to a local port (in this example, Local Port 1030), you can display connection information, which is described in “[Displaying Connection Information](#)” on page 120.

```
FKXK2220          TCP/IP for z/OS Connection Management
Active Connections 7
CLIENT - - - - - - - - - - -> SERVICE POINT
                                NMPIPL10
                                9.42.45.10
*                               TCP/IP STACK

Local                           Client
Port   Client Address           Port   Send       Receive
1030 201.2.10.11              23    13        36
23     201.2.10.11             1030   14         72
4022   9.42.45.101             1029   94         11875
1064   9.42.45.217             4022   3709       2689
4022   9.42.45.190             1027   105235     11358680
1028   9.42.45.190             4022   106994     11337389
1024   127.0.0.1               1025   17         1054

Command ===>
F1=Help      F2=Main Menu   F3=Return    F4=Commands  F5=Refresh   F6=Roll
F7=Backward  F8=Forward    F9=Filters   F12=Cancel
```

Figure 106. Connection Management: Connections for Selected Stack

## Displaying Connection Information

To see the information for a connection, including the LU and application, and a list of commands that you can issue for the connection, select a connection by placing the cursor next to it and press F4 (Commands). If you select local port 1030 on the TCP/IP for z/OS Connection Management panel (FKXK2220), which is shown in [Figure 106 on page 120](#), and press F4, a panel that is similar to the one in [Figure 107 on page 121](#) is displayed.

```

FKXXK2221      TCP/IP for z/OS Connection Management
                SERVICE
CLIENT - - - - - > POINT - - - - - > CONNECTION
                NMPIPL10          1030
                201.2.10.11      9.42.45.10      23
Local : IP Address      201.2.10.11
Port  : Port           1030
1030  : Connection ID   00009B7F      - 1. Ping
23    : LU              NTC7001        - 2. Tracerte
4022  : APPL            NT70TS02        - 3. Drop
1064  : Send            13              - 4. ARP Cache
4022  : Receive         36              - 5. SNMP Commands
1028  : Send Window     32720           - 6. Display Conn
1024  : Conn ResourceName TCPIP          - 7. Quiesce Port
      :                  :              - 8. Resume Port
      :                  :              - 9. Conn Details
      :                  :              - 10. Connection Status
      :                  :              - 11. Packet Trace
      : F1=Help   F4=LU Cmds   F5=APPL Cmds   F6=Roll   F12=Cancel
      : .....
Command ==>
F1=Help      F2=Main Menu   F3=Return      F6=Roll
F7=Backward  F8=Forward    F9=Filters      F12=Cancel

```

Figure 107. Connection Management: Connection Information and Commands

From the list of commands that are displayed, type the number of the command and press Enter. For example, if you type 9 (Conn Details) to see connection details and press Enter, a panel similar to the one in Figure 108 on page 121 is displayed. If you type 11 (Packet Trace) and press Enter, the PKTTRACE Control panel (FKXXK2A22) is displayed.

**Note:** For options 5, 6, 9, and 10, the resources must support SNMP requests.

```

CNMKWIND OUTPUT FROM TCPIP CONNECTION DETAILS          LINE 0 OF 22
*----- Top of Data -----*
ibmMvsTcpConnectionInSegs.1.4.201.2.10.11.23.1.4.201.2.10.11.1030 = 14
ibmMvsTcpConnectionOutSegs.1.4.201.2.10.11.23.1.4.201.2.10.11.1030 = 13
ibmMvsTcpConnectionInOctets.1.4.201.2.10.11.23.1.4.201.2.10.11.1030 = 492
ibmMvsTcpConnectionOutOctets.1.4.201.2.10.11.23.1.4.201.2.10.11.1030 = 492
ibmMvsTcpConnectionAge.1.4.201.2.10.11.23.1.4.201.2.10.11.1030 = Wrong Type (sh
ibmMvsTcpConnectionLastActivity.1.4.201.2.10.11.23.1.4.201.2.10.11.1030 = 67150
ibmMvsTcpConnectionResourceName.1.4.201.2.10.11.23.1.4.201.2.10.11.1030 = TCPIP
ibmMvsTcpConnectionResourceId.1.4.201.2.10.11.23.1.4.201.2.10.11.1030 = 39807
ibmMvsTcpConnectionSockOpt.1.4.201.2.10.11.23.1.4.201.2.10.11.1030 = "C0 00 "
ibmMvsTcpConnectionPolicyAction.1.4.201.2.10.11.23.1.4.201.2.10.11.1030 =
ibmMvsTcpConnectionPolicyRule.1.4.201.2.10.11.23.1.4.201.2.10.11.1030 =
ibmMvsTcpConnectionServerResrcId.1.4.201.2.10.11.23.1.4.201.2.10.11.1030 = 26
ibmMvsTcpConnectionApplName.1.4.201.2.10.11.23.1.4.201.2.10.11.1030 = NT70TS02
ibmMvsTcpConnectionLuName.1.4.201.2.10.11.23.1.4.201.2.10.11.1030 = NTC7001
ibmMvsTcpConnectionLogMode.1.4.201.2.10.11.23.1.4.201.2.10.11.1030 = NSX32702
ibmMvsTcpConnectionProto.1.4.201.2.10.11.23.1.4.201.2.10.11.1030 = "02 "
ibmMvsTcpConnectionTtlsPolStat.1.4.201.2.10.11.23.1.4.201.2.10.11.1030 = 1
ibmMvsTcpConnectionTtlsConnStat.1.4.201.2.10.11.23.1.4.201.2.10.11.1030 = 1
ibmMvsTcpConnectionTtlsSslProt.1.4.201.2.10.11.23.1.4.201.2.10.11.1030 = 4
ibmMvsTcpConnectionTtlsNegCipher.1.4.201.2.10.11.23.1.4.201.2.10.11.1030 =
TO SEE YOUR KEY SETTINGS, ENTER 'DISPFK'
CMD==>

```

Figure 108. Connection Details Output

Press F3 to return to the Commands panel. To see the commands that can be issued for the LU or the application, press F4 (LU Cmds) or F5 (APPL Cmds), respectively. To close any of the command panels, press F12.

## Determining If a Connection Is Stopped

To determine if a connection has stopped, press F5 to refresh the connection information in the TCP/IP for z/OS Connection Management panel (FKXXK2220), shown in Figure 106 on page 120. Check the send and receive columns. If, after you refresh the panel, the send and receive columns for the connection have not changed, the connection might be stopped.

To drop the connection, press F4 to display the command panel (Figure 107 on page 121) and issue the command to drop the connection. The following actions are taken:

1. A VARY TCPIP DROP command is issued for the connection.

**Note:** For the DROP option to work properly, the operator issuing the DROP command must be authorized to issue the VARY TCPIP DROP command.

2. The connection list is refreshed.
3. A message similar to the following message is displayed:

```
FKX611I SESSION 1006 WAS SUCCESSFULLY DROPPED
```

## Setting Connection Management Filters

Displaying the connection management information can result in a large volume of information for you to review to find a particular resource or set of resources. With the connection management filters, you can limit the display to only the resources of particular interest. You can set connection management filters from a panel; see [“Using the Connection Management Filters Panel” on page 122](#). You can also set connection management filters from the command line; see [“Using a Command” on page 123](#).

### Using the Connection Management Filters Panel

To display the TCP/IP for z/OS Connection Management Filters panel (FKXKCSF0), which is shown in Figure 109 on page 122, press PF9 (Filters). The current filter settings are displayed on the panel.

```
FKXKCSF0      TCP/IP for z/OS Connection Management Filters      ID
The current filter settings are shown.
Type the data to be displayed in one or more fields.
Client IP Address
*
Port          Logical Unit      Application Name
*             *                 *
Type an action code to define the logical operator for the search:
2 1 OR
2 2 AND

Command ==>
F1=Help          F3=Return      F6=Roll
                  F9=Defaults    F12=Cancel
```

Figure 109. Connection Management Filters Panel

You can filter the connections by client IP address, port, logical unit, and application name. A blank or an asterisk (\*) in an input field indicates that that data is not to be filtered.

The logical operator setting is used for the filtering criteria when several input fields are specified. If 1 (OR) is specified, connections that match any of the specified criteria are displayed. If 2 (AND) is specified, only the connections that match all of the specified criteria are displayed.

To see connections with IP addresses that begin with 201, type 201.\* in the **IP Address** field, and press Enter.

**Note:** Use of wild cards is the same as for the TCPCONN command; for more information, see the online help for TCPCONN.

The EZL919I message that is displayed indicates that the filters have been saved. Press F3 to return to the connections panel, which now lists the filtered connections, as displayed in Figure 110 on page 123. Out of the connections originally displayed, only two connections met the filter criteria.

```
FKXK2220          TCP/IP for z/OS Connection Management
Active Connections  2
CLIENT - - - - - - - - - - -> SERVICE POINT
                                NMPIPL10
*                               9.42.45.10
                                TCP/IP STACK
Local Client
Port   Client Address      Port Send      Receive
1030   201.2.10.11         23    59           105
23     201.2.10.11        1030  37           141

Command ==>
F1=Help       F2=Main Menu   F3=Return     F4=Commands   F5=Refresh    F6=Roll
F7=Backward   F8=Forward    F9=Filters    F12=Cancel
```

In the Display Packet Control panel (FKXK2A24) that is displayed here, as called from the IPSTAT FKXK2220 or FKXK221C panel (see [Figure 110 on page 123](#)), the IP addresses and ports are carried over from the selected connection. In this example, the **Infc Name** field shows ALL, which indicates that all packets from all interfaces for this call are to be collected. From the Display Packet Control panel (FKXK2A24), you can use any of the trace display and formatting functions that are provided by the IPTRACE dialog.

|               |                        |                 |                 |               |                 |
|---------------|------------------------|-----------------|-----------------|---------------|-----------------|
| FKXK2A24      | Display Packet Control |                 |                 |               | LOCAL<br>NMPIPL |
| Stack: TCPIP  |                        | Infc Name: ALL  |                 |               |                 |
| LAddr *       |                        |                 |                 |               |                 |
| RAddr *       |                        |                 |                 |               |                 |
| Portnum: *    | LPort: *               | RPort: *        | Protocol: _     | ALL (default) |                 |
|               |                        |                 | -               | TCP           |                 |
|               |                        |                 | -               | UDP           |                 |
| Start Time: * |                        |                 | -               | ICMP          |                 |
| End Time: *   |                        |                 | -               | OSPF          |                 |
|               |                        |                 | -               | (Number)      |                 |
| MaxRecs: 1    | 1-Last 100             | Truncate: 65535 |                 |               |                 |
|               | 2-First                |                 |                 |               |                 |
| Command ==>   |                        |                 |                 |               |                 |
| F1=Help       | F2=Save Packets        | F3=Return       | F4=View Packets | F6=Roll       |                 |
|               | F8=Extended Opts       |                 | F10=Analyze     | F12=Cancel    |                 |

Figure 112. Display Packet Control Panel (FKXK2A24)

When you exit the IPTRACE panels, you are asked whether you want to stop the trace. If you specify **Y** (Yes), the packet trace filter is stopped. If you specify **N** (No), the packet trace filters are left active, and you must either stop them manually or stop them by using the IPTRACE panels.

## Using TRACERTE

**Note:** This function no longer requires AON. The information about this function is included in both *IBM Z NetView IP Management* and *IBM Z NetView User's Guide: Automated Operations Network*.

The TRACERTE command is used to trace the routes of data packets to a specified IP host from the IP stack on the host on which the NetView program is running. Use this command to determine connectivity with or routing to a particular endpoint, roundtrip times between the NetView and target hosts, and routers along the way. The TRACERTE command is useful in problem determination, for example, in troubleshooting lost packets.

**Note:** The TRACERTE command is also available from NetView management console menus.

To access TRACERTE functions, select option 3 in the MVS TCP/IP Commands Menu panel or enter the TRACERTE or AON 3.2.3 command. The panel that is shown in [Figure 113 on page 125](#) is displayed. For information about the fields on this panel, press F1 to see the online help.



FKXK2300 MVS TCP/IP Automation: Trace Route from a service point CNM01

Host Name or IP Address:

-----

Service Point Name \_\_\_\_\_ (? for Selection list)

Max 30

Try 3

Port 33434

Wait 5

Debug 2 ( 1=Y , 2=N )

Skip DNS 1 ( 1=Y , 2=N )  
lookup

Command ==>

F3=Ret F4=Fndprev F5=Rptfnd F6=Roll F7=Back F8=Forward F12=Cancel

*Figure 113. TCP/IP Automation: Trace Route Panel*

The resource name can be an IP host name or an IP address. If a stack name is not specified, the stack definitions in the CNMPOLCY member are searched for the specified IP address or host name.

Figure 114 on page 125 shows output from a TRACERTE command for a workstation with an IP address of 1.23.45.678:

FKXKLWN2 OUTPUT FROM TCP/IP 390 TRACE ROUTE Line 1 of 6

Trace route to 1.23.45.678 (1.23.45.678)

|   |               |       |       |       |
|---|---------------|-------|-------|-------|
| 1 | (1.23.45.6)   | 2 ms  | 0 ms  | 0 ms  |
| 2 | (1.23.45.66)  | 3 ms  | 3 ms  | 3 ms  |
| 3 | (1.23.45.67)  | 8 ms  | 9 ms  | 11 ms |
| 4 | (1.23.45.678) | 13 ms | 11 ms | 10 ms |

Command ==>

F3=Ret F4=Fndprev F5=Rptfnd F6=Roll F7=Back F8=Forward F12=Cancel

*Figure 114. TCP/IP Automation: Trace Route Output*

For the syntax of and detailed information about the TRACERTE commands, see the online help.

## Issuing Commands

You can issue line mode TSO or UNIX commands from the NetView program without logging on to TSO. To do that, select option 4 from the FKXK2000 panel to display the [Figure 115 on page 126](#) panel.

```

FKXK2400      TCP/IP Automation: Issue Command to Service Point      CNM01

Service Point Name nmpipl10      (? for Selection list)

Command Interface: 1      ( 1 = UNIX   2 = TSO )

Type in the remote command below and press Enter.

0.....1.....2.....3.....4.....5.....6.....7
netstat conn _____
netstat home _____
netstat _____
tracerte pquigley _____
time _____
netstat ? _____
netstat sock _____
netstat up _____
netstat tcp tcp32 _____
netstat byte _____

Command ==>
F1=Help  F2=Main Menu  F3=Return      F6=Roll
F11=Right  F12=Cancel

```

Figure 115. TCP/IP Automation: Issue Command to Service Point Panel

You can also access the panel displayed in Figure 115 on page 126 by using the AON 3.2.4 command. To easily issue a command, for example, move to the first command line (netstat conn) and press Enter. A panel similar to the one displayed in Figure 116 on page 126 is displayed.

```

FKXKLWN2      OUTPUT FROM COMMAND      Line 1 of 15
COMMAND:netstat conn WAS ISSUED TO SERVICE POINT: nmpipl10
MVS TCP/IP Netstat V3R2

Active Transmission Blocks
User ID  Conn  Local Socket      Foreign Socket      State
-----  ---  -
INTCLIEN 1000  *..TELNET          *..*                Listen
INTCLIEN 1006  1.23.45.67..TELNET  1.23.45.678..1234   Established
OMVS     1004  *..10007           *..*                Listen
SNMPD32  UDP   *..161             *..*                UDP
SNMPD32  1001  *..1024            *..*                Listen
SNMPQ32  UDP   *..162             *..*                UDP
SNMPQ32  UDP   *..1024            *..*                UDP
SNMPQ32  1002  *..1025            *..*                Listen
NSSERV   1003  *..6000            *..*                Listen

Command ==>
F3=Ret  F4=Fndprev  F5=RPTfnd  F6=Roll  F7=Back  F8=Forward  F12=Cancel

```

Figure 116. Output from Command Issued to Service Point Panel

## Using SNMP Management

**Note:** This function no longer requires AON. The information about this function is included in both *IBM Z NetView IP Management* and the *IBM Z NetView User's Guide: Automated Operations Network*.

To use SNMP management functions, select option **5** from the MVS TCP/IP Automation: Commands Menu displayed in the Figure 102 on page 117 or issue the NVSNMP or SNMP command. The NVSNMP command provides a panel interface. The SNMP command, which provides the WALK, BULKWALK, GET, GETBULK, GETNEXT, INFORM, and TRAP operations, is intended more for use in application programs.

When you issue the NVSNMP command, the panel that is shown in Figure 117 on page 127 is displayed. For information about the fields on the panel, press F1 to see the online help.

Figure 117. TCP/IP SNMP Menu Panel

If you specify a community name, it is used for the resulting SNMP request. If you do not specify a community name, the stack definitions are searched for the community name. If no community name is found, the z/OS default name `public` is used. If a community name is not defined for the stack, then the default name defined to z/OS Communications Server IP is used. The community name can be defined in the TCP390 definition for the associated stack where the SNMP request is being issued. For more information, see the TCP390 definition in the *IBM Z NetView Administration Reference*.

## Get Command

Figure 118. TCP/IP SNMP Commands Panel: Get Command

Chapter 18. TCP/IP for z/OS **127**

```
FKXK2519          TCP/IP for 390 SNMP Commands          NTV70
```

```
Resource: LOCAL  
Password(Community)  
TCP/IP Stack: LOCAL      Comm .....  
                          : Options:                :  
MIB Variables:           :                               :  
-----                 : - Debug              :  
-----                 : - Retry 2_            :  
-----                 : - Timeout 3_         :  
-----                 : - Bulk               :  
-----                 :     Max Repetitions 10 :  
-----                 :     non-Repetitions 0_ :  
-----                 :                               :  
-----                 : Additional options:   :  
-----                 : -----             :  
-----                 : -----             :  
-----                 : -----             :  
-----                 : .....
```

```
Command ==>  
F1=Help       F2=Main Menu  F3=Return                     F6=Roll  
                                           F12=Cancel
```

Figure 119. TCP/IP SNMP Commands Panel: Options

The options panel shows the SNMP options and the system settings. These settings can be defined in the policy definitions. Type an X next to each option that you want to override. If the system definition is different from the SNMP default, the fields are automatically selected.

**Note:** If **Bulk** is selected using F9 (Options), the command is changed to GETBULK.

## Set Command

If you select the Set command for option **1** in the TCP/IP SNMP Menu Panel, the panel that is shown in Figure 120 on page 128 is displayed.

```
FKXK251S          TCP/IP for 390 SNMP Commands          CNM01  
Resource: LOCAL  
Password(Community)  
TCP/IP Stack: LOCAL           Command: Set  
MIB Variable                                           Type  
Value  
sysName.0_-----  
NMPIPL10 Test system -----  
-----  
-----  
-----  
-----  
-----  
-----  
  
Command ==>  
F1=Help      F2=Main Menu   F3=Return     F6=Roll  
              F9=Options    F12=Cancel
```

Figure 120. TCP/IP SNMP Commands Panel: Set Command

For the **Set** command, the **Type** and **Value** fields are added to the panel. **Type** is used to override the definition type of the MIB variable. **Value** is used to specify the new value of the MIB variable. To specify optional parameters for this command, press F9 (Options) to display the SNMP options panel.

## Walk Command

If you select the Walk command for option **1** in the TCP/IP SNMP Menu Panel, the panel that is shown in [Figure 121](#) on [page 129](#) is displayed. This command returns the values of all variables in the MIB tree that is identified by the specified MIB variable.

```
FKXK2510          TCP/IP for SNMP Commands          NTV70

Resource: 1.23.45.678
Password(Community)
TCP/IP Stack: LOCAL

Command: WALK

MIB Variables:
-----

Command ==>
F1=Help      F2=Main Menu  F3=Return  F6=Roll
F9=Options   F12=Cancel
```

Figure 121. TCP/IP SNMP Commands Panel: Walk Command

For a **Walk** command, only one MIB variable can be specified.

**Note:** If **Bulk** is selected using F9 (Options), the command is changed to BULKWALK.

Using the Group Command

Figure 122 on page 129 is the resulting panel if you select **2** (the Group Menu option) in the TCP/IP SNMP Menu panel.

```
FKXK2520          TCP/IP for 390 SNMP Groups          NTV9D

More: +

Resource: LOCAL
Password (Community)
TCP/IP Stack: LOCAL

Groups:
ExtGroup      LIST+      UDPTable      TABLE      atm           WALK
system        WALK      ip            WALK      3172sys       WALK
sys2          LIST      ipforward     WALK      3172ifTrap    WALK
sysOR         TABLE    ipAddrTable   WALK      3172channel   WALK
snmp          WALK      ipRouteTable  WALK      3172lan       WALK
IFTable       WALK      ipNoTab       LIST      3172blk       WALK
IFXTable      WALK      icmp          WALK      3172dblk      WALK
atTable       WALK      tcp           WALK      3172device    WALK
UDP           WALK      tcpConnTable  WALK      MvsTcpSystem  WALK
UDPnotab      LIST      tcpNoTab      LIST      MvsTcpIf      WALK

COMMAND ==>
ENTER=Get Group Data
F1=Help      F2=Main Menu  F3=Return  F4=Description  F6=Roll
F7=Backward F8=Forward  F9=Options  F12=Cancel
```

Figure 122. TCP/IP SNMP Groups Panel

The SNMP Group panel displays the groups that are defined in DSIPARM sample FKXSNMP. To display more information about the group, tab to the group and press F4. To display the SNMP options, press F9. For more information about creating MIB groups, see [Appendix D, “Customizing the SNMP Group Definition File \(FKXSNMP\),”](#) on page 381.

Figure 123 on page 130 displays the description of the group UDPnotab, and has a type of LIST. Note the four MIB variables that are displayed when you use the UDPnotab group.

```

FKXK2524      TCP/IP for 390 SNMP Groups Description      NTV70

Group: UDNotab      Type: LIST

Abstract:
  UDP group MIBs without the UDP Table

MIB Variables:
  udplnDatagrams.0
  udpNoPorts.0
  udplnErrors.0
  udpOutDatagrams.0

Command ==>
F1=Help      F2=Main Menu  F3=Return      F6=Roll
F9=Options  ENTER=Get Group Data  F12=Cancel

```

*Figure 123. TCP/IP SNMP Groups Description Panel: LIST*

Figure 124 on page 130 displays the description of the group system and has a type of WALK.

```

FKXK2524      TCP/IP for 390 SNMP Groups Description      NTV70

Group: system      Type: WALK

Abstract:
  System group MIB variables for SNMP v1 or SNMP v2 including OR Table
  The definition of this group can be found in:
  RFC1907 for v2 or RFC1450 for v1.
MIB Variables:      FULL Name:SYSTEM

Command ==>
F1=Help      F2=Main Menu  F3=Return      F6=Roll
F9=Options  ENTER=Get Group Data  F12=Cancel

```

*Figure 124. TCP/IP SNMP Groups Description Panel: WALK*

Figure 125 on page 130 displays the description of the group UDPTable and has a type of TABLE.

```

FKXK2524      TCP/IP for 390 SNMP Groups Description      NTV70

Group: UDPTable      Type: TABLE

Abstract:
  UDP group MIBs with the UDP Table

MIB Variables:      INDEX MIB: 1.3.6.1.2.1.7.5
  udpLocalAddress
  udpLocalPorts

Command ==>
F1=Help      F2=Main Menu  F3=Return      F4=Description      F6=Roll
F9=Options  ENTER=Get Group Data  F12=Cancel

```

*Figure 125. TCP/IP SNMP Groups Description Panel: TABLE*

## Using Extended SNMP Groups

If you need more flexibility than SNMP Groups provides, use extended SNMP Groups. These groups are different in that you can use them to code generic MIB objects (using the LIST+ statement), which are to be appended with group variables obtained from a pop-up panel when this group is selected.

To use extended SNMP groups, select option **2** on the SNMP Menu panel. In the TCP/IP SNMP Groups Panel that is displayed, move the cursor to LIST+ Group name (ExtGroup) and press Enter.

The panel in [Figure 126 on page 131](#) is displayed.

**Note:** To use the LIST+ function you must first customize the FKXSNMP member in DSIPARM offline. For information about customizing the FKXSNMP member, see [Appendix D, “Customizing the SNMP Group Definition File \(FKXSNMP\),” on page 381](#).

```
FKXK252V      TCP/IP for 390 SNMP Group Extensions      NTV9D

Resource: LOCAL
Password (Community)
TCP/IP Stack: LOCAL

Groups :
ExtGroup : PLEASE ENTER AN INTERFACE NUMBER
system :
sys2 : 2
sysOR :
snmp :
IFTable :
IFXTable :
atTable :
UDP :
UDPnotab :
WALK

COMMAND ==>
F1=Help      F2=Main Menu  F3=Return  F6=Roll      F12=Cancel
```

*Figure 126. TCP/IP SNMP Group Extensions Panel*

On the SNMP Group Extensions panel, type 2 to display details for interface 2 and press Enter. Listed MIB variable information for interface adapter number 2 is collected and displayed as [Figure 127 on page 131](#).

```
CNMKWIND Output from  SNMP GET to LOCAL                      LINE 0 OF 20
*-----TOP of DATA -----*
START OF DATA
ifName.2 = LOOPBACK
ifLinkUpDownTrapEnable.2 = 1
::::::::::::::::::
END OF DATA

TO SEE YOUR KEY STTINGS, ENTER 'DISPLAY'
CMD==>
```

*Figure 127. SNMP GET to LOCAL Output*

## Managing IP Servers

To manage TSO server sessions, select option 6 from the TCP/IP for z/OS Automation: Commands Menu displayed in [Figure 102 on page 117](#). When you select option 6, the panel displayed in [Figure 128 on page 132](#) is displayed:

```
FKXK2600                      TCP/IP for 390 Servers                      NTVE1
Select an option:
  1=Start  2=Stop

  Domid  Service  Server  Type  Submit  PPI  PPI  Status
        Point                               JCL  Receiver  Buffer
-  NTVE1  NMP217   UNIX   UNIX  n/a     CNMEUNIX  0     ACTIVE
-  NTVE1  NMP217   NV2TS1  TSO   CNMSJTSO $E100001  0     ACTIVE
-  NTVE1  NMP217   NV2TS2  TSO   CNMSJTSO $E100002  0     ACTIVE
-  NTVE1  NMP217   NV2TS3  TSO   CNMSJTSO $E100003  0     ACTIVE

Command ==>
F1=Help      F2=Main Menu  F3=Return      F5=Refresh    F6=Roll
F7=Backward  F8=Forward
```

Figure 128. IP Servers Panel

In the panel displayed in [Figure 128 on page 132](#), three TSO servers for NMP217 and the UNIX server for NMP217 are displayed. Use the following steps to start or stop a server:

1. Move the cursor to a server.
2. Type 1 to start the server or 2 to stop the server.
3. Press F5 to refresh the panel.

**Note:** You can start or stop multiple servers at the same time.

A single CNMEUNIX server is used for each service point that requires UNIX system services. Therefore, starting or stopping any of the service point entries that use the CNMEUNIX server also starts or stops the single CNMEUNIX server. The status of the server is reflected on all service point entries in the service point list. The status of STARTING or STOPPING is displayed only on the service point entry for which a start or stop command is entered.

## Using IP Resource Management

**Note:** This function no longer requires AON. The information about this function is included in both *IBM Z NetView IP Management* and the *IBM Z NetView User's Guide: Automated Operations Network*.

This section provides information about managing IP resources that are defined using the IP Resource Management panels.

**Note:** Some of the IP Resource Management panels use SNMP requests to collect data. The Community name used for those requests is defined on the TCP390 definition for the associated stack where the request is being issued. For more information, see the TCP390 definition statement in the *IBM Z NetView Administration Reference*.

### IP Resource Management Main Panel

You can control monitoring of IP Resources from the IP Resource Management main panel, displayed in [Figure 129 on page 133](#). To display this panel, enter AON 3.2.7 or issue the IPMAN command.



You can select the following monitoring functions in the IP Resource Management main panel:

- ADD/START
- DISPLAY/CHANGE
- DELETE
- START
- STOP

|                                                                   |                  |                                    |                   |            |            |         |
|-------------------------------------------------------------------|------------------|------------------------------------|-------------------|------------|------------|---------|
| FKXK2700                                                          |                  | TCP/IP for 390 Resource Management |                   |            |            |         |
|                                                                   |                  |                                    |                   |            | REFRESH: 0 |         |
| Select an IP Management Active Monitoring command and press ENTER |                  |                                    |                   |            |            |         |
| 1=ADD/START                                                       | 2=DISPLAY/CHANGE | 3=DELETE                           | 4=START           | 5=STOP     |            |         |
| Resource                                                          | Resource Type    | TCP/IP Stack                       | Actmon Definition | M          | Status     |         |
| -----                                                             |                  |                                    |                   |            |            |         |
| -                                                                 | **NEW**          | IPHOST                             |                   |            |            |         |
| -                                                                 | **NEW**          | IPPORT                             |                   |            |            |         |
| -                                                                 | LOOP1026         | IPPORT                             | NMP101            |            | R          | DOWN    |
| -                                                                 | NMPIPL10         | IPHOST                             | NMP101            |            | A          | NORMAL  |
| -                                                                 | LOOP1024         | IPPORT                             | NMP101            |            | A          | NORMAL  |
| -                                                                 | TN3270           | IPPORT                             | NMP101            |            | A          | NORMAL  |
| -                                                                 | NMPIPL10         | TCP390                             | NMPIPL10          |            | A          | NORMAL  |
| -                                                                 | NMP101           | TCP390                             | NMP101            |            | A          | NORMAL  |
| -                                                                 | NMP217           | TCP390                             | NMP217            |            | A          | NORMAL  |
| -                                                                 | NMPIPL10V6       | TCP390                             | NMPIPL10V6        |            | R          | UNKNOWN |
| -                                                                 | TELNETAS         | IPTELNET                           |                   |            | N          | NORMAL  |
|                                                                   |                  |                                    |                   |            |            |         |
| Command ==>                                                       |                  |                                    |                   |            |            |         |
| F1=Help                                                           | F2=Main Menu     | F3=Return                          | F4=Commands       | F5=Refresh | F6=Roll    |         |
| F7=Backward                                                       | F8=Forward       | F9=Display Opts                    | F10=Connections   | F12=Cancel |            |         |

Figure 129. TCP/IP Resource Management Main Panel

Note the templates that are named **\*\*NEW\*\***, one for IPHOST and one for IPPORT. These are the most used resource types. Select either template to add a resource by using the required fields for the type in the FKXK2760 panel, described in [“IP Resource Management Add Panel” on page 134](#).

The following commands can be issued directly from the IP Resource Management main panel by typing the command number next to the resource name:

- 1** Display the Add/Change Panel in add mode. When the resource is added, monitoring is started. For additional information, see [“IP Resource Management Add Panel” on page 134](#).
- 2** Display the Add/Change panel in change mode. For additional information, see [“IP Resource Management Change Panel” on page 135](#).
- 3** Deletes the entry only from the in-store control file and ends all monitoring.
- 4** Start monitoring for the resource.
- 5** Stop monitoring for the resource.

**Note:** For IP Port Monitoring, all ports under a defined stack are always actively monitored at the start of the NetView program. They cannot be stopped or started individually.

The following fields are displayed for each IP resource defined in the control file:

#### Resource

The name of the resource you are monitoring.

**Resource Type**

The resource type can be TCP390, IPHOST, IPINFC, IPNAMESRV, IPPORT, IPTELNET, IPTN3270, or OSAPORT.

**TCP/IP Stack**

When the name in this field is the same as the resource name, it is a stack. This status indicates whether the TCP/IP stack specified on the TCPNAME parameter of the TCP390 statement is active. It does not indicate whether the IP address as specified on the IPADDR statement or the host name as specified on the HOSTNAME statement match the primary interface address or host name currently in use by the TCP/IP stack.

**Actmon Definition**

If a reference to an ACTMON definition exists, that name is displayed in this field.

**M**

The monitoring field contains the current active monitoring status. This list shows the monitoring status:

**A**

Active monitoring

**R**

Recovery monitoring

**N**

No monitoring.

**Status**

The status field contains the status. These are the possible status indicators:

- NORMAL (GREEN)
- DOWN (RED)
- DEGRADED (PINK)
- THRESH (YELLOW)
- UNKNOWN (BLUE)

**REFRESH**

The IP Resource Manager main panel can be automatically refreshed, by changing the value of this field, from 0 (no refresh) to 59 minutes.

**IP Resource Management Add Panel**

Select ADD (option 1) on the IP Resource Management main panel to display the following panel ([Figure 130 on page 135](#)):

| FKXK2760                                                                      |              | TCP/IP for 390 Resource Management                                                           |                       |
|-------------------------------------------------------------------------------|--------------|----------------------------------------------------------------------------------------------|-----------------------|
| Update the allowable fields, Press F4<br>to ADD resource and Start Monitoring |              | CMD Options<br>X - Fixed Field      R-Required Field<br>1 - Change Field<br>2 - Delete Field |                       |
| CMD                                                                           | KEYWORD      | VALUE                                                                                        |                       |
| R                                                                             | RESNAME      |                                                                                              |                       |
| X                                                                             | RESTYPE      | IPPORT                                                                                       |                       |
| X                                                                             | OPTION       | IP390                                                                                        |                       |
| R                                                                             | SP           | NMP217                                                                                       |                       |
| -                                                                             | PORT         | 8008                                                                                         |                       |
| -                                                                             | PROTOCOL     | TCP                                                                                          |                       |
| -                                                                             | TCPNAME      | T530EENV                                                                                     |                       |
| -                                                                             | FORMAT       | PORT                                                                                         |                       |
| -                                                                             | ACTMON       | IPPORT                                                                                       |                       |
| -                                                                             | DESC         | "NetView Web Browser Socket"                                                                 |                       |
| -                                                                             | STATUS       | NORMAL                                                                                       |                       |
| -                                                                             | INTVL        | 00:10                                                                                        |                       |
| Command ==>                                                                   |              |                                                                                              |                       |
| F1=Help                                                                       | F2=Main Menu | F3=Return                                                                                    | F4=SUBMIT UPDATE      |
| F7=Backward                                                                   | F8=Forward   | F9=Add a field                                                                               | F6=Roll<br>F12=Cancel |

Figure 130. TCP/IP Resource Management Add Panel

With the Add panel, you can add resources dynamically into the in-store control file. When the resource is added, proactive monitoring is started for the resource. There is a delay before the monitoring field (M column) is updated on the main panel. Use F5 to refresh the panel until the change is displayed.

The IP Resource Management Add panel contains the following fields:

#### Command

A fixed field or a field that is updated using one of the following command options:

- 1**  
Change Value for keyword.
- 2**  
Delete keyword and its value.
- R**  
Indicates field is required. This option is set by the program.
- X**  
Indicates the field cannot be changed. This option is set by the program.

#### Keyword

Specifies the keyword name as set in the control file.

#### Value

Specifies the current value of the keyword under most circumstances. The value is cleared for ADD operations if a new value is required.

#### Notes:

- Keywords marked with an X cannot be updated. In cases where multiple field relationships exist, not all required keywords are marked with an R.
- Values for keywords are not syntax checked. Entering incorrect data can cause unpredictable results.

Changes are validated before a page forward or backward attempt. When R required fields are accepted their command is changed to X or fixed field, automatically.

To submit the changes, press F4. To add a new keyword-value pair press F9 to display the Add keyword window.

## IP Resource Management Change Panel

Selecting Option 2 (change) on the IP Resource Management main panel displays the panel displayed in [Figure 131 on page 136](#).

```

FKXK2760          TCP/IP for 390 Resource Management

Update the allowable fields, Press F4      CMD Options
to Change the settings                    X - Fixed Field      R-Required Field
                                          1 - Change Field
                                          2 - Delete Field

CMD  KEYWORD      VALUE
-----
X    RESNAME      NMPIPL10
X    RESTYPE      IPHOST
X    OPTION       IP390
-    SP           NMPIPL10
-    ACTMON       ALLHOSTS
-    IPADDR       1.23.45.67
-    HOSTNAME     NMPIPL10
-    INTVL        00:15
-    STATUS       NORMAL
-    FORMAT       PING

Command ==>
F1=Help      F2=Main Menu  F3=Return      F4=SUBMIT UPDATE      F6=Roll
F7=Backward  F8=Forward   F9=Add a field  F12=Cancel

```

Figure 131. TCP/IP Resource Management Change Panel

## IP Resource Management Filters Panel

The IP Resource Management Filters panel can be displayed by entering IPMANSSF with no parameters or by pressing F9 on the IP Resource Management main panel.

```

FKXK2710          IP Resource Management Filters          NTVE1

The current filter and sort settings are displayed. Change the settings
and press ENTER to activate.

Type the data to be displayed in one or more fields.
Resource      Resource      TCP/IP      Status
Name          Type          Stack Name
*             *             *             *
Type an action code to define the logical operator for the search:
1  1 - OR    2 - AND

Type numbers from 1 to 4 to define the column order for sorting.
Resource      Resource      TCP/IP      Status
Name          Type          Stack Name
1             2             3             4
Type an action code to define the sort order.
1  1 - Ascending order  2 - Descending order

Command ==>
F1=Help          F3=Return      F6=Roll
                  F9=Defaults   F12=Cancel

```

Figure 132. IP Resource Management Filters Panel

Use this panel to select the resources that you want to display. The settings selected are saved on a task basis and apply in future queries. You can specify filter criteria for exact names or names starting with specific characters, if the criteria is followed by an asterisk (\*). The following fields can be filtered:

### Resource Name

Specify criteria for the resource name.

### Resource Type

Specify criteria for the resource type.

### TCP/IP

Specify criteria for the TCP/IP stack name.

### Status

Specify criteria based on resource status.

For filter options, specify the relationship between these options as 1 for OR or 2 for AND.

For sort options, specify 1–4 in any order. The four fields are then sorted based on the specified sort order. Indicate whether these sorts are to be done in ascending or descending order by entering a 1 or 2 in the last input field.

Pressing Enter causes the filter and sort entries to be validated and saved for use by the IP Resource Manager.

Pressing F9 restores the default filter and sort settings.

Press F3 to exit or F12 to cancel.

## IP Resource Management Command Window

To display the COMMANDS window, press F4 from the IP Resource Management main panel or the SNMP Details panel. The information displayed depends on the resource type of the resource that has been selected.

```
FKXK2740          TCP/IP for 390 Resource Management

Select a command and press Enter

Resource          Resource Type : COMMANDS (based on Resource Type) :
-----          -
- **NEW**          IPHOST      : 1- Ping                      :
- **NEW**          IPPORT      : 2- Tracerte                 :
- LOOP1026         IPPORT      : 3- Connection Management    :
- NMPIPL10         IPHOST      : 4- SNMP                     :
- LOOP1024         IPPORT      : 5- Display Timers          :
- TN3270           IPPORT      : 6- SNMP Details Panel      :
- NMPIPL10         TCP390      :                             :
/ NMP101           TCP390      :                             :
- NMP217           TCP390      :                             :
- NMPIPL10V6       TCP390      :                             :
- TELNETAS         IPTELNET    :                             :

Command ==>
F1=Help      F2=Main Menu  F3=Return

F6=Roll
F12=Cancel
```

Figure 133. TCP/IP Resource Management Panel: Commands Window

## Using SNMPView

In general, the data displayed by SNMPView is retrieved from SNMP MIB variables. Any data field, with the underscore (\_) next to it, can be set by performing the following steps:

1. Enter a non-blank character in place of the underscore.
2. Change the MIB data to be displayed.
3. Press Enter.

You can use the SNMPView function to display system wide and interface specific MIB data for your TCP/IP stacks and network resources. However, the SNMPView function does not support IPv6 addressing.

To navigate through the panels, and display information about a resource, use option 8 on the MVS TCP/IP Commands Menu shown in [Figure 102 on page 117](#). Type 8 and press Enter. The panel [Figure 134 on page 138](#) is displayed.

```

FKXK2800          TCP/IP for 390 SNMP Resource View          NTVE1

Host Name or IP Address:  (blank: Use Local Stack Name)
-----
X MVS Stack          _ IP Resource
TCP/IP Stack Name :   LOCAL

_ Go Directly to the Interface List
_ Go to the details for Interface ___ IP Address -----
_ Go to the Connection List for IP Address -----

Community Name :

Command ==>
F1=Help          F2=Main Menu    F3=Return
F6=Roll
F12=Cancel

```

Figure 134. TCP/IP SNMP Resource View Panel

## IP Resource Data

To display IP Resource MIB data for NMPIPL25, fill in the text fields in [Figure 134 on page 138](#) in the following way:

1. Type NMPIPL25 in the Host Name or IP Address field.
2. Delete the X from the default MVS Stack selection.
3. Type X to select IP Resource view.
4. Press Enter. A selection list is displayed.
5. Type / in front of the selection you want and press Enter.

```

FKXK2800          TCP/IP for 390 SNMP Resource View          B63NV

Host Name or IP Address: NMPIPL25-----
(blank: Use Stack Name) -----

MVS Stack          X IP Resource
TCP/IP Stack Name: NMPIPL25 (? for Selection list)

_ Go Directly to the Interface List
_ Go to the details for Interface ___ IP Address -----
_ Go to the Connection List for IP Address -----

Community Name:

Command ==>
F1=Help          F2=Main Menu    F3=Return
F6=Roll
F12=Cancel

```

Figure 135. TCP/IP SNMP Resource View Sample Panel

When you press Enter from the Resource View panel, displayed in [Figure 135 on page 138](#), System MIB data is displayed on the Resource View: System panel, displayed in [Figure 136 on page 139](#).

Figure 136. TCP/IP SNMP Resource View: System Panel

Figure 137. TCP/IP SNMP Resource View: System Panel with Commands Panel

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```

FKXK281P      TCP/IP for 390 SNMP Resource View: System      NTV70

IP DATA for:
NMPIPL25

- Gateway Forwarding      = YES(1)      Reassemble Timeout      = 60
- Default TTL             = 64           Reassemble Requests     = 0
  Total Datagrams Received = 7617        Reassemble Success      = 0
  IP Header Discards      = 0           Reassemble Fail         = 0
  Invalid Address Discards = 0           Fragments Success       = 0
  Forwarded Datagrams      = 0           Fragments Fail          = 0
  Unknown Protocol Discards = 0         Fragments Create        = 0
  All Other Discards      = 0           Routing Discards        = 0
  Delivered Datagrams      = 8619
  Output Requests         = 7657
  Output Discards         = 0
  Discard No Route Avail. = 0

Set Community Name:

Command ==>
F1=Help      F2=Main Menu  F3=Return      F5=Refresh      F6=Roll
              F12=Cancel

```

Figure 138. TCP/IP SNMP Resource View: System Panel

Press F3 to return to the panel displayed in [Figure 137 on page 139](#). Type 6, and press Enter. The panel [Figure 139 on page 140](#) is displayed.

```

FKXK281T      TCP/IP for 390 SNMP Resource View: System      NTV70

TCP DATA
NMPIPL25

Retransmit Algorithm      = vanj(4)
Min. Retransmit Timeout   = 500
Max. Retransmit Timeout   = 120000
Maximum TCP Connections   = -1
Active Open Connections   = 1
Passive Open Connections  = 1
Failed Connect Attempts   = 0
Establish Resets          = 0
Current Established Conn   = 2
Segments Received         = 40
Segments Sent             = 41
Retransmitted Segments    = 0
Bad TCP CHKSUMS           = 0
RST Flagged Segments      = 0

UDP DATA

Datagrams Recieved        = 5587
No Destination Ports      = 998
Error Datagrams Received  = 0
Datagrams Sent            = 6627

Command ==>
F1=Help      F2=Main Menu  F3=Return      F5=Refresh      F6=Roll
              F12=Cancel

```

Figure 139. TCP/IP SNMP Resource View: System Panel

Press F3 to return to panel displayed in [Figure 137 on page 139](#). Choose option **7** and press Enter. The Resource View: System panel with Commands window is displayed, as displayed in [Figure 140 on page 141](#).



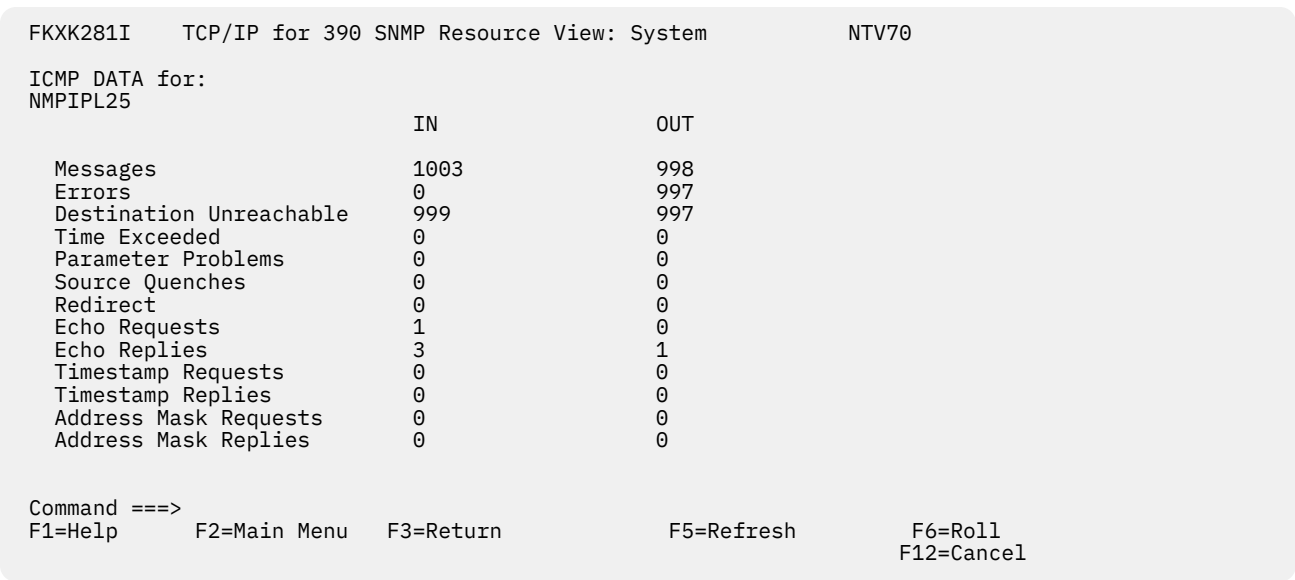


Figure 140. TCP/IP SNMP Resource View: System Panel with Commands Window

Return to the panel displayed in [Figure 136 on page 139](#) and press F11. The panel displayed in [Figure 141 on page 141](#) is displayed.

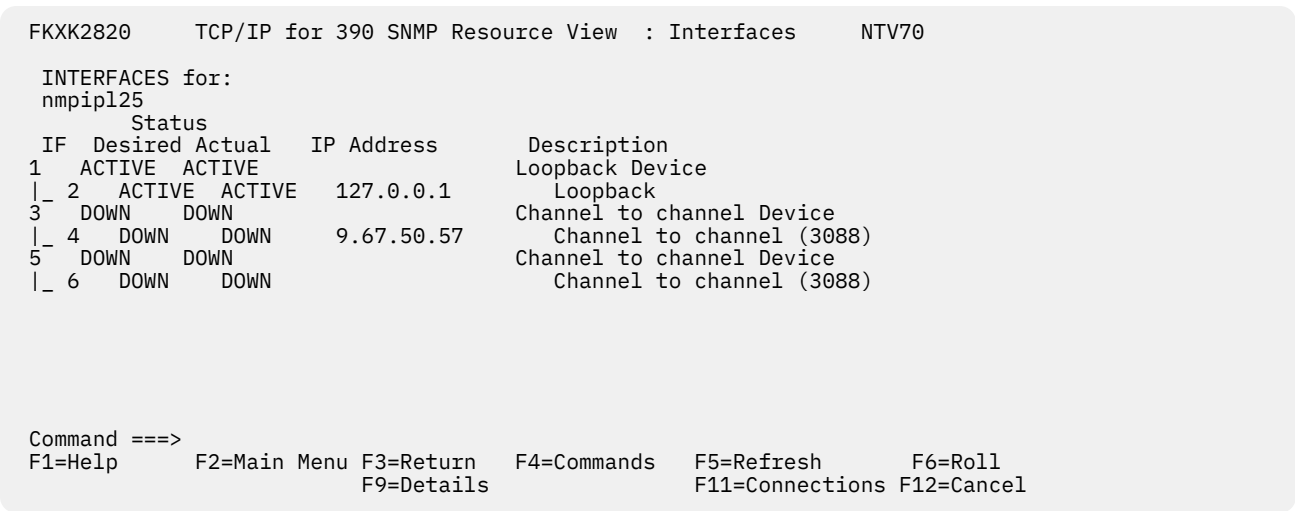


Figure 141. TCP/IP SNMP Resource View: Interfaces Panel

From the Interfaces panel, press F4. The SNMP Resource View: Interfaces panel is displayed with the Commands window as displayed in [Figure 142 on page 142](#).



```

FKXK282D      TCP/IP for 390 SNMP Resource View: Interfaces      NTV9D
INTERFACE DETAILS for:
resources.xyz.com
IF Name       : N/A
IP ADDRESS    : 34.34.130.1 Physical Address: '123412341234'h
Description:   XYZ PCI Token-Ring Family Adapter

_ Desired Status: ACTIVE      Actual Status: ACTIVE

Interface Type      = iso88025TokenRing(9)

Max Datagram Size   = 4056           Interface Speed      = 1600000
Last Status Change  = 0               TCP Connections      = N/A
Network Mask        = 34.34.130.1    Reassemble Max Size   = 65535
IF Specific         = 0.0
Connector Present   = N/A           Link Traps Enabled     = N/A
High Speed          = N/A           Accept Any Packets     = N/A

Set Community Name:

Command ==>
F1=Help      F2=Main Menu  F3=Return      F5=Refresh      F6=Roll
              F8=Statistics F11=Connections F12=Cancel

```

Figure 144. TCP/IP SNMP Resource View: Interfaces Panel

## MVS Stack Data

To display MVS stack data for NMPIPL27, perform the following steps:

1. Type NMPIPL27 in the Host Name or IP Address field.
2. If the current default is IP Resource, then delete the X from the IP Resource selection.
3. Type X to select **MVS Stack** view.
4. In the **TCP/IP Stack Name** field, type ? and press Enter. A selection list is displayed.
5. Type / in front of the selection you want and press Enter.

The TCP/IP SNMP Stack View: System panel, displayed in [Figure 145 on page 143](#).

```

FKXK28M0      TCP/IP for 390 SNMP Stack View: System      NTV9D

SYSTEM DATA for MVS Stack:
NMPIPL27

Proc Name = TCP38      ASID = 60      _ Primary IF Index = 4
                                   Primary IP ADDRESS = 1.23.45.67
_ System Name = SNMPBASE - Unspecified

UP Time      = 2019-4-17,10:2:28.8

Services     = Physical  DL/Subnet  Internet  E to E  Appl

Description  = Sysname: OS/390 Nodename: F79MVS Release: 08.00 Vers
              ion: 02 Machine: 9672

Set Community Name:

Command ==>
F1=Help      F2=Main Menu  F3=Return      F4=Commands      F5=Refresh      F6=Roll
              F8=Page 2    F9=Page 3      F11=Interfaces   F12=Cancel

```

Figure 145. TCP/IP SNMP Stack View: System Panel

From the panel shown in [Figure 145 on page 143](#), press F11. The TCP/IP Stack View: Interfaces panel is displayed with the interfaces defined for your TCP/IP stack.

```

FKXK282M      TCP/IP for 390 SNMP Stack View: Interfaces      NTV9D

INTERFACES for:
NMPIPL27
      Status
IF GWY Desired Actual   IP Address      Description
1  N ACTIVE  ACTIVE      127.0.0.1    Loopback Device
|_ 2  N ACTIVE  ACTIVE      127.0.0.1    Loopback
3  N ACTIVE  ACTIVE      1.23.45.67   Channel to channel Device
|_ 4  Y ACTIVE  ACTIVE      1.23.45.67   Channel to channel (3088)

Command ==>
F1=Help   F2=Main Menu   F3=Return   F4=Commands   F5=Refresh   F6=Roll
          F9=Details   F11=Connections F12=Cancel

```

Figure 146. TCP/IP SNMP Stack View: Interfaces Panel

To display interface details, tab to an interface, and press F9. The interface detail panel, shown in [Figure 147 on page 144](#), is displayed.

```

FKXK28MK      TCP/IP for 390 SNMP Stack View: Interfaces      NTV9D

INTERFACE DETAILS for:
NMPIPL27
IF Name       : TCPLNK9D      IP ADDRESS:  1.23.45.67
Device Base Number: 270      on IF:  3
Description:  Channel to channel (3088)

_ Desired Status:  ACTIVE      Actual Status:  ACTIVE

Interface Type      = channel(70)
Link Type           = ctc(2)
Max Datagram Size   = 32760
Last Status Change  = 2787
Network Mask        = 255.0.0.0
IF Specific         = 0.0
Connector Present   = NO(2)
High Speed          = 3,500,000 - 4,499,999

Interface Speed      = 450000
TCP Connections      = N/A
Reassemble Max Size = 0
Link Traps Enabled   = YES(1)
Accept Any Packets   = NO(2)

Set Community Name:

Command ==>
F1=Help   F2=Main Menu   F3=Return   F4=Gateway   F5=Refresh   F6=Roll
          F8=Statistics   F9=Gateway   F11=Connections F12=Cancel

```

Figure 147. TCP/IP SNMP Stack View: Interfaces (Detail) Panel

## Running IP Traces

**Note:** This function no longer requires AON. The information about this function is included in both *IBM Z NetView IP Management* and the *IBM Z NetView User's Guide: Automated Operations Network*.

Use the IPTRACE command to start and view diagnostic traces to help resolve TCP/IP problems. The following traces are available:

- IP packet trace is used for IP data flow problems and copies IP packets as they are received or sent.
- OSA packet trace is used for Open Systems Adapter (OSA) data flow problems and copies OSA packets as they are received or sent.
- Component trace is used to trace data processing problems between the client and the server.

If you want to use an external writer, it must be established before you use IP trace. The trace data is written to the writer. See *z/OS MVS Diagnosis: Tools and Service Aids* for more information about creating source JCL for an external writer, or for more details about trace functions.

# Issuing the IPTRACE Command

Issue the IPTRACE command from the NetView command line.

If you issue the IPTRACE \* command, all known stacks are displayed in the TCP/IP for 390 IPTrace Control Center panel (FKXK2A00), which is shown in [Figure 148 on page 145](#). From this panel, you must select the stack that you want to trace by tabbing to the stack and pressing Enter. Active stacks are highlighted in green. Inactive stacks are displayed in blue and cannot be selected.

|                         |              |                                       |                 |
|-------------------------|--------------|---------------------------------------|-----------------|
| FKXK2A00                |              | TCP/IP for 390 IPTrace Control Center |                 |
| Service Point/<br>Stack | Proc<br>Name | NetView<br>Domain                     | Trace<br>Status |
| NMP101                  | TCPIP        | LOCAL                                 | PKT/ACT OSA/ACT |
| NMP217                  | TCPIP        | NTVE1                                 | OSA/ACT         |
| Command ===>            |              |                                       |                 |
| F1=Help                 | F2=Main Menu | F3=Return                             | F6=Roll         |
| F7=Backward             | F8=Forward   |                                       | F12=Cancel      |

Figure 148. TCP/IP IPTrace Control Center Panel

If you issue IPTRACE with no parameters, information about the local stack is displayed in the IPTrace Control Center panel (FKXK2A01), which is shown in [Figure 149 on page 145](#). This panel is also displayed if you selected a stack on the TCP/IP for 390 IPTrace Control Center panel (FKXK2A00).

|                       |                       |           |                        |            |                       |  |
|-----------------------|-----------------------|-----------|------------------------|------------|-----------------------|--|
| FKXK2A01              |                       |           | IPTrace Control Center |            | NTVE1                 |  |
| Service Point: NMP217 |                       |           | Stack: TCPIP           |            | Domain: Local         |  |
| Status/Owner          |                       |           | Start                  | For        | Writer                |  |
| _ CTRACE              | SYSTCPIP              | NONE/NA   | NA                     | NA         | *NONE*                |  |
| _ PKTTRACE            | SYSTCPDA              | NONE/NA   | NA                     | NA         | *NONE*                |  |
|                       | ACTIVE TRACE SESSIONS |           |                        |            |                       |  |
| _ OSATRACE            | SYSTCPOT              | NONE/NA   | NA                     | NA         | *NONE*                |  |
|                       |                       |           |                        |            |                       |  |
| Command ==>           |                       |           |                        |            |                       |  |
| F1=Help               | F2=Main Menu          | F3=Return |                        | F5=Refresh | F6=Roll<br>F12=Cancel |  |

Figure 149. IPTrace Control Center Panel

This panel provides information about tracing on the selected stack. To select a trace, type any non-blank character next to the trace that you want and press Enter.

- If you select PKTTRACE (IP packet trace), see [“IP Packet Tracing”](#) on page 146.
- If you select OSATRACE (OSA packet trace), see [“OSA Packet Tracing”](#) on page 159.
- If you select CTRACE (component trace), see [“Component Tracing”](#) on page 162.

## IP Packet Tracing

The NetView program can collect trace data through two interfaces:

### instance trace

Allows multiple concurrent traces, each with a set of trace criteria. This type of trace is available with z/OS Communications Server V2.1. Each individual trace can be referred to as a *trace instance*.

### global trace

Allows only one trace at a time, with one set of trace criteria. If a second trace, with its set of trace criteria, is started while a trace is already running, the second trace replaces the first trace.

A global trace and one or more trace instances can run simultaneously.

FKXK2C01

Packet Trace Control

NTVF9

| Owner/Autotask  | Status  | Start Date/Time     | Description   |
|-----------------|---------|---------------------|---------------|
| - GLOBAL        | ACTIVE  | 2019-08-20 08:52:37 |               |
| - USER1/AUTTRA1 | ACTIVE  | 02/12/19 07:00:00   | STATUS FOR AP |
| - USER1/AUTTRA3 | DORMANT |                     |               |
| - USER1/AUTTRA4 | DORMANT |                     |               |
| - USER1/AUTTRA2 | DORMANT |                     |               |
| - USER2/AUTTRA7 | ACTIVE  | 02/12/19 12:24:12   | SLOW ATMS     |

Command ==>

F1=Help

F3=Return

F6=Roll

F7=Backward

F8=Forward

F9=Create Instance

F12=Cancel

Figure 150. Packet Trace Control Panel

From this panel, you can make the following selections:

- If you press F9 (Create Instance) or select a packet trace instance such as USER1/AUTTRA1, you can control a new or existing packet trace instance; see [“Starting and Stopping IP Packet Trace Instances”](#) on page 146.
- If you select GLOBAL, you can start or stop a global IP packet trace; see [“Starting and Stopping a Global IP Packet Trace”](#) on page 147.

### Starting and Stopping IP Packet Trace Instances

When you press F9 or select a packet trace instance on the Packet Trace Control panel (FKXK2C01), the Packet Trace Details panel (FKXK2C02) is displayed. You can set filters for a new packet trace instance or you can view information for or control an active packet trace instance, as follows:

- If you select an active packet trace instance, details about the trace instance are displayed. The information that is displayed cannot be changed.
- If you select a dormant packet trace instance, details about the trace instance are displayed and can be changed.
- If you press F9, you can set the filters for a new packet trace instance.

The following options are available from the Packet Trace Details panel (FKXK2C02), depending on the status of the trace. On the command line, enter the option number that you want:

- 1=START: If the trace is stopped (DORMANT status), this option restarts the collection of packets. If you are starting a new packet trace instance, this is the only valid option.
- 2=STOP: If the trace is active (collecting packets), this option stops the collection. When a trace is stopped, you can view the packets that were collected, but no new packets are collected.
- 3=DISPLAY: This option displays the collected packets; see [“Viewing IP Packet Trace Data” on page 148](#).
- 4=END: This option closes the trace, deletes all collected data, and releases storage.

### Starting and Stopping a Global IP Packet Trace

The PKTTRACE Control panel (FKXK2A22), which is shown in [Figure 151 on page 147](#), is displayed as follows:

- When you are running z/OS Communications Server V2.1 or later and select GLOBAL on the Packet Trace Control panel (FKXK2C01)
- When you are running an earlier version of z/OS Communications Server and select PKTTRACE on the IPTrace Control Center panel (FKXK2A01)
- When you select command **11. Packet Trace** on the TCP/IP for z/OS Connection Management panel (FKXK2221), which is shown in [Figure 107 on page 121](#)

|                                            |              |                             |                     |            |                |                |          |           |              |
|--------------------------------------------|--------------|-----------------------------|---------------------|------------|----------------|----------------|----------|-----------|--------------|
| FKXK2A22                                   |              | PKTTRACE Control - SYSTCPDA |                     |            |                | LOCAL          |          |           |              |
| Service Point: NMP217                      |              | Stack: TCPIP                |                     |            |                | Task: AUTOPKTS |          |           |              |
| PKTS: ACTIVE                               |              | SYSTCPDA: ACTIVE            |                     |            |                | GTF: NO        |          |           |              |
| Start Time: 2019-04-28-07:57:45            |              |                             |                     |            |                | Writer: *NONE* |          |           |              |
| Options: 1-START/ADD 2-STOP 3-VIEW PACKETS |              |                             |                     |            |                |                |          |           |              |
| Infrc/Link                                 |              | Stat                        | Prot                | IP         | Address/Prefix | Src Port       | Port Num | Dest Port | Record Count |
| -                                          | ALL          |                             | *                   | *          |                | *              | *        | *         |              |
| -                                          | TCPIPLINK6   | OFF                         | *                   | *          |                | *              | *        | *         | 0            |
| -                                          | TCPIPLINK    | OFF                         | *                   | *          |                | *              | *        | *         | 0            |
| -                                          | EZASAMEMVS   | OFF                         | *                   | *          |                | *              | *        | *         | 0            |
| -                                          | EZ6SAMEMVS   | OFF                         | *                   | *          |                | *              | *        | *         | 0            |
| -                                          | VIPLC9020A3C | OFF                         | *                   | *          |                | *              | *        | *         | 0            |
| -                                          | VIPLC9020A3D | OFF                         | *                   | *          |                | *              | *        | *         | 0            |
| -                                          | VIPLC9020A3E | OFF                         | *                   | *          |                | *              | *        | *         | 0            |
| -                                          | VIPLC9020A3F | OFF                         | *                   | *          |                | *              | *        | *         | 0            |
| Command ==>                                |              |                             |                     |            |                |                |          |           |              |
| F1=Help                                    | F2=Main Menu | F3=Return                   | F4=Stop SYSTCPDA    | F5=Refresh | F6=Roll        |                |          |           |              |
| F7=Backward                                | F8=Forward   | F9=Assist                   | F10=PKTS Management | F12=Cancel |                |                |          |           |              |

Figure 151. PKTTRACE Control Panel

Starting and stopping a global IP packet trace involves the following tasks:

- [“Managing the z/OS Communications Server IP Packet Trace” on page 147](#)
- [“Managing the NetView IP Packet Trace Collection Process” on page 148](#)

### Managing the z/OS Communications Server IP Packet Trace

If SYSTCPDA is not shown as ACTIVE on the PKTTRACE Control panel (FKXK2A22), type any character next to at least one interface (for example, TCPIPLINK6 as shown in [Figure 151 on page 147](#)) and press F4 (Start SYSTCPDA) to start packet tracing by the SYSTCPDA component of z/OS Communications Server. After packet tracing starts, SYSTCPDA is shown as ACTIVE, the status of the interface changes to ON, and the options change.

Instead of tracing by interface name, you can select **ALL** to start tracing for all the interfaces on the stack. Status and record counts are reported by and displayed in the PKTTRACE Control panel (FKXK2A22) for

each interface on the stack. If you select **ALL**, status and record counts are not totaled and are not displayed in the Stat and Record Count columns of the ALL line in the FKXK2A22 panel.

To set optional filters or see more detail for an interface, press F9 (Assist) on the PKTTRACE Control panel (FKXK2A22) to go to the PKTTRACE Control Assistant panel (FKXK2A29). On that panel, make any changes that are needed and press F3 to return.

On the PKTTRACE Control panel (FKXK2A22), note the Options line:

```
Options: 1-START/ADD 2-STOP 3-VIEW PACKETS
```

To start an IP packet trace filter for an interface, type 1 next to the interface name that you want to trace, and press Enter. To stop an IP packet trace filter, type 2 next to the interface name, and press Enter. To view the packets that were traced, type 3 next to the interface name, and press Enter; see [“Viewing IP Packet Trace Data”](#) on page 148.

### Managing the NetView IP Packet Trace Collection Process

To manage the packet collection process (the PKTS task), press F10 (PKTS Management). The NetView PKTS Management panel (FKXK2A23), which is shown in [Figure 152 on page 148](#), is displayed. Use this panel to start and stop the collection of packet data. Make any changes that are needed, and press F3 to return to the PKTTRACE Control panel (FKXK2A22).

**Note:** Instead of using the NetView PKTS Management panel, you can use the PKTS command or statements in the CNMSTUSR or CxxSTGEN member; see information about defining and controlling packet trace data collection in *IBM Z NetView IP Management*.

| FKXK2A23                 | PKTS Management | LOCAL               |
|--------------------------|-----------------|---------------------|
| Start with Storage Size: | M               | PKTS Status: ACTIVE |
| - Stop                   |                 |                     |
| - Stopcoll               |                 |                     |
| Define TCPName: TCPIP    | OPID: AUTOPKTS  |                     |
| - Purge                  |                 |                     |
| Intfname: *              |                 |                     |
| LAddr: *                 |                 |                     |
| RAddr: *                 |                 |                     |
| LPort: *                 | RPort: *        | Portnum: *          |
| Start Time: *            |                 |                     |
| End Time: *              |                 |                     |
|                          | Protocol 1      | 1-All               |
|                          |                 | 2-TCP               |
|                          |                 | 3-UDP               |
|                          |                 | 4-ICMP              |
|                          |                 | 5-OSPF              |
|                          |                 | 6 (Number)          |
| Command ==>              |                 |                     |
| F1=Help                  | F3=Return       | F6=Roll             |
|                          |                 | F12=Cancel          |

Figure 152. NetView PKTS Management Panel for IP Packet Trace

### Viewing IP Packet Trace Data

#### Display Packet Control panel (FKXK2A24)

Use the Display Packet Control panel (FKXK2A24), which is shown in [Figure 153 on page 149](#), to specify additional filtering criteria. This panel is displayed by typing 3 next to the interface to be traced on the PKTTRACE Control panel (FKXK2A22).



```

FKXK2A24                Display Packet Control                LOCAL
                                                                NMPIPL10

Stack: TCPIP                                Infc Name: ALL

LAddr *

RAddr *

Portnum: *          LPort: *          RPort: *          Protocol: - ALL (default)
                                                                - TCP
                                                                - UDP
                                                                - ICMP
                                                                - OSPF
                                                                - (Number)

Start Time: *
End Time:  *

MaxRecs: 1  1-Last  100          Truncate: 65535
          2-First

Command ==>
F1=Help  F2=Save Packets  F3=Return  F4=View Packets          F6=Roll
          F8=Extended Opts          F10=Analyze          F12=Cancel

```

*Figure 153. Display Packet Control Panel*

The interface name in the **Infc Name** field is carried over from the **Infc/Link** field on the PKTTRACE Control panel (FKXK2A22) but can be changed to search for packets that are associated with any valid interface. You can also use an asterisk (\*) at the end of the interface name as a wildcard to see packets that are from several interfaces that have similar names; for example, to see packets for all interfaces that have names that start with TCPIP, you can specify TCPIP\* in the Infc Name field.

To save packets in CTRACE and Sniffer trace format, press F2 (Save Packets). The Display Packet Control - Save Packets pop-up panel (FKXK2S24) will be displayed, which is shown in [Figure 156 on page 151](#).

The NetView program issues an MVS START command to run the MVS procedure defined in PKTS.SNFPROC to copy or convert the CTRACE data set to the Sniffer format data set. The NetView user that issues the request must have the authority to issue the NetView MVS command.

To display packet trace data, set any display options and press F4 (View Packets). For displaying packets, you can make only one selection in the Protocol field. Pressing F4 opens the PKTTRACE SUMMARY panel (FKXK2A26); see [Figure 154 on page 150](#).

**Note:** You can request a more specific trace report by pressing F8 (Extended Options), which opens the Display Packets Control Extended Options panel (FKXK2A25). On that panel, you can set options for detailed packet data.

### **PKTTRACE SUMMARY panel (FKXK2A26)**

The PKTTRACE SUMMARY panel (FKXK2A26) provides summary packet trace information for the packets requested. To navigate the panel, press F11 (Right) to scroll to the right or F8 (Forward) to scroll forward.

To save packets in CTRACE format and Sniffer trace format, press F2 (Save Packets). The PKTTRACE Summary - Save Packets pop-up panel (FKXK2S26) will be displayed, which is shown in [Figure 155 on page 150](#).

To see commands that are available for a specific packet, press F9 (Commands). A list of commands that you can use for the selected packet, such as Ping, TraceRte, and Hostnames, is displayed.

```

FKXK2A26      PKTTRACE  SUMMARY                                NTV70
                                                         More:+
DP   Nr hh:mm:ss.mmmmmm IpId   Seq_num   Ack_num   Wndw  Flags
OU1582418 12:42:23.053840 911E protocol=UDP
                        ESP Spi: 3082015D Seq: 02010104
OU1582417 12:42:23.053296 911C protocol=UDP
                        ESP Spi: 308201C8 Seq: 02010104
IU1582416 12:42:23.053281 911D protocol=UDP
                        30470201 0104076E *.> 0G.....n*
OU1582415 12:42:23.052033 911A protocol=UDP
                        ESP Spi: 308201EC Seq: 02010104
IU1582414 12:42:23.052017 911B protocol=UDP
                        30470201 0104076E *.> 0G.....n*
IU1582413 12:42:23.050783 9119 protocol=UDP
                        30470201 0104076E *.> 0G.....n*
OU1582412 12:42:23.050075 9118 protocol=UDP
                        ESP Spi: 308201BC Seq: 02010104
IU1582411 12:42:23.049256 9117 protocol=UDP
                        30470201 0104076E *.> 0.....n*

Command ==>
F1=Help      F2=Save Packets F3=Return      F4=Details   F5=Refresh   F6=Roll
F7=Backward  F8=Forward      F9=Commands   F11=Right    F12=Cancel

```

Figure 154. PKTTRACE SUMMARY Panel

```

FKXK2S26      PKTTRACE  SUMMARY - Save Packets                NTV70
                                                         More:+
DP   Nr hh:mm:ss.mmmmmm IpId   Seq_num   Ack_num   Wndw  Flags
IT  710 05:40:57.570752 0046 2754826990 846188506 32767 ACK

: .....:
: CTRACE format data set name:      :
:  NETVIEW.NMPIPL10.D030619.T083012.CTRACE :
: .....:
: Create Sniffer File (Y/N)  N      :
: Sniffer format data set name:      :
:  NETVIEW.NMPIPL10.D030619.T083012.SNIFFER :
: .....:
: Press Enter to save packets to the data set(s) :
: F1=Help      F3=Return      F6=Roll      F12=Cancel :
: .....:
Command ==>

```

Figure 155. PKTTRACE Summary - Save Packets Panel

### Display Packet Control - Save Packets pop-up panel (FKXK2S24)

Use the Figure 156 on page 151 to save packet traces in CTRACE format and Sniffer trace format. If you have specified a PKTS.SAVEHLQ in CNMSTYLE, the data set names for CTRACE and Sniffer trace formats are generated by the NetView program, and are displayed in the fields. You can change the names as required by overtyping the fields.

Set the **Create Sniffer File** field to Y to run the process to generate a Sniffer trace format data set. If the packet traces are saved successfully in CTRACE format and Sniffer trace format, the DSI633I message is displayed.

```

FKXK2S24          Display Packet Control - Save Packets          LOCAL
                                                             NMPIPL10

Stack: TCPIP                      Infc Name: ALL

LAddr *

RAddr *

: .....:
: CTRACE format data set name:      :
:  NETVIEW.NMPIPL10.D030619.T083012.CTRACE  :
: .....:
: Create Sniffer File (Y/N)  N      :
: Sniffer format data set name:      :
:  NETVIEW.NMPIPL10.D030619.T083012.SNIFFER  :
: .....:
:   Press Enter to save packets to the data set(s)      :
: F1=Help          F3=Return          F6=Roll          F12=Cancel      :
: .....:
Command ==>

```

Figure 156. Display Packet Control - Save Packets panel

### Analyzing and Saving IP Packet Traces

If you press F10 (Analyze) in the Display Packet Control panel (FKXK2A24), the analysis function scans the TCP, UDP, and ICMP trace sessions in the collected trace data and analyzes the TCP sessions for a set of error conditions. To keep data from being lost, stop the trace collection before starting analysis. The amount of data that is analyzed is based on the options that are selected on the Display Packet Control panel (FKXK2A24).

**Note:** The UDP and ICMP sessions are listed so that you can view the session report for the individual sessions, but these sessions are not analyzed.

From the resulting display, you can request a list of the sessions that contain specific types of errors, and, from that list, you can request the detailed information for a specific session. The detailed data for a session contains all the packets in the collected trace data that is associated with the selected session.

For the analysis function, the rules for the Protocol field on the Display Packet Control panel (FKXK2A24) are as follows:

- The TCP, UDP, and ICMP protocols are supported.
- More than one protocol can be selected, and any combination of the supported protocols is allowed.
- Selecting ALL, or selecting ALL in combination with any of the supported protocols, processes all the supported protocols.
- The OSPF and numeric protocols are not supported. If these protocols are specified, either singly or in combination with other protocols (including ALL), an error message is displayed.

If you select only UDP or ICMP as the protocol on the Display Packet Control panel (FKXK2A24), when you press F10 (Analyze), the trace analysis panel for the selected protocol is displayed. See [“Viewing UDP and ICMP Sessions” on page 156](#).

Otherwise, when you press F10 (Analyze), the Packet Trace Analysis panel (FKXK2B10), which is shown in [Figure 157 on page 152](#), lists all the sessions that were found, based on the options that you selected.

|                                 |           |                       |   |                 |            |
|---------------------------------|-----------|-----------------------|---|-----------------|------------|
| FKXK2B10                        |           | Packet Trace Analysis |   | LOCAL<br>NMP217 |            |
| TCP Sessions 2                  |           | UDP Sessions 7        |   | ICMP Sessions 0 |            |
| TCP Sessions with error flags 2 |           |                       |   |                 |            |
| Unacknowledged Syms             | 0         | Window Probes         | 0 |                 |            |
| Retransmissions                 | 0         | Reset Flags           | 0 |                 |            |
| Duplicate Acks                  | 2         | Delayed Acks          | 0 |                 |            |
| Zero Window Size                | 0         |                       |   |                 |            |
| Command ==>                     |           |                       |   |                 |            |
| F1=Help                         | F3=Return | F4=Sessions           |   | F6=Roll         | F12=Cancel |

Figure 157. Packet Trace Analysis Panel

From this panel, you can select the TCP Sessions with error flags option to display the TCP sessions that have error flags, or select a specific TCP session error type (Retransmissions, Duplicate Acks, Zero Window Size, Window Probes, Reset Flags, or Delayed Acks) to display all sessions that have the selected error flag. After you make a selection (for example, Duplicate Acks), press F4 (Sessions) to display a list of the sessions that meet your specification on the Packet Trace Analysis TCP Sessions panel (FKXK2B22). For more information about TCP session analysis, see [“Analyzing TCP Sessions” on page 152](#).

The following actions are also available from the Packet Trace Analysis panel (FKXK2B10):

- Place the cursor on TCP Sessions and press F4 (Sessions) to go to the Packet Trace Analysis All TCP Sessions panel (FKXK2B20), which lists all TCP sessions and the number of packets and error flags for each. For information about filtering and navigation on this panel, see the online help.
- Place the cursor on UDP Sessions or ICMP Sessions and press F4 (Sessions) to go to the trace analysis panel for the selected protocol. For more information about UDP and ICMP sessions, see [“Viewing UDP and ICMP Sessions” on page 156](#).

### Analyzing TCP Sessions

The Packet Trace Analysis TCP Sessions panel (FKXK2B22), which is shown in [Figure 158 on page 153](#), displays the sessions with the error type that you specified on the Packet Trace Analysis panel (FKXK2B10). The Trace Flag Type field shows the error type that you selected, in this case, Duplicate Ack. Two lines are displayed for each session with packets that match the requested error flag type. The first line contains the number of packets in the session and the local IP address and port. The second line contains the remote IP address and port for the session.

```

FKXK2B22          Packet Trace Analysis TCP Sessions          NTVF9

Trace Flag Type: Duplicate Ack

  Packets  Flags  IP Address-Ports (Local and Remote)
-   90      2      9.42.45.101-1033
-   140     1      9.42.45.101-1035
-   96      2      9.42.45.101-1038
-   70      2      9.42.45.101-23
-   14      1      9.42.45.196-1026
-   24      1      9.42.45.101-1034
-   22      1      9.42.45.196-21
-   22      1      9.42.45.101-1037
-   22      1      9.42.45.196-21
-   22      1      9.42.45.101-1039
-   22      1      9.42.45.196-23

Command ==>
F1=Help          F3=Return    F4=Select      F6=Roll
                  F9=Filters          F12=Cancel

```

Figure 158. Packet Trace Analysis TCP Sessions Panel

In this panel, you can navigate through the session list by using commands to move to the beginning or the end of the session list or you can specify the number of sessions to scroll; for more information, see the online help. To filter the session list, press F9 (Filters); you can filter by local IP address and port, by local IP address, by local port, by remote IP address and port, by remote IP address, or by remote port. From the filtered session list that is displayed, you can return to the unfiltered list by pressing F9 (Unfilter).

If you select a session on the Packet Trace Analysis TCP Sessions panel (FKXK2B22) and press F4 (Select), the Session Analysis panel (FKXK2B51) is displayed. This panel is shown in [Figure 159 on page 153](#). Note the following information on this panel about the selected session, all of which can be useful in diagnosing a connection or network problem:

- The resolved host names for the local and remote IP addresses
- The connection status (for this example, ESTABLISHED)
- The number of errors of each type for both inbound and outbound data
- The variations in window sizes for inbound and outbound data

```

FKXK2B51          Session Analysis          LOCAL

Local IP 9.42.45.101
Port 1026 Host Name nmp101.tivlab.raleigh.ibm.com

Remote IP 9.42.45.10
Port 23 Host Name nmpipl10.tivlab.raleigh.ibm.com

Total Packets Summarized 30 Status ESTABLISHED

  Flags      Inbound  Outbound  Window Size  Inbound  Outbound
-----
Retransmissions 0        0        Largest      32768    32768
Duplicate Acks  0        1        Average      32759    32753
Reset           0        0        Smallest     32750    32707
Window Size 0   0        0
Window Probes  0        0
Delay Ack      1        2

Command ==>
F1=Help          F3=Return    F4=Select      F6=Roll
                  F8=Packages  F9=Actions     F10=Report     F12=Cancel

```

Figure 159. Session Analysis Panel

From the Session Analysis panel (FKXK2B51), shown in [Figure 159 on page 153](#), you can continue your analysis as follows:

- To go to the Session Analysis Packets panel (FKXK2B53), press F8 (Packets); see [“Displaying Session Packets” on page 154](#).
- To go to the Session Analysis - Actions panel (FKXK2B52), press F9 (Actions); see [“Issuing Commands for Sessions” on page 155](#).
- To display a full z/OS Communications Server session report, press F10 (Report). For more information, see the z/OS Communications Server library.

#### Displaying Session Packets

The Session Analysis Packets panel (FKXK2B53), which is shown in [Figure 160 on page 154](#), displays the packets for the selected connection. To navigate the panel, press F11 (Right) to scroll to the right or F8 (Forward) to scroll forward.

| FKXK2B53 Session Analysis Packets |    |   |            |            |        |      |          |                 |           | LOCAL  |
|-----------------------------------|----|---|------------|------------|--------|------|----------|-----------------|-----------|--------|
|                                   |    |   |            |            |        |      |          |                 |           | More:+ |
| Packet Summary                    |    |   |            |            |        |      |          |                 |           |        |
| TcpHdr                            | IO | F | Seq        | Ack        | RcvWnd | Data | Delta    | Time            | TimeStamp |        |
| S                                 | 0  |   | 2258808397 | 0          | 32768  | 0    | 0.000000 | 13:15:56.397877 |           |        |
| A                                 | S  | I | 2414361815 | 2258808398 | 32768  | 0    | 0.017182 | 13:15:56.415059 |           |        |
| A                                 |    | 0 | 2258808398 | 2414361816 | 32768  | 0    | 0.000071 | 13:15:56.415130 |           |        |
| AP                                |    | I | 2414361816 | 2258808398 | 32768  | 3    | 0.042128 | 13:15:56.457258 |           |        |
| A                                 |    | 0 | 2258808398 | 2414361819 | 32765  | 0    | 0.277782 | 13:15:56.735040 |           |        |
| AP                                |    | 0 | 2258808398 | 2414361819 | 32765  | 3    | 0.344952 | 13:15:57.079992 |           |        |
| AP                                |    | 0 | 2258808401 | 2414361819 | 32765  | 3    | 0.000511 | 13:15:57.080503 |           |        |
| AP                                |    | I | 2414361819 | 2258808404 | 32765  | 0    | 0.000861 | 13:15:57.081364 |           |        |
| AP                                |    | I | 2414361819 | 2258808404 | 32765  | 3    | 0.011026 | 13:15:57.092390 |           |        |
| AP                                |    | 0 | 2258808404 | 2414361822 | 32765  | 0    | 0.245366 | 13:15:57.337756 |           |        |
| AP                                |    | I | 2414361822 | 2258808404 | 32765  | 6    | 0.000851 | 13:15:57.338607 |           |        |
| AP                                |    | 0 | 2258808404 | 2414361828 | 32762  | 18   | 0.000643 | 13:15:57.339250 |           |        |
| AP                                |    | I | 2414361828 | 2258808422 | 32750  | 3    | 0.016542 | 13:15:57.355792 |           |        |
| AP                                |    | 0 | 2258808422 | 2414361831 | 32765  | 3    | 0.000281 | 13:15:57.356073 |           |        |
| AP                                |    | 0 | 2258808425 | 2414361831 | 32765  | 3    | 0.000100 | 13:15:57.356173 |           |        |
| AP                                |    | I | 2414361831 | 2258808425 | 32765  | 9    | 0.000855 | 13:15:57.357028 |           |        |

Command ==>

F1=Help

F7=Backward F8=Forward

F3=Return

F4=Packet Detail

F11=Right

F6=Roll

F12=Cancel

*Figure 160. Session Analysis Packets Panel*

To see the details of a packet, select a packet on the Session Analysis Packets panel (FKXK2B53 or FKXK2B54) and press F4. The Packet Detail window, which is shown in [Figure 161 on page 155](#), displays the full content of the selected packet. To return (to FKXK2B53 or FKXK2B54), press F3.

```

CNMKWIND OUTPUT FROM Packet Detail                               LINE 0 OF 48
*----- Top of Data -----*
z/OS TCP/IP Packet Trace Formatter, Copyright IBM Corp. 2000, 2009; 2009.028

**** 2019/02/12
RcdNr Sysname Mnemonic Entry Id Time Stamp Description
-----
971 NMP101 PACKET 00000004 13:15:56.415130 Packet Trace
To Interface : TCPIPLINK Device: QDIO Ethernet Full=52
Tod Clock : 2019/02/12 13:15:56.415130 Intfx: 5
Segment # : 0 Flags: Out
Source : 9.42.45.101
Destination : 9.42.45.196
Source Port : 1029 Dest Port: 23 Asid: 002C TCB: 00000000
IpHeader: Version : 4 Header Length: 20
Tos : 00 QOS: Routine Normal Service
Packet Length : 52 ID Number: 523A
Fragment : Offset: 0
TTL : 64 Protocol: TCP CheckSum: BB0D F
Source : 9.42.45.101
Destination : 9.42.45.196

TCP
Source Port : 1029 ( ) Destination Port: 23 (telnet)
Sequence Number : 2258808398 Ack Number: 2414361816
Header Length : 32 Flags: Ack
Window Size : 32768 CheckSum: C87B FFFF Urgent Data Pointer:
Option : NOP
Option : NOP
Option : Timestamp Len: 10 Value: 1EE2BE99 Echo: 1EE2BE98

Ip Header : 20 IP: 9.42.45.101, 9.42.45.196 Offset: 0
000000 45000034 523A0000 4006BB0D 092A2D65 092A2DC4

Protocol Header : 32 Port: 1029, 23 Offset: 14
000000 04050017 86A2AE4E 8FE83CD8 80108000 C87B0000 0101080A 1EE2BE99 1EE2BE98
No packets required reassembly

=====
TO SEE YOUR KEY SETTINGS, ENTER 'DISPFK'
CMD==>

```

Figure 161. Packet Detail for the Packet Selected in Session Analysis Packets

### Issuing Commands for Sessions

The Session Analysis - Actions panel (FKXK2B52), which is shown in [Figure 162 on page 155](#), displays commands that you can issue for a selected connection.

```

FKXK2B52 Session Analysis - Actions LOCAL NMPIPL10
Local IP: 9.30.240.21 *****
Port: 175 Host Name: NMPIPL - Command
Remote IP: 9.30.238.6 1. Drop Connection
Port: 2339 Host Name: HSSVMA 2. Save Session Packets in
Total Packets Summarized: 400 NetView Format
3. Save Session Packets in
CTTRACE format and
SNIFFER format
Flags Inbound Outbound
-----
Retransmissions 0 0
Duplicate Acks 0 0
Reset 0 0
Window Size 0 0 F1=Help F3=Return
Window Probes 0 0 F6=Roll F12=Cancel
Delayed Acks 0 0 *****
Selective Acks 0 0

Command ==>

```

Figure 162. Session Analysis - Actions Panel

The following actions are available from the Session Analysis - Actions panel (FKXK2B52):

- To drop the active connection for this session, enter 1 (Drop Connection). The Session Analysis Drop Command Confirmation panel (FKXK2B5A) is displayed. This panel provides information about the connection that is to be dropped. Press F4 (Confirm Drop) to confirm that you want to drop the connection, or press F3 (Return) to cancel the drop request.
- To save the session packets in NetView format, enter 2. The Saved Packet Trace Details panel (FKXK2B62) is displayed. This panel provides details about the packet trace that you want to save. Provide a description (for example, Print server problem) and press F4 to save the trace. The FKX475I message indicates that the trace was saved. You can retrieve the trace later; see [“Retrieving Saved IP Packet Traces”](#) on page 157.
- To save the session packets in CTRACE format and Sniffer trace format, enter 3. The Session Analysis Data Set Name Input panel (FKXK2B55) is displayed so that you can specify the name of a data set in which to save the packets. If the save is successful, the DSI633I message is displayed.

### Viewing UDP and ICMP Sessions

If you selected only **UDP** or **ICMP** as the protocol before pressing F10 (Analyze) on the Display Packet Control panel (FKXK2A24) or if you placed the cursor on **UDP Sessions** or **ICMP Sessions** before pressing F4 on the Packet Trace Analysis panel (FKXK2B10), the sessions are displayed in the Packet Trace Analysis UDP Sessions or the Packet Trace Analysis ICMP Sessions panel (FKXK2B23), which is shown in [Figure 163 on page 156](#).

|             |                                             |                                            |
|-------------|---------------------------------------------|--------------------------------------------|
| FKXK2B23    | Packet Trace Analysis UDP Sessions          | NTVF9                                      |
|             | Packets IP Address-Ports (Local and Remote) |                                            |
| - 2         | 9.42.45.101-9850                            |                                            |
|             | 9.42.4.253-53                               |                                            |
| - 4         | 9.42.45.101-9851                            |                                            |
|             | 9.42.4.253-53                               |                                            |
| - 4         | 9.42.45.101-9860                            |                                            |
|             | 9.42.4.253-53                               |                                            |
| - 4         | 9.42.45.101-9861                            |                                            |
|             | 9.42.4.253-53                               |                                            |
| Command ==> |                                             |                                            |
| F1=Help     | F3=Return<br>F9=Filters                     | F4=Session Report<br>F6=Roll<br>F12=Cancel |

Figure 163. Packet Trace Analysis UDP Sessions Panel

In this panel, the IP addresses and ports that are shown under **IP Address-Ports** are the connection partners for UDP sessions. For ICMP sessions, only IP addresses are shown.

**Note:** ICMP is a connectionless protocol that does not have actual sessions, but the Communications Server IP Trace Session Report collects the ICMP packets between two hosts and displays them in a session report; this is known as an ICMP session.

In this panel, you can navigate through the session list by using commands to move to the beginning or the end of the session list or you can specify the number of sessions to scroll; for more information, see the online help. To filter the session list, press F9 (Filters); you can filter by local IP address and port, by local IP address, by local port, by remote IP address and port, by remote IP address, or by remote port. From the filtered session list that is displayed, you can return to the unfiltered list by pressing F9 (Unfilter).

Select a session and press F4 to see the session report for that session, such as the UDP session report that is shown in the PACKET DETAIL window in [Figure 164 on page 157](#).



```

CNMKWIND OUTPUT FROM  PACKET DETAIL                                LINE 0 OF 50
*----- Top of Data -----*
BNH773I NUMBER OF PACKETS: 2 , MISSED BUFFERS: 0 , TCPNAME: TCPIP
z/OS TCP/IP Packet Trace Formatter, Copyright IBM Corp. 2000, 2009; 2009.028

**** 2019/04/16
      No packets required reassembly

=====
Interface Table Report
Index Count Link                      Address
   5      2 TCPIPLINK                9.42.45.101

UDP Sessions Report
      1 sessions found
-----
2 UDP packets summarized
Local IP address:                    9.42.45.101
Local port, app:                      9850
Remote IP address:                    9.42.4.253
Remote port, app:                      53 domain
Protocol:                             UDP
First TimeStamp:      2019/04/16 13:26:29.545709
Last TimeStamp:       2019/04/16 13:26:29.546448
Duration:              00:00:00.000739
Statistics:              Inbound,    Outbound
Packets:                1,          1
Fragments:              0,          0 (  0%), (  0%)
Bytes sent:              298,        49
Time spent sending:      00:00:00.0, 00:00:00.0 ( 100%), (  0%)
Time spent fragments:    00:00:00.0, 00:00:00.0 (  0%), (  0%)
Throughput:              454.711,    74.768 K/s
Largest segment size:    298,        49
Average segment size:    298,        49
Smallest segment size:   298,        49
Direction change:        1,          0
UDP Flags IO              Delta      Time DatLn  RcdNr  Inf  Ip_id  Froff  Drop
      0 00:00:00.000000 13:26:29.545709 49 2848 5 738C 0 0
      D I 00:00:00.000739 13:26:29.546448 298 2849 5 0000 0 0
16384 bytes of storage were used for the session report
TO SEE YOUR KEY SETTINGS, ENTER 'DISPFK'
CMD==>

```

Figure 164. UDP Session Report (Packet Detail for the Selected UDP Session)

The following actions are available from this window:

- To view the contents of a packet, move the cursor to a packet summary that is highlighted in green and press F4.
- To save the UDP or ICMP packet data, press F9.

### Retrieving Saved IP Packet Traces

Use the LISTTRC command, which has no parameters, from the NetView command line to see a list of all of the traces that are saved. The Saved Packet Traces panel (FKXK2B61), which is shown in [Figure 165 on page 158](#), displays any packet traces that you saved. This panel shows the print server problem trace that was saved in “[Issuing Commands for Sessions](#)” on [page 155](#).



```

CNMKWIND OUTPUT FROM  FMTPACKT Session Detail Report                LINE 0 OF 269
*----- Top of Data -----*
BNH773I NUMBER OF PACKETS: 221 , MISSED BUFFERS: 0 , TCPNAME: TCPIP
z/OS TCP/IP Packet Trace Formatter, Copyright IBM Corp. 2000, 2009; 2009.028

**** 2019/09/09
      No packets required reassembly

=====
Interface Table Report
Index Count Link                Address
   2    221 LOOPBACK            9.42.45.101

UDP Sessions Report
   1 sessions found
-----
221 UDP packets summarized
Local IP address:                9.42.45.101
Local port, app:                 161 snmp
Remote IP address:              9.42.45.101
Remote port, app:               5823
Protocol:                       UDP
First TimeStamp:                2019/09/09 10:59:02.698435
Last TimeStamp:                 2019/09/09 10:59:02.800651
Duration:                       00:00:00.102216
Statistics:                     Inbound, Outbound
Packets:                        0,      221
Fragments:                      0,      0 ( 0%), ( 0%)
Bytes sent:                      0,    10657
Time spent sending:              00:00:00.0, 00:00:00.1 ( 0%), ( 100%)
Time spent fragments:            00:00:00.0, 00:00:00.0 ( 0%), ( 0%)
Throughput:                      0,    101.887 K/s
Largest segment size:           0,      83
Average segment size:           0,      48
Smallest segment size:          0,      45
Direction change:               0,      0
UDP Flags IO                    Delta      Time  DatLn  RcdNr  Inf  Ip_id  Froff  Drop
      0 00:00:00.000000 10:59:02.698435    47  41638   2   73F6    0    0
      0 00:00:00.000588 10:59:02.699023    47  41640   2   73F8    0    0
      0 00:00:00.000426 10:59:02.699450    47  41642   2   73FA    0    0
TO SEE YOUR KEY SETTINGS, ENTER 'DISPFK'
CMD==>

```

Figure 167. FMTPACKT Session Detail Report

## OSA Packet Tracing

When you select OSATRACE, the OSATRACE Control panel (FKXK2A30), which is shown in [Figure 168](#) on [page 159](#), is displayed.

```

FKXK2A30      OSATRACE Control SYSTCP0T ACTIVE      for NVDomain: LOCAL
                                           z/OS      : V1R12
Service Point/Stack: TVT2007 TCPNAME: TCPIP7
OPKTS: ACTIVE      On Task: AUTOOPKT GTF: NO

Start Time:                                           Writer: *NONE*

Options: 1-START  2-STOP  3-VIEW PACKETS

  OSA Port  Stat/ Length Data      Record      Time  Discard  Nofilter
           Auth
- OSA1      OFF   224   1024      2147483647 10080 EXCEPTION NONE
           UNKNOWN 0      0      0      0
- OSAA      ON    224   1024      2147483647 10080 EXCEPTION ALL
           LOGICAL 0      172   12    164
           NEW    224   1024      2147483647 10080 EXCEPTION NONE

Command ==>
F1=Help          F3=Return  F4=Stop SYSTCP0T F5=Refresh  F6=Roll
F7=Backward     F8=Forward  F9=Filters      F10=PKTS Management F12=Cancel

```

Figure 168. OSATRACE Control Panel

## Starting and Stopping an OSA Packet Trace

Starting and stopping OSA packet traces involves the following tasks:

- “Managing the z/OS Communication Server OSA Packet Trace” on page 160
- “Managing the NetView OSA Packet Trace Collection Process” on page 160

### Managing the z/OS Communication Server OSA Packet Trace

If SYSTCPOT is not shown as ACTIVE on the OSATRACE Control panel (FKXK2A30), type any character next to at least one interface (for example, OSAA) and press F4 (Start SYSTCPOT) to start packet tracing by the SYSTCPOT component of z/OS Communications Server. After OSA packet tracing starts, SYSTCPOT is shown as ACTIVE, the status of the interface changes to ON, and the options change.

**Note:** The OSA SNMP subagent must be active to control OSA traces.

To set optional filters or to see more detail for an interface, press F9 (Filters) on the OSATRACE Control panel (FKXK2A30) to go to the OSATRACE Filters panel (FKXK2A31), which is shown in [Figure 169 on page 160](#). Make any changes that are needed, press F4 to update the filters, and then press F3 to return.

|                              |          |         |                            |        |                     |
|------------------------------|----------|---------|----------------------------|--------|---------------------|
| FKXK2A31                     | OSATRACE | Filters | SYSTCPOT                   | ACTIVE | for NVDomain: LOCAL |
| Service Point/Stack: TVT2007 |          |         | Proc: TCPIP7               |        | z/OS : V1R12        |
| OSA Port Name: OSAA          |          |         | Clear Filters: NO (YES/NO) |        |                     |

| Protocol | Ethernet Type | Port  | Device ID | VLAN ID | Mac Address |
|----------|---------------|-------|-----------|---------|-------------|
| TCP      |               | 00020 |           |         |             |
|          |               | 00021 |           |         |             |
|          |               |       |           |         |             |
|          |               |       |           |         |             |
|          |               |       |           |         |             |
|          |               |       |           |         |             |
|          |               |       |           |         |             |

Command ==>  
F1=Help  
F3=Return  
F4=Update Filters  
F6=Roll  
F12=Cancel  
F8=IP Addresses

Figure 169. OSATRACE Filters Panel

To set optional IP address filters for the interface from the OSATRACE Filters panel (FKXK2A31), press F8 (IP Addresses) to go to the OSATRACE Filters panel (FKXK2A32). After you make your changes, press F3 until you return to the OSATRACE Control panel (FKXK2A30).

On the OSATRACE Control panel (FKXK2A30), note the Options line:

Options: 1-START 2-STOP 3-VIEW PACKETS

To start an OSA packet trace filter for an OSA port name, type 1 next to the OSA port name that you want to trace, and press Enter. To stop an OSA packet trace filter, type 2 next to the OSA port name, and press Enter. To view the packets that were traced, type 3 next to the OSA port name, and press Enter; see “Viewing OSA Packet Trace Data” on page 161.

### Managing the NetView OSA Packet Trace Collection Process

To manage the packet collection process (the PKTS task), press F10 (PKTS Management). The NetView PKTS Management panel (FKXK2A33), which is shown in [Figure 170 on page 161](#), is displayed. Use this panel to start and stop the collection of packet data. Make any changes that are needed, and press F3 to return to the OSATRACE Control panel (FKXK2A30).

**Note:** Instead of using the NetView PKTS Management panel, you can use the PKTS command or statements in the CNMSTUSR or CxxSTGEN member; see information about defining and controlling packet trace data collection in *IBM Z NetView IP Management*.

|                                       |                                            |                                     |
|---------------------------------------|--------------------------------------------|-------------------------------------|
| FKXK2A33                              | NetView PKTS Management<br>for PSOURCE=OSA | PKTS Status: ACTIVE<br>Domain:LOCAL |
| Start with Storage Size: M            |                                            |                                     |
| _ Stop                                |                                            |                                     |
| _ Stopcoll                            |                                            |                                     |
| Define TCPName: TCPIP7 OPID: AUT00PKT |                                            |                                     |
| _ Purge                               |                                            |                                     |
| Portname *                            |                                            |                                     |
| Time: Start *                         |                                            |                                     |
| End *                                 |                                            |                                     |
|                                       |                                            |                                     |
| Command ==>                           |                                            |                                     |
| F1=Help                               | F3=Return                                  | F6=Roll<br>F12=Cancel               |

Figure 170. NetView PKTS Management Panel for OSA Packet Trace

### Viewing OSA Packet Trace Data

Use the Display OSA Packet Control panel (FKXK2A34), which is shown in Figure 171 on page 161, to specify additional filtering criteria. This panel is displayed by typing 3 next to the OSA port name to be traced on the OSATRACE Control panel (FKXK2A30).

|                                           |                            |                       |
|-------------------------------------------|----------------------------|-----------------------|
| FKXK2A34                                  | Display OSA Packet Control | LOCAL                 |
| Service Point/Stack: TVT2007 Proc: TCPIP7 |                            |                       |
| OSA Port Name: OSAA                       |                            |                       |
| Time: Start *                             |                            |                       |
| End *                                     |                            |                       |
|                                           |                            |                       |
| MaxRecs: 1                                | 1-Last 100                 | Truncate: 65535       |
|                                           | 2-First                    |                       |
|                                           |                            |                       |
| Command ==>                               |                            |                       |
| F1=Help                                   | F3=Return F4=View Packets  | F6=Roll<br>F12=Cancel |
| F8=Extended Options                       |                            |                       |

Figure 171. Display OSA Packet Control Panel

To display packet trace data, set any display options and press F4 (View Packets). Pressing F4 opens the OSA TRACE PACKETS SUMMARY panel (FKXK2A36); see Figure 172 on page 162.

**Note:** You can request a more specific trace report by pressing F8 (Extended Options), which opens the Display OSA Packets Control Extended Options panel (FKXK2A35). On that panel, you can set options for detailed packet data.

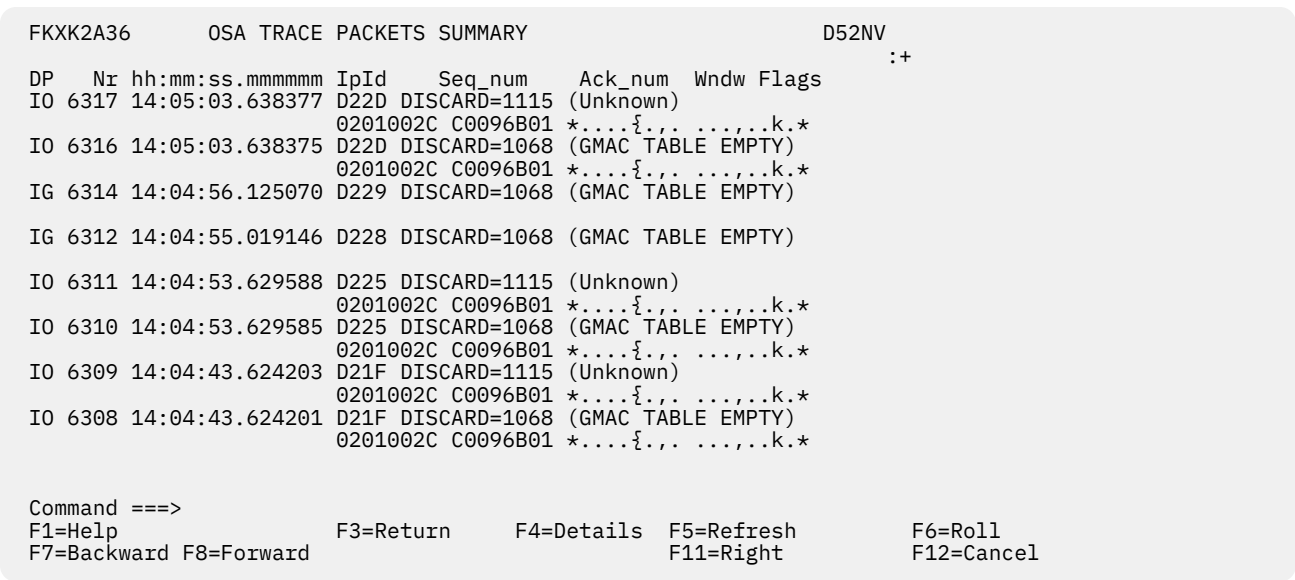


Figure 172. OSA TRACE PACKETS SUMMARY Panel

The OSA TRACE PACKETS SUMMARY panel (FKXK2A36) provides summary packet trace information for the packets requested. To navigate the panel, press F11 (Right) to scroll to the right or F8 (Forward) to scroll forward.

From the OSA TRACE PACKETS SUMMARY panel (FKXK2A36 or FKXK2A37), select a packet and press F4 (Details). The details of the selected packet are displayed in the Packet Detail window, which is shown in [Figure 173 on page 162](#).

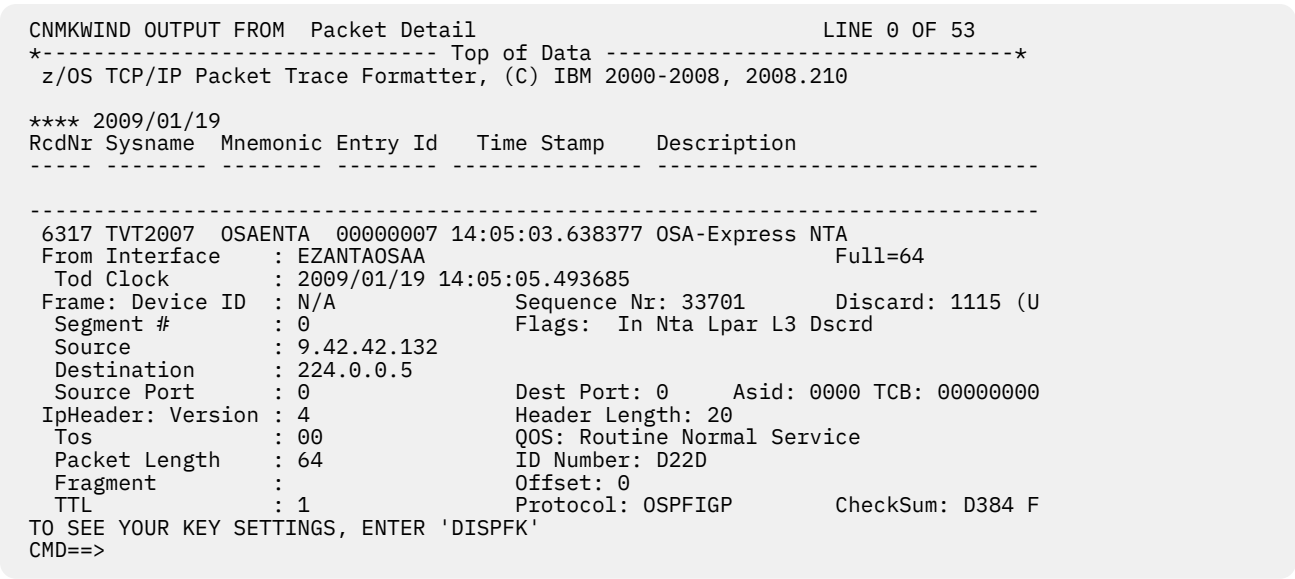


Figure 173. Packet Detail for the Selected OSA Packet

### Component Tracing

When you select CTRACE for a service point with no scheduled tracing, the TCP/IP for 390 CTRACE Control panel (FKXK2A12), which is shown in [Figure 174 on page 163](#), is displayed. For a service point with active or delayed tracing, the TCP/IP for 390 CTRACE Control panel (FKXK2A10), which is shown in [Figure 178 on page 166](#), is displayed. For information about the fields on the panel, press F1 to see the online help.

|                                             |                               |                    |                               |
|---------------------------------------------|-------------------------------|--------------------|-------------------------------|
| FKXK2A12                                    | TCP/IP for 390 CTRACE Control | SYSTCPIP           | A55NV                         |
| Service Point/Stack: TVT2009                |                               | Proc: TCPIP        | Domain: LOCAL                 |
| Delay Start Until: YYYY-MM-DD-HH.MM.SS      |                               | On Task: JOEF_____ |                               |
| Set a Timer for HH.MM.SS                    |                               | Writer: CTTCP__    |                               |
| _ LISTS (IPAddrS, Ports, Job Names, ASIDs ) |                               |                    |                               |
| Command ==>                                 |                               |                    |                               |
| F1=Help                                     | F2=Main Menu                  | F3=Return          | F4=Start Trace<br>F10=Options |
|                                             |                               |                    | F6 =Roll<br>F12=Cancel        |

Figure 174. CTRACE Control Panel (FKXK2A12)

Before you begin a trace, provide the following information on the TCP/IP for 390 CTRACE Control panel (FKXK2A12):

- You can specify IP addresses, IP ports, job names, and address space identifiers (ASIDs) that are to be traced. To do that, select the LISTS field; see [“Specifying IP Addresses, IP Ports, Job Names, and Address Space Identifiers for the Component Trace”](#) on page 163.
- You must specify options for the trace. To do that, press F10; see [“Specifying Options for the Component Trace”](#) on page 164.
- You must decide if the trace is to begin immediately or is to be scheduled for a later date or time. This cannot be done until the trace options are specified. See [“Starting or Scheduling the Component Trace”](#) on page 165.

### Specifying IP Addresses, IP Ports, Job Names, and Address Space Identifiers for the Component Trace

When you select the LISTS field, the TCP/IP for 390 CTRACE Control panel (FKXK2A11) displays the IP addresses and submasks component trace filters currently set for this stack. If the CTRACE status is ACTIVE, all component trace filters currently set are collected and displayed. When the CTRACE status is in DELAY mode, all global variables for component trace filters that are set for a delayed start are displayed. When the trace is inactive, this panel looks similar to the one in [Figure 175 on page 164](#), and you can edit the IPADDRS/Mask(Prefix) fields. For information about the fields on the panel, press F1 to see the online help.

Figure 175. CTRACE Control Panel (FKXK2A11)

Figure 176. CTRACE Control Panel (FKXK2A1A)

**Note:** If you select **ALL**, all options except Serial, Storage, and Timer are selected. These options degrade performance if you choose all of them simultaneously.



```

FKXK2A14          TCP/IP for 390 CTRACE Control  SYSTCPIP          A55NV

OPTIONS:

-  ALL          -  ALLMIN        -  ACCESS        -  AFP          -  ARP
-  CLAW         -  CONFIG       -  C_SOCKET     -  DLC          -  EID
-  ENGINE       -  FIREWALL    -  ICMP         -  IN           -  INIT
-  INTERNET     -  IOCTL       -  IPADDR       -  IPSEC        -  LATCH
-  LCS          -  MESSAGE     -  MINIMUM      -  MISC         -  ND
-  NONE         -  OETCP       -  OEUDP        -  OPCMDS       -  OPMSG
-  PASAPI       -  PFS         -  PFSMIN       -  PING         -  POLICY
-  PORT         -  QUEUE       -  RAW          -  ROUTE        -  RW
-  SERIAL       -  SMTP        -  SNMP         -  SOCKAPI      -  SOCKET
-  STORAGE     -  SYSTEM      -  TC           -  TCP          -  TCPMIN
-  TELNET       -  TELNVTAM    -  TIMER        -  TN           -  UD
-  UDP          -  VTAM        -  VTAMDATA     -  WORKUNIT     -  XCF

Command ==>
F1=Help      F2=Main Menu      F3=Return
F6 =Roll
F12=Cancel

```

OPTIONS:

|   |          |   |           |   |          |   |          |   |        |
|---|----------|---|-----------|---|----------|---|----------|---|--------|
| - | ALL      | - | ALLMIN    | - | ACCESS   | - | AFP      | - | ARP    |
| - | CLAW     | - | CONFIG    | - | C SOCKET | - | DLC      | - | EID    |
| - | ENGINE   | - | FIREWALL  | - | ICMP     | - | IN       | - | INIT   |
| - | INTERNET | - | IOCTL     | - | IPADDR   | - | IPSEC    | - | LATCH  |
| - | LCS      | - | MESSAGE   | - | MINIMUM  | - | MISC     | - | ND     |
| - | NONE     | - | OETCP     | - | OEUDP    | - | OPCMDS   | - | OPMSG  |
| - | PASAPI   | - | PFS       | - | PFSMIN   | - | PING     | - | POLICY |
| - | PORT     | - | QUEUE     | - | RAW      | - | ROUTE    | - | RW     |
| - | SERIAL   | - | SMTP      | - | SNMP     | - | SOCKAPI  | - | SOCKET |
| - | STORAGE  | - | SYSTEM    | - | TC       | - | TCP      | - | TCPMIN |
| - | TELNET   | - | TELNV TAM | - | TIMER    | - | TN       | - | UD     |
| - | UDP      | - | VTAM      | - | VTAMDATA | - | WORKUNIT | - | XCF    |

Command ==>  
F1=Help      F2=Main Menu      F3=Return      F6 =Roll  
F12=Cancel

Figure 177. CTRACE Control Panel (FKXK2A14)

## Starting or Scheduling the Component Trace

After you select options on the TCP/IP for 390 CTRACE Control panel (FKXK2A14), you can start a trace on the TCP/IP for 390 CTRACE Control panel (FKXK2A12) in any of these ways:

- To start the trace immediately, press F4 (Start Trace).
- To delay the start of the trace, specify the date or date and time that tracing is to begin in the **Delay Start Until** field. The time must be specified in a 24-hour format. If the date portion YYYY-MM-DD is omitted and the input HH.MM.SS is earlier than the current time, the trace starts on the next day at the input time. For example, if the current time on the system clock is 9:33 a.m. and you specify 07.45.00 as the time to begin component tracing, the trace is scheduled to begin the following day at 7:45 a.m.
- To specify how long the trace is to run, specify a time in the **Set a Timer for** field. If you leave the **Set a Timer for** field blank, the trace runs until it is manually stopped.

If you want the trace to run on another task, enter any valid operator ID (as defined in DSIOPF) in the **On Task** field, which is valid only for delayed traces.

**Note:** If you specify another operator ID, that operator must be logged on at the time the trace runs.

The **Writer** field contains the source JCL to create an external writer where trace data is stored. The writer must be established before the trace runs. For more information about creating source JCL for an external writer, see *z/OS MVS Diagnosis: Tools and Service Aids*. The writer name for component tracing can be customized. Copy the following statement to the CNMSTUSR or CxxSTGEN member, and then change it to the appropriate name. For information about changing CNMSTYLE statements, see *IBM Z NetView Installation: Getting Started*. If the NetView program has been started, issue the RESTYLE COMMON command to pick up the change.

```
COMMON.EZLTCPcTRACEwriter = new_CTTCP // AON TCP component writer name
```

After you finish setting options, press F4 to begin or schedule the trace. The trace can take several minutes to run because of system processing. The trace might also begin a couple of minutes past the specified time, depending on system processing. When an immediate trace is successfully scheduled, a message similar to the following message is displayed in the IPTrace Control Center panel (FKXK2A01):

FKX400I CTRACE SCHEDULED FOR SP TVT2009 BY OPERATOR JOEF

For a delayed trace, the FKX401I message is displayed.

## Displaying an Active or Delayed Component Trace

To view the details of an active or delayed component trace, select CTRACE from the IPTrace Control Center panel (FKXK2A01). The TCP/IP for 390 CTRACE Control panel (FKXK2A10), which is shown in [Figure 178 on page 166](#), is displayed.

```
FKXK2A10      TCP/IP for 390 CTRACE Control SYSTCPIP      A55NV
Service Point/Stack: TVT2009      Proc: TCPIP      Domain: LOCAL
Start Time: 2007-10-04-14:19:18      Duration: NA
Timer ID: NONE      Writer: CTTCP

Active Options:
ALL

_  LISTS (IPAddrS, Ports, Job Names, ASIDs )

Command ==>
F1=Help      F2=Main Menu      F3=Return      F4=Stop Trace      F6 =Roll
F12=Cancel
```

*Figure 178. CTRACE Control Panel (FKXK2A10)*

## Stopping an Active or Canceling a Delayed Component Trace

To stop an active trace or to cancel a delayed trace, press F4 (Stop Trace) on the TCP/IP for 390 CTRACE Control panel (FKXK2A10), which is shown in [Figure 178 on page 166](#). When you press F4, the system can take several minutes to respond because of system processing.

When an active trace is successfully stopped, the FKX403I message is displayed. When a delayed trace is successfully canceled, the DSI205I message is displayed.

---

## Part 4. AON Customization



---

## Chapter 19. Introducing AON customization

The Automated Operations Network (AON) provides a comprehensive set of programs that can be customized and extended to provide network automation. The following components of AON provide consistent automation across multiple network protocols:

- SNA Automation (AON/SNA)
- TCP/IP Automation (AON/TCP)

AON intercepts alerts and messages that indicate problems with network resources. AON attempts to recover failed resources and monitor resources until they recover. After a resource has recovered, the components of AON keep a record of the resource failures to track recurring network problems.

To automate network functions, AON provides facilities for:

- Online access to centralized information
- Automated problem determination and recovery actions
- User-written routines and user-tailored panels
- Online tutorials and product information through the operator interface.

The following sections describe the programs that enable AON to automate z/OS networks.

---

### Centralized access to information

AON enables users to gain access to network information from panels and issue commands to multiple systems from a single terminal. This centralized control is provided by:

- Dynamic Display Facility (DDF)
- Focal-point services
- NetView operator interface

### Dynamic Display Facility (DDF)

The DDF is the central focus of resource status information. It displays network resources currently being acted on by AON in an exception-based hierarchical panel display. As you add new resources to your network, DDF can automatically reflect those resources without requiring you to change AON control file entries. AON also updates message notifications held by the command facility (NCCF).

The DDF enterprise-wide resource display provides real-time, exception-oriented status monitoring. An operator at the focal-point NetView can view the status of multiple networks from the DDF panels. AON dynamically updates DDF panels with resource status information. Multiple programs can update DDF with resource status without concern for the sequence or priorities posted by other programs. Many operators can view DDF and receive updates without having an impact on the network.

AON provides default settings for DDF to represent the status of your network resources. Default DDF settings display only resources that require operator intervention. The mark function enables operators to take responsibility for a failed resource that requires operator intervention. DDF is updated to indicate who is working on the problem. When failed components recover, DDF automatically removes them from the display.

DDF displays are color-coded to indicate the status of the network components. The color indicates the severity of a failure. If AON detects a problem with a resource, it displays the resource in red, pink, or yellow (default color definitions). Typically, a failure displayed in red is more severe than a failure displayed in yellow. Green (or not displayed) indicates that the resource status is normal.

You can customize DDF settings, including colors. For example, by changing a control file, you can make DDF display all resources instead of displaying only those resources that require intervention.

For more information about customizing and using DDF, see Chapter 20, “Understanding Dynamic Display Facility (DDF) design,” on page 177 and subsequent chapters.

## NetView Operator Interface

With the NetView operator interface to AON, users can:

- Issue commands
- Receive responses for functions provided by AON and other NetView facilities

Operators can enter commands either in line mode (on any NetView command line) or in full-screen mode. Full-screen mode provides easy-to-use command pop-up windows.

Additionally, NetView management console (NMC) operators can access an AON window to display automated resource recovery information and control file information for SNA resources.

## Focal point services

You can specify one domain as the focal point in a multidomain network. AON routines forward notification messages from multiple hosts to a single focal-point host. This enables a network operator to receive all the network notifications at a single console.

To support focal points, you define routes, called *gateways*, from one NetView program to another. AON forwards automation notifications, commands, and responses through these gateways.

To implement focal-point services, you define a hierarchical domain structure composed of a focal-point domain and distributed domains. You can also define a backup for the focal-point host, for situations when the primary focal-point host is unavailable. Through focal-point services, you can:

- Enable operators to send commands and receive responses through the gateways. This eliminates the need for personal NetView-NetView task (NNT) sessions for each operator.
- Manage the Resource Access Control Facility (RACF®) passwords for gateway automation operators
- Set up and initiate automated operator NNT sessions
- Display the status of gateway automation operators and user NNT sessions

## Automated recovery

---

This section describes the AON generic recovery processes. AON provides generic automated recovery processes used by the components of AON, which enable you to focus on other areas of operations. AON automation provides the following benefits:

- Fast recovery
- Great network availability
- Automatic responses to predefined messages
- Reminder notification of unavailable resources or resources that are in manual recovery mode
- Limited operator intervention
- AON categorization recovery as critical or non-critical
- More time for the operations staff to work in other areas where automation is not currently available

Although AON can help you automate and manage several different types of networks, the tasks required to automate these networks are similar in nature. There are different methods of getting resource status in each type of network, but the basic structure of the automation is essentially the same.

AON includes routines that perform similar tasks on different types of networks. This means that the automation process for systems network architecture (SNA), Advanced Peer-to-Peer Networking (APPN), or Transmission Control Protocol/Internet Protocol (TCP/IP) resources is the same, although the actual programs that perform the automation steps might be different. For example, if a resource fails in a SNA or Internet Protocol network, information about the resource (such as resource type, connectivity, and

status) must be gathered before automated recovery can continue for that resource. Each of these network types has a different program that gathers resource information, but they are called and processed in the same manner by the AON automation routine.

The benefits of having common automation routines are:

- Code reliability
- Transfer of skill across networks (because the process is the same for the different types of networks)
- Easier problem determination

AON automates network operations such as resource recovery. This section provides an overview of how AON facilitates recovery, and describes the tools it uses to accomplish automation of network operators, including:

- Automation table
- Automated tasks
- Logging
- Providing generic failure and recovery routines
- Sending messages to the notification operators when further action is required
- Sending records to the automation log for tracking purposes

AON supports network resource recovery by monitoring critical resources and taking automated action based on tailored criteria. Each automation component recovers specific types of resources. For example, SNA automation recovers SNA (VTAM) resources. AON reacts to adverse conditions of network resources and notifies operators of these conditions, when appropriate.

Recovery criteria can be set based on resource type, resource naming convention, explicit resource name, or network-wide settings. You can select various parameters and options to control when and how recovery takes place.

The AON control file contains automation criteria. There are two ways to update the control file, dynamically or statically:

- You can update the control file dynamically through the POLICY command interface or through the AON operator interface. These changes remain in effect until you reload the control file or until the current tasks that were started by the NetView program end.
- You can update the control file statically with a system editor, such as the Interactive System Productivity Facility (ISPF). After editing the control file, you can use the POLICY command to reload the control file, reinitialize AON, or recycle your NetView environment.

## **Automation and status logs**

To improve problem determination productivity, AON uses automation logs and status logs to record the automation process and status of the network resources.

The status log tracks:

- Last 10 failures of a resource, with the time stamp
- Current automation status and threshold exceptions
- Last operator notified about a resource or one that acted on a resource

Users can issue the AON DSPSTS command to display status log information. AON provides a facility (DBMAINT) to maintain this VSAM status file.

The AON automation log records all automation activity. When a resource becomes unavailable or when the resource becomes available again, AON writes availability records to the log. These records indicate whether the action was caused by automation, the Help Desk function, or by an operator. You can issue the AON NLOG command to view the automation log.

## NetView log

AON writes entries to the NetView log for all significant events relative to the control file. This includes operator messages, internal events, and errors within the control file.

## Automation notification

AON provides you with a notification policy that can be customized to do the following types of operator notification:

- Issue a message.
- Generate an alert or resolution to the hardware monitor.
- Update the DDF.
- Send an event to the event server.
- Send an email request.
- Generate a beeper request.

Operators using NetView management console can request Alert History to view alerts generated by AON. AON also sets the Automation in Progress status so you can see that automation is attempting to recover the failed resource. Failed resources that cannot be recovered can be seen in the Operator Intervention Network view (OIV).

## Automation table

The NetView program enables you to issue commands based on any incoming message or Management Services Unit (MSU). You can specify criteria that must be satisfied before running a program. The criteria can include the presence of specific message text data, resource names, other message attributes, or specific MSU data.

AON provides a predefined message table structure (included in DSITBL01) with each component adding prescribed messages, or MSUs, and criteria. When any of these prescribed messages and conditions occur, the AON generic failure or recovery routines are started to take action. Actions are defined in the AON option definition tables. In some cases, the automation table directly drives a command list.

## Automation operators

The multiple automation operator design enables AON to divide its workload among several, separately defined automated operators by using the concurrent processing capability of NetView program. All AON automation operators are started during initialization and must be active at all times.

The automation operators are identified by unique names corresponding to their responsibilities. For example, some of the automation operators supplied with AON include:

### **AIPOPER**

Sets and resets the AIP (Automation In Progress) operator status bit in RODM. This bit causes a display pattern to be placed on the object in NetView management console. RODM AIP operators issue the commands necessary to update resource objects in RODM views with the AIP operator status. These operators are also used in the management of the OIV processing.

### **ALRTOPER**

Sends alerts and resolutions to NetView over an LU 6.2 session.

### **BASEOPER**

Provides backup for other automation operators.

### **GATOPER**

The outbound gateway operator for automation notification forwarding.

### **INFOPER**

Serializes the updates to the inform log.

### **MSGOPER**

Formats and issues AON notifications and DDF updates.



**NETOPER**

Initiates routines based on the NetView automation table and AON generic failure and recovery routines.

**OIVOPER**

An optional operator task used by the Operation Intervention View (OIV) function. When enabled, automatically deletes resources from the OIV at specified intervals. Only resources with the display status of satisfactory (129) are removed.

**TCPOPER**

Used for TCP/IP automation

**TRAPOPER**

Used for trap automation processes

**WKSTOPER**

Sends and receives commands and responses between AON and a workstation with the interface installed.

**X25OPER**

Used by X25 automation processes

For more information, refer to the *IBM Z NetView Administration Reference*.

## Notification operators

Notification operators are operators identified (in NTFYOP control file entries) to receive *messages* generated by AON. Notifications are necessary to understand and operate the network. They must be routed to the notification operators or the automation operators.

In a distributed network, notification operators are defined at the focal-point domain. Notification operators can also be defined at the distributed domain. If they are defined at both, notifications are forwarded to notification operators at both the focal point and distributed domains.

If the host that creates the automation notification is not the focal-point domain, the message is forwarded to the focal-point domain. In a single NetView environment, the domain is its own focal point.

When the notification arrives or is generated at the focal-point host, this notification is sent to the notification operators and can be "frozen" on the operator's screen. AON enables you to specify the types of messages that are frozen on the operator's screen. When notifications are frozen, operators can use the DM command to remove them from the screen.

The notification operators are defined in the control file by operator name, operator description, and message class. For example:

```
NTFYOP OPER1,OPER='OPERATOR 1',CLASS=10
```

The class of the notification is compared to the class of each defined notification operator. Each notification operator with a class matching the class of the automation notification receives the notification. For example, all AON notifications have a class of 90. Therefore, any notification operators defined with CLASS=90 receive all notifications.

## Thresholds (SNA Only)

AON tracks resource failures and recoveries in a status file. If the number of errors exceeds a defined threshold over a period of time (which is defined in the THRESHOLDS control file entry), AON alerts the notification operator that the resource is experiencing multiple errors and continues recovery attempts. For critical thresholds, AON stops recovery attempts.

AON uses three types of threshold definitions:

**Critical**

AON uses critical thresholds to stop automated recovery. AON deactivates the resource and notifies the notification operators. Automated recovery is stopped until an operator reactivates the resource.

### **Frequent**

AON uses frequent thresholds to indicate recurring errors that might warrant an operator's attention. When a resource exceeds a frequent threshold, automated recovery continues, but AON sends a message to the notification operators and makes an entry in the automation log.

### **Infrequent**

AON uses infrequent thresholds to indicate that a resource is experiencing random or intermittent errors. Automated recovery continues, but AON sends a message to the notification operators and makes an entry in the automation log. When a resource fails and one of these thresholds is exceeded, AON sends a notification to the defined AON notification operator.

## **Large-scale thresholding**

AON uses the LSTHRESH (large-scale thresholding) control file entry to count network-wide events of a particular type, and then sets a threshold on the number of times the event can happen over specific period of time.

## **User exits**

AON is designed to fulfill most automated recovery needs. However, an installation can have particular automation requirements that must be met. AON provides user exits that you can code for specific recovery processing, monitored intervals, threshold checking, and SNA resource information gathering.

You set user exit values in the control file. The user exits are run from certain AON common routines.

For more information about user exits, see [Chapter 28, “AON user exits,” on page 333](#).

## **System console**

AON uses the NetView write-to-operator with reply (WTOR) facility to communicate with and solicit responses from the system operator. Likewise, a system operator can run AON routines and commands from the system console by preceding the command with the subsystem designation assigned to the NetView program.

## **Database Maintenance Facility (DBMAINT)**

AON provides a database maintenance facility (DBMAINT) that performs database maintenance for the NetView hardware monitor and session monitor. You can also use DBMAINT to purge records from the AON status file. This database maintenance includes the deletion of records before a specified date and the reorganization of the VSAM database.

DBMAINT uses the ENVIRON SETUP entry in the AON control file to correctly perform database maintenance. The ENVIRON SETUP control file entry must match the way the VSAM database was allocated, or errors occur.

You can use the DBMAINT facility from a full-screen operator interface panel or you can schedule it to run on an automation operator station task. The latter can be achieved by building a Timer entry in the AON control file that runs DBMAINT at a specific time and date. This process automates database and file maintenance. It also enables you to schedule database maintenance during non-peak hours of operation.

## **Tailored routines and displays**

---

You can expand the AON functions to meet additional automation needs unique to your network systems. You can do this by:

- Tailoring the EZLCFG01 control file
- Using the common routines provided with AON

## Tailoring the EZLCFG01 Control File

The NetView program loads the DSIPARM member EZLCFG01 control file during initialization. This file contains values such as the notification operator IDs, the automation operator IDs, threshold values for resources, monitoring values, and recovery values. In most cases you can change the control file values without restarting the NetView program, because the new values become effective as soon as the AON control file is reloaded.

The data in the control file is organized by keywords, and therefore is independent of the program language or operating system. You can set values for individual resources, resource types, or system-wide defaults. This reduces the need for generating a set of variables to contain the information needed for each individual resource. You can input your own data in the AON control file to have a common depository for information used by programs on your network.

When you first install AON, system programmers customize control file entries for their network needs.

For more information, see the *IBM Z NetView Automation Guide*.

## Using the Common Routines Provided with AON

AON includes routines that you can use to code network automation extensions. These routines provide easy-to-use generic functions for expanding automation capabilities beyond those supplied and supported by AON. You can use these generic functions to reduce development time when you create procedures or extend those provided.

Common routines perform automated actions such as transferring information and checking the control file. User-written programs can call common routines from the message table, control file, or an extended routine to accomplish a required task.

For more information about common AON routines, see [Chapter 26, “Coding common routines,” on page 273](#).

## Cross-domain logon

---

With AON cross-domain logon (CDLOG), you can log on to all or a select group of domains. CDLOG enables you to select which domains you want.

If CDLOG is unable to establish a session with the domain you selected, it displays the message received when attempting to start the session. After trying to set up sessions with other NetView domains, CDLOG indicates the status of all the domains in the domain list.

To use the AON/SNA Help Desk for resources in another domain, you must define the CDLOG entry. The SNA Help Desk determines which domain owns a resource. For the help desk to do this, however, you must be able to log on to all the domains known to your NetView program. Automatic logon has two types of logon procedures:

- Automatic mode
- Semiautomatic mode

Automatic mode takes the operator ID and password from the control file and logs on to the selected domains. Semiautomatic mode requires operator action to select domains and provide passwords.

The AON remote gateway (RGWY command) requires CDLOG definitions to start RMTCMD sessions. Several AON functions use the remote gateway sessions.



---

## Chapter 20. Understanding Dynamic Display Facility (DDF) design

The Dynamic Display Facility (DDF) enables you to identify and focus on a specific problem in the network. The DDF functions enable you to alter or extend your DDF installation. This chapter describes the structure and processing of DDF:

- Understanding the hierarchical display of DDF
- Defining dependencies
- Defining the priority and color of the resources
- Updating status
- Defining panels
- Defining multiple systems
- Implementing DDF
- Defining the contents of DDF
- Understanding the flow of DDF
- Starting and stopping DDF
- Loading panels
- Loading tree structures
- Using the definition procedure

**Note:** Most of the examples used in this chapter are specific to Systems Network Architecture (SNA) resources, but the concepts apply to all AON automation components. To customize DDF to suit your environment, modify one or more members in DSIPARM and DDF panels in CNMPNL1. Sample members and panels are provided. Those samples contain references to domain CNM01. You cannot substitute system symbolics within any of those samples. Instead, you can use the EZLEISP1 utility shipped in CNMCLST. For more information, refer to the *IBM Z NetView Installation: Configuring Additional Components*.

---

### Setting up the Dynamic Display Facility for AON

The Dynamic Display Facility (DDF) enables you to identify and focus on a specific problem in the network. DDF uses colors to represent the status of various network resources. DDF can track the error, warning, action, or informational states of network resources. DDF uses tree structures to implement the hierarchical display of status information. Although DDF can display the status of all network resources, you can tailor DDF to display only resources that have an error.

To facilitate the installation of DDF, the distribution tape provides samples to set up a typical DDF implementation. This implementation reflects DDF updates for a single domain (no focal point implementation).

You can use three types of definitions to set up DDF. Each of these types has its own file or files. The definition types are:

- Entries in the control file
- Tree structures (one file per domain)
- Panel characteristics (one file per panel)

The sample control file and panels are set up to run for domain CNM01. If your domain ID is different, change all occurrences of CNM01 to your domain ID in the following members:

- All members whose name starts with EZL (such as EZLCFG01 and EZLTREE)

- All members whose name starts with FKVPN (for AON/SNA)
- All members whose name starts with FKXPN (for AON/TCP)

## Understanding the Hierarchical Status display

---

DDF uses colors to represent the status of various network resources. DDF can track the error, warning, action, or informational states of network resources. Although DDF can display the status of all network resources, you can tailor DDF to display only resources that have an error. This helps you focus on network problems rather than the entire network and save central processing unit (CPU) time.

DDF displays the status of resources (active or inactive). Each type of status is displayed in a different color. For example, inactive resources are red and active resources are green. For more information, see [“Defining default colors for status components \(EMPTYCOLOR\)”](#) on page 207.

You can tailor DDF to display the status of all resources of a particular type. If you use this design, DDF displays each resource type in a different color. For example, DDF might display a Network Control Program (NCP) problem in red and a line problem in pink. See [“Displaying Network Status on a single panel”](#) on page 223 for instructions about tailoring DDF to display by resource type.

DDF presents a hierarchical status display. [Figure 179 on page 179](#) illustrates a sequence of sample SNA-specific DDF panels, moving from the domain to the resource level.

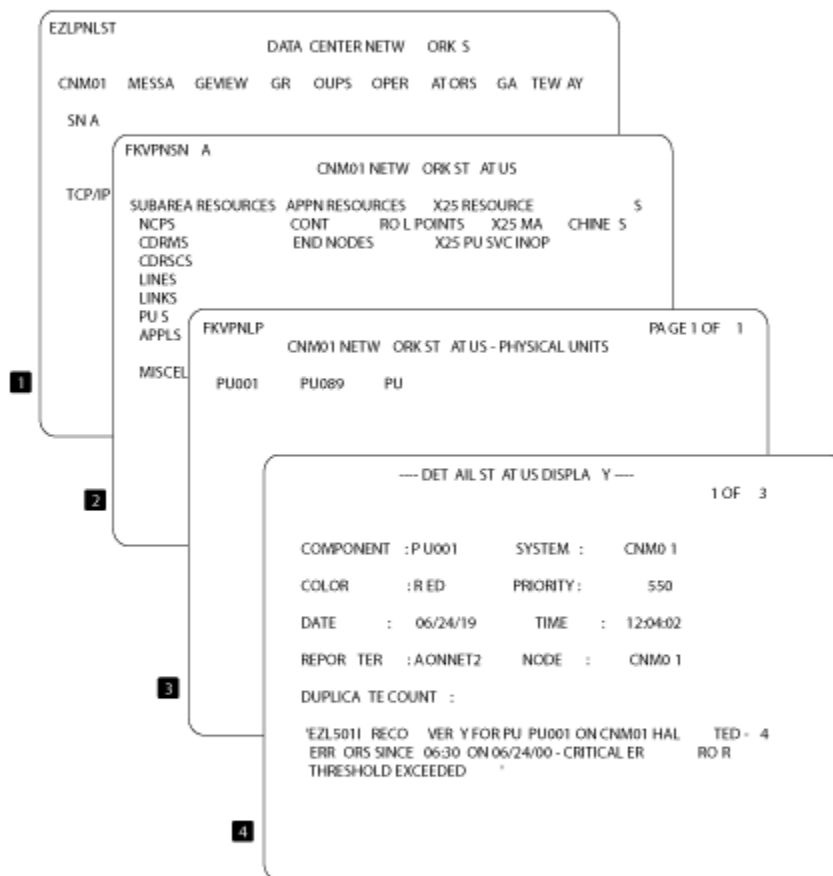


Figure 179. Sample DDF panel flow of SNA Resources

In Figure 179 on page 179 you can see:

- 1 The Data Center Networks panel is the first DDF panel. In Figure 179 on page 179, **SNA**, and **TCP/IP** represent the SNA and TCP/IP resources in the domain, CNM01. To see the types of SNA resources defined to CNM01, move the cursor to **SNA** and press the down key, **F8**.

**2**

All the monitored SNA resource types within network CNM01 are displayed on the CNM01 Network Status panel. To see the names of the physical units defined to CNM01, move the cursor to **PUS** and press the down key, **F8**.

**3**

The next panel shows the names of the physical units that are experiencing problems. DDF shows each monitored resource in the color of its status. For example PU001 is shown in red if the status is INACTIVE, which requires operator intervention for recovery. To see the detail record, move the cursor to **PU001** and press the Detail key, **F2**.

**4**

The status details record for PU001 is shown on the Detail Status Display. The physical unit PU001 requires operator intervention because a critical threshold has been reached and automation has stopped.

When you select a problem for analysis, press **F2** (the DDF MARK command) to inform others that you are working on the problem. Also, before you resolve a problem, check DDF to determine whether someone is already working on the problem.

AON uses many different programs to update resource status in DDF regardless of the sequence or priorities posted by other programs. As AON automation resolves the problems, DDF removes the resources from the display.

## Defining dependencies

DDF uses tree structures to implement the hierarchical display of status information. A tree structure starts with the system name as the root node and a level number of 1. The leaves of the tree represent the monitored resources, and the level numbers reflect which resources depend on each other. DDF generic values can also represent the leaves. Figure 180 on page 180 shows the tree structure provided with the AON/SNA component and includes member EZLTREE in the DSIPARM. This example also shows the order of dependency.

```

/* NETWORK: CNM01 */
1 CNM01
  2 SYSTEM
    3 GATEWAY
    3 GROUPS
      4 CALIF
        5 LA
        5 SANFRAN
        5 SANDIEGO
      4 NEWYORK
      4 ATLANTA
    3 OPID
    4 OPER1
  3 NETWORK
    4 RESOURCE
      4 SNA
        5 SNA
          6 NCP
          6 LINE
          6 LINKSTA
          6 CDRM
          6 CDRSC
          6 PU
          6 LU
          6 SESSION
          6 APPL
          6 ERR
        5 APPN
          6 CP
          6 EN
        5 X25
          6 X25MCH

```

Figure 180. DDF Tree Structure



Color in the tree structure is based on the order of dependencies. To consolidate the status of monitored network resources at the root node, specify the DDF parameter PROPAGATE UP. Figure 180 on page 180 shows how DDF reflects any color changes for LINE on LINE, SNA, NETWORK, SYSTEM, and CNM01. The color of the root node reflects the most important or critical status in a network operations center. If all the monitored resources are green, or active and not displayed, DDF shows the root node in green on the Data Center Network panel.

## Defining the priority and color of the resources

If more than one panel of information is available, DDF places 1 of  $n$  field in the upper right corner of the first Detail Status Display. On any series of detail status displays, DDF displays the most critical status color first. If more than one display has the same status color, DDF displays the most recent first.

Figure 181 on page 181 shows three of these panels. This example shows that DDF has 14 status display panels.



Figure 181. Sample Detail Status Displays

In Figure 181 on page 181, notice that the pink and turquoise panels precede the green status panel. During DDF initialization, DDF assigns colors to specific priority ranges as defined in DSIPARM member EZLINIT. DDF orders Detail Status Displays by priority and assigns the following priority ranges to these colors:

**Red**

Priority 100 - 199

**Pink**

Priority 200 - 299

**Yellow**

Priority 300 - 399

**Turquoise**

Priority 400 - 499

**Green**

Priority 500 - 599

When a new status is added, DDF issues a DDFADD command with the appropriate priority. For example, when the status descriptor with the priority 350 is added DDF displays yellow.

Status descriptors are connected to the status component in ascending order of priority. Therefore, if DDF assigns two status descriptors for PU with priorities of 120 and 150, DDF displays the one with the priority of 120 first. The PU is displayed in red. If a status component has multiple status descriptors with equal priorities, the status descriptors are chained off the status component in order of arrival time. Use the DDFDEL command to delete status descriptors.

## Updating status

To define the status and types used to set priority and color, place the definitions in the control file. When an automation event occurs, AON logging routines scan the control file for the DDF entry for that status or type. DDF uses the information from the control file and issues a DDFADD request. Table 1 on page 182 shows the status, color, and priority as provided with the AON base control file entries. The TCP/IP components add more DDF status types.

| Table 1. Examples of DDF Status Defaults (EZLCFGDS) |                |          |
|-----------------------------------------------------|----------------|----------|
| Resource Status                                     | Color Defaults | Priority |
| ACT*                                                | GREEN+         | 550      |
| CON*                                                | GREEN+         | 550      |
| IIN*                                                | RED            | 160      |
| INA*                                                | RED            | 150      |
| INO*                                                | RED            | 160      |
| NEV*                                                | YELLOW         | 360      |
| PAC*                                                | PINK           | 260      |
| PCT*                                                | PINK           | 260      |
| UP                                                  | GREEN+         | 550      |
| STOP                                                | RED            | 150      |
| RES*                                                | TURQUOISE      | 460      |
| RCV*                                                | GREEN          | 550      |
| REAC*                                               | PINK           | 270      |

| Table 1. Examples of DDF Status Defaults (EZLCFGDS) (continued)                                                                      |                |          |
|--------------------------------------------------------------------------------------------------------------------------------------|----------------|----------|
| Resource Status                                                                                                                      | Color Defaults | Priority |
| <b>Note:</b> A plus sign (+) indicates that the status of resources with the default of green are displayed with a REQ=NOADD option. |                |          |

If the DDF status contains a CLEAR=Y, REQ=NOADD, and DDF posts a resource at that status, DDF removes the existing descriptor, and does not add the new descriptor. Thus, DDF no longer displays the resource name.

If DDF uses resource status for color determination, code both Virtual Telecommunications Access Method (VTAM) status and automation status. Use wildcard characters to reduce the number of entries required. The NETSTAT command for DDF update uses VTAM status when you define DDFREFRESH as Y in the ENVIRON SETUP DDF control file. The current network status output from NETSTAT primes DDF.

## Dynamic updates

DDF uses automation status for dynamic updates during network automation activity. If you define the NETSTAT CHKAUTO parameter as Y, AON checks the automation flags for the resource before adding them to DDF. If this check determines that automation is off for the resource, DDF does not add the resource.

For example, when DDF sets PU001 to an INACTV state, the logging routines scan the control file for the INA\* entry for DDF and generate a DDFADD request with a priority of 150. PU001 is displayed in red on the DDF status panel. After DDF sets PU001 to an ACTIVE state, the logging routines scan the control file for the active state DDF entry and generate a DDFDEL request. DDF then deletes PU001 from the DDF status panel.

## Problem resources

If a program detects a warning message for LINE02 on CNM01, DDF issues a DDFADD command to add a status descriptor for LINE02. The status display for LINE02 on system CNM01 now indicates a problem with LINE02. If you specify the DDF PROPAGATE UP parameter, the CNM01 field also reflects this.

Occasionally, a more serious problem might arise. The routine that detects this problem updates DDF with a status descriptor of a lower priority number. Because DDF links status descriptors in order of priority, the LINE02 status now reflects the status of the more serious problem. After you resolve the more serious problem, the program detecting the resolution of the problem issues a DDFDEL command to remove LINE02 from the display.

## Defining status panels

AON defines DDF status statements for status panels, in the CNMPNL1 data set and embeds the definitions in the EZLPNLS member with %INCLUDE. However, the DDF formats and internally builds the Detail Status Display.

If you define status components in the panel definitions, also define them in the corresponding tree structure. However, not all the status components defined in the tree structure require a corresponding entry on the Detail Status Display. In [Figure 180 on page 180](#), the NETWORK status component is only a pseudo entry and is not defined on any DDF Detail Status Display.

You can customize the DDF status panels to reflect any environment. For example, you can define a panel to show the status of all the networks on all CPUs within the network operations center. The network operator then views the panel to determine the status of any of the network resources in the complex.

## Defining multiple systems

You can define multiple NetView domains to DDF. You must first implement AON focal-point services for the target system DDF status update on the focal-point DDF. For more information on focal-points, refer to *IBM Z NetView Administration Reference*.

In a multidomain environment, define a tree structure for each domain in the EZLTREE member of the DSIPARM data set on the designated focal-point DDF. Provide a unique root name for each systems tree structure, and match the focal-point root name with the ENVIRON SETUP SYSNAME entry in the control file. Also, code an entry for the SYSNAME parameter.

Because each root name is unique in a multiple systems environment, DDF uniquely addresses any status component on a system defined to the focal-point DDF. By adding the status component as a prefix to the root component name, DDF performs this addressing as the following example shows:

```
ROOT_COMPONENT.STATUS_COMPONENT (for example, CNM01.PU)
```

Similarly, AON routines add a prefix to DDF status descriptors shipped from the target system to the focal-point DDF with the root name of the target system.

## Implementing DDF

When defining resources to DDF, you can explicitly define every network, gateway, domain, and resource in the DDF tree and panels. However, this technique has a disadvantage. As the number of resources tracked by AON increases, DDF maintenance grows more complex. To simplify this maintenance, AON provides a generic implementation of DDF that requires you to define only the group identifier, not the individual resources. For example, if you display resources using the generic definitions, the DDF tree requires only resource types. If you use this method, it is not necessary to define all network resources or all resource types known to this system (for example, to the PU level).

### Using generic implementation

Generic implementation takes advantage of three functions within DDF:

- Storing detail records by priority
- Adding detail records using the default status component
- Displaying records using the status descriptor number

Use the default status component when a request to ADD, DELETE, or QUERY for the actual resource name fails. For example, if you add a component such as CNM01.PU001(RESOURCE) when PU001 is not in the tree, DDF attempts to add the component to the generic component of RESOURCE.

The relative position of the detail record in the detail record chain under a component determines the status descriptor number. Each component consists of detail records that were added directly to that component and all detail records with subordinate levels. For example, consider the following tree:

```
/*  SYSTEM: CNM01    */
1 CNM01
  2 SYSTEM
    3 GATEWAY
    3 NETWORK
    4 RESOURCE
```

The DDFADD command for this tree is:

```
DDFADD CNM01.PU001(RESOURCE), IN=/PU001/
```

When you issue the DDFADD command, DDF adds the entry to the resource because PU001 is not defined in the tree.

When you request the detail records for the generic resource, DDF displays the detail records for all components added using the generic component RESOURCE and PU001. The DDFQRY command is:

```
DDFQRY CNM01.RESOURCE
```

Alternatively, you can code the panel as shown in this example:

```
SF(CNM01.RESOURCE,04,05,16,N, , ,nn)
```

When you request the detail records for NETWORK, DDF not only displays the detail records added directly to NETWORK and detail records added directly to generic component RESOURCE.

```
DDFQRY CNM01.NETWORK
```

Alternatively, you can code the panel s shown in this example:

```
SF(CNM01.NETWORK,04,05,16,N, , ,nn)
```

DDF arranges detail records in priority order under each component.

Complete the following steps to define your network to DDF using the generic component:

1. Define the generic components to the tree as the following example shows:

```
1 CNM01
2 SYSTEM
3 NETWORK
4 RESOURCE
```

2. Use the generic component in the tree to display the detail records at the specified positions on the panel. The empty status text field, coded as ST( ), causes DDF to display the INFO field of the DDFADD command for the detail record. If a descriptor with a descriptor number of 11 does not exist, DDF shows only 10 records and the status text field displays is empty.

With this method, resources display in order of the priority of the condition of the resource. For example, if you use the DDF statements in the EZLCFG01 sample, DDF assigns the highest priority to red and the lowest to green. The following example shows a panel displaying a status of 12 network resources. This panel displays the highest priority error in the upper left position and the resources are listed in rows across the panel:

```
/* DEFINE CNM01 NETWORK STATUS PANEL */
P(EZLPNL1,24,80,SYSTEM,SYSTEM, , , )
TF(01,27,57,WHITE,NORMAL)
TT(CNM01 NETWORK STATUS)
SF(CNM01.NETWORK,04,05,16,N, , ,01)
ST( )
SF(CNM01.NETWORK,04,25,36,N, , ,02)
ST( )
SF(CNM01.NETWORK,04,45,56,N, , ,03)
ST( )
SF(CNM01.NETWORK,04,65,76,N, , ,04)
ST( )
SF(CNM01.NETWORK,06,05,16,N, , ,05)
ST( )
SF(CNM01.NETWORK,06,25,36,N, , ,06)
ST( )
SF(CNM01.NETWORK,06,45,56,N, , ,07)
ST( )
SF(CNM01.NETWORK,06,65,76,N, , ,08)
ST( )
SF(CNM01.NETWORK,08,05,16,N, , ,09)
ST( )
SF(CNM01.NETWORK,08,25,36,N, , ,10)
ST( )
SF(CNM01.NETWORK,08,45,56,N, , ,11)
ST( )
SF(CNM01.NETWORK,08,65,76,N, , ,12)
ST( )
TF(24,01,48,T,NORMAL)
TT(PF1=HELP 2=DETAIL 3=END 4=DIS 5=CY 6=ROLL 7=UP)
TF(24,51,79,T,NORMAL)
```

```
TT(      10=LF 11=RT 12=TOP)
EP
```

This technique gives you the advantage of first seeing the list of resources with the most critical problems. DDF determines resource panel position according to the order and priority of the problems. AON ships examples of components using this feature in the following members of CNMPNL1:

- EZLPNLS
- EZLPNLTY
- EZLPNLST
- EZLPNL1
- FKVPNL21

With DDF, you can define generic values for your environment. Multiple generic values can apply to one message added to DDF to help organize network data in DDF. For example, you can test the RESTYPE message processing field to see whether it contains a valid generic value for DDF on this system. AON checks the ENVIRON DDF DDFGENERIC statements in the control file to verify generic values.

The DDFGENERIC RESTYPE VALUE=(*restype*,...) statement defines valid generic values in the control file. Samples provided with AON categorize DDF by resource type. For each valid generic value, define a panel which can display status descriptors added under this generic. In this case, SF (CNM01.PU,04,05,16,N,...) displays components for the PU generic. SF (CNM01.NETWORK,14,05,16,N,...) also displays components for the PU generic because the tree hierarchy shows that NETWORK is higher than PU.

## Specific implementation of DDF

If you define your network to DDF with the specific method, DDF displays resource names in a fixed position on the DDF panels. When DDF records a detail record for the resource, a resource name changes color. When no detail records exist for the resource, DDF shows the resource name in the status text field as blue, which is considered an empty color.

**Note:** When using the specific implementation of DDF, do not use REQ=NOADD in any DDF status definition in the control file.

Because large networks require excessive effort and storage to define and maintain in DDF, reserve-specific implementation for small networks.

Complete the following steps to define your network to DDF using the specific component:

1. Define resources to the tree as shown in the following example:

```
/* NETWORK: CNM01
1 CNM01
  2 SYSTEM
    3 GATEWAY
    3 NETWORK
      4 RESOURCE
        4 NCP01
        4 LINE01
        4 LINE02
        4 PU001
        4 PU02
        4 630-S
```

2. Define resources to the panel as shown in the following example:

```
/* DEFINE CNM01 NETWORK STATUS PANEL */
P(EZLPNL,24,80,SYSTEM,SYSTEM, , , )
TF(01,27,47,WHITE,NORMAL)
TT(CNM01 NETWORK STATUS)
SF(CNM01.NCP01,04,05,16,N, , )
ST(NCP01)
SF(CNM01.LINE01,06,05,16,N, , )
ST(LINE01)
SF(CNM01.LINE02,08,05,16,N, , )
ST(LINE02)
SF(CNM01.PU001,10,05,16,N, , )
```

```

ST(PU001)
SF(CNM01.PU02,12,05,16,N, ,)
ST(PU02)
SF(CNM01.630-S,14,05,16,N, ,)
ST(630-S)
TF(24,01,48,T,NORMAL)
TT(PF1=HELP 2=DETAIL 3=END          6=ROLL 7=UP 8=DN)
TF(24,51,79,T,NORMAL)
TT(9=DEL 10=LF 11=RT 12=TOP)
EP

```

If you define the code as shown in the previous example, the DDF panel is similar to [Figure 182 on page 187](#). The resources are hard-coded on the panel, so they are displayed at all times regardless of the status of the resources. This implementation differs from the default DDF implementation, which displays only resources with an exception status.

```

EZLPNL
                                CNM01 NETWORK STATUS

NCPS
CDRMS
CDRSCS
LINES
LINKS
PUS
APPLS
MISCELLANEOUS RESOURCES
ALL RESOURCES

                                06/24/19 14:52
====>
PF1=HELP 2=DETAIL 3=END          6=ROLL 7=UP 8=DN          10=LF 11=RT 12=TOP

```

*Figure 182. Sample panel using specific implementation of DDF*

## Defining the contents of DDF

[Table 2 on page 187](#) shows the elements that constitute DDF:

| Table 2. Defining DDF Contents |            |                                                                                                                                                                                                             |
|--------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name                           | Type       | Purpose                                                                                                                                                                                                     |
| EZLTDDF                        | Task       | Initializes DDF and maintains status information.                                                                                                                                                           |
| DDF                            | Command    | Enters a DDF operator session.                                                                                                                                                                              |
| DDFADD                         | Command    | Adds status information.                                                                                                                                                                                    |
| DDFDEL                         | Command    | Deletes status information.                                                                                                                                                                                 |
| DDFQRY                         | Command    | Queries information.                                                                                                                                                                                        |
| DDFTREE                        | Command    | Dynamically loads a tree member from the DSIPARM data set.                                                                                                                                                  |
| DDFPANEL                       | Command    | Dynamically loads a panel member from the CNMPNL1 data set.                                                                                                                                                 |
| EZLINIT                        | Input file | Contains the initialization parameters defined with the statements described in <a href="#">“Defining initialization statements (EZLINIT)” on page 200</a> . The EZLINIT member is in the DSIPARM data set. |

| Table 2. Defining DDF Contents (continued) |                    |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name                                       | Type               | Purpose                                                                                                                                                                                                                                                                                                                                                                                                               |
| EZLTREE                                    | Input file         | Contains the tree structures as described in “ <a href="#">Defining the panel hierarchy (EZLTREE)</a> ” on page 197. This member can consist of a list of %INCLUDE statements that reference other members that contain tree structures. The format is %INCLUDE <i>member</i> , where <i>member</i> is the name of the member to include.<br><br>The EZLTREE member is in the DSIPARM data set.                       |
| EZLPNLS                                    | Input file         | Contains the DDF panel parameters defined with the statements described in “ <a href="#">Defining the panel statements (EZLPNLS)</a> ” on page 209. This member can consist of a list of %INCLUDE statements that reference other members that contain panel definitions. The format is %INCLUDE <i>member</i> , where <i>member</i> is the name of the member to include.<br><br>EZLPNLS is in the CNMPNL1 data set. |
| <i>panel_name</i>                          | Input file         | Contains the definition of a single panel. Name the member the panel name as defined with the PANEL statement.                                                                                                                                                                                                                                                                                                        |
| <i>tree_name</i>                           | Input file         | Contains the definition of a single tree structure. Name the member the root component name.                                                                                                                                                                                                                                                                                                                          |
| DDF                                        | Control file entry | Defines status priority.                                                                                                                                                                                                                                                                                                                                                                                              |
| ENVIRON DDF                                | Control file entry | Defines DDF setup defaults for the system.                                                                                                                                                                                                                                                                                                                                                                            |
| DDFGENERIC                                 | Control file entry | Defines valid values for implementing a generic set of values for DDF.                                                                                                                                                                                                                                                                                                                                                |

## Starting and stopping DDF

During AON initialization, the EZLTDDF task loads the members that define panel format, navigation, and tree structures. The EZLINIT member defines parameters that are common to all DDF panels as well as basic initialization specifications such as the screen size, default function keys, and the initial screen displayed when a DDF session starts.

The EZLPNLS member defines parameters unique to specific panels, and the EZLTREE member defines the tree structures. These two members consist of either tree structure and panel definitions for all DDF panels or %INCLUDE statements that point to members containing definitions of single panels or tree structures.

Using %INCLUDE statements enables you to reload all definition members besides EZLINIT, EZLTREE, and EZLPNLS without the DDFADD command. For more information, see “[Loading panels](#)” on page 189 and “[Loading tree structures](#)” on page 189. To change panels or tree structure specifications defined in the EZLTREE and EZLPNLS members, stop DDF and restart it to load the new definitions.

To start DDF, issue one of the following commands from a NetView console:

- STARTEZL DDF
- START TASK=EZLTDDF

To stop DDF, issue one of the following commands from a NetView console:

- **STOPEZL DDF**
- STOP TASK=EZLTDDF



When you stop DDF, it loses all existing status descriptors because the computer stores them only in memory. If you define DDFREFRESH=YES in the ENVIRON SETUP statement of the EZLCFG01 member, you can use the NETSTAT command to access VTAM displays and to prime DDF with current network exception conditions when it initializes. If you do not set DDFREFRESH to YES, DDF updates the file when it detects new failures, and then processes and logs the MONIT intervals.

## Loading panels

---

Without restarting DDF, you can dynamically load panels using either of the following two methods:

- Use the DDFPANEL command. For more information about the DDF panel command, see the following section, [“Loading tree structures” on page 189](#) and [“Loading a panel member \(DDFPANEL\)” on page 249](#).
- Use the PANEL statement. The PANEL statement parameters search for a panel that is not defined in the EZLPNLS member. DDF locates a member in the CNMPNL1 data set with the same name as the requested panel name. For more information about the PANEL statement and its parameters, see [“Defining new panels \(PANEL\)” on page 209](#).

## Loading tree structures

---

To dynamically load tree structures with the DDFTREE command, a member must exist in the DSIPARM data set that is named the same as the panel name or the root component in the tree structure. DDF makes the panels loaded with the DDFPANEL command resident only. All others are loaded when an operator issues commands that calls for them. DDF does not use %INCLUDE statements.

For example, you can change the tree structure for the root component CNM01 and the panel named EZLPNL1. You can also define these elements in members other than the EZLTREE or EZLPNLS members. You can then use the following commands to load the new definitions:

- **DDFTREE EZLTREES,ADD**
- **DDFPANEL EZLPNL1,ADD**

These commands enable you to load a few panels during initialization and add or delete panel subsets when required. This can significantly decrease the number of panels that DDF recognizes at any one time.

When DDF loads a new tree and replaces the existing tree, DDF copies status descriptors with similar leaf names, in both trees, to the new tree. For more information, see [“Loading tree members \(DDFTREE\)” on page 252](#).

## Using the definition procedure

---

When you start a DDF session, use the following procedure to define the panels that DDF displays:

1. Define the component hierarchy as described in [“Defining the panel hierarchy \(EZLTREE\)” on page 197](#). The EZLTREE tree contains tree structure definitions. Perform either or both of the following procedures:
  - Place all tree structure definitions in this member.
  - Use %INCLUDE statements to point to other members that contain definitions for entire tree structures.

If you use %INCLUDE statements when you add and delete trees, use the same name for referenced members and the root component. Complete the trees, starting at level 1. DDF deletes trees by the root name and adds them by the DSIPARM member name.

To load some tree structures during initialization and after DDF is started use the EZLTREE member to define those tree structure entries that you load during initialization. Then, use the DDFTREE command to load additional tree structures as needed. After you start DDF, put the tree structures in

separate members. Name each member after the root component for which you define the tree structure.

The following example shows a typical tree structure definition for AON/SNA:

```
/* NETWORK : CNM01 */
1 CNM01
2 SYSTEM
3 GATEWAY
3 GROUPS
4 CALIF
5 LA
5 SANFRAN
5 SANDIEGO
4 NEWYORK
4 ATLANTA
3 OPID
4 OPER1
3 NETWORK
4 RESOURCE
4 SNA
5 SA
6 NCP
6 LINE
6 LINKSTA
6 CDRM
6 CDRSC
6 PU
6 LU
6 SESSION
6 APPL
6 ERR
5 APPN
6 CP
6 EN
5 X25
6 X25MCH
6 X25PU
```

In this example, CNM01 is the root component. Embed this definition in the EZLTREE member or in a separate member named CNM01. [“Defining the panel hierarchy \(EZLTREE\)” on page 197](#) describes tree structures in detail.

Define the initialization and common panel specifications as described in [“Defining initialization statements \(EZLINIT\)” on page 200](#). The EZLINIT member defines parameters that are common to all DDF panels and basic initialization specifications such as:

- Screen size
- Initial screen shown when you start the DDF session
- Maximum operator logon limit
- Temporary error limit value
- Default function key definitions
- Detail function key definitions
- Detail function key descriptions
- Default priorities and colors

Define the panels as described in [“Defining the panel statements \(EZLPNLS\)” on page 209](#). After you initialize DDF, use the EZLPNLS member to load the panel definitions. This member can contain the following items:

- Definitions for all panels.
- %INCLUDE statements that point to separate members containing panel definitions.
- Combination of both panel definitions and %INCLUDE statements.
- Subset of panel entries that are loaded during initialization so that you can load additional panel definitions when needed. For more information, see [“Loading panels” on page 189](#).

The default 3270 screen size for DDF is 24 rows by 80 columns. The VTAM LOGMODE used for any DDF (NetView) 3270 work station must specify the default screen size of 24 x 80. The alternate screen size in the LOGMODE can specify any size supported by NetView program. DDF uses the 3270 command Erase/Write to present its screens to the user, which uses the default screen size specified in the LOGMODE.

The following example shows a typical menu panel definition named EZLPNLST. Embed this definition in the EZLPNLS member or in a separate member named EZLPLNST.

```
P(EZLPNLST,24,80, , , , , )
TF(01,02,09,T,NORMAL)
TT(EZLPNLST)
TF(02,25,57,WHITE,NORMAL)
TT(DATA CENTER NETWORKS)
SF(CNM01.NETWORK,04,05,10,N, ,EZLPNL2)
ST(CNM01)
SF(CNM01.SYSTEM,04,17,30,N, ,EZLPNL01)
ST(MESSAGEVIEW)
SF(CNM01.GROUPS,04,35,40,N, ,EZLPNLGR)
ST(GROUPS)
SF(CNM01.OPID,04,48,57,N, ,EZLPNLW0)
ST(OPERATORS)
SF(CNM01.GATEWAY,04,65,72,N, ,EZLPNLG)
ST(GATEWAY)
TF(24,01,48,T,NORMAL)
TT(PF1=HELP 2=DETAIL 3=END          6=ROLL 7=UP 8=DN)
TF(24,51,79,T,NORMAL)
TT(          10=LF 11=RT 12=TOP)
PFK4()
PFK5()
PFK9()
EP
```

Figure 183. Menu panel definition

Table 3 on page 191 provides a description for each statement in the preceding example.

| Table 3. Menu panel definitions                                                                                                                                                                                  |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Statement and Description                                                                                                                                                                                        |  |
| P(EZLPNLST,24,80, , , , , )                                                                                                                                                                                      |  |
| This is the panel definition statement. The panel name is EZLPNLST, the length of the panel is 24, and the width of the panel is 80.                                                                             |  |
| TF(01,02,09,WHITE,NORMAL)                                                                                                                                                                                        |  |
| This is the text location statement used to define constant panel fields. This field starts on line 01, in position 02, and ends in position 09. The color of the field is white and the highlighting is normal. |  |
| TT(EZLPNLST)                                                                                                                                                                                                     |  |
| The text data statement (EZLPNLST) that specifies data that goes in the field that was just defined.                                                                                                             |  |
| TF(01,25,57,WHITE,NORMAL)                                                                                                                                                                                        |  |
| Text location statement for another constant field.                                                                                                                                                              |  |
| TT(DATA CENTER NETWORKS)                                                                                                                                                                                         |  |
| Text data statement for the field just defined. Notice that TF and TT are grouped in pairs.                                                                                                                      |  |

| Table 3. Menu panel definitions (continued)                                                                                                                                                                                                                                     |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Statement and Description                                                                                                                                                                                                                                                       |  |
| SF (CNM01.NETWORK,04,05,10,N, ,EZLPNL2)                                                                                                                                                                                                                                         |  |
| This statement defines the location of the status component field. The status component is CNM01, this field starts on line 04 in position 05 and ends in position 10. The highlight level is normal, and the next panel displayed when <b>F8</b> (Down) is pressed is EZLPNL2. |  |
| ST (CNM01)                                                                                                                                                                                                                                                                      |  |
| This text is displayed in the SF field. The field name is CNM01. Notice also that SF and ST are grouped in pairs.                                                                                                                                                               |  |
| SF (CNM01.SYSTEM,04,17,30,N, ,EZLPNL01)                                                                                                                                                                                                                                         |  |
| Status field definition.                                                                                                                                                                                                                                                        |  |
| ST (MESSAGEVIEW)                                                                                                                                                                                                                                                                |  |
| Status text definition.                                                                                                                                                                                                                                                         |  |
| SF (CNM01.GATEWAY,04,65,72,N, ,EZLPNLG)                                                                                                                                                                                                                                         |  |
| Status field definition.                                                                                                                                                                                                                                                        |  |
| ST (GATEWAY)                                                                                                                                                                                                                                                                    |  |
| Status text definition.                                                                                                                                                                                                                                                         |  |
| TF (24,01,39,T,NORMAL)<br>TT (1=HELP 2=DETAIL 3=RET 6=ROLL 8=DN)<br>TF (24,40,79,T,NORMAL)<br>TT (10=LF 11=RT 12=TOP)                                                                                                                                                           |  |
| Here, TF and TT are used to display the function key definitions, which these are the defaults defined in EZLINIT. You can define function keys unique to this panel.                                                                                                           |  |
| EP                                                                                                                                                                                                                                                                              |  |
| This is the end panel statement used to indicate that this is the end of definitions for this panel.                                                                                                                                                                            |  |

Figure 184 on [page 193](#) shows the panel that is displayed when you run the statements listed in [Table 3](#) on [page 191](#).

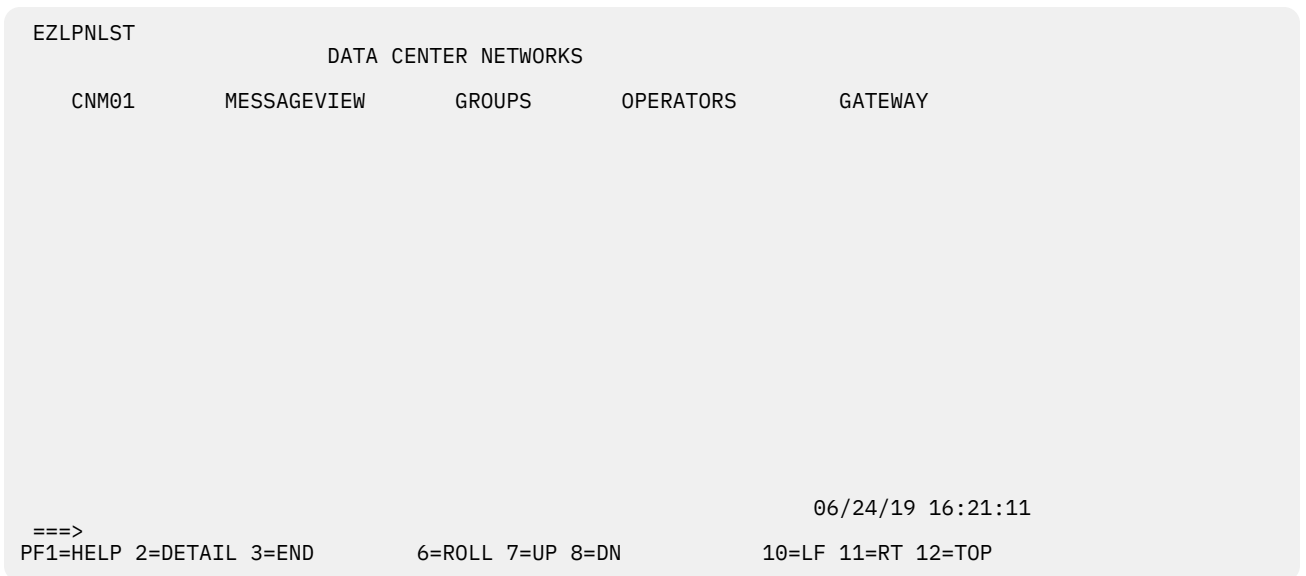


Figure 184. Data Center Networks (EZLPNLST) panel

## Modifying the Control File for DDF

The EZLCFG01 sample control file requires minimal changes to set up DDF. You must, however, customize the definition of your NetView domain. Customize ENVIRON SETUP SYSNAME=CNM01 to reflect the system name you choose for AON. Use the NetView domain ID. The value is referred to as the SYSNAME in this document. The SYSNAME value is used during panel and tree installation.

In the sample implementation, the ENVIRON DDF statement is defined as follows:

```
ENVIRON DDF DDF=STATUS,DDFREFRESH=NO,DDFGENERIC=(RESTYPE),DDFAUTO=NO
```

### Where:

#### **DDF=STATUS**

Causes the resource status to determine the color of the indicators in DDF.

#### **DDFREFRESH=NO**

Causes DDF to not be taken from current VTAM status only at DDF initialization. In some networks, the displays required to do this can cause an increase in CPU usage during initialization. If you want to do this, change this definition to DDFREFRESH=YES.

#### **DDFGENERIC=(RESTYPE)**

Causes DDF to group the resource status descriptors and display them by resource type.

#### **DDFAUTO=NO**

Causes DDF to display status changes for the resources regardless of whether the automation flag is on. If you do not want failures of non-automated resources posted to DDF, change this definition to DDFAUTO=YES.

Other control file definitions that influence DDF are the DDF and DDFGENERIC statements. Use the AON defaults for these definitions.

## Modifying the EZLTREE tree structure

Tree structures define resources into hierarchical groups, with the domain as the highest component in the hierarchy.

The sample tree provided with AON enables DDF to display current problems in your networking environment, with minimal maintenance. Change domain CNM01 in the EZLTREE member to reflect your SYSNAME value (domain ID).

## Modifying DDF panels

---

The sample panels that are copied into the CNMPNL1 data set during installation establish a typical implementation of DDF. To activate DDF, edit the following panels, changing all occurrences of CNM01 to your domain ID (SYSNAME value on ENVIRON SETUP statement):

- EZLPNL1
- EZLPNL2
- EZLPNLG
- EZLPNLG1
- EZLPNLL
- EZLPNLO1
- EZLPNLO2
- EZLPNLOA
- EZLPNLOB
- EZLPNLST
- EZLPNLTY

If you want to use the sample DDF group, you must change the domain ID in the following panels:

- EZLPNLAT
- EZLPNLC1
- EZLPNLCA
- EZLPNLGR
- EZLPNLLA
- EZLPNLNY
- EZLPNLSD
- EZLPNLSF

To use the same operator signout panels, change the domain ID in the following panels:

- EZLPNLW0
- EZLPNLW1
- EZLPNLW2

## Adding AON/TCP and SNA to the main DDF panel

---

Complete the following sections to add the AON components to the main DDF panel.

### Adding TCP to the main DDF panel

To display a TCP/IP selection on the main DDF panel, take one of the following actions:

- If you are using the EZLPNLST member as your main DDF panel (the first panel to display), add the following statements:

```
SF (CNM01.TCPIP,10,10,17,N, ,FKXPNLST)
ST (TCPIP)
```

These statements are also in the FKXPNLST data set and you can copy them from there.

- If you are using the EZLPNLTY panel as your main DDF panel, add the following statements:

```
SF (CNM01.TCPIP,10,10,17,N, ,FKXPNT7)  
ST (TCPIP)
```

These statements are also in the FKXPNTY data set and you can copy them from there.

**Note:** Remember to update CNM01 to your domain ID.

## Adding SNA to the main DDF panel

To use AON/SNA, add AON/SNA to the main DDF panel by taking one of the following actions:

- If you are using the EZLPNLT panel as your main DDF panel (the first panel to display), add the following statements:

```
SF (CNM01.NETWORK,06,10,15,N, ,FKVPNSNA)  
ST (SNA01)
```

These statements are also in the FKVPNLST data set and you can copy them from there.

- If you are using EZLPNTY as your main DDF panel, add the following statements:

```
SF (CNM01.NETWORK,06,10,15,N, ,FKVPL1)  
ST (SNA)
```

These statements are also in the FKVPNTY data set and you can copy them from there.

## Editing EZLPNLS

---

To customize the panel list, copy the following members at the end of EZLPNLS:

- FKXPNLS (TCP)
- FKVPNLS (SNA)





# Chapter 21. Defining Dynamic Display Facility (DDF) statements

Table 4 on page 197 lists the members where Dynamic Display Facility (DDF) statements are defined. This figure also explains the purpose of each member and shows where to find more information about each member.

| Table 4. Members Containing DDF Statements |                                                  |                                                                                  |
|--------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------|
| Member                                     | Purpose                                          | Location of Heading                                                              |
| EZLTREE                                    | Defines panel hierarchy                          | <a href="#">“Defining the panel hierarchy (EZLTREE)” on page 197</a>             |
| EZLINIT                                    | Defines initialization statements                | <a href="#">“Defining initialization statements (EZLINIT)” on page 200</a>       |
| EZLPNLS                                    | Defines panel statements                         | <a href="#">“Defining the panel statements (EZLPNLS)” on page 209</a>            |
| EZLCFG01                                   | Defines generics and status color relationships. | Refer to the <i>IBM Z NetView Administration Reference</i> for more information. |

## Defining the panel hierarchy (EZLTREE)

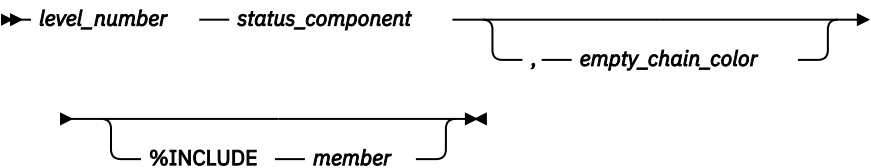
### Purpose

The hierarchy for status color changes is defined in the EZLTREE member in the DSIPARM data set. When the status changes for a component, the corresponding color change extends up or down the tree to the higher or lower-level components. The level number assigned to each component determines the level. The entry in the EZLINIT member or the individual DDFADD command requests determines the type of propagation.

The following syntax diagram shows the level number entry.

### Format

#### level\_number



### Parameters

#### empty\_chain\_color

Defines a color for the empty chain. You can use red, blue, turquoise, pink, green, yellow, or white. If DDF does not associate a status descriptor with a status component, the status component is displayed in this color on the DDF status panel. This entry is optional and can also be coded in the EMPTYCOLOR entry of the EZLINIT member. See [“Defining default colors for status components \(EMPTYCOLOR\)” on page 207](#) for more details.

***level\_number***

Defines a valid number 1 - 99. A tree starts with a root level of 1.

***status\_component***

Defines the resource for which status information is displayed. AON uses the resource name as defined in the control file. The status component entry for the root matches the ENVIRON SETUP SYSNAME entry in the control file.

***%INCLUDE member***

Includes members that contain EZLTREE statements, where *member* is the member name. Each included member must start with level 1 and must contain an entire tree structure.

**Usage**

The level numbers define the order of dependence. For an example, see [Figure 185 on page 199](#). In that figure, NCP is defined to be dependent on NETWORK. Therefore, when a change occurs for an NCP, it is reflected on NCP, NETWORK, SYSTEM, and CNM01.

You cannot use duplicate status components within the same tree. Not all status components that are defined in the tree must have a corresponding panel entry or a status descriptor associated with them. Each root name must be unique to avoid addressing conflicts. DDF can address each status component defined in the tree as follows:

```
root_component.status_component
```

**Examples**

Two separate AON/SNA trees, CNM01 and CNM02, representing two different networks are defined [Figure 185 on page 199](#). In this example, CNM01 is the focal point and CNM02 is the remote domain. In the EZLTREE member on CNM01, define a tree for both CNM01 and CNM02.

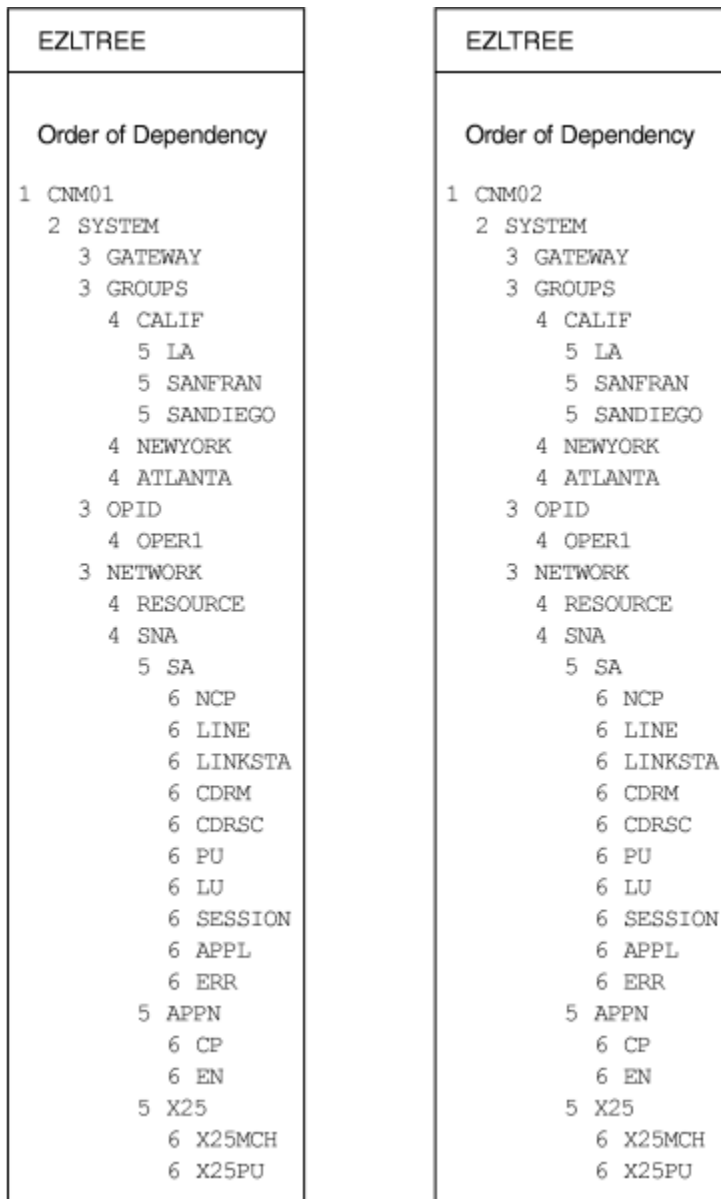


Figure 185. DDF Tree Structure

Both trees start with level number 1 and each has a unique root name. They can have similar status component names, such as LINE, PU, and NCP. The corresponding entry for the root component in the control file on system CNM01 is:

```

ENVIRON SETUP,
        NETVIEW=NET,
        SYSNAME=CNM01,
        :

```

Do not define the root component name CNM02 on the ENVIRON SETUP SYSNAME keyword in the control file for CNM01. Instead, define it in a similar way in the ENVIRON SETUP SYSNAME keyword in the control file for CNM02. Assuming that AON notification forwarding is established between CNM02 and CNM01, DDF adds a CNM02 prefix to all status components when status information is forwarded to CNM01. For example, it adds CNM02.PU01 for PU01. The EZLTREE member in the CNM02 system contains only the tree for CNM02.

## Defining initialization statements (EZLINIT)

The EZLINIT member in the DSIPARM data set defines the DDF initialization parameters. Use the defaults supplied in the EZLINIT member in DSIPARM.

### Defining the screen size (SCREENSZ)

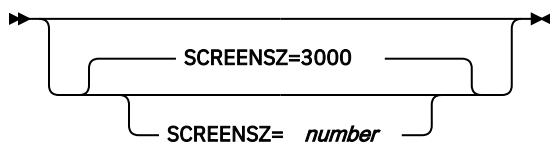
#### Purpose

The SCREEN SIZE parameter defines the screen buffer size. This entry is optional. If you do not specify this parameter, AON uses a program default value of 3000.

The following syntax diagram shows the screen size parameter.

#### Format

##### SCREENSZ



#### Parameters

##### *number*

Values can be in the range of 3000–9999.

#### Usage

For a larger screen buffer size, increase the *number* parameter.

#### Examples

```
SCREENSZ = 4000
```

### Linking the chain detail records (CHAIN)

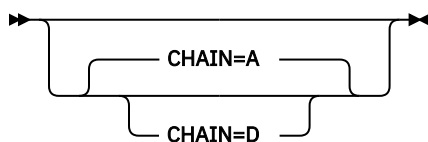
#### Purpose

The CHAIN parameter defines whether the detail status is linked in descending order (last to most recent) or in ascending order (most recent to last).

The following syntax diagram shows the chain parameter.

#### Format

##### CHAIN



#### Parameters

##### **A**

Link detail records within the same priority in ascending order.

***D***

Link detail records within the same priority in descending order.

## Examples

CHAIN = A

## Defining the initial screen (INITSCRN)

## Purpose

The INITSCRN parameter defines the initial panel that is displayed by DDF.

### Format

**INITSCRN**

►► INITSCRN — = — *panel* ◄◄

## Parameters

***panel***

A valid alphanumeric name with maximum length of eight characters.

## Usage

If you change the initial panel as defined in the EZLPNLS member of the DSIPARM data set, also change the name in the INITSCRN entry.

## Examples

INITSCRN = EZLPNLST

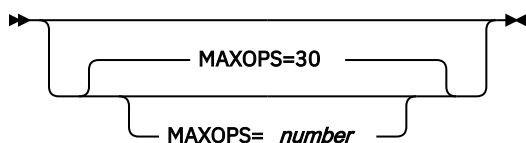
### Defining the number of operators (MAXOPS)

## Purpose

The MAXOPS parameter defines the maximum number of logged on operators that can use DDF. This entry is optional. If you do not specify this parameter, the default is 30.

### Format

# MAXOPS



## Parameters

***number***

Values range from 1 to 99.

## Usage

If the number of operators attempting to use DDF is more than the number defined in the MAXOPS parameter, AON denies the additional operators access to DDF. The dynamic update facility keeps an internal count of the operators that are logged on.

## Examples

```
MAXOPS = 35
```

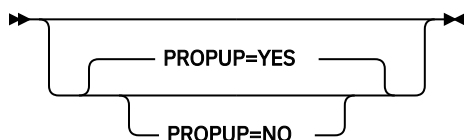
## Propagating status upward (PROPUP)

### Purpose

The PROPAGATE UP parameter defines whether status word text is sent up the status tree as a system default. This entry is optional. If you do not specify this parameter, the default is YES.

### Format

#### PROPUP



### Usage

Specify a value of YES. This parameter can be overridden with individual DDFADD requests. See [“Adding status descriptors \(DDFADD\)”](#) on page 244 for more information.

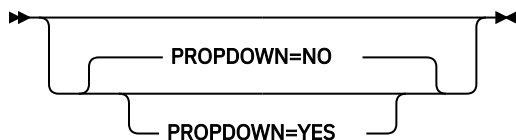
## Propagating status downward (PROPDOWN)

### Purpose

The PROPAGATE DOWN parameter defines whether status information is sent down the status tree as a system default. If you do not specify a value, the default is NO. This entry is optional.

### Format

#### PROPDOWN



### Usage

Specify a value of NO. This parameter can be overridden with individual DDFADD requests. See [“Adding status descriptors \(DDFADD\)”](#) on page 244 for more information. This entry is optional. If not coded, the default is NO.

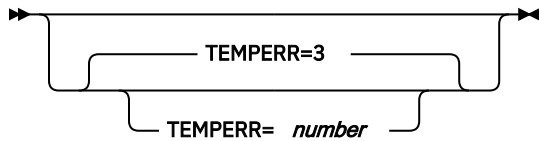
## Defining the temporary error limit (TEMPERR)

### Purpose

The TEMPERR parameter defines the maximum number of temporary input and output errors you can receive when trying to display a DDF panel. This entry is optional. If you do not specify a value, the default is 3.

### Format

#### TEMPERR



### Parameters

#### *number*

Values range from 3 to 99.

### Usage

Use the value 5, which is provided in EZLINIT member in the DSIPARM data set.

### Examples

```
TEMPERR = 5
```

## Defining the default function key definitions (PFKnn)

### Purpose

The PFKnn parameter defines the default function key settings. You cannot change F3.

### Format

#### PFK

```
PFK nn = command var
```

### Parameters

#### *nn*

Values range from 1 to 24.

#### *command*

The command is issued when the function key is pressed.

#### *var*

A variable or any text data passed with this command. Use the following variables as part of the command:

#### **&CO or &COLOR**

Identifies to the color of the detail entry.

#### **&COMP**

Identifies the component.

**&COMPAPPL**

Identifies the generic component within parentheses (for example, PU01(PU)). The generic component is shown only if the component was added using one.

**&DA or &DATA**

Identifies the actual message text.

**&DATE**

Identifies the date that the detail entry was added.

**&HI or &HIGHLIGHT**

Identifies the highlight level of the detail entry.

**&IN or &INFO**

Identifies the detail entry word text that is displayed on the status screen.

**&PR or &PRIORITY**

Identifies the priority of the detail entry.

**&ROOT**

Identifies the root or system.

**&RV or &REFVALUE**

Identifies the reference value of the detail entry.

**&SENDERID**

Identifies the reporter who submitted the detail entry.

**&SNODE or &SENDERNODE**

Identifies the node of the reporter who submitted the detail entry.

**&SYSDATE**

Identifies the system date.

**&SYSTIME**

Identifies the system time.

**&TIME**

Identifies the time that the detail entry was added.

**Usage**

You can redefine all of the function keys except F3.

**Examples**

In this example, the DIS PU01 command is issued when F4 is pressed while the cursor is placed on the PU01 entry on the status screen:

```
PF4=DIS &INFO
```

**Defining the detail function key for the detail display (DPFKnn)****Purpose**

The DPFKnn parameter defines the function keys that are unique to the detail panel.

**Note:** The function keys defined are active only when the detail panel is displayed and, therefore, override the default settings defined with the PFKnn statement.

**Format****DPFK**

➤ DPFK — nn — = — command var ➤



## Parameters

### ***command***

The command issued when the defined function key is pressed.

### ***nn***

Values range from 1 to 24.

### ***var***

The variables passed along with the command when the defined function key is pressed.

The following variables can be used as the command:

### **&CO or &COLOR**

Identifies the color of the detail entry.

### **&COMP**

Identifies the component.

### **&COMPAPPL**

Identifies the generic component within parentheses (for example, PU01(PU)). The generic component is shown only if the component was added by using the value of 1.

### **&DA or &DATA**

Identifies actual message text.

### **&DATE**

Identifies the date the detail entry was added.

### **&HI or &HIGHLIGHT**

Identifies the highlight level of the detail entry.

### **&IN or &INFO**

Identifies the detail entry word text that is displayed on the status panel.

### **&PR or &PRIORITY**

Identifies the priority of the detail entry.

### **&ROOT**

Identifies the root or system.

### **&RV or &REFVALUE**

Identifies the reference value of the detail entry.

### **&SENDERID**

Identifies the reporter who submitted the detail entry.

### **&SNODE or &SENDERNODE**

Identifies the node of the reporter who submitted the detail entry.

### **&SYSDATE**

Identifies the system date.

### **&SYSTIME**

Identifies the system time.

### **&TIME**

Identifies the time the detail entry was added.

## Usage

You can tailor all of the functions except F3.

## Describing function keys on the detail panel, part 1 (DPFKDESC1)

### Purpose

The DPFKDESC1 parameter defines the first part of the function key description that is displayed at the bottom of the detail panel. This text is concatenated with the text defined with the DPFKDESC2 statement.

## Format

### DPFKDESC1

➤ DPFKDESC1 — = — *text* ➤

## Parameters

### *text*

Up to 40 characters of data.

## Examples

```
DPFKDESC1='PF1=HELP PF2=END PF3=RETURN'
```

## Describing function-key text on the detail panel, part 2 (DPFKDESC2)

## Purpose

The DPFKDESC2 parameter defines the second part of the PF key description that is displayed at the bottom of the detail display. This text is concatenated with the text defined with the DPFKDESC1 statement.

## Format

### DPFKDESC2

➤ DPFKDESC2 — = — *text* ➤

## Parameters

### *text*

Up to 40 characters of data.

## Examples

```
DPFKDESC2='PF6=ROLL PF7=UP PF8=DOWN'
```

## Defining default colors (DCOLOR)

## Purpose

The DCOLOR parameter defines the appropriate color for a status descriptor that is outside of the defined priority and color ranges. This entry is optional. If you do not specify a value, the default is green.

## Format

### DCOLOR

➤ \_\_\_\_\_ ➤

└── DCOLOR=GREEN ─┘

└── DCOLOR= *color* ─┘

## Parameters

### *color*

Blue, red, pink, green, turquoise, yellow, and white are valid colors.

## Usage

Use the value WHITE, which is provided in the EZLINIT member in DSIPARM, and which does not conflict with existing status and color definitions.

## Examples

```
DCOLOR = WHITE
```

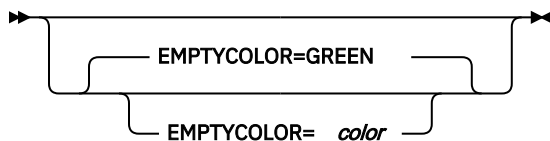
## Defining default colors for status components (EMPTYCOLOR)

### Purpose

The EMPTY COLOR parameter defines the color that is displayed for a status component that has no status descriptor associated with it. This entry is optional. If you do not specify a value, the default is green.

### Format

#### EMPTYCOLOR



### Parameters

#### *color*

Blue, red, pink, green, turquoise, yellow, and white are valid colors.

## Usage

Use the color BLUE, which is provided in the SEZLSENU member of EZLINIT, it does not conflict with existing status or color definitions. This parameter can be overridden in the EZLTREE member.

## Examples

```
EMPTYCOLOR = BLUE
```

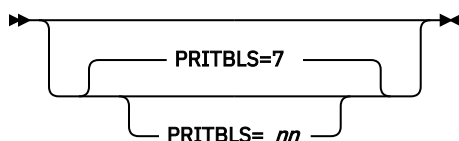
## Defining priority and color ranges (PRITBLSZ)

### Purpose

The PRITBLSZ parameter defines the number of priority and color ranges defined by the PRIORITY entries. This entry is optional. If you do not specify a value, the default is 7.

### Format

#### PRITBLS



## Parameters

***nn***

Valid number greater than 7

## Usage

Use the number 12, which is provided with the EZLINIT member in DSIPARM.

## Examples

```
PRITBSZ = 12
```

## Defining color and priority ranges (PRIORITY)

### Purpose

The PRIORITY parameter defines the relationship between colors and priority ranges. This entry is optional. If you do not specify values, the defaults are:

#### Color

##### Priority

#### Red

001 - 099

#### Pink

100 - 199

#### Yellow

200 - 299

#### Blue

300 - 399

#### Turquoise

400 - 499

#### White

500 - 599

#### Green

600 - 699

### Format

#### PRIORITY

→ PRIORITY — = — *nnn* — , — *mmm* — , — *color* →

## Parameters

### ***color***

Red, green, yellow, turquoise, pink, blue, and white are valid colors.

### ***mmm***

Valid number 001 - 999 and equal to or greater than the value specified in *nnn*.

### ***nnn***

Valid number 001 - 999.

## Usage

Use the values provided with the EZLINIT member in DSIPARM.

## Examples

```
PRIORITY=100,199,RED
PRIORITY=200,299,PINK
PRIORITY=300,399,YELLOW
PRIORITY=400,499,TURQUOISE
PRIORITY=500,599,GREEN
PRIORITY=600,699,BLUE
DCOLOR=WHITE
EMPTYCOLOR=BLUE
```

## Defining the panel statements (EZLPNLS)

The EZLPNLS member in the CNMPNL1 data set defines the DDF status panels. The structure of each panel definition is as follows:

1. Beginning panel definition (PANEL)
2. Status component definition (STATUSFIELD and STATUSTEXT)
3. Text fields and data definition (TEXTTEXT and TEXTFIELD)
4. Function key definition unique to this panel (PFKnn)
5. End panel definition (ENDPANEL)

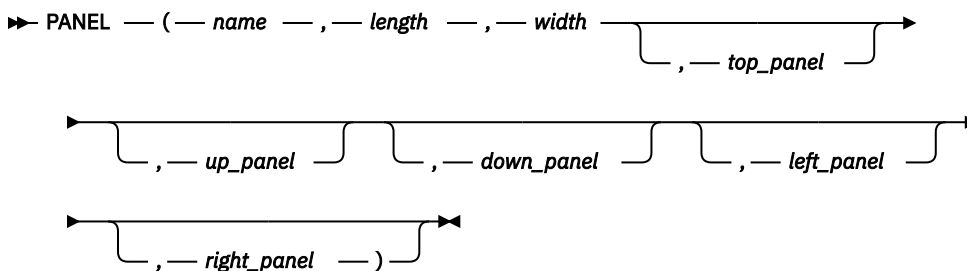
## Defining new panels (PANEL)

### Purpose

The panel entry identifies the start of a new panel and its general attributes.

### Format

#### PANEL



### Parameters

#### *down\_panel*

Defines the panel that displays when the DOWN command is issued or the appropriate function key is pressed.

#### *left\_panel*

Defines the panel that displays when left panel function key is pressed or LEFT command is issued.

#### *length*

Defines the number of lines or rows on the screen. You must specify a numeric value.

#### *name*

Identifies a user-defined panel name up to 8 characters in length.

#### *right\_panel*

Defines the panel that displays when the right panel function key is pressed or the RIGHT command is issued.

**top\_panel**

Defines the panel that displays when the TOP command is issued or the appropriate function key is pressed.

**up\_panel**

Defines the panel that displays when the UP command is issued or the appropriate function key is pressed.

**width**

Defines the number of columns on the screen. You must specify a numeric value.

**Usage**

The default initial panel name supplied with AON is called EZLPNLST. If you change this name, also change the definition in the EZLINIT member for the INITSCRN entry to reflect this. If you have more data than can be displayed on a single screen, you can define continuation panels with the *left\_panel\_name*, *right\_panel\_name*, and *down\_panel\_name* parameters.

The parameters are positional.

**Examples**

This example defines EZLPNSNA as the panel name:

```
PANEL (EZLPNSNA,24,80,EZLPNL1,EZLPNL1, , , )
```

The length of the panel is 24 and the width is 80. The panel named EZLPNL1 is displayed when the TOP command is used and the panel named EZLPNL1 is displayed when the UP command is used. No entries are defined for the DOWN, LEFT, or RIGHT commands.

**Locating the status component (STATUSFIELD)****Purpose**

The STATUSFIELD entry defines the location of the status component on a panel and the panels that display when the UP and DOWN commands are used. A STATUSFIELD entry is always accompanied by a STATUSTEXT entry.

**Format****STATUSFIELD**

```
➤ STATUSFIELD  — ( — root_cmpt  — . ————— , — start_line  ➔
                  |—————|
                  | status_cmpt |
                  |—————|

➤ ————— , — end_position ————— ➔
  |—————|
  | — start_position ————— |
  |—————|

➤ —————
  |—————| ————— |—————|
  | — up_panel ————— | | — down_panel ————— |
  |—————|

➤ ————— ) ➔
  |—————|
  | — status_descriptor_number ————— |
```

**Parameters****down\_panel**

Defines the panel that is to be displayed if the Down function key is pressed.

***end\_position***

Defines the column number in which the definition of the component ends. This is governed by the length of the text that is defined in the STATUSTEXT definition. For example, if you define LINE, the length of the STATUSTEXT is four and the end position is the start position plus three. See [“Defining the text area \(STATUSTEXT\)”](#) on page 213 for more information. You must have at least 2 bytes between the end position of one field and the beginning position of the next field for attribute bytes.

***highlight***

Defines the type of highlighting. You can use normal, flash, reverse, or underscore. Define highlighting as normal, although it can be overridden with individual DDFADD requests.

***root\_cmp***

Defines the root name as defined in the root node of the tree structure. You must specify the root (as opposed to the status component alone), because different systems can have status components with the same name defined in their respective tree structures. For example, the status component LINE is often used. Because the root and status component must always be unique, each status component in a tree structure can be uniquely identified by adding the root component entry as a prefix.

***start\_line***

Defines the line number where the status component is displayed. The entry is numeric and within the range specified in the length parameter in the panel definition statement.

***start\_position***

Defines the actual column number on the start line. The start line is specified above where the component is to be placed. There must be a minimum of 2 spaces between the ending position of one field and the beginning position of the next field to allow for attribute type. For example, if the end position of the status field is in column 10, the start position of the next STATUSFIELD must be in column 13.

**Note:** You cannot start the component in position one because it always needs one leading byte.

***status\_cmp***

Defines the status component name as defined in the EZLTREE member. The maximum length is 8 characters.

***status\_descriptor\_number***

Defines the number of the status descriptor. Use a numeric value in the range of 1–999. Status descriptors are chained with the status component in ascending order of priority. A panel can be designed with the same status-component name defined in every STATUSFIELD entry. The status descriptor number specifies the status descriptor displayed in each field. For example, if you specified two STATUSFIELD entries for the same status component with status descriptor numbers 1 and 2, the status descriptor with the higher priority is displayed in the status field with status descriptor number 1. The next higher priority status descriptor is displayed in the status field with status descriptor number 2.

You can add a letter that denotes the type of information to be displayed as a prefix for the status descriptor. If you do not add a prefix, you can provide text with individual DDFADD requests using the INFO keyword. See [“Adding status descriptors \(DDFADD\)”](#) on page 244 for more information. Valid description prefixes are:

**C**

Displays the name of the component

**D**

Displays the date the record was added

**M**

Displays the message text

**P**

Displays the priority of the record

**R**

Displays the name of the root

- S** Displays the ID of the requester
- T** Displays the time the record was added
- U** Displays the number of duplicate records
- V** Displays the reference value of the requester
- X** Displays the domain of the requester

#### ***up\_panel***

Defines the panel that is to be displayed if the Up function key is pressed.

#### **Usage**

When designing a panel for any status component, make the end position greater than or equal to the start position; otherwise, unpredictable results can occur during DDF initialization.

The component start position is column 02.

Parameters are positional.

#### **Examples**

In this example, the LINES on CNM01 status component start on line 04 in column 10, end in column 14, have normal highlighting:

```
STATUSFIELD(CNM01.LINES,04,10,14,NORMAL, ,)
```

No entries are defined for the UP or DOWN commands.

#### **Examples**

In this example, the status component starts on line 02 in column 04, ends in column 11, and has normal highlighting:

```
SF(CNM01.SYSTEM,02,04,11,N,,EZLPNSNA)
```

No entries are defined for the UP panel and the EZLPNSNA entry is defined for the DOWN command.

#### **Examples**

In this example, three STATUSFIELD entries are defined for the same CNM01.PU status component:

```
SF(CNM01.PU,02,04,11,NORMAL, , ,01)
SF(CNM01.PU,03,04,11,NORMAL, , ,02)
SF(CNM01.PU,04,04,11,NORMAL, , ,03)
```

The highest priority status descriptor is displayed in the first entry, the next in the second, and so on.

#### **Examples**

In this example, three STATUSFIELD entries are defined for the same CNM01.RESOURCE status component:

```
SF(CNM01.RESOURCE,04,02,05,NORMAL, , ,P01)
SF(CNM01.RESOURCE,04,08,15,NORMAL, , ,C01)
SF(CNM01.RESOURCE,04,18,79,NORMAL, , ,M01)
```



The first status descriptor is displayed, with its priority in column 02, its component name in column 08, and its message text starting in column 18.

## Defining the text area (STATUSTEXT)

### Purpose

The STATUSTEXT entry defines the text data displayed in the STATUSFIELD entry. The value of this entry is normally the name of the status.

### Format

#### STATUSTEXT

►► STATUSTEXT — ( — *text* — )-►►

### Parameters

#### *text*

Defines the data displayed for the status component defined in STATUSFIELD entry. The maximum length is 76 alphanumeric characters. Use the name specified for the status component, such as CNM01.LINE. The length of text determines the end position that is coded in the STATUSFIELD entry.

### Usage

Each STATUSFIELD entry must have a STATUSTEXT entry associated with it.

The STATUSTEXT text can be overridden by individual DDFADD requests using the INFO keyword. See [“Adding status descriptors \(DDFADD\)” on page 244](#) for more information.

### Examples

In this example, the resource type LINE, on CNM01 displays as LINE on the status display panel:

```
STATUSFIELD(CNM01.LINE,04,10,13,NORMAL, , )
STATUSTEXT(LINE)
```

**Note:** The end position in the STATUSFIELD reflects the length of resource line names.

### Examples

In this example, the text is not specified for the STATUSTEXT entry for LINE:

```
STATUSFIELD(CNM01.LINE,10,15,25,NORMAL, , )
STATUSTEXT( )
```

You can use 12 characters for any text by the associated STATUSFIELD entry. Provide this text with individual DDFADD requests by using the INFO keyword.

## Defining text location (TEXTFIELD)

### Purpose

This entry defines the location and attributes of fields that remain constant such as panel headings, field names, and function key designations.

## Format

### TEXTFIELD

➤ TEXTFIELD — ( — *start\_line* — , — *start\_position* — , — *end\_position* — , — *color* — , — *highlight* — ) ➤

## Parameters

### *start\_line*

Defines the line number on which the text field is displayed. The entry is numeric and within the range specified in the length parameter in the panel definition statement.

### *start\_position*

Defines the column number on which the text field is placed.

### *end\_position*

Defines the column number in which the data specified in entry TEXTTEXT ends. See [“Defining displayed text \(TEXTTEXT\)”](#) on page 214 for more details.

### *color*

Defines the color of the text. You can use blue, red, green, yellow, pink, turquoise, or white for the color to be displayed in the TEXTTEXT entry text.

### *highlight*

Defines the type of highlighting. You can use normal, flash, reverse, or underscore to determine the TEXTTEXT entry text display.

## Usage

When designing a panel, for any TEXTFIELD, make the end position of the TEXTFIELD greater than, or equal to, the start position. Otherwise, unpredictable results can occur during DDF initialization.

The parameters are positional.

## Examples

In this example, the TEXTFIELD is defined as being on line 1, starting in column 25, ending in column 57, white in color, and with normal highlighting:

```
TEXTFIELD(01,25,57,WHITE,NORMAL)
```

## Defining displayed text (TEXTTEXT)

### Purpose

The TEXTTEXT entry defines the data displayed in the corresponding TEXTFIELD entry.

### Format

#### TEXTTEXT

➤ TEXTTEXT — ( — *text* — ) ➤

## Parameters

### *text*

Defines the data displayed for the TEXTFIELD entry. The length of the data determines the end position that is coded in the TEXTFIELD entry. Maximum length is 72 alphanumeric characters.

## Usage

Each TEXTFIELD entry must have a TEXTTEXT entry associated with it.

## Examples

In this example, the text DATA CENTER NETWORKS displays on the status display panel in white:

```
TEXTFIELD(01,25,57,WHITE,NORMAL)
TEXTTEXT(DATA CENTER NETWORKS)
```

## Examples

In this example, all the function key settings will be displayed on line 24 of the Status Display panel:

```
TF(24,01,48,TURQUOISE,NORMAL)
TEXTTEXT(PF1=HELP 2=DETAIL 3=END 6=ROLL 7=UP 8=DN)
TF(24,51,79,T,N)
TT(10=LF 11=RT 12=TOP)
```

Notice that two entries are coded for the same line number.

## Defining function keys on the status panel (PFKnn)

### Purpose

The PFK $nn$  entry defines all of the function keys that are unique to this panel.

### Format

#### PFK

►► PFK —  $nn$  — ( — *command* — — *var* — ) ►►

### Parameters

#### $nn$

Values can range from 1 to 24.

#### *command*

The command is issued when the defined function key is pressed.

#### *var*

Variable can have the following values for this command:

#### &COMP

Identifies the component.

#### &COMPAPPL

Identifies the generic component within parentheses (for example, PU01(PU)). The generic component is shown only if the component was added using one.

#### &ROOT

Identifies the root or system.

#### &SYSDATE

Identifies the system date.

#### &SYSTIME

Identifies the system time.

#### &IN or &INFO

Identifies the detail entry word text that is displayed on the status panel.

#### &DATE

Identifies the date that the detail entry was added.

**&TIME**

Identifies the time that the detail entry was added.

**&SENDERID**

Identifies the reporter who submitted the detail entry.

**&SNODE or &SENDERNODE**

Identifies the node of the reporter who submitted the detail entry.

**&DA or &DATA**

Identifies the actual message text.

**&RV or &REFVALUE**

Identifies the reference value of the detail entry.

**&PR or &PRIORITY**

Identifies the priority of the detail entry.

**&CO or &COLOR**

Identifies to the color of the detail entry.

**&HI or &HIGHLIGHT**

Identifies the highlight level of the detail entry.

**Usage**

You can redefine all of the function keys except F3.

The definitions are active only when the status panel is displayed. The default settings defined with the initialization `PFKnn` statement are overridden.

F3 is always RETURN.

**Examples**

In this example, the `DIS PU01` command is issued when F4 is pressed with the cursor placed on the PU01 entry:

```
PFK4(DIS &INF0)
```

**Defining the end of a panel (ENDPANEL)****Purpose**

The ENDPANEL entry identifies the end of a panel.

**Format****ENDPANEL**

```
➤ ENDPANEL ➤
```

**Including additional members %INCLUDE****Purpose**

The %INCLUDE entry includes additional members that contain panel definitions statements.

**Format****%INCLUDE**

```
➤ %INCLUDE — member ➤
```

## Parameters

### *member*

Use this keyword to specify the name of a member that defines a panel using the panel statements described in this section. The member must reside on a data set referenced by DSIPARM.



---

## Chapter 22. Implementing Dynamic Display Facility (DDF)

This chapter describes the implementation of Dynamic Display Facility (DDF) and shows samples. You can use these samples to customize the DDF displays for your environment. These samples:

- Display network status on multiple panels
- Display network status on a single-panel
- Define a MessageView display
- Implement DDF in a focal point environment
- Use operator MARK panels
- Store a DDF update under multiple components
- Group resources in DDF

Each sample in this chapter shows how DDF panels are displayed when the sample is implemented, how DDF processes resource status, how you define the sample, and how to use it.

---

### Displaying Network Status on multiple panels

When AON is shipped, a multiple panel display is sent as the default. This is the most useful display implementation. [Figure 186 on page 220](#) shows a multiple panel display with numbered panels. The panels are described in the following section.

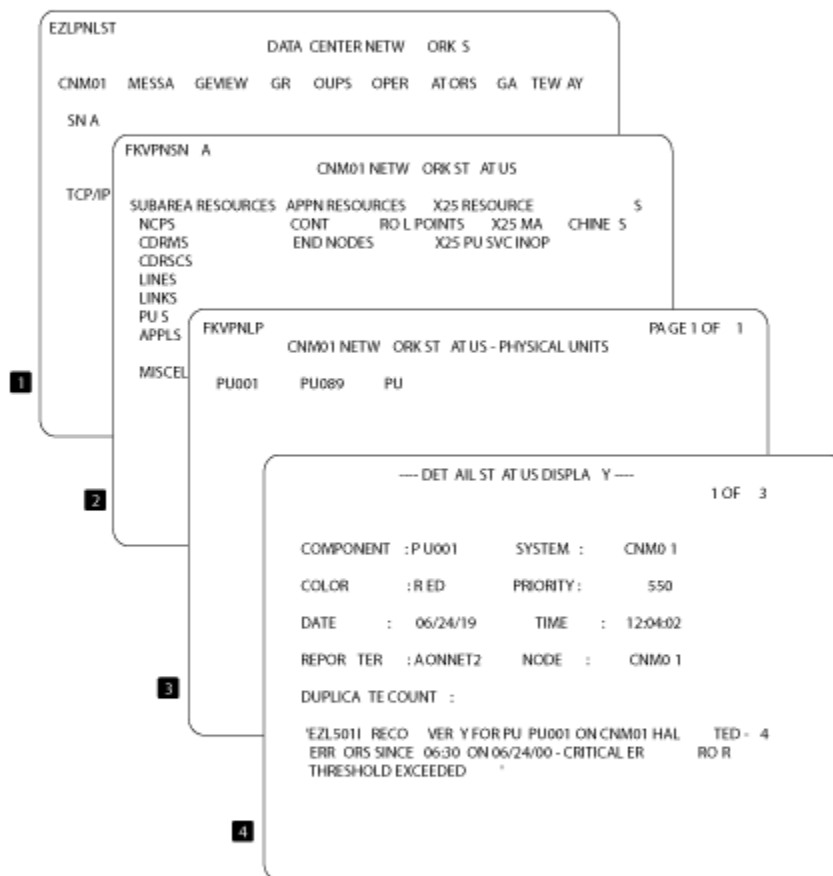


Figure 186. Displaying Network Status on Multiple Panels

**1**

The Data Center Network panel is the first DDF panel. In Figure 179 on page 179, **SNA** and **TCP/IP** represent the SNA, and TCP/IP resources in the domain CNM01. To see the types of SNA resources defined to CNM01, move the cursor to **SNA** and press the down key, **F8**.



**2**

All the monitored SNA resource types within network CNM01 are displayed on the CNM01 Network Status panel. To see the names of the physical units that require operator attention defined in CNM01, move the cursor to **PUS** and press the down key, **F8**.

**3**

The next panel shows the names of the physical units that are experiencing problems. DDF shows each monitored resource in the color of its status. For example, PU001 is shown in red if the status is INACTIVE and requires operator intervention for recovery. To see the detail record, move the cursor to **PU001** and press the detail key, **F2**.

**4**

The status details record for PU001 is shown on the Detail Status Display. The physical unit, PU001, requires operator intervention because a critical threshold has been reached and automation has stopped.

Multiple programs can update DDF at the same time with resource status without concern for the sequence or priorities posted by other programs. As AON automation resolves the problems, the resources are removed from DDF.

Use the DDF MARK function to show that a problem is being addressed. To mark a problem, display the Detail Status Display for the problem and press **F2**.

The operator ID is appended to the message at the bottom of the detail panel. When other users view DDF, their panel shows that someone is working on that particular problem. The MARK function eliminates the possibility of duplicated effort. The MARK and UNMARK commands are explained in [“Using Operator MARK panels”](#) on page 233.

## Understanding the Multiple Panel Display function

The AON member EZLTREE contains entries for specific VTAM resource types, including NCP, LINES, and CDRMs when AON/SNA is installed. When a network problem occurs, AON adds a problem descriptor to DDF with the appropriate VTAM resource type. These problems propagate up the hierarchy that is defined in the EZLTREE member. AON arranges the problems in EZLTREE in order of severity. The highest level component reflects the most severe problem.

DDF displays network problems in a series of panels. AON provides a high-level system panel, an intermediate resource type panel, and a unique panel for each resource type. The high-level panel (EZLPNLST) shows the status of the problem with the highest priority in the system. The intermediate panel (FKVPNSNA) shows the color of the status of problem with the highest priority for unique resource types. Finally, the unique resource type panels show all resource names for a given resource type in order of the priority of the problem.

## Defining Multiple Panel displays

The following SNA-specific example requires these CNMPNL1 and DSIPARM members:

### **EZLTREE**

Tree

### **EZLPNLST**

Main DDF menu panel

### **FKVPNSNA**

Resource Type menu panel

### **FKVPNLA**

APPLs panel

### **FKVPNLN**

NCP panel

### **FKVPNLC**

CDRM panel

**FKVPNLD**

CDRSC panel

**FKVPNLL**

LINES panel

**FKVPNLM**

LINKS panel

**FKVPNLP**

PU panel

**FKVPNLR**

Resource (MISC--unrecognized resource type)

**FKVPNL21**

All Resources menu panel

**EZLPNLG**

Gateway panel

**EZLPNL01**

MESSAGEVIEW panel (First)

**EZLPNL02**

MESSAGEVIEW panel (First Down)

**EZLPNL0A**

MESSAGEVIEW panel (Right)

**EZLPNL0B**

MESSAGEVIEW panel (Right Down)

**FKVCFGDS**

Control file DDF status statements

**FKVPNAC**

AON Control Point panel

**FKVPNAE**

APPN End Nodes panel

**FKVPNA1**

All APPN Resources panel

**FKVPNLX1**

X25 All Resources panel

**FKVPNLX2**

X25 Machines panel

**FKVPNLX3**

X25 Inop SVCs panel

The following sections describe how to customize the samples.

**Updating the EZLTREE member**

Update the EZLTREE tree root name CNM01 to the name that is coded in your ENVIRON SETUP SYSNAME parameter. Refer to the following:

```

/*  NETWORK : CNM01                                */
1  CNM01
2  SYSTEM
3  GATEWAY
3  GROUPS
4  CALIF
5  LA
5  SANFRAN
5  SANDIEGO
4  NEWYORK
4  ATLANTA
3  OPID
4  OPER1

```

```

3 NETWORK
4 RESOURCE
4 SNA
5 SA
6 NCP
6 LINE
6 LINKSTA
6 CDRM
6 CDRSC
6 PU
6 LU
6 SESSION
6 APPL
6 ERR
5 APPN
6 CP
6 EN
5 X25
6 X25MCH
6 X25PU

```

### Updating the EZLCFG01 member

To update the EZLCFG01 member:

1. In the ENVIRON SETUP definition in EZLCFG01, customize the SYSNAME value to reflect your domain ID. The SYSNAME can be any five-character name used to identify this NETWORK domain. Refer to the *IBM Z NetView Administration Reference* for more information about tailoring the ENVIRON SETUP statement.
2. Set the DDF parameter in the ENVIRON DDF statement to DDF=STATUS if you want DDF to assign problem priority based on resource status. The DDF panel display shipped with AON is assigned by status.

### Updating the EZLPNLS member

To update the EZLPNLS member:

1. For AON/SNA, ensure %INCLUDE statements for the panels listed previously are present and not commented out.
2. Edit each of the panel samples and change all occurrences of CNM01 to the SYSNAME defined in the ENVIRON SETUP SYSNAME definition in EZLCFG01.

### Updating the EZLINIT member

Ensure that the INITSCRN parameter is set to EZLPNLST.

## Displaying Network Status on a single panel

In addition to viewing network status on multiple panels, you can view network status on a single panel. [Figure 187 on page 224](#) shows a single-panel display. The panels are described in the definition list that follows the figure.

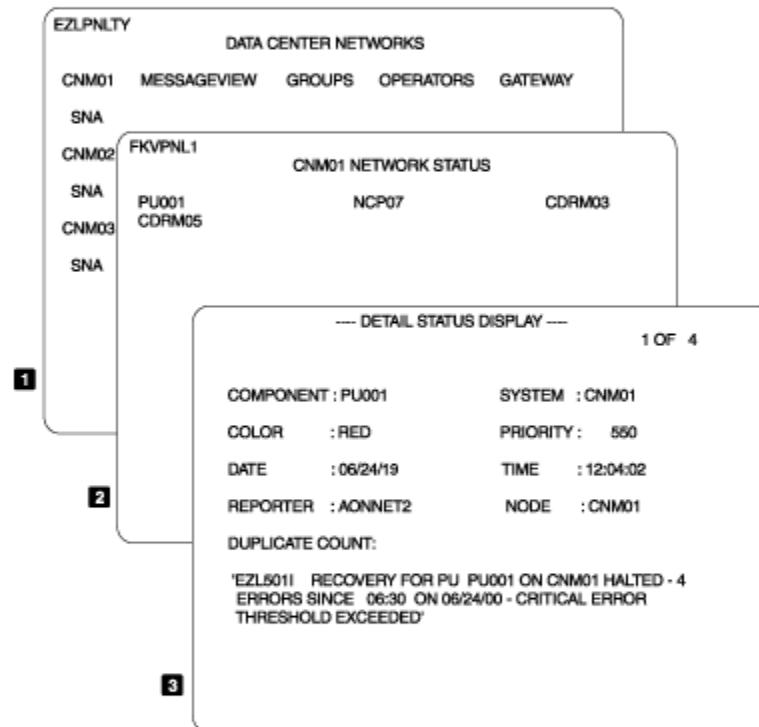


Figure 187. Displaying Network Status on a single panel

**1**

The Data Center Network panel is the main panel from which you can display a number of unique networks. This is the first panel displayed when you start DDF. The NetView domains that send notifications to this DDF are CNM01, CNM02, and CNM03. For this example, suppose that CNM01 is

highlighted in red, indicating an error status. The operator moves the cursor to CNM01 and presses F8 to move down in the hierarchy and display the CNM01 Network Status panel.

**2**

All resources requiring operator attention are color-coded by status. All resources in the network are displayed on this panel. You can customize the keys on this panel in EZLINIT [“Defining initialization statements \(EZLINIT\)”](#) on page 200.

The resource name that is displayed in the upper left corner has the highest priority of current problems. In this example, PU01 has the problem with the highest priority. PU01 is in red because the status is INACTIVE and requires operator intervention for recovery. The operator uses the Tab key to move the cursor to PU01 and presses F2 to display the Detail Status Display.

**3**

The Detail Status Display describes a problem for status descriptor PU01. PU01 requires operator intervention because a critical threshold was reached and automation stopped. To see only SNA resource failures (EZLPNL1), move the cursor to SNA and press F8.

The Detail Status display has function keys for the DDF commands, MARK (F2) and UNMARK (F10). With MARK, an operator can assign a status error to his ID from the Detail Status display. The MARK and UNMARK commands are explained in [“Using Operator MARK panels”](#) on page 233.

## Understanding how single panel displays work

The AON member EZLTREE contains entries for specific VTAM resource types, including NCP, LINES, and CDRMS. In EZLTREE, all these VTAM resource types are defined one level lower than the AON component NETWORK. When a problem occurs in the network, AON adds a problem descriptor to DDF, using the appropriate VTAM resource type. When a problem is added using the resource type, it propagates up the tree so that the component NETWORK contains all network-related problems.

AON provides a single panel that shows, in priority order, all the resource names in the network that have problems. The panel uses the component name NETWORK to display all the network-related problems.

The priority of the problem is determined by what is coded on the ENVIRON DDF statement. If DDF=STATUS is coded, the VTAM status determines the priority of the problem. Resources that are inactive have the highest priority and are red. Resources with pending status have a lower priority and are pink or turquoise. Active resources are not displayed.

When DDF=TYPE is coded, the resource type that has the problem determines the priority. For example, if an NCP is inactive or pending, the problem has high priority and is red. If a PU has an error condition (inactive or pending), the problem has a lower priority and is yellow.

## Defining single panel displays

The values shipped in the samples are the default DDF values. The required members are:

### **EZLTREE**

Tree

### **EZLCFGDS**

Control File DDF status definitions

### **EZLPNLTY**

Network View Panel

### **EZLPNL1**

Resource View Panel

### **EZLPNLG1**

Gateway View Panel

### **EZLPNL03**

MessageView Panel First

### **EZLPNL04**

MessageView Panel Right

## **EZLPNLOC**

MessageView Panel Down

## **EZLPNLOD**

MessageView Panel Right

## **FKVPNL1**

SNA Network View panel

### **Updating the EZLTREE member**

Update the EZLTREE tree root name CNM01 to the SYSNAME defined in the ENVIRON SETUP statement of the control file. Refer to the following example:

```
/* NETWORK : CNM01 */
1 CNM01
2 SYSTEM
3 GATEWAY
3 GROUPS
4 CALIF
5 LA
5 SANFRAN
5 SANDIEGO
4 NEWYORK
4 ATLANTA
3 OPID
4 OPER1
3 NETWORK
4 RESOURCE
4 SNA
5 SA
6 NCP
6 LINE
6 LINKSTA
6 CDRM
6 CDRSC
6 PU
6 LU
6 SESSION
6 APPL
6 ERR
5 APPN
6 CP
6 EN
5 X25
6 X25MCH
6 X25PU
```

### **Updating the EZLCFG01 member**

To update the EZLCFG01 member:

1. In the ENVIRON SETUP definition in EZLCFG01, customize the SYSNAME value to reflect your domain ID. The SYSNAME can be any five character name used to identify this NETWORK domain.
2. Set the DDF parameter in the ENVIRON DDF statement to:
  - DDF=STATUS if you want DDF to assign problem priority based on resource status.
  - DDF=TYPE if you want DDF to assign problem priority based on resource type.

### **Updating the EZLPNLS member**

To update the EZLPNLS member:

1. For AON/SNA, ensure that the %INCLUDE statements for the panels listed previously are present and not commented out.
2. Edit each of the panel samples and change all occurrences of CNM01 to the SYSNAME defined in the ENVIRON SETUP SYSNAME definition in EZLCFG01.

### **Updating the EZLINIT member**

In the EZLINIT member, change the INITSCRN parameter to EZLPNLTY.

## Updating the EZLPNL1 member

```
/* DEFINE CNM01 NETWORK STATUS PANEL */
P(EZLPNL1,24,80,EZLPNLTY,EZLPNLTY, , , )
TF(01,02,09,T,NORMAL)
TT(EZLPNL1)
TF(02,27,57,WHITE,NORMAL)
TT(CNM01 NETWORK STATUS)
SF(CNM01.NETWORK,04,05,16,N, , ,01)
ST( )
SF(CNM01.NETWORK,04,25,36,N, , ,02)
ST( )
SF(CNM01.NETWORK,04,45,56,N, , ,03)
ST( )
SF(CNM01.NETWORK,04,65,76,N, , ,04)
ST( )
:
SF(CNM01.NETWORK,20,25,36,N, , ,30)
ST( )
SF(CNM01.NETWORK,20,45,56,N, , ,31)
ST( )
SF(CNM01.NETWORK,20,65,76,N, , ,32)
ST( )
TF(24,01,52,T,NORMAL)
TT(PF1=HELP 2=DETAIL 3=END 4=DIS 5=CY 6=ROLL 7=UP 8=DN)
TF(24,53,79,T,NORMAL)
TT( 9=DEL 10=LF 11=RT 12=TOP)
EP
```

The tree shows NETWORK to be a level higher than any descriptors added using RESOURCE or under a resource type generic, for example CNM01.PU01(PU). The problem descriptors are propagated up to the status field of CNM01.NETWORK. All descriptors for the NETWORK are displayed on this panel.

If your network has more than 32 resources in an exception condition, duplicate this panel, customize the panel name, and add 32 to the descriptor count on each status field definition. This number causes descriptors 33 through 64 to display on this panel. Update the PANEL definition statement in EZLPNL1 to reflect a RIGHT panel with your new panel name (see [“Defining the panel statements \(EZLPNLS\)”](#) on page 209). Customize the PANEL statement of your new panel to point LEFT to EZLPNL1.

## Defining a MessageView display

The following sample scenarios demonstrate MessageView. MessageView illustrates the use of status descriptor information types (refer to STATUSFIELD). Every detail descriptor is displayed on the panel along with its date, time, resource name, and message text. This panel provides a quick view of the DDF current descriptors. [Figure 188 on page 228](#) shows a sample MessageView display.

|                         |          |                        |                                                  |
|-------------------------|----------|------------------------|--------------------------------------------------|
| EZLPNL01                |          | AON: MessageView       |                                                  |
|                         |          | Resource               | Message text                                     |
| 07/03/19                | 08:20:25 | NTV9D-0                | 1 'EZL563E ERROR ACCESSING CNM01 OUTBOUND GATEWA |
| 07/03/19                | 08:20:25 | NTV74-0                | 1 'EZL563E ERROR ACCESSING CNM01 OUTBOUND GATEWA |
| 07/03/19                | 08:40:59 | GULLIVER               | 1 'EZL507I REMINDER: IPHOST GULLIVER ON NMPNETV1 |
| 07/03/19                | 08:29:21 | DUMMYSRV               | 1 'EZL507I REMINDER: IPNAMESERV DUMMYSRV ON NMPN |
| 07/03/19                | 08:24:56 | IPL10INF               | 1 'EZL507I REMINDER: IPINFC IPL10INFC ON NMPNETV |
| 07/03/19                | 08:24:56 | GULL1                  | 1 'EZL507I REMINDER: IPINFC GULL1 ON NMPNETV1 HA |
| 07/03/19                | 08:24:56 | TNGUY                  | 1 'EZL507I REMINDER: IPTN3270 TNGUY ON NMPNETV1  |
| 07/03/19                | 08:22:17 | NMPIPL27               | 1 'EZL506I SP NMPIPL27 ON NMPIPL27 INACTIVE - RE |
| 07/03/19                | 08:22:16 | NMPIPL25               | 1 'EZL506I SP NMPIPL25 ON NMPIPL25 INACTIVE - RE |
| 07/03/19                | 08:22:11 | NMP217                 | 1 'EZL506I SP NMP217 ON NMP217 INACTIVE - RECOVE |
| 07/03/19                | 08:22:09 | NMP190                 | 1 'EZL506I SP NMP190 ON NMP190 INACTIVE - RECOVE |
| 07/03/19                | 08:21:58 | MNOSH1                 | 1 'EZL506I SP MNOSH1 ON MNOSH1 INACTIVE - RECO   |
| 07/03/19                | 08:37:41 | NCP01                  | 1 'EZL507I REMINDER: NCP NCP01 ON CNM01 HAS BEEN |
| 07/03/19                | 08:26:04 | TESTPU                 | 1 'EZL507I REMINDER: PU TESTPU ON CNM01 HAS BEEN |
| 07/03/19                | 08:25:47 | TA1T1046               | 1 'EZL507I REMINDER: LU TA1T1046 ON CNM01 HAS BE |
| 07/03/19                | 08:25:08 | LINE1                  | 1 'EZL507I REMINDER: LINE LINE1 ON CNM01 HAS BEE |
|                         |          | 07/03/19 08:47:37      |                                                  |
| ====>                   |          |                        |                                                  |
| PF1=HELP 2=DETAIL 3=RET |          | 6=ROLL 7=UP 8=DN 9=DEL |                                                  |
|                         |          | 11=RT 12=TOP           |                                                  |

Figure 188. Message View panel

The following example shows how the MessageView panel is defined.

```

/* MESSAGEVIEW DISPLAY */
/*****
P(EZLPNL01,24,80,EZLPNLST,EZLPNLST,EZLPNL02, ,EZLPNL0A)
TF(01,02,09,T,NORMAL)
TT(EZLPNL01)
TF(01,27,60,Y,NORMAL)
TT(AON: MessageView)
TF(03,22,29,T,N)
TT(Resource)
TF(03,34,45,T,N)
TT(Message text)
SF(CNM01.SYSTEM,04,02,09,N, , ,D01)
ST( )
SF(CNM01.SYSTEM,04,12,19,N, , ,T01)
ST( )
SF(CNM01.SYSTEM,04,22,29,N, , ,C01)
ST( )
SF(CNM01.SYSTEM,04,32,79,N, , ,M01)
:
TF(24,01,52,T,NORMAL)
TT(PF1=HELP 2=DETAIL 3=RET 6=ROLL 7=UP 8=DN)
TF(24,53,79,T,NORMAL)
TT( 9=DEL 11=RT 12=TOP)
PFK4()
PFK5()
EP

```

In this case, the D01 status descriptor locates the date of the first descriptor in the chain. T01 locates the time, C01 locates the component (usually resource name) and M01 locates the message text of the same descriptor.

## Implementing DDF in a focal point environment

You can customize DDF to display the status of multiple networks at a focal point AON. You can implement notification forwarding from the distributed host to the focal point host.

In a focal point environment, DDF must be defined the same for each domain. The control file for each domain in a focal point environment reflects either ENVIRON DDF,DDF=STATUS or ENVIRON DDF,DDF=TYPE if this code is defined on the focal points.

### Viewing a focal point environment

Figure 189 on page 229 shows focal point implementation.



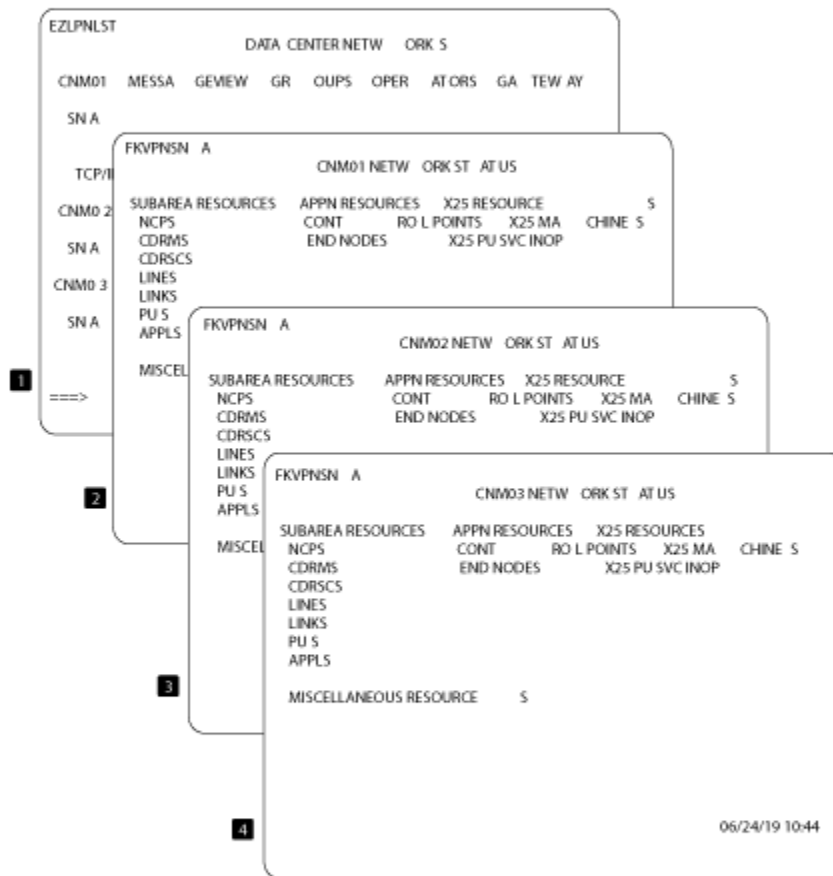


Figure 189. Focal point implementation

For an explanation of each panel in Figure 189 on page 229, match the numeric callouts (for example, **1**) for the following paragraphs with the callouts for each panel in Figure 189 on page 229.

- 1** The Data Center Networks panel is the main panel from which you can display a number of unique networks. This is the first panel that is displayed when DDF is invoked. The NetView domains sending

notifications to this DDF are CNM01, CNM02, and CNM03. This example shows which panels are displayed when you select any of the domains from the Data Center Network panel.

**2**

All the monitored resource types within domain CNM01 are displayed on the CNM01 Network Status panel. This panel is tailored to display only resources in domain CNM01.

**3**

All the monitored resource types within domain CNM02 are displayed on the CNM02 Network Status panel. This panel is tailored to display only resources in domain CNM02.

**4**

All the monitored resource types within domain CNM03 are displayed on the CNM03 Network Status panel. This panel is tailored to display only resources in domain CNM03.

## **Understanding how a focal point environment works**

AON forwards DDF updates to the focal point. At the focal point, define a tree structure for each domain in the EZLTREE member, so that DDF can determine how to store the update. This update works the same as it did on the originating domain.

To display the update, create unique panels for each domain. Use unique panel names and change the root name coded inside the panels to the root name specified on the originating system.

## **Defining how a focal point environment works**

The values shipped in the sample in your DSIPARM data set contain the default DDF values.

## **Updating the EZLTREE member**

1. Copy EZLTREE to a new member, EZLTRE01.
2. Create new tree members for CNM02 and CNM03.
3. Copy EZLTRE01; change the member name and 01 root name.
4. Edit and change EZLTREE, so that EZLTREE contains only the includes for the new members. When completed, the members look like this:

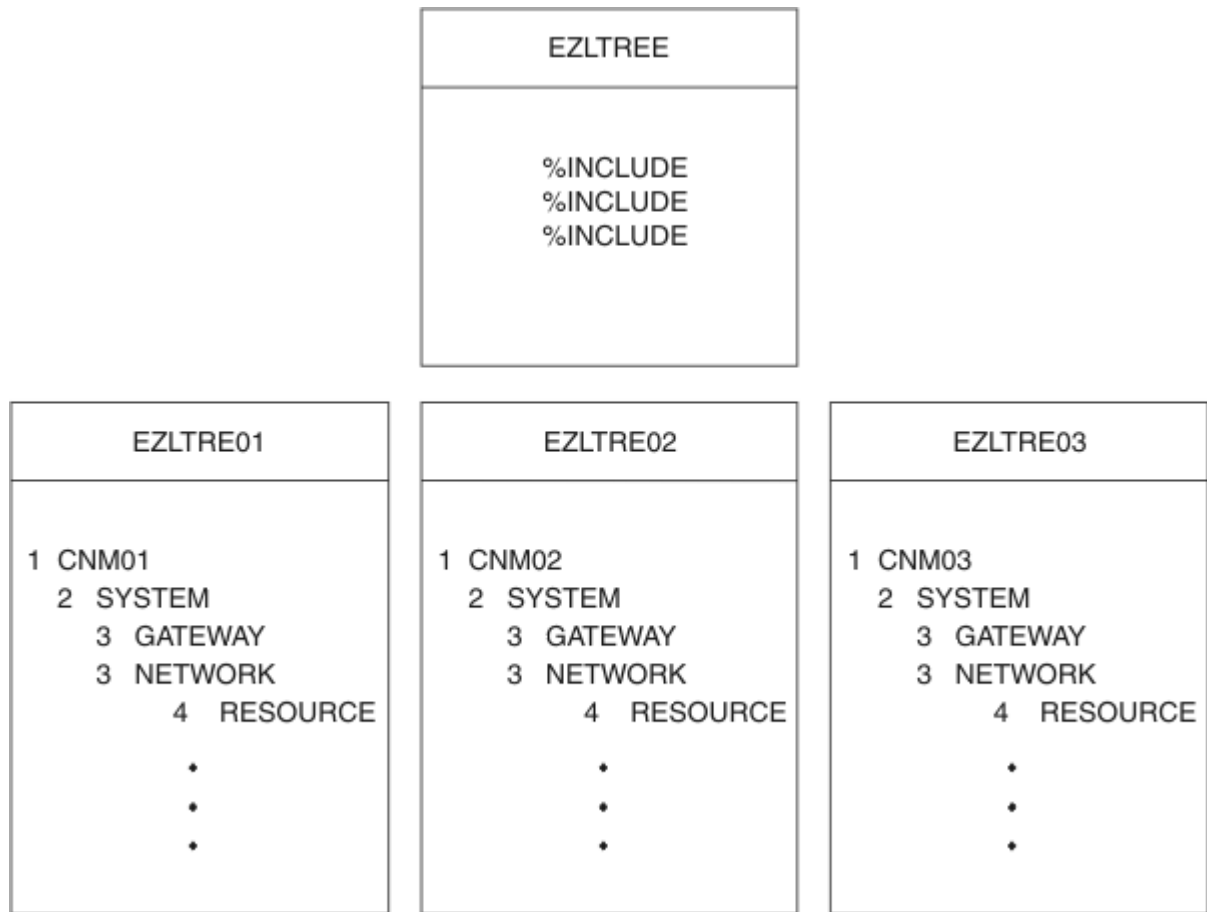


Figure 190. EZLTREE members

## Updating the EZLPNLS member

Create unique copies of existing panels for each domain. This requires some naming convention for the panels. In this example, the naming convention used is xxxxP, where xxxx is the domain ID. Depending on how you have chosen to display the network status, you must select one of the following sets of panels:

- AON/SNA multiple-panel display
- Single-panel display

For an AON/SNA multiple-panel display, select one of the following sets of panels:

### **EZLTREE**

Tree

### **EZLPNLS**

Main Panel

### **FKVPSNA**

Resource Type menu panel

### **FKVPNLA**

APPLs menu panel

### **FKVPNLN**

NCP menu panel

### **FKVPNLC**

CDRM menu panel

### **FKVPNLD**

CDRSC menu panel

**FKVPNLL**

LINES menu panel

**FKVPNLM**

LINKS menu panel

**FKVPNLP**

PU menu panel

**FKVPNLR**

Resource (MISC—unrecognized resource type)

**FKVPNL21**

All Resources menu panel

**EZLPNLG**

Gateway menu panel

**EZLPNL01**

MESSAGEVIEW menu panel First

**EZLPNL02**

MESSAGEVIEW menu panel Down

**EZLPNL0A**

MESSAGEVIEW menu panel Right

**EZLPNL0B**

MESSAGEVIEW menu panel Down

**FKVPNAC**

AON Control Point panel

**FKVPNAE**

APPN End Nodes panel

**FKVPNA1**

All APPN Resources panel

**FKVPNLX1**

X25 All Resources panel

**FKVPNLX2**

X25 Machines panel

**FKVPNLX3**

X25 Inop SVCs panel

For single-panel display, select one of the following sets of panels:

**EZLTREE**

Tree

**EZLPNLTY**

Network View Panel

**EZLPNL1**

Resource View Panel

**EZLPNLG1**

Gateway View Panel

**EZLPNL03**

Message View Panel First

**EZLPNL04**

Message View Panel Right

**EZLPNL0C**

Message View Panel Down

**EZLPNL0D**

Message View Panel Right

## FKVPNL1

### SNA Resource View Panel

To update the EZLPNLS member:

1. Create unique panels for CNM02.

Copy all panels for your display method, changing the panel name from EZLPNLxx to CNM02Pxx. For example, change EZLPNL1 to CNM02P1. Within each copied panel, change all occurrences of CNM01 to CNM02 and change all occurrences of EZLPNLxx to CNM02P (the EZLPNLTY and EZLPNLST— names must remain unchanged inside the panel). Commands assigned to DDF function keys might not be valid for resources in remote networks. Remove function key definitions that are not appropriate.

2. Create unique panels for CNM03.

Copy all panels for your display method, changing the panel name from EZLPNLxx to CNM03Pxx. For example, change EZLPNL1 to CNM03P1. Within each copied panel, change all occurrences of CNM01 to CNM03, and change all occurrences of EZLPNLxx to CNM03P (except EZLPNLTY and EZLPNLST— these names must remain unchanged inside the panel). Commands assigned to DDF function keys might not be valid for resources in remote networks. Remove any function key definitions that are not appropriate.

3. Add pointers from the Main Data Centers panels to the new panels.

For multiple panel display, add the following to EZLPNLST:

```
ST(CNM02.NETWORK,04,09,18,N, ,CNM02P2)
SF(CNM02)
ST(CNM03.NETWORK,06,09,18,N, ,CNM03P2)
SF(CNM03)
```

For single-panel display, add the following to EZLPNLTY:

```
ST(CNM02.NETWORK,04,09,18,N, ,CNM02P1)
SF(CNM02)
ST(CNM03.NETWORK,06,09,18,N, ,CNM03P1)
SF(CNM03)
```

When DDF is started, the CNM01 network status is primed. As the gateways to CNM02 and CNM03 are established, the DDF status for those networks is forwarded to CNM01. If CNM02 or CNM03 are lost, the descriptors are deleted for that network and a COMMLOST entry is added to indicate that the network status is no longer current.

## Using Operator MARK panels

AON provides two commands for tagging each DDF detail panels with an operator ID. These commands are MARK (**F2**) and UNMARK (**F10**). With these commands, an operator can select network problems from DDF for analysis.

[Figure 191 on page 234](#) and [Figure 192 on page 235](#) illustrates operator MARK panels.

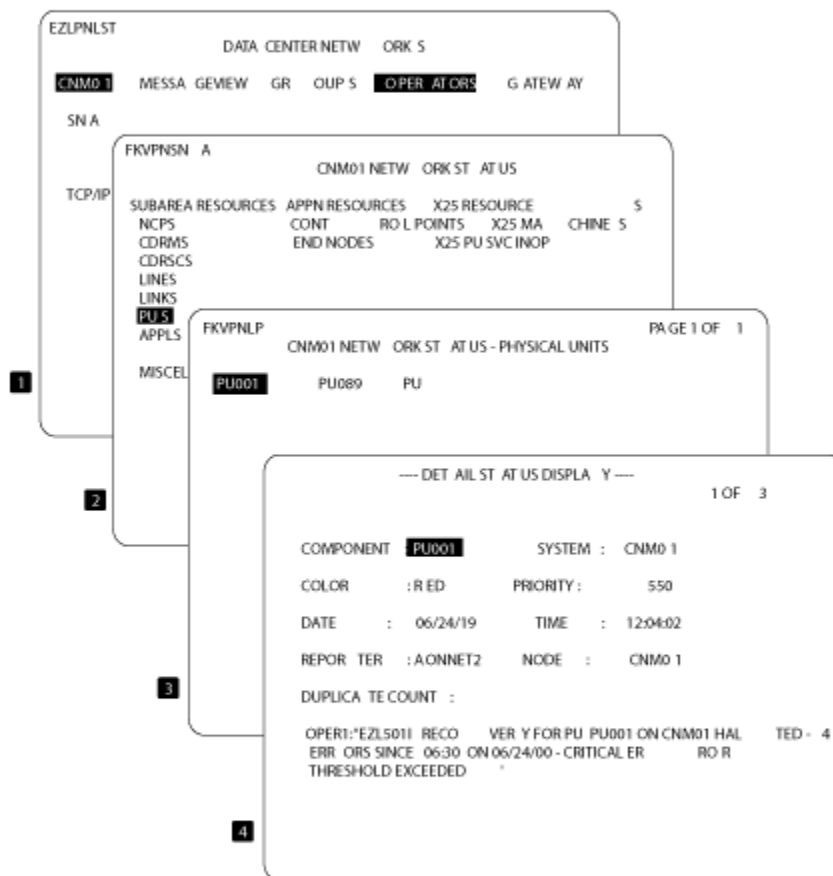


Figure 191. Operator MARK panel 1

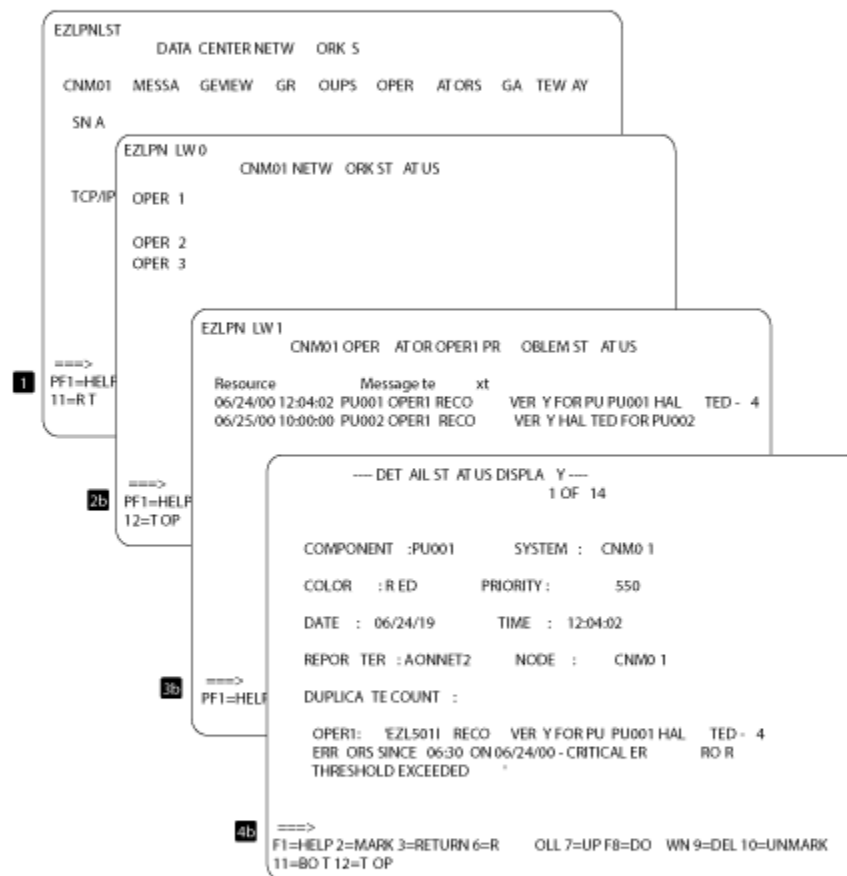


Figure 192. Operator MARK panel 2

For an explanation of each panel in Figure 191 on page 234, match the numeric callouts for the panels (for example, **1**) with the following paragraphs:

- 4b** To mark a problem, display the Detail Status Display for the problem and then press **F2**. This action appends the operator's ID to the message at the bottom of the detail panel shown in **4b**.

## **1 and 2b**

Marking a DDF problem also causes DDF to display the problem in reverse video, both on the Detail Status Display and on the upper level panels, as shown in **1** and **2b**.

If problems are being handled by an operator, they are marked. Other operators can choose unmarked problems for analysis; thereby, preventing duplicate effort.

## **Understanding Operator MARK panels**

Because the status is displayed in reverse video, operators know at a glance the most severe problems are being addressed. In addition, because the problem is time-ordered, operators know which problems have been outstanding the longest.

From these DDF panels, operators use function keys to recycle the resource to which the cursor points, display those resources, mark descriptors, unmark descriptors, and delete descriptors. NetView commands work from these command lines.

## **Defining Operator MARK panels**

To define operator mark panels update the EZLTREE member used for your DDF with the operator ID appropriate for your DDF installation. To add operator IDs for the DDF MARK function, update EZLTREE:

```
/* NETWORK : CNM01 */
1 CNM01
2 SYSTEM
3 GATEWAY
3 GROUPS
4 CALIF
5 LA
5 SANFRAN
5 SANDIEGO
4 NEWYORK
4 ATLANTA
3 OPID
4 OPER1
4 OPER2
4 OPER3
3 NETWORK
4 RESOURCE
4 SNA
5 SA
6 NCP
6 LINE
6 LINKSTA
6 CDRM
6 CDRSC
6 PU
6 LU
6 SESSION
6 APPL
6 ERR
5 APPN
6 CP
6 EN
5 X25
6 X25MCH
6 X25PU
```

When DDF deletes descriptors at the NETWORK level, it also deletes them from the panel of the operator. Define each operator ID having independent panels in the EZLTREE member.

## **Grouping resources in DDF**

Group resources in DDF by site, application, customer base, company, or by any other criterion. [Figure 193 on page 237](#) shows a sample DDF group.



```

EZLPNLST
                                DATA CENTER NETWORKS

      CNM01      MESSAGEVIEW      GROUPS      OPERATORS      GATEWAY

      SNA

      TCP/IP

                                06/24/19 16:21:00

====>
PF1=HELP 2=DETAIL 3=END      6=ROLL 7=UP 8=DN      10=LF 11=RT 12=TOP

```

Figure 193. DDF panel Showing GROUPS

To view the resources in this DDF group, move the cursor to GROUPS and press Enter. [Figure 194 on page 237](#) shows the resources defined to GROUPS. This sample DDF group is by geographical area.

```

EZLPNLGR
                                CNM01 NETWORK GROUP STATUS

      CALIFORNIA

      NEW YORK

      ATLANTA

                                06/24/19 09:46:00

====>
PF1=HELP 2=DETAIL 3=END      6=ROLL 7=UP 8=DN      10=LF 11=RT 12=TOP

```

Figure 194. Sample DDF Group panel

## Grouping resources for specific requirements

The DDFGROUP control file entry groups DDF resources for display. For example, you can group dissimilar DDF resources by geographic location.

Using DDFGROUP statements, AON supports wild cards in DDFGROUP lists and places resources in multiple groups. During initialization, the EZLEAC11 program creates DDFRES control file entries from the DDFGROUP control file entries. To determine which groups are defined for a given resource, check only the DDFRES control file entries. DDF resources are updated in both the applicable DDFGENERIC and DDFGROUP matches.

The syntax of the DDFGROUP control file entry is:

```

DDFGROUP groupname,
LIST=(res1,res2,...resn)
LIST=(res1,res2,...resn)
LIST=(res1,res2,...resn)
LIST=(res1,res2,...resn)

```

**groupname**

The name of this group of DDF resources.

**Note:** You must use *groupname* in EZLTREE and the panel used to display members of this group. Multiple DDFGROUP statements for the same *groupname* are valid.

**LIST**

The list of resources in the group. Wild cards are supported in resource names. You can create multiple lists. Each complete definition list must be on a single line.

Refer to the *IBM Z NetView Administration Reference* for information about DDFGROUP syntax.

The EZLEAC11 program creates the DDFRES control file entries during startup in this format:

```
DDFRES res1, GROUP=(group_name1, group_name2, ... group_namen)
```

**Note:** AON creates all DDFRES control file entries.

The following sample list illustrates the format of DDFGROUP control file entries that create a group of resources according to city:

```
DDFGROUP LA, LIST=(SALA*, GWATLA, GWLANY, GWSFLA, GWSDLA)
DDFGROUP SANDIEGO, LIST=(SASD*, GWATSD, GWSDNY, GWSFSD, GWSDLA)
DDFGROUP SANFRAN, LIST=(SASF*, GWATSF, GWSFNY, GWSFSD, GWSFLA)
DDFGROUP NEWYORK, LIST=(SANY*, GWATNY, GWSFNY, GWSDNY, GWLANY),
DDFGROUP NEWYORK, LIST=(IMSNY1, IMSNY2, IMSNY3)
DDFGROUP ATLANTA, LIST=(SAAT*, GWATNY, GWATSF, GWATSD, GWATLA),
LIST=(CICSAT1, CICSAT2)
```

Some of the DDFRES control file entries that the EZLEAC11 program creates from the previous example are:

```
DDFRES SALA*, GROUP=(LA)
DDFRES GWATLA, GROUP=(LA, ATLANTA)
DDFRES GWLANY, GROUP=(LA, NEWYORK)
DDFRES GWSDLA, GROUP=(LA, SANDIEGO)
DDFRES GWSFLA, GROUP=(LA, SANFRAN)
```

In this example, SALA\* is defined in group LA and SALACICS is defined in SANDIEGO. Therefore, when SALACICS is added to DDF, it is displayed in the SANDIEGO group only, not in the LA group. However, if SALACICS is explicitly defined in LA, then SALACICS is added to DDF in both SANDIEGO and LA.

In this example, the Gateway nodes (GWsa1sa2) are defined in both of the cities they are connecting. Each city has a subarea associated with it by naming convention; therefore defining SAsa\_id\* is sufficient to assign all resources in the city to the city status panel. As a result, gateways are displayed in several city panels as shown in [Figure 195 on page 238](#). NEWYORK and ATLANTA are also responsible for associated applications shown in [Figure 195 on page 238](#).

|          |                              |             |
|----------|------------------------------|-------------|
| EZLPNLNY | NEW YORK CITY NETWORK STATUS | PAGE 1 OF 1 |
| GWATNY   | GWLANY                       |             |

|          |                             |             |
|----------|-----------------------------|-------------|
| EZLPNLAT | ATLANTA CITY NETWORK STATUS | PAGE 1 OF 1 |
| GWATLA   | GWATNY                      |             |

Figure 195. NEWYORK and ATLANTA Groups

## Defining DDF groups

The following sample definitions are based on the DDFGROUP control file entries shown in [Figure 195 on page 238](#). These definitions are located in DSIPARM.

Before you create DDF groups, consider your network and how you want the DDF panels to display. Refer to the *IBM Z NetView Administration Reference* to code the DDFGROUP control file entries.

### Updating the EZLTREE DSIPARM member

To update EZLTREE, add DSIPARM member *group\_name* to the EZLTREE member. Use a level 3 element that is not subordinate to NETWORK, so the same message is not displayed in MESSAGEVIEW multiple times (one for each resource type and DDF group). The following example tree is valid:

```
/* NETWORK : CNM01 */
1 CNM01
2 SYSTEM
3 GATEWAY
3 GROUPS
4 CALIF
5 LA
5 SANFRAN
5 SANDIEGO
4 NEWYORK
4 ATLANTA
3 OPID
4 OPER1
3 NETWORK
4 RESOURCE
4 SNA
5 SA
6 NCP
6 LINE
6 LINKSTA
6 CDRM
6 CDRSC
6 PU
6 LU
6 SESSION
6 APPL
6 ERR
5 APPN
6 CP
6 EN
5 X25
6 X25MCH
6 X25PU
```

Figure 196. Example EZLTREE member

### Updating DDF Menus

To merge the new DDF panels into existing DDF panels, update the main panel menu, EZLPNLST, where you see CNM01, MESSAGEVIEW, and GATEWAY.

EZLPNLST is shown in [Figure 197 on page 240](#).

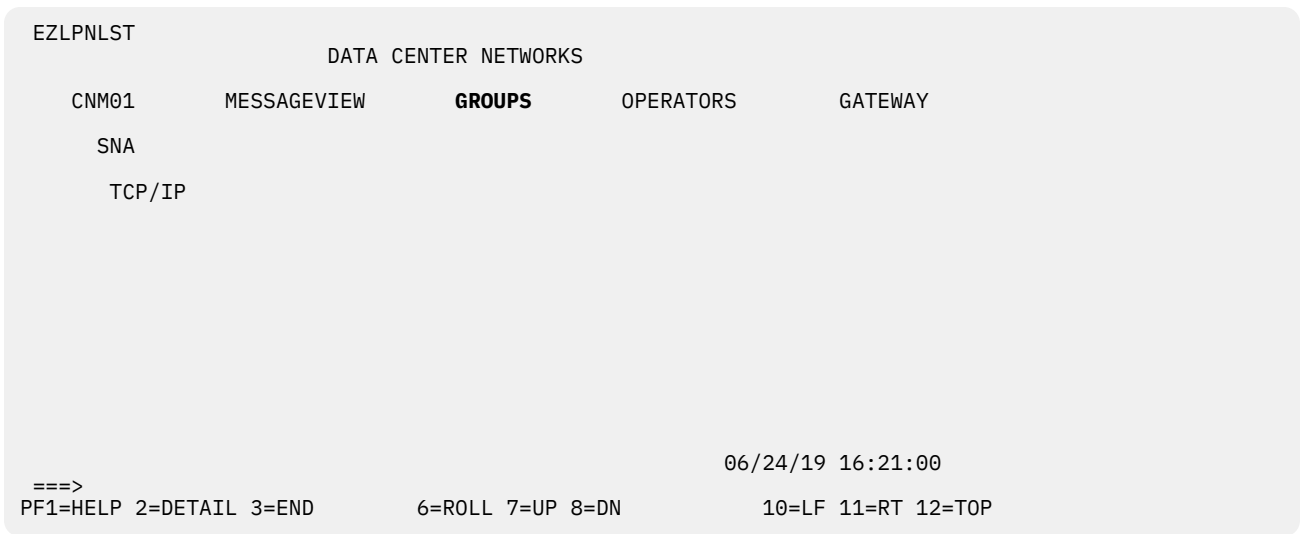


Figure 197. EZLPNLST panel with DDF Group

EZLPNLGR displays the Network Group Status panel shown in [Figure 198 on page 240](#):

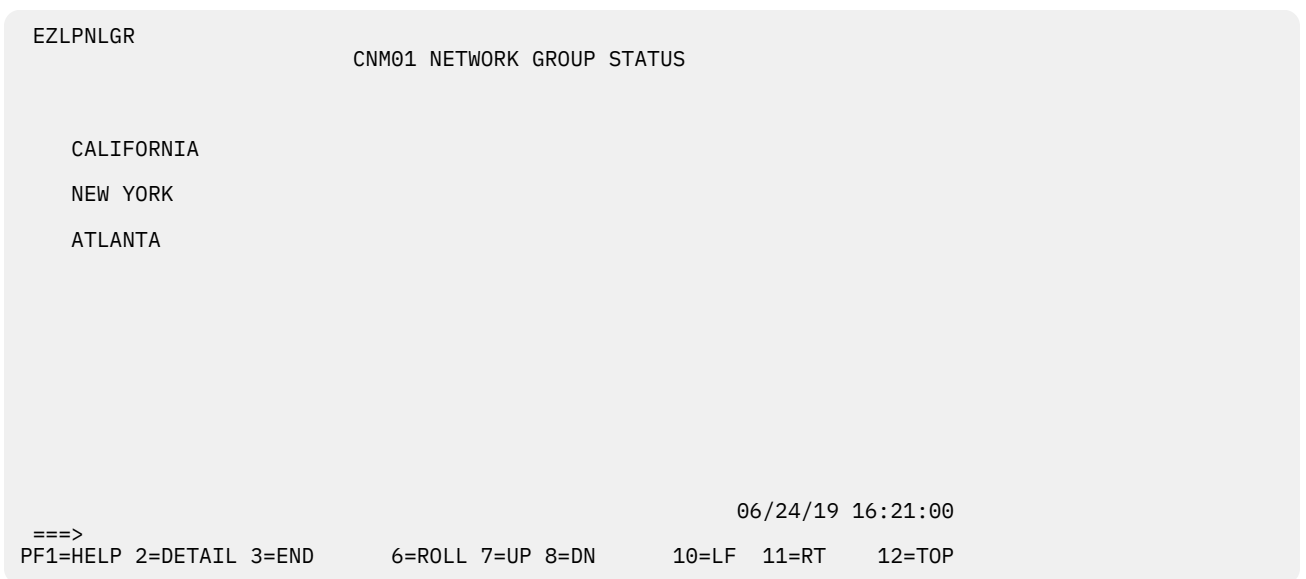


Figure 198. EZLPNLGR panel Showing a List of Cities

To view the CALIFORNIA subgroups, use the EZLPNLCA indicator panel shown in [Figure 199 on page 241](#).

```

EZLPNLCA
                                CALIFORNIA NETWORK STATUS SUMMARY

LOS ANGELES
SAN FRANCISCO
SAN DIEGO

ALL CALIFORNIA RESOURCES

                                06/24/19 16:21:00
====>
PF1=HELP 2=DETAIL 3=END      6=ROLL 7=UP 8=DN      10=LF 11=RT 12=TOP

```

Figure 199. EZLPNLCA panel for CALIFORNIA Group

### Creating generic display panels for DDF groups

Creating a generic panel for each new group, enables operators to see resources saved under them. The following is a generic panel:

```

/*****
/* DEFINE CNM01 NETWORK STATUS PANEL                               */
/* NEWYORK          STATUS DISPLAY                                */
/*                                                           */
/*****
/* APAR#          DATE          DESCRIPTION                      */
/* -----          -          -                               */
/*                                                           */
/*****
P(EZLPNLNY,24,80,EZLPNL2,EZLPNL2, ,)
TF(01,02,09,T,NORMAL)
TT(EZLPNLNY)
TF(02,21,59,WHITE,NORMAL)
TT(CNM01 NEW YORK CITY NETWORK STATUS)
TF(01,65,77,BLUE,NORMAL)
TT(PAGE 1 OF 1)
SF(CNM01.NEWYORK,04,05,16,N, , ,01)
ST( )
SF(CNM01.NEWYORK,04,25,36,N, , ,02)
ST( )
SF(CNM01.NEWYORK,04,45,56,N, , ,03)
ST( )
:
TF(24,01,52,T,NORMAL)
TT(PF1=HELP 2=DETAIL 3=END 4=DIS 5=CY 6=ROLL 7=UP 8=DN)
TF(24,53,79,T,NORMAL)
TT( 9=DEL 10=LF 11=RT 12=TOP)
EP
/*****

```

The generic group panels for ATLANTA, SANDIEGO, LA, and SANFRAN look just like this one for CALIFORNIA except all occurrences of NEWYORK would be changed to the correct group name and the name of the panel would be updated.

### Updating the EZLPNLST CNMPNL1 member

To improve DDF performance, include new panels created for DDF groups in the EZLPNLS member of the CNMPNL1 data set for preloading. The DDF group sample panels provided with AON in the CNMPNL1 data set are:

- EZLPNLAT
- EZLPNLCA

- EZLPNLCI
- EZLPNLGR
- EZLPNLLA
- EZLPNLNY
- EZLPNLSF

---

## Chapter 23. Issuing Dynamic Display Facility (DDF) commands

AON routines use these commands to update the DDF status information from within DDF. If additional status information is required, use the following commands to update the DDF status information:

### **DDF**

Displays DDF or a particular DDF panel

### **DDFADD**

Adds a status descriptor to a component

### **DDFCLEAR**

Clears DDF

### **DDFDEL**

Deletes a status descriptor from DDF

### **DDFPANEL**

Loads a panel member

### **DDFQRY**

Queries the status of specific status descriptors

### **DDFTREE**

Loads a tree member

### **MARK**

Marks a DDF resource with an operator ID

### **UNMARK**

Removes an operator ID from a DDF resource

---

## Using the Dynamic Display Facility (DDF)

### **Purpose**

The DDF command shows a color-coded status panel for the resources that are currently being acted upon by AON or require operator intervention for recovery. You can also use the DDF command to display a specific DDF panel.

### **Format**

#### **DDF**

►► DDF —————►  
          └─ *panel\_name* ─┘

### **Parameters**

#### ***panel\_name***

The name of the panel displayed in the upper left corner of the screen. Use this parameter to view a specific DDF panel.

### **Usage**

- The workstation from which you enter the DDF command must support 3x79 Terminal Extended Attributes, which are blue, red, pink, green, turquoise, yellow, and white for color and flash, reverse video, and underscore for highlighting.

- If you enter **DDF** without the *panel\_name* parameter, the main DDF panel is displayed. (The panel name for this panel is EZLPNLST.)
- The DDF command operates in full-screen mode only.

### Examples

To display the panel named FKVPNSNA when AON/SNA is installed, type the following:

```
DDF FKVPNSNA
```

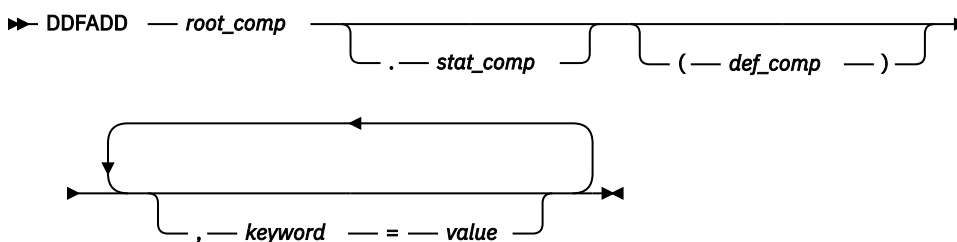
## Adding status descriptors (DDFADD)

### Purpose

The DDFADD command issues a request to the EZLTDDF task to add a status descriptor to a status component with data supplied by a calling program, a command list, or NetView command facility (NCCF) operator. The date and time are provided automatically.

### Format

#### DDFADD



**Note:** You can specify more than one *keyword=value* pairs.

### Parameters

#### *def\_comp*

The element in the DDF tree that organized DDF descriptors into groups. This element is usually a resource type, service point name, or group name defined in DDFGROUP control file entries. For example, you can define CNM01.LINE99(LINE) where CNM01 is the *root\_comp*, LINE99 is the *stat\_comp*, and LINE is the *def\_comp*.

#### *keyword=value*

The status information that is necessary to add a status descriptor. In general, the only keywords that are needed are reference value, priority, and data. Valid values are:

**COLOR = {Red | Blue | Turquoise | Green | Pink | Yellow | White}**

Specifies color. If this is not specified, color is based on priority.

#### **Data =text**

Specifies that text can be user or message data with a maximum length of 240 characters. Data can be composed of one or more lines. Each line of data can be delimited to produce a formatted display. Delimiters can be any unique character that does not occur in the text. Delimiters between lines must be doubled. For example:

```
DA=#LINE1-DATA ##### LINE2-DATA ##### LINE3-DATA #
```

Causes the following to display on the status descriptor data field:



```
LINE1-DATA  
LINE2-DATA  
LINE3-DATA
```

**DataType = {OTH|MSG}**

MSG specifies that message data is to be displayed on the status descriptor. OTH specifies that the data field contains user-defined data.

**HighL = {Normal|Blink|Reverse|Underscore|Panel}**

If this is not specified, highlighting is determined by the panel definition.

**Info= text**

Specifies the information displayed on DDF status panel in the STATUSTEXT field with a maximum length of 80 alphanumeric characters. The text must be delimited. Delimiters can be any unique character that does not occur in the text. The length of the text field must be within the range specified by the start and end positions in the corresponding STATUSFIELD entry. See [“Locating the status component \(STATUSFIELD\)”](#) on page 210 for more information.

**Priority= value**

Specifies a valid number within the range specified in the EZLINIT file. If the field is not specified, a program default priority of 999 is used.

**PropDwn = {Yes|No}**

Specifies propagation downward. If this is not specified, propagation down is governed by entry in the EZLINIT file.

**ProplvLD = {\*|root\_comp.stat\_comp}**

An asterisk (\*) specifies that the status is to be propagated to the leaves of the tree. If a status component is specified, the status is only propagated down the leaves until, and including, the specified status component.

**ProplvLU = {\*|root\_comp.stat\_comp}**

An asterisk (\*) specifies that the level to which the status is to be propagated is the root of the tree. If a status component is specified, the status is only propagated up to, and including, the specified status component.

**PropUp = {Yes|No}**

Specifies propagation upward. If this is not specified, propagation up is governed by the entry in the EZLINIT file.

**RefValue= value**

Specifies reference value, which must be alphanumeric, with a maximum length of 20 (for example RV=NCP01).

**root\_comp**

The root component name that is defined in the root node of the tree structure. The root component is required because different systems can have status components with the same name defined in their respective tree structures. The root component must be unique, therefore, each status component in a tree structure can be uniquely identified by prefixing it with the root component entry.

**stat\_comp**

A valid resource name or resource type, as defined in the EZLTREE member, which can contain up to 8 characters.

**Usage**

If the data supplied in the DDFADD command exactly matches an existing descriptor in the chain, the new descriptor is not added.

With AON, status changes are logged in the automation log. The AON common logging routines provide the data for the DDFADD command and issue the DDFADD command. Any resource that is monitored by AON is automatically updated. User applications can also call the common logging routines to add DDF descriptors. The logging routines refer to the control file for status and priority information. See [Chapter 20, “Understanding Dynamic Display Facility \(DDF\) design,”](#) on page 177 for more details.

## Examples

```
DDFADD CNM01.NCP001(NCP),RV=NCP001,PR=550,DT=MSG,  
DA='EZL509I NCP001 IS UNAVAILABLE (REPORTED BY AUTNET1)'
```

In this example, a status descriptor is added for NCP001 on CNM01 with a priority of 550. NPC001 is green on the status panel because priority 550 is in the green color range.

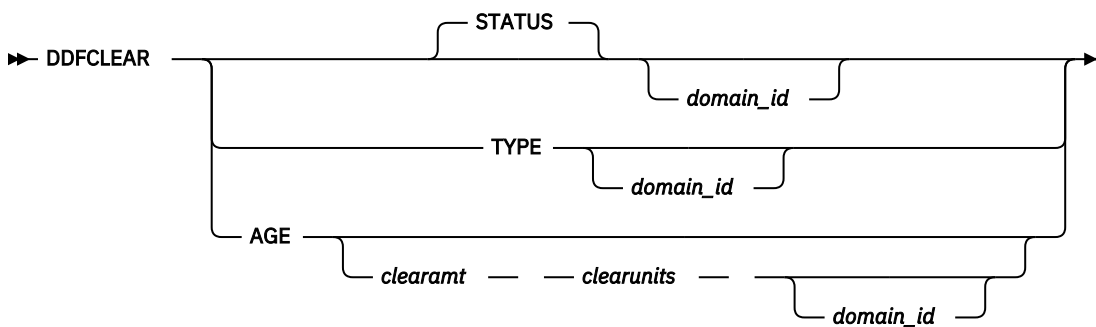
## Clearing DDF (DDFCLEAR)

### Purpose

The DDFCLEAR command deletes DDF elements based on resource. DDFCLEAR is a synonym for EZLEADCL.

### Format

#### DDFCLEAR



### Parameters

#### AGE

Deletes elements from DDF based on the age of the DDF element. A date and time stamp are stored into DDF when an element is added to DDF. This date and time are compared to the current date and time. If the difference exceeds the amount of DAYS or HOURS specified, the record is deleted. If you specify AGE with no other parameters, then the following command is issued:

```
DDFCLEAR AGE 5 DAYS current_domain_id
```

#### clearamt

The number of days or hours to be used for an age threshold for deleting DDF records in long-lived NetView systems. This number must be a whole number. If you are deleting records by hours of age, the number is in the range of 0 – 24. If you are deleting by days, any number is valid up to 999. Because all DDF elements are kept in storage, they do not survive a recycle of NetView program. For this reason, the AGE option is useful only in environments where the NetView program has continuous availability for more than a week. If you issue DDFCLEAR AGE with no other parameters, the default value used is five days.

#### clearunits

Either DAYS or HOURS. All records whose DDF date is older than *clearamt* DAYS or *clearamt* HOURS is deleted. If no value is specified, DAYS is the default. If you specify HOURS, also specify a *clearamt* value because the parameters are positional.

#### domain\_id

The NetView domain ID to be evaluated for extraneous elements. The domain ID must match one of the level 1 values in EZLTREE. For a single domain DDF, this is usually the current domain ID. Check the value of the control file entry ENVIRON SETUP SYSNAME to determine this value. For a focal-point

DDF representing several NetView domains, this value can be the EZLTREE root name or the ENVIRON SETUP SYSNAME value set up on any of the distributed systems. If no domain ID is provided, the current NetView domain ID is used.

## STATUS

Deletes records based on VTAM status. If you issue DDFCLEAR with no parameters, the default command issued is:

```
DDFCLEAR STATUS current_domain_id
```

STATUS = DDF records are deleted based on the VTAM status of the resource in the DDFname, in the format *domainid.resname(restype\_generic)*. The resource is displayed with a D NET command. If the resource cannot be displayed, the DDF element is *not* deleted. If the resource cannot be displayed, AON assumes that the resource is unavailable such as NCPs and switched major nodes.

If the resource status matches a DDF status, that DDF element is deleted. If the resource has REQ=NOADD defined for it, the wild cards in the ENVIRON DDF statements are evaluated. For example, AON default DDF statements provide for status of ACT\* and CON\* as REQ=NOADD. If DDFCLEAR were run and some LUs were found among the DDF elements that now had an active status (because of activating the PU,SCOPE=U), the LUs are deleted from DDF.

## TYPE

Deletes records based on resource type. TYPE = DDF records are deleted based on the VTAM resource type of the resource in the DDFname (usually in the format *domainid.resname(restype\_generic)*). The resource is displayed with a D NET command. If the resource cannot be displayed, the DDF element is *not* deleted. If the resource is assumed to be nondisplayable, a problem is indicated such as NCPs and switched major nodes. If the resource type matches a DDF type that has REQ=NOADD defined for it, the DDF element is deleted. Wild cards in the ENVIRON DDF statements are evaluated.

For example, AON default DDF statements provide for type of LU as REQ=NOADD. If DDFCLEAR were run and some LUs were found among the DDF elements that had *any* status, the LUs are deleted from DDF.

## Usage

Refer to the *IBM Z NetView Administration Reference* for more information about the ENVIRON DDF control file entries.

## Examples

To clear all DDF records based on status, issue:

```
DDFCLEAR
```

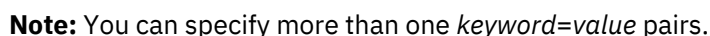
## Deleting status descriptors (DDFDEL)

---

### Purpose

The DDFDEL command deletes status descriptors. Each status descriptor in the chain is compared against the data specified on the DDFDEL command. If each specified element in the command matches the data, the status descriptor is removed from the chain. The parameters supplied with the DDFDEL command can include wildcard characters. An asterisk (\*) takes the place of a single character or, if used at the end of the field, indicates that the rest of the string is automatically matched. If a keyword is not specified on the command, it is considered to be matched by any other value.

## DDFDEL



***def\_comp***

The element in the DDF tree that organized DDF descriptors into groups. This element is usually a resource type, service point name, or group name defined in DDFGROUP control file entries. For example, you can define CNM01.LINE99(LINE) where CNM01 is the *root\_comp*, LINE99 is the *stat\_comp*, and LINE is the *def\_comp*.

***keyword=value***

Describes the status information that is necessary to delete a status descriptor. Valid values are:

**Color = {Red | Blue | Turquoise | Green | Pink | Yellow | White}**

Specifies a color.

**Data =text**

Specifies user or message data with a maximum length of 240 characters.

**DataType** = {OTH|MSG}

MSG specifies that message data in the status descriptor is in message format. OTH specifies that the data field contains user-defined data.

```
HighL = {Normal|Blink|Reverse|Underscore|Panel}
```

Specifies highlighting. If highlighting is specified in the DDFDEL command, it must be the same as that of the existing status descriptors, otherwise they are not deleted. Let highlight default, because it is considered to be matched by any other value.

**INfo= *text***

Displays the information in the STATUSTEXT field, with a maximum length of 80 alphanumeric characters. The text must be delimited. Delimiters can be any unique character that does not occur in the text.

**PRiority= value**

Specifies the priority, which must be a valid number within the range specified in the EZLINIT PRI entry.

**PropDwn** = {Yes | No}

Specifies propagation downward. If this is not specified, propagation down is governed by entry in the EZLINIT file.

**RefValue= *value***

Specifies a reference value, which must be alphanumeric, with a maximum length of 20 characters (for example RV=NCP001).

**Source= *value***

Specifies the operator under whom the DDFADD request was issued. This is the REPORTER field on the Detail Status Display panel.

***root\_comp***

The root component name that is defined in the root node of the tree structure. The root component is required because different systems can have status components with the same name defined in their respective tree structures. Because the root component must always be unique, each status

component in a tree structure can be uniquely identified by using a prefix with the root component entry.

#### ***stat\_comp***

Valid resource name or resource type that is defined in the EZLTREE member up to 8 characters.

#### **Usage**

Use a wildcard character (\*) with a single DDFDEL command to delete multiple status descriptors.

#### **Examples**

In this example, any status descriptor for NCP001 on CNM01 with a reference value of NCP001 is deleted. Because no other keywords are used, no other criteria are used to determine a match:

```
DDFDEL CNM01.NCP001,RV=NCP001
```

#### **Examples**

In this example, any status descriptors for NCP001 on CNM01 that have a reference value in the range of 100–190 are deleted:

```
DDFDEL CNM01.NCP001,RV=1*0
```

#### **Examples**

In this example, any status descriptors for NCP001 on CNM01 that have a reference value beginning with R and a priority beginning with the number 3 are deleted.

```
DDFDEL CNM01.NCP001,RV=R*,PR=3*
```

## **Loading a panel member (DDFPANEL)**

---

#### **Purpose**

The DDFPANEL command dynamically loads a panel member from the DSIPARM data set or deletes a panel member.

#### **Format**

##### **DDFPANEL**

➡ DDFPANEL — *panel* — ADD  
DELETE ➡

#### **Parameters**

##### ***panel***

The name of the member that contains the panel to be loaded. The panel name and the member name must match.

#### **Usage**

When DDF is started, only the EZLPNLS panel definitions are loaded. Panels loaded with the DDFPANEL command are not reloaded during EZLTDDF initialization. If you add panels dynamically, restart DDF, Add the panels again. If you update a panel and want the updated version to be displayed by DDF, use the DDFPANEL command to replace the current copy of the panel.

## Examples

The following command loads member NEWPANEL into memory, which gives you access to the panel defined in this member:

```
DDFPANEL NEWPANEL,ADD
```

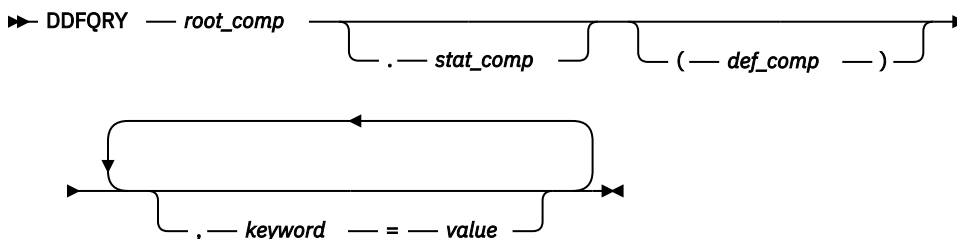
## Querying status descriptors (DDFQRY)

### Purpose

The DDFQRY command queries the status of specific status descriptors. Parameters specified with the DDFQRY command can include wildcard characters. Asterisks (\*) take the place of a single character or, if used at the end of the field, indicate that the rest of the string are automatically matched. If a keyword is not provided on the command, any value is considered a match.

### Format

#### DDFQRY



**Note:** You can specify more than one *keyword=value* pairs.

### Parameters

#### *root\_comp*

The root component name that is defined in the root node of the tree structure. The root component is required because different systems can have status components with the same name defined in their respective tree structures. Because the root component must be unique, each status component in a tree structure can be uniquely identified by using a prefix with the root component entry.

#### *stat\_comp*

Valid resource name or resource type (*def\_comp* value) that is defined in the EZLTREE member. You can use up to 8 characters.

#### *def\_comp*

The element in the DDF tree that organized DDF descriptors into groups. This element is usually a resource type, service point name, or group name that is defined in DDFGROUP control file entries. For example, define CNM01.LINE99(LINE) where CNM01 is the *root\_comp*, LINE99 is the *stat\_comp*, and LINE is the *def\_comp*.

#### *keyword=value*

Describes the status information that can be used to query status descriptors. Valid values are:

##### **RefValue = value**

Specifies reference value, which must be alphanumeric, with a maximum length of 20 characters (for example, RV=NCP001).

##### **INfo = text**

Specifies information text, which must be alphanumeric, with a maximum length of 80 characters. The text must be delimited. Delimiters can be any unique character that does not occur in the text (for example, IN='TAPE001').

**PRIority = value**

Specifies the priority, which must be a valid number within the range specified in the EZLINIT entry PRIORITY (for example, PR=330).

**COLOR = {Red | Blue | Turquoise | Green | Pink | Yellow | White}**

Specifies color. Status descriptors with the specified color are displayed if all other keyword values match.

**HighL = {Normal|Blink|Reverse|Underscore|Panel}**

Specifies highlighting. Status descriptors with the specified highlighting are displayed if all other keyword values match.

**SOurce = value**

Specifies the operator ID that issued the DDFADD command. The ID can be a maximum length of 8 alphanumeric characters. This value is displayed in the status descriptor field.

**DataType = {OTH|MSG}**

Status descriptors with the specified data type value are displayed if all other keyword values match.

**DAta = text**

Status descriptors with the same data text are displayed if all other keyword values match.

**Usage**

Use a wildcard character with a single DDFQRY command to view multiple status descriptors. AON uses DDFQRY to query DDF status information so that it can be forwarded from a distributed domain to a focal-point domain.

**Examples**

In this example, status descriptors for NCP001 on CNM01 with a reference value of NCP001 are displayed:

```
DDFQRY CNM01.NCP001,RV=NCP001
```

Because no other keywords are used, an automatic match is assumed.

**Examples**

In this example, status descriptors for NCP001 on CNM01 that have a reference values in the range of 100 – 190 are displayed:

```
DDFQRY CNM01.NCP001,RV=1*0
```

**Examples**

Display all status descriptors saved with a *def\_comp* value of NCP:

```
DDFQRY CNM01.NCP
```

**Examples**

Display all status descriptors for the system CNM01:

```
DDFQRY CNM01
```

## Loading tree members (DDFTREE)

---

### Purpose

The DDFTREE command dynamically loads a tree member from the DSIPARM data set or deletes a tree member from storage.

### Format

#### DDFTREE

➤ DDFTREE — *tree* — — ADD —  
— DELETE — ➤

### Parameters

#### *tree*

The name of the member that contains the tree structure to be loaded

### Usage

The tree member can refer to other tree members with the %INCLUDE keyword. When a new tree is loaded to replace the existing tree, status descriptors that have similar *leaf* names in both trees are copied to the new tree.

When DDF is started, only the EZLPNLS panel definitions are loaded. Panels loaded with the DDFPANEL command are not reloaded during EZLTDDF initialization. If you add panels dynamically and restart DDF, add the panels again.

The DDFTREE command adds trees at the 01 or root level. Trees added exist with other trees in the EZLTREE member, unless the tree added has the same root as one that exists. If the tree being added exists in DDF, the current tree is replaced by the new one. You can replace only complete trees.

The member name of the tree and root of the tree must be the same for the DDFTREE command to replace a tree.

### Examples

The following command loads member NEWTREE into memory, which gives you access to the tree structure defined in this member:

```
DDFTREE NEWTREE,ADD
```

## Assigning a problem to an operator in DDF (MARK)

---

### Purpose

The MARK command assigns a DDF entry to an operator. Specify the data used to identify the DDF entry and the ID of the operator to be assigned to the entry.

### Format

➤ MARK — *root\_comp* — . — *rv* — ( — *opid* — ) ➤



## Parameters

### *opid*

The ID of the operator to whom this entry is to be assigned

### *root\_comp*

The root component name that is defined in the DDF entry to be assigned

### *rv*

The resource name that is displayed in the RefValue field of the DDF entry to be assigned

## Usage

The *root* and *rv* parameters are required to issue this command from a command line. If the *opid* parameter is not specified, DDF assigns the entry to the operator ID issuing the MARK command.

If the entry to be signed out is already signed out no action is taken. Before attempting to reassign the entry to another operator, issue the UNMARK command. When assigned, a resource remains assigned during recovery monitoring, as long as the operator who marked the resource is logged on. If the operator assigned to the resource logs off, the next recovery monitoring timer causes the resource to be unassigned.

## Examples

To assign the DDF entry for NCP001 under the domain CNM01 to the operator OPER1, issue:

```
MARK CNM01.NCP001(OPER1)
```

## Removing an operator assignment in DDF (UNMARK)

---

## Purpose

The UNMARK command removes a DDF assignment entry from the specified operator. The data used to identify the DDF entry and the ID of the operator to be removed must be specified.

## Format

### UNMARK

➤ UNMARK — *root\_comp* — . — *rv* — ( — *operator\_id* — ) ➤

## Parameters

### *operator\_id*

The operator ID assigned to this DDF entry

### *root\_comp*

The root component name that is defined in the DDF entry to be unmarked

### *rv*

The resource name that is displayed in the RefValue field of the DDF entry to be unmarked

## Usage

- The *root* and *rv* parameters are required. If you do not specify the *operator\_id* parameter, the UNMARK command uses your operator ID as the default.
- If you issue the UNMARK command for a DDF entry that is not assigned to an operator, the UNMARK command takes no action.
- Each automation function has a syntax for specifying the name of the resource to be displayed (RV).

## Examples

To remove the operator ID, OPER1, from the DDF entry NCP001 under CNM01:

```
UNMARK CNM01.NCP001(OPER1)
```

---

## Chapter 24. Creating customized procedures

Before you begin to write automation procedures for your network, become familiar with writing and interpreting NetView command list programs. Refer to *IBM Z NetView Programming: REXX and the NetView Command List Language* for more information.

AON automation functions enable you to build and expand network automation. AON automation functions provide selected recovery routines and the ability to expand the recovery using common routines. To extend the automation capability functions of AON automation you can write programs or extend control file definitions. You can write extensions to:

- Check the status of network components using VTAM commands before taking action.
- Perform variable substitution for commands and replies with information obtained from the incoming message or from additional solicited information.
- Check automation settings for a resource, resource type, or network before taking action.
- Check specific error codes and take corrective action depending on the error code.
- Coordinate multiple independent messages into a single automated response.
- Create specialized operator dialogs.
- Check thresholds for events in a specified interval.

Interface user-written extensions with the common routines in AON when possible. Common routines provide the following benefits:

- Less code must be written, reducing the development time
- Portability of the code if all installation unique information is maintained in the control file
- Consistent network management interface

Also note the following functions and features:

- Messages with HDRMTYPE='Y' are sent directly to the system console and cannot use the automation by the program operator interface (POI) to the NetView program. To circumvent this, use the SSI interface.
- Unsolicited VTAM messages sent to the operating system console (such as IST931I) are also sent to the NetView program by the POI if the PPOLOG start option is in effect. Message automation occurs when messages enter the NetView address space through the SSI task or through an operator task.
- You must define a subsystem interface (SSI) address space for the NetView component on which AON is running. This enables AON to submit jobs for log maintenance, and to respond to console messages other than those from the NetView program. When you define the SSI, check your message processing facility list (MPFLST) in PARMLIB and ensure that all messages you want to automate can pass to and from the NetView program.

---

### How programs use AON control file routines

The common routines enable a user-written program the capability to interface to the control file, status file, and log file with a minimum amount of coding. Functions, such as determining whether automation is available from the control file, are supported through the common routines.

To start a program that performs automated functions, use one of the following methods:

- Issue the program from the NetView automation table.
- Have the operator enter the program name or a synonym for the program.
- Have a higher level program call a lower-level program.
- Issue a program with a timer.

- Issue a program with the NetView EXCMD command.

Starting a program from the NetView automation table is commonly used for passive monitoring.

How a program is started and what function the program performs determines which AON routines the program calls. This section contains information that is applicable to all programs written to use the AON common routines, but the format and description is most applicable to programs which are started from the NetView automation table. The basic AON program:

- Performs program initialization processing.
- Checks recovery automation.
- Performs program unique processing such as THRESHOLDING and MONIT.
- Notifies operators of intervention required.
- Logs automation activity.

See [Figure 200 on page 256](#) for the basic AON program flow.

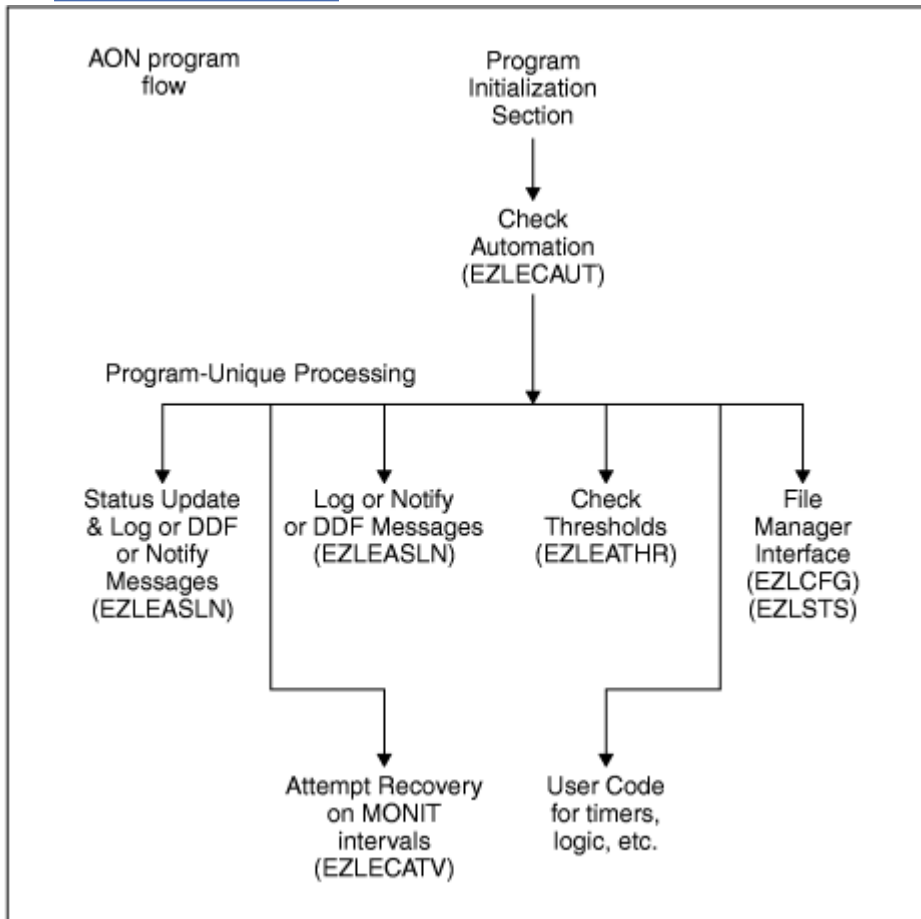


Figure 200. Basic AON Command List Flow

## Performing Program Initialization Processing

Initialization processing:

- Identifies the program with an &IDENT=name variable.
- Declares the common global variables (CGLOBALS) and task common global variables (TGLOCALS) that are used.
- Issues debugging messages, if AONTRACE is turned on.
- Validates the program was properly started.
- Forces the program to be run under a specific operator.

- Saves the NetView message parameters in case the program uses the NetView WAIT statement.

These tasks are not required in user-written programs. However, to perform complex processing logic or to simplify the coding process, you might need to perform several of these housekeeping routines.

The identify step simplifies the coding of the message logging and notification routines. Declaring task global variables (TGLOBS) and common global variables (CGLOBS) is necessary if you must use any of them. The AONTRACE function is useful in problem determination. Validating that the command was properly issued is useful to stop the operator from accidentally running the program. The validation can also be accomplished through NetView command authorization. If you use the NetView WAIT statement and still want to access the original message text or control information, save the variables.

See the coding in “Sample AON extended module” on page 258, the supplied extended programs, or the *IBM Z NetView Programming: REXX and the NetView Command List Language* for further information about coding the initialization sections.

## Determining if automation is available

This step must always be performed by programs that are started from the automation table. The step is accomplished by calling the EZLECAUT common routine. The routine checks the RECOVERY policy definitions to make sure the automation flags enable automation. These checks eliminate the risk of automating messages for resources that should not be automated or for which automation is turned off.

## Performing program-unique logic

The program unique logic can be any combination of AON common routines and user-written code. The requirements to be met and the functions to be performed determine the structure and content of the program. Several possible common routines and user-written code sections are:

- Activating resources
- Starting resource monitoring with or without reactivation attempts
- Logging messages and sending notifications
- Checking threshold values
- Interfacing with the file manager
- Writing user code for:
  - Issuing inquiries and checking answers
  - Setting common global variables (CGLOBS) and task global variables (TGLOBS) to control processing
  - Setting timer delays for reinitiating processing

### Activating resources

You can activate VTAM resources by issuing an AON common routine, EZLEVACT.

The routine is necessary when an INACTIVE message is received and a user-written program is handling the message. VTAM commands or NetView commands can be used to assist in this.

### Recovery and reminder monitoring

You can use recovery and reminder monitoring by issuing an EZLECATV or EZLESRMD AON common routine. These routines use the MONIT policy definitions. The EZLECATV routine starts monitoring the resource status and, if necessary, tries to recover the resource. The EZLESRMD routine monitors the resource and notifies operators, but does not start recovery.

### Updating status, DDF, logs, or sending notifications

You can update status, DDF, logs, or send notifications by starting the EZLEASLN common routine. The EZLEASLN common routine also generates alerts, based on your NOTIFY policy definitions.

This routine logs a message to the automation log (NLOG), DDF, and the NetView log (NETLOG). If you specify STATUS=YES, the status file is updated with the current automation status and acting operator. If you specify NOTIFY=YES on the EZLEASLN invocation, an operator notification is sent to all notification operators.

### Checking thresholds

You can check thresholds, defined with the THRESHOLD policy definition, by starting an EZLEATHR AON common routine.

This routine is used if it is necessary to track and maintain a threshold count. A threshold is used to change the recovery action based upon the threshold level that was exceeded. The common routine also records the current date and time as a new error to the status file.

### Interfacing with the file manager

You can interface with the file manager by using the file manager interface commands.

The commands are used if you must access uniquely defined entries in the control file or want to directly inquire or update the status file.

Several file manager interface commands exist. See [Chapter 25, “Using AON Command Processors,” on page 261](#) for more information.

### User code for issuing inquiries and checking answers

You can write user code to issue inquiries and check answers by using NetView or z/OS commands to check the value or status of some non-VTAM owned resource, system component, or subsystem before continuing.

Refer to the WAIT command in *IBM Z NetView Programming: REXX and the NetView Command List Language* for more information.

### User code for setting CGLOBALs and TGLOBALs to control processing

You can write user code for setting CGLOBALs and TGLOBALs to control processing by using NetView commands to set a flag indicating progress, message counts, and other various indicators that must be kept from one occurrence of a message or event to the next.

Refer to the discussion of CGLOBALs and TGLOBALs in *IBM Z NetView Programming: REXX and the NetView Command List Language* for more information.

### User code for setting timer delays for reinitiating processing

You can write user code for setting timer delays to reinitiate processing by using NetView commands.

Use this code when a program must reissue itself or another program after a given time period to perform additional processing. One example is in the active monitoring of failed network resources.

Refer to the discussion of AT, AFTER, and DELAY commands in the *IBM Z NetView Command Reference Volume 1 (A-N)* for more information.

## Sample AON extended module

This section describes the AON extended module that tries to recover the resource name that is passed to it. The module uses a subset of the AON common routines. The notes following the example refer to portions of the module and the AON common routines that are described in [“How programs use AON control file routines” on page 255](#).

```
/******  
/*  
/*      DESCRIPTION:  THIS CLIST WILL ATTEMPT TO RE-ACTIVATE THE FAIL /*  
/*                   RESOURCE. /*  
/*      WHERE CALLED: INVOKED FROM THE MESSAGE TABLE BY USER CODED /*  
/*                   MESSAGE TABLE ENTRY SENDING RESOURCE NAME ONL /*  
/*  
/*      ITS PURPOSE IS TO REACTIVATE THE RESOURCE AUTOMATICALLY. IT /*  
/*      ALSO INFORMS NCCF ALERT OPERATORS IF THE RESOURCE CAN NOT BE /*  
/*      ACTIVATED. IT WILL ALSO CHECK THRESHOLDS SET BY THE CUSTOMER /*  
/*      TO INFORM THE ALERT OPERATORS OR TERMINATE RECOVERY. /*  
/*  
/*-----*/  
  
trace off  
  
Parse source . invok ident .  
parse upper arg ArgString
```

```

Parms = ArgString

/*-----*/
/* Entry Trace Call */
/*-----*/
1 GLOBALV GETC EZLTRACED EZLTRACE0.'Opid()|',/* Trace Variable */
  'NETOPER NETOPER2 BASEOPER'
If Substr(EZLTRACED,1,4) <> 'NONE' Then /* If not NONE ... */
Do
2 EZLTRACE ENTRY 'ident argstring /* Entry trace */

  interpret 'trace' EZLTRACER
End
OperTrace = Substr(Value('EZLTRACE0.'Opid()),1,2)

Return_Code = 0

/* If opid is valid then process, else route to BASEOPER autotask */
if opid() = Netoper2 | opid() = Netoper | opid() = Baseoper then
  Call PROCESS
Else
  Do /* route recovery */
    /* 'EZLEASCD AON Baseoper EZLEAGEN 'Parms */
    3 'AONCMD 'Baseoper' EZLEAGEN 'Parms
    Return_Code = 9
  End /* route recovery */

  Call Leave_Now

PROCESS:
  Resname = Parms
  4 'EZLEAGRN 'Resname
  'GLOBALV GETT RESTYPE RESSTAT '
  /* Issue EZL509I - resource UNAVAILABLE message */
  5 'EZLEASLN NOTIFY=Y,STATUS=Y,||| ,
    'AON,INACTV,EZLEAGEN,EZL509,'date("U")',||| ,
    substr(time(),1,5),'Resname','Restype','domain()'
  Call ChkAuto
  Return

/*-----*/
/* PERFORM AUTOMATION */
/*-----*/

CHKAUTO:
/* Call the Check_Auto routine to check the RECOVERY Policy */
6 'EZLECAUT' Resname Restype
if Rc = 0 then /* don't auto resource? */
  Call EXIT0
7 'EZLEACKT' Resname /* Ck for exist timer */
if Rc = 1 then /* If timer exists? */
  Call EXIT1
if substr(Resstat,1,3) = 'ACT' then /* if resource is ACT */
  Call EXIT0
if substr(Resstat,1,3) = 'CON' then /* If resource is CONCT*/
  Call EXIT0

/* Call Thresholding routine to check the INFREQ/FREQ/CRIT Thresholds*/
8 'EZLEATHR' Resname,'Restype',OPTION=SNA'
if Rc = 0 then /* thresh ok? */
  Call RECOVERY /* attempt recovery */
else
  'VARY NET,ID='Resname',INACT,F' /* inactivate it */
Return

/*-----*/
/* ATTEMPT RECOVERY OF THE RESOURCE. */
/*-----*/

RECOVERY:
9 "EZLECATV" Resname,"Restype",0,"date('U')","substr(time(),1,5)
Call EXIT0

EXIT0:
  Return_Code = 0
  Call Leave_Now

EXIT1:
  Return_Code = 1

```

```

Call Leave_Now
/*-----*/
/* Exit Trace Call */
/*-----*/
Leave_now:
/* If Domain E/E trace is 'ON' or Oper E/E trace 'ON' ... */
If Substr(EZLTRACED,1,2) = 'ON' |, /* If Domain on ..*/
  (Substr(EZLTRACED,1,3) = 'OFF' & OperTrace = 'ON') Then
    'EZLTRACE EXIT 'Ident Return_code /* Call Exit Trace */
Exit Return_code

```

**1**

The common global variables and task global variables are defined.

**2**

EZLTRACE is called to perform AON standard entry tracing.

**3**

This section of code ensures the module is running under an automated operator, and if not, sends the command to the NETOPER or BASEOPER automation operator to run.

**4**

Common routine EZLEAGRN is called to get resource information.

**5**

An EZL509I message is logged regarding the availability of the resource.

**6**

The common routine EZLECAUT is called to verify that automation is enabled. The resource type and status are set from this module. If the recovery flag is on, the recovery module runs.

**7**

The existence of a timer for this resource is checked. If a timer exists with timer ID equal to the resource name, recovery is already in effect for this resource through another means. If recovery is already in effect, the program (command list) exits.

**8**

The resource is analyzed to determine whether a critical threshold has been exceeded. If it has, the resource is forced into the inactive state and the module exits.

**9**

EZLECATV is called to initiate recovery for the resource and continue recovery monitoring based on MONIT intervals.



---

## Chapter 25. Using AON Command Processors

AON provides command processors to interface with the control file, status file, or automation log. These command processors update and retrieve data contained in the control file, status file, or automation log. AON provides the following command processors:

- [“Control File Interface Command \(EZLCFG\)” on page 261](#)
- [“Log File Interface Command \(EZLLOG\)” on page 266](#)
- [“Status File Interface Command \(EZLSTS\)” on page 268](#)

---

### Control File Interface Command (EZLCFG)

---

#### Purpose

Use the EZLCFG command to load, display, and modify the control file entries. You can also use the command to display the status of the control file function or to perform a syntax check on the control file member. Before you can display panels and modifications, you must load the control file into storage. After they are loaded, the panels and modifications affect an in-storage version of the control file, providing a temporary update facility. To permanently change the control file member, edit the DSIPARM data set, and reload.

**Note:** There are certain restrictions when you update the control file. Command updates to entries in the control file cannot be performed online if the control file entry is longer than 200 characters. This is a NetView restriction. To code an entry that is longer than 200 characters, change the configuration file and reload it.

The control file entries are keyword-oriented and, therefore, independent of the NetView command list language or the operating system. Modifications or displays using the EZLCFG command capitalize upon the use of these keywords to retrieve entries for individual resources, resource types (such as all NTFYOPS), or system-wide defaults. This keyword structure decreases the need to generate command list variables that contain information for each resource.

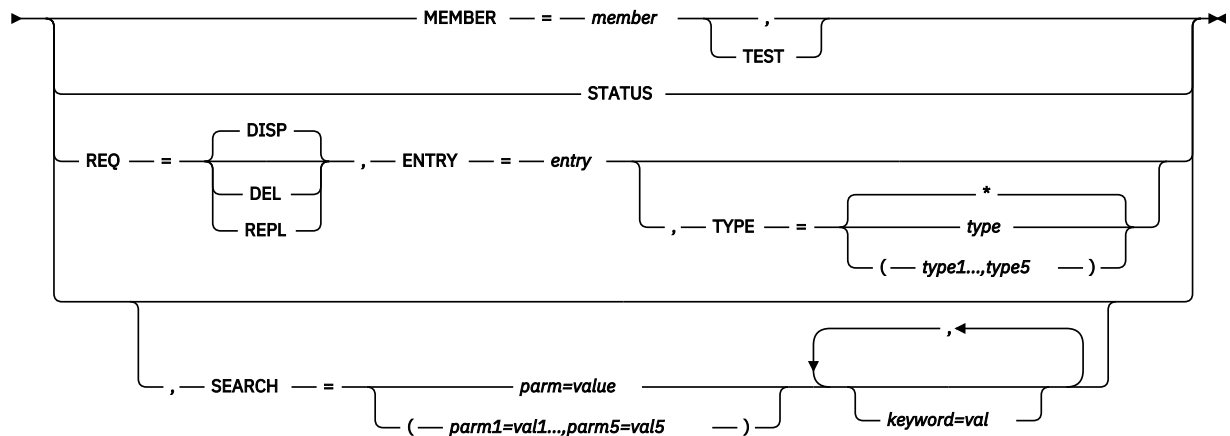
The EZLCFG command functions both as an operator command and a command list interface to the control file.

**Note:** Use of the EZLCFG command is discouraged. Use the POLICY command. The EZLCFG command is still available for migration purposes. For additional information about the POLICY command refer to the *IBM Z NetView Automation Guide*.

## Format

### EZLCFG

➔ EZLCFG ➔



**Note:** You can specify more than one *keyword=value* pairs.

## Parameters

### ENTRY

Specifies the entry field of the control file. The entry-name must be 1 to 32 characters, without embedded blanks, commas, or quotation marks.

### MEMBER

Specifies the member name containing the control file entries. This member is loaded into storage for subsequent use. The member must exist in a data set that is concatenated with the data sets in the DSIPARM DD statement.

Using EZLCFG MEMBER=member might cause problems since all policy files to be loaded are defined in the CNMSTYLE member and are loaded using the POLICY command.

To load or test the control file, enter EZLCFG MEMBER=member[,TEST].

### REQ

Specifies the type of request. DISP (display) is the default request if not entered. DEL (delete) and REPL (replace) must be entered when performing that function. REPL adds an entry if one does not exist.

### SEARCH

Valid only when used with REQ=DISP. Searches all entries specified by ENTRY and TYPE for a match where *parms=value* defines the data to be associated with the specified ENTRY and TYPE fields. The *parm* can be any character data, without embedded blanks, commas, or quotation marks. The character (=) and the *value* must immediately follow the *parm*. The *value* can have embedded quotation marks, commas, and blanks as long as single quotation marks or parentheses are around the *value*. The *parms=value* field has a restriction of 70 characters.

When multiple search arguments are specified, an implied logical 'OR' operation is performed. Entries that match one of the search arguments are shown.

AON defines a number of ENTRY, TYPE and *parms=value* fields. The JOB=jobname parameter in the resource control file entry is an example of the *parms=value* field.

### STATUS

Displays the name of the logical policy file loaded, or a message stating that the control file function is inactive.

For display, delete, replace, or add requests:

## TEST

Enables the user to perform simple syntax checking of a control file member. Entries are not verified.

**Note:** MEMBER is the only operand allowed with TEST.

## TYPE

The type field on the control file. The default TYPE field is an asterisk (\*) which is allowed only for DISP (display) commands. The specification of asterisk (\*) returns all the type fields associated with a given entry name. For example, all resource entries or all NTFYOP entries.

You can specify up to five types of DISP, DEL, and REPL requests when performing the command. Only the first type found in the list of types is affected by the command.

When specified, *type* must be 1 - 32 characters, without embedded blanks, commas, or quotation marks.

## Usage

The following wildcard characters can be used:

**\***

Multiple character wildcard

**%**

Single character wildcard

For example, both PU0\* and PU%% matches for PU01. SEARCH=(AUTO=\*) matches for entries containing the AUTO parameter. ENTRY=ENVI\* matches for all entries starting with ENVI.

When an EZLCFG REQ=DISP command is entered that requests a certain ENTRY with 1 or more specific TYPEs, it searches for those types in the order they were specified in the command. When the first match is found, the information is returned to the requestor as a multiline message. If there are no matches, it performs a final search with a type-name of DEFAULTS for that ENTRY. If there is still not a match, a message is returned to the requestor. If the *type-name* of DEFAULTS is found, that information is returned to the requestor.

When performing a display, if a specific TYPE is found, it is treated as a COMPLETE entry. Only that specific entry is displayed.

To replace an existing entry, type the entire new entry. The REPLACE command is treated as a COMPLETE entry. The existing entry and the replacement entry do not merge. This replacement process has a limitation on the length of the REPLACE command. The maximum length command is 240 characters. When you attempt to replace a control file entry, the replacement entry can be no greater than 200 characters. To do a large replace command, use the INPUT 3 NetView command to expand your command buffer.

KEYWORD=*value* operands in the control file can be no longer than a single line.

**Note:** The control file member must be placed in storage using the PIPE INSTORE command. Before placing the item in storage, the SUBSYM stage must be started to resolve system symbolics. Failure to perform these actions before the EZLCFG call results in the stored version being used. Therefore, without starting the SUBSYM stage, any changes to the control file on the DASD are ignored.

All policy definitions are loaded into NetView storage using PIPE INSTORE. If you change your policy files in DSIPARM, you can use POLICY REQ=LOAD to reload you new policy definitions and remove the existing policy definitions.

The POLICY command supports all functions that EZLCFG performs except the search function. Use the POLICY command whenever possible.

## Messages

The following messages are issued during the successful operation of the EZLCFG command.

For the test function (TEST):

```
EZL026I TEST OF THE CONTROL FILE MEMBER "member-name" WAS UNSUCCESSFUL
```

For the status function (STATUS):

```
EZL005I MEMBER member-name CURRENTLY BEING USED FOR THE CONTROL FILE
```

For delete and replace commands:

```
EZL001I REQUEST "request" WAS SUCCESSFUL
```

For DISPLAY commands:

```
EZL111I AUTOMATION CONFIGURATION DISPLAY - ENTRY= entry-name  
EZL112I ACTIVE TYPE= type-name , DESIRED TYPE= type-name1 ...  
EZL113I DATA IS parms=value  
EZL002I END
```

**Note:** Multiple EZL113I messages might be displayed after an EZL112I command and where the *type-name* is an asterisk (\*). When the type is omitted or specified as an asterisk (\*), the DESIRED TYPE is not displayed on the EZL112I message. For example, the following can occur:

```
EZL111I AUTOMATION CONFIGURATION DISPLAY - ENTRY= NTFYOP  
EZL112I ACTIVE TYPE= NETOP1  
EZL113I DATA IS OPER='OPER 1'  
EZL113I DATA IS CLASS=(10,40)  
EZL112I ACTIVE TYPE= NETOP2  
EZL113I DATA IS CLASS=(10)  
EZL002I END
```

The following messages are issued if the entry or member name was not found or the request failed.

For load function (MEMBER=):

```
EZL042I MEMBER member-name NOT FOUND
```

For the test function (TEST):

```
EZL027I THE FOLLOWING ERRORS ENCOUNTERED IN PROCESSING MEMBER member-name  
EZL023A FIELD "KEYWORD='VALUE(' " CONTAINS UNBALANCED PARENTHESIS  
EZL029I ENTRYA TYPEA,KEYWORD='VALUE('  
EZL028I member-name ERROR DISPLAY  
EZL004I PROCESSING FAILED FOR "EZLCFG MEMBER=member-name" COMMAND  
EZL026I TEST OF THE CONTROL FILE MEMBER "member-name" WAS UNSUCCESSFUL
```

For the status function (STATUS):

```
EZL040I CONTROL FILE INACTIVE
```

For the delete and DISPLAY commands:

```
EZL041I UNABLE TO FIND type name
```

**Note:** The REPLACE command adds an entry for a message if one does not exist.

Any messages beginning with EZL0 other than those previously documented must be considered an error situation.

## Examples

This example contains the command that displays the RECOVERY flag for PU01. The control file entry is:

```
RECOVERY PU01,AUTO=Y,NOAUTO=(TUESDAY,10:00,12:00)
```

The command is:

```
EZLCFG REQ=DISP,ENTRY=RECOVERY,TYPE=PU01
```

The response is:

```
EZL111I AUTOMATION CONFIGURATION DISPLAY - ENTRY= RECOVERY
EZL112I ACTIVE TYPE= PU01          , DESIRED TYPE= PU01
EZL113I DATA IS AUTO=Y
EZL113I DATA IS NOAUTO=(TUESDAY,10:00,12:00)
EZL002I END
```

In this example, a RECOVERY flag exists for the PU01 resource. The operator or command list processes the command to display the entry and the associated response is a multi-line message.

## Examples

This example shows a command to display the RECOVERY flag for PU0. The control file entries are:

```
RECOVERY DEFAULTS,AUTO=Y,NOAUTO=(MONDAY,10:00,12:00)
RECOVERY PU01,AUTO=Y,NOAUTO=(TUESDAY,10:00,12:00)
```

The command is:

```
EZLCFG REQ=DISP,ENTRY=RECOVERY,TYPE=PU01
```

The response is:

```
EZL111I AUTOMATION CONFIGURATION DISPLAY - ENTRY= RECOVERY
EZL112I ACTIVE TYPE= PU01          , DESIRED TYPE= PU01
EZL113I DATA IS AUTO=Y
EZL113I DATA IS NOAUTO=(TUESDAY,10:00,12:00)
EZL002I END
```

The previous example causes no change in the displayed data. Although a DEFAULTS flag exists, data from the different RECOVERY types (PU01 and DEFAULTS) do not merge.

## Examples

This example contains a command to display the RECOVERY flag for PU01. The control file entry is:

```
RECOVERY DEFAULTS,AUTO=Y,NOAUTO=(MONDAY,10:00,12:00)
```

The command is:

```
EZLCFG REQ=DISP,ENTRY=RECOVERY,TYPE=PU01
```

The response is:

```
EZL111I AUTOMATION CONFIGURATION DISPLAY - ENTRY= RECOVERY
EZL112I ACTIVE TYPE= DEFAULTS     , DESIRED TYPE= PU01
EZL113I DATA IS AUTO=Y
EZL113I DATA IS NOAUTO=(MONDAY,10:00,12:00)
EZL002I END
```

The previous example shows the EZL112I message with the found *type-name* (ACTIVE) as DEFAULTS, but the requested *type-name* (DESIRED) is PU01. This occurs because there was no RECOVERYPU01 flag, therefore the EZLCFG command automatically searches for a *type-name* of DEFAULTS. For this example, a RECOVERY DEFAULTS entry existed; therefore that information was displayed.

Assuming a DEFAULTS entry did not exist, the command would have resulted in an EZL041I UNABLE TO FIND TYPES... message.

## Examples

This example contains a command to display the RECOVERY flag for PU01 or PU, depending on which one exists. The control file entry is:

```
RECOVERY PU,AUTO=Y
```

The command is:

```
EZLCFG REQ=DISP,ENTRY=RECOVERY,TYPE=(PU01,PU)
```

The response is:

```
EZL111I AUTOMATION CONFIGURATION DISPLAY - ENTRY= RECOVERY  
EZL112I ACTIVE TYPE= PU , DESIRED TYPE= PU01  
EZL113I DATA IS AUTO=Y  
EZL002I END
```

The EZL112I message shows that the *type-name* found (ACTIVE) is PU, the requested types (DESIRED) were PU01 and PU. The EZLCFG command searched first for the RECOVERY PU01 flag; when none was found, the command searched for the RECOVERY PU flag.

## Log File Interface Command (EZLLOG)

The EZLLOG function is used to update and view the automation log. The syntax of the EZLLOG function is:

### Format

#### EZLLOG

```
➤ EZLLOG — ID — = — resource — , — FUNC — = — function — , — STATUS — = — ➤  
  
➤ status — , — OPID — = — operator-id — , — DOMAIN — = — domain-id — , — ➤  
  
➤ FROM — = — clist — , — DESC — = — message_text — ➤  
                    └── operid ─┘
```

### Parameters

#### DESC

Description text to be written in log record. The format is

```
DESC=msgno text
```

Where the variables are:

#### ***msgno***

Number of the message that generated the log entry.

#### ***text***

Free-form text describing the log event.

The maximum length is 240 bytes.

#### DOMAIN

Originating domain ID

#### FROM

Name of the command list or operator that generated the record to the log.

#### FUNC

Feature that wrote this record. The function entries in the automation log indicate which function in network automation wrote the record.

This field can be any 4 characters.

**ID**

Resource associated with this log record

**OPID**

Requesting operator ID.

**STATUS**

Automation status of the resource. For a list of the status codes, see [“Status File Interface Command \(EZLSTS\)” on page 268.](#)

**Usage**

The log records are:

- Availability records
- Information records
- Errors and debug records

Use availability records to create an availability trend report. AON writes records to the log when a resource becomes unavailable or when the resource becomes available again.

Information records describe the activities of the automation process. These are the AON tracking records.

Errors and debug records are written when an error has occurred (for example, wait timeout) or when the debugging feature is in effect.

The EZLLOG and EZLALOG commands, and the EZLTLOG task are:

**EZLALOG**

This command is run as a data services command. The EZLALOG command writes the record to the automation log file and returns messages to the requestor indicating the success or failure of the request.

**EZLLOG**

This command can be issued from the operator's terminal, program, or another command processor. EZLLOG performs syntax checking necessary before sending the request to the automation log task (EZLTLOG). Such errors as, valid length and required parameters are checked. If it detects errors, it returns a message explaining the error to the originator of the request.

**EZLTLOG**

This task handles the necessary interfaces to perform the NetView facilities. When EZLTLOG receives the internal function request (IFR) from EZLLOG, it issues the appropriate command, in this case EZLALOG.

To start the automation log task, the following command must be entered on the operator command line of a NetView screen or from the NetView initial startup program using STARTEZL:

```
STARTEZL LOG
```

The following messages are displayed when the task has been initiated:

```
DSI166I EZLTLOG IS ACTIVATED BY operator_ID
DSI530I EZLTLOG : DST IS READY AND WAITING FOR WORK
```

**Examples**

The command list writes a record to the automation log indicating that a resource has recovered. This example is for illustrative purposes and might not reflect current AON code:

```
'GLOBALV GETC DOMAINID'
ReqDomid = Domainid

Ezlstatus = 'ACTIV'
```

```

Ez1msgtxt = ez1msg('EZL504','N',Clnm,Resname,Restype)

if Nlogname = '' then
  Nlogname = SUBSTR(Resname,1,8)

if Nlogfunc = '' then
  if LENGTH(Ez1tower) <= 4 then
    Nlogfunc = Ez1tower
  else
    Nlogfunc = SUBSTR(Ez1tower,1,4)

if pos('/',clnm) > 0 then
  parse var clnm clnm '/' .

if length(clnm) > 8 then
  clnm = substr(clnm,1,8)

'TRAP AND SUPPRESS MESSAGES ONLY EZL006I EZL009I EZL01* EZL020I '||,
'EZL030I DSI002I CNM421I'
Logcmd = 'EZLLOG '||,
'ID='Nlogname',FUNC='Nlogfunc',OPID='Reqopid',DOMAIN='||,
Reqdomid',FROM='Clnm',STATUS='Ez1status',DESC='Ez1msgtxt

Logcmd = LEFT(Logcmd,240)
Logcmd

```

The previous command results in the following entry in the automation log:

```
EZL504I HOST AVOSTIN IS AVAILABLE (REPORTED BY NTC0PUN6)
```

If you page to the left in the automation log, you see this:

```
13:44:31 EZLERECV AUTMSG2 NV6K NORMAL AVOSTIN CNM01
```

## Status File Interface Command (EZLSTS)

### Purpose

Use the EZLSTS command to add, display, change, and delete the status file records. The records are maintained in a VSAM data set. The EZLSTS command interfaces with the VSAM file to maintain control information vital to AON. The critical information maintained is:

- Automation status
- Whether an error threshold has been exceeded
- Time and date information for error conditions. All entries use GMT timestamps.

The EZLSTS command functions as an operator command and as a command list interface to the status file.

**Note:** Restrict the use of EZLSTS as an operator command, because this command can modify any field on the status file, and incorrect use can lead to unpredictable results.

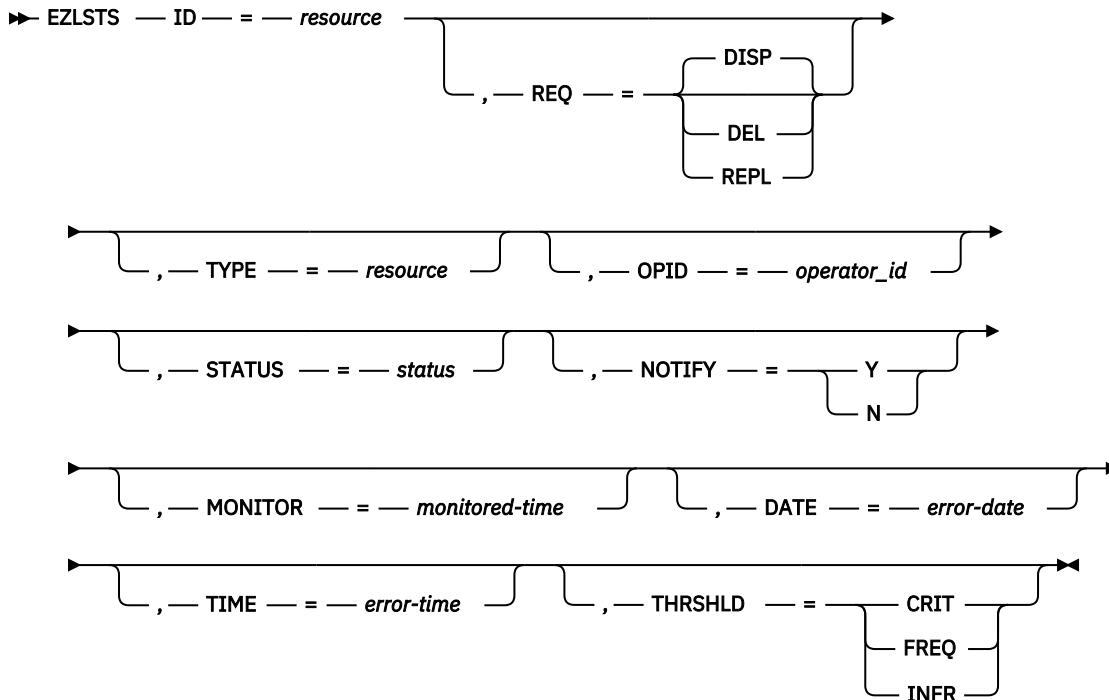
Every field defined in the status file is updated by certain automation routines. User-written command lists must use common routines to perform the updates, to ensure proper updating of all the fields. Use the EZLSTS command only to display information.

### Format

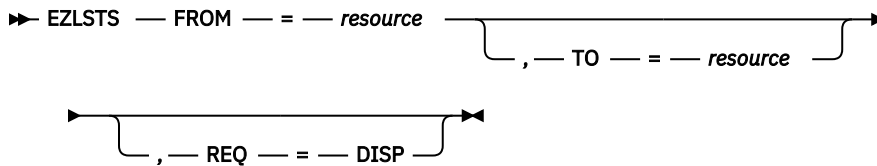
Use the following syntax for adding, updating, displaying, or deleting a single entry:



## EZLSTS



Use the following for displaying multiple records:



### Parameters

#### DATE=*error-date*

Specifies the date of this error time stamp being recorded, in MM/DD/YY format. Date is used for threshold purposes.

#### FROM=*resource*

Specifies the 16-character resource ID that is the starting key when displaying multiple status file records. This resource is the resource name for resource records.

#### ID=*resource*

Specifies the 16-character resource ID that is the key to the status file record. This ID is the resource name for resource records.

#### MONITOR=*monitored-time*

Specifies the time a monitor or status change command was issued in HH:MM format. Time is in military time (00:00 to 23:59).

#### NOTIFY={Y|N}

Specifies whether a notification message was sent to the operator for a specified condition.

#### OPID=*operator\_id*

The NetView operator ID which requested or is performing the status update. If not supplied, the EZLSTS command uses the current operator ID. This is only applicable when the STATUS parameter is used.

**REQ={DISP|DEL|REPL}**

Specifies the type of request. DISP (display) is the default request if not entered. DEL (delete) and REPL (replace) must be entered when performing that function. REPL adds a record if one does not exist.

**STATUS=status**

Specifies the resource status. The parameter can be any valid status value. The following is a list of valid status codes generated by AON automation:

**ACTIVE**

Resource is active.

**ERRORS**

Command list error conditions.

**INACTV**

Resource is inactive.

**INRCVY**

Resource is in recovery.

**IUSER**

Resource deactivated by user.

**MANUAL**

Manual intervention is required.

**NCPDMP**

NCP memory dump has a reply to (AON/SNA only).

**NCPRLD**

NCP reload has a reply to (AON/SNA only).

**RCVAUT**

Resource recovered by AON automation.

**RCVSYs**

Resource recovered by the system.

**REACTV**

Reactivation process enabled.

**RECVRY**

Resource currently being recovered by automation.

**REMIND**

Reminder processing enabled.

**REMTRM**

Reminders halted; reminder intervals exceeded.

**TRACKS**

Command list tracking records.

**TREACT**

Recovery halted; reactivation intervals exceeded.

**TTHRS**

Resource recovery ended by automation thresholds.

**TUSER**

Resource recovery ended by user.

**THRESHLD={CRIT|INFR|FREQ}**

Specifies the threshold which has been exceeded. Used for operator monitoring of thresholds.

**TIME=error-time**

Specifies the time of this error time stamp being recorded, in HH:MM format. Time is used for threshold purposes.

**TO=resource**

Specifies the 16-character resource ID that is the ending key when displaying multiple status file records. If not specified, the value defaults to the same key as the FROM parameter. This resource is the resource name for resource records.

**TYPE=resource**

Specifies the 10-character resource type for display and summary purposes. This value is the resource type for the specific resource. For example, physical units are represented by PU and NCPs are represented by NCP. Other values can also be defined.

**Usage**

If a user-written command list performs a replace and update function, only those fields that need replacing must be specified. No change occurs to the other fields on the status file. The EZLSTS command has a maximum length of 240 characters, so a REPLACE/UPDATE command is limited to the 240 characters.

**Messages**

The following messages are issued during the successful operation of the EZLSTS command. For the DELETE and REPLACE commands, a message such as the following is issued:

```
EZL001I REQUEST "request" WAS SUCCESSFUL FOR "resource"
```

For the DISPLAY commands, the following messages are issued for PU01:

```
EZL150I STATISTICS DISPLAY REQUESTED FOR PU01
EZL151I ID= PU01 , TYPE= PU , STATUS= ACTIV
EZL152I LAST UPDATE BY OPERATOR AUTNET2
EZL153I LAST THRESHOLD EXCEEDED - INFR
EZL155I OPERATOR NOTIFIED: Y , TIMERSET:
EZL156I LAST STATUS CHANGE DATE= 01/06/19 , TIME= 13:41, OPID= AUTNET2
EZL157I LAST MONITORED DATE= 01/06/19 , TIME= 13:41
EZL160I ERROR COUNT DATE TIME
EZL161I 01 01/06/19 13:35
EZL161I 02 01/06/19 13:40
EZL002I END
```

**Note:** For DISPLAY commands, EZL160I and EZL161I messages that contain error time stamp information can be replaced with EZL159I if error timestamp information does not exist.

Messages are issued if the entry or member name is not found. For the DELETE and DISPLAY commands, a message such as the following is issued:

```
EZL041I UNABLE TO FIND RECORD "resource"
```

**Note:** The REPLACE command adds an entry if one does not exist, so successful message would result.

Any message beginning with EZL0, other than those previously documented, is an error.

**Examples**

A command that displays the status record for PU01 follows:

```
EZLSTS REQ=DISP,ID=PU01
```

The response follows:

```
EZL150I STATISTICS DISPLAY REQUESTED FOR PU01
EZL151I ID= PU01 , TYPE= PU , STATUS= ACTIV
EZL152I LAST UPDATE BY OPERATOR AUTMSG
EZL153I LAST THRESHOLD EXCEEDED - INFR
EZL155I OPERATOR NOTIFIED: Y , TIMERSET:
EZL156I LAST STATUS CHANGE DATE= 05/06/19 , TIME= 09:17, OPID= AUTNET2
EZL157I LAST MONITORED DATE= 05/06/19 , TIME= 09:17
```

```
EZL160I  ERROR COUNT      DATE      TIME
EZL161I           01      05/06/19    14:21
EZL002I  END
```

The operator or command list processes the command to display the entry, and the associated response is a multiline message.

### Examples

The following command adds or replaces the PU01 status value:

```
EZLSTS REQ=REPL,ID=PU01,STATUS=INACT
```

The response follows:

```
EZL001I REQUEST "REPL" WAS SUCCESSFUL FOR "PU01  "
```

## Chapter 26. Coding common routines

Common routines provide easy-to-use generic functions for expanding automation capabilities beyond those provided and supported by AON. These generic functions can be used to reduce development time when creating procedures or extending those provided.

Transferring information and checking the control file are examples of tasks where common routines are applicable. A user-written routine calls one of these routines from the message table, control file, or extended routine to perform a specific task.

AON also provides user exits that can be called by the common routines. User exits perform functions such as syntax or threshold checking.

Table 5 on page 273 lists the common routines described in this section.

| Table 5. AON Common Routines |                                                                                            |
|------------------------------|--------------------------------------------------------------------------------------------|
| Name                         | Usage                                                                                      |
| CGLOBAL                      | <a href="#">“Using the Common Global Variable Command Processor (CGLOBAL)” on page 274</a> |
| EXIST                        | <a href="#">“Querying Command Availability (EXIST)” on page 275</a>                        |
| EZLEATDF                     | <a href="#">“Calculating Time (EZLEATDF)” on page 276</a>                                  |
| EZLE1UFW                     | <a href="#">“Forwarding User Messages (EZLE1UFW)” on page 277</a>                          |
| EZLEACKT                     | <a href="#">“Checking the Timer (EZLEACKT)” on page 277</a>                                |
| EZLEAGEN                     | <a href="#">“Recovering Generic Resources (EZLEAGEN)” on page 278</a>                      |
| EZLEAGRN                     | <a href="#">“Getting Resource Information (EZLEAGRN)” on page 279</a>                      |
| EZLEASLN                     | <a href="#">“Updating the Status File and Logging Messages (EZLEASLN)” on page 281</a>     |
| EZLEATHR                     | <a href="#">“Checking Thresholds (EZLEATHR)” on page 285</a>                               |
| EZLECALL                     | <a href="#">“INFORM Action (EZLECALL)” on page 288</a>                                     |
| EZLECATV                     | <a href="#">“Using Active Monitoring and Recovery (EZLECATV)” on page 288</a>              |
| EZLECAUT                     | <a href="#">“Checking Automation (EZLECAUT)” on page 289</a>                               |
| EZLEFAIL                     | <a href="#">“Processing Generic Failures (EZLEFAIL)” on page 290</a>                       |
| EZLEMCOL                     | <a href="#">“Setting Panel Message Color (EZLEMCOL)” on page 296</a>                       |
| EZLEMSG                      | <a href="#">“Formatting Panel Messages (EZLEMSG)” on page 297</a>                          |
| EZLENFRM                     | <a href="#">“Driving the Inform Policy (EZLENFRM)” on page 298</a>                         |
| EZLENTFY                     | <a href="#">“Notify Policy List (EZLENTFY)” on page 298</a>                                |
| EZLERAIP                     | <a href="#">“Setting the AIP User Status Bit (EZLERAIP)” on page 299</a>                   |
| EZLERCMD                     | <a href="#">“Routing Commands over Cross-Domain Sessions (EZLERCMD)” on page 301</a>       |
| EZLERECV                     | <a href="#">“Recovering Resources (EZLERECV)” on page 302</a>                              |
| EZLERGWY                     | <a href="#">“Routing Commands To Other NetView Domains (EZLERGWY)” on page 304</a>         |
| EZLEROUT                     | <a href="#">“Routing NNT Cross-Domain Logon Information (EZLEROUT)” on page 305</a>        |
| EZLERTVE                     | <a href="#">“Retrieving AON Information (EZLERTVE)” on page 306</a>                        |
| EZLESRMD                     | <a href="#">“Issuing Resource State Reminders (EZLESRMD)” on page 308</a>                  |

| Table 5. AON Common Routines (continued) |                                                                                     |
|------------------------------------------|-------------------------------------------------------------------------------------|
| Name                                     | Usage                                                                               |
| EZLESTOP                                 | <a href="#">“Stopping Cross-domain Sessions (EZLESTOP)” on page 308</a>             |
| EZLESTRT                                 | <a href="#">“Starting Cross-domain Sessions (EZLESTRT)” on page 309</a>             |
| EZLEVACT                                 | <a href="#">“Activating VTAM Resources (EZLEVACT)” on page 311</a>                  |
| EZLEVINA                                 | <a href="#">“Deactivating VTAM Resources (EZLEVINA)” on page 312</a>                |
| EZLEVMOV                                 | <a href="#">“Moving VTAM Resources (EZLEVMOV)” on page 312</a>                      |
| EZLSMSU                                  | <a href="#">“Sending MSUs to an MS Transport Application (EZLSMSU)” on page 313</a> |
| EZLTRACE                                 | <a href="#">“Running Entry and Exit Traces (EZLTRACE)” on page 314</a>              |
| FKVESYNC                                 | <a href="#">“SNA Resource Automation (FKVESYNC)” on page 322</a>                    |
| FKXECNVT                                 | <a href="#">“SNMP RFC Conversion (FKXECNVT)” on page 316</a>                        |
| FKXETRA1                                 | <a href="#">FKXETRA1 program syntax</a>                                             |
| IPCMD                                    | <a href="#">“TCP/IP Command Support (IPCMD)” on page 318</a>                        |

## Using the Common Global Variable Command Processor (CGLOBAL)

### Purpose

The CGLOBAL routine displays the names and value that are associated with common global variables. Running the CGLOBAL routine displays the EZL016I message that contains the name of the common global variable and the EZL017I message that contains the value that is associated with the common global variable.

### Format

#### CGLOBAL

➤ CGLOBAL — *varname* ➤

### Parameters

#### *varname*

The specific name of a global variable or a name pattern with wildcard characters (\* or %)

### Examples

Issuing **CGLOBAL domainid** returns the following messages:

```
EZL016I NAME = DOMAINID
EZL017I VALUE = CNM01
EZL002I END
```

The EZL017I message shows that CNM01 is the value associated with the common global variable domainid.

### Examples

Issuing **CGLOBAL \*t** returns the following messages:

```
EZL001I C *T
EZL016I NAME = CNMSTYLE.NPDA.ALT_ALERT
```

```

EZL017I VALUE = DOMAIN
EZL016I NAME = CNMIP.DNSTIMEOUT
EZL017I VALUE = 5
EZL016I NAME = FKXIPSTAT
EZL017I VALUE = DETAIL
EZL016I NAME = CNMSTYLE.NLDM.LUCOUNT
EZL017I VALUE = 4000
EZL016I NAME = CNMSTYLE.NRM.STATUS.INACT
EZL017I VALUE = UNKNOWN
EZL016I NAME = CNMSTYLE.REXEC.PORT
EZL017I VALUE = 512
EZL016I NAME = CNMSTYLE.RSH.PORT
EZL017I VALUE = 514
...
EZL002I END

```

The EZL016I messages show all of the command global variables that end with the character T. The EZL017I messages show the values that are associated with each command global variable.

## Querying Command Availability (EXIST)

### Purpose

EXIST determines whether the specified command is available and sets a return code as its response. Only libraries in the DSICLD or STEPLIB concatenation are searched. The command must be a valid command list, REXX program, or NetView command processor.

### Format

#### EXIST

➤ EXIST — *command* ➤

### Parameters

#### *command*

Name of a command

### Return codes

- 0** Valid command
- 4** Operator is not authorized to issue the command
- 8** Error (incorrect command name or no command specified)
- 16** Not found

### Examples

The following REXX fragment is an example of how you can use EXIST:

```

/* Check to see if BLOG command is installed, and call if found */
'EXIST BLOG'
ExistRC = RC
if ExistRC = 0 then
  do
    'BLOG'
  end

```

## Calculating Time (EZLEATDF)

---

### Purpose

The EZLEATDF routine calculates the difference between two date and time stamps and returns the values in TGLOBALs in terms of days, hours, minutes, and time. Keep the parameters for the EZLEATDF routine in the same order as they are shown in the following syntax diagram.

### Format

#### EZLEATDF

➤ EZLEATDF — — *start\_date* — — *start\_time* — — *end\_date* — — *end\_time* ➤

### Parameters

#### *end\_date*

Later date (in *mm/dd/yy* format)

#### *end\_time*

Later time (in *hh:mm* format)

#### *start\_date*

Earliest date (in *mm/dd/yy* format)

#### *start\_time*

Earliest time (in *hh:mm* format)

### Return codes

0

Program processed successfully.

5

Parameters are not valid.

### Flags

#### TIMEDIFF

The number of hours and minutes between start and end (after the difference in days is calculated)

#### DAYSDIFF

The number of days between start date and end date

#### HOURLDIFF

Total hours and minutes between start date and end date (in *hh:mm* format)

#### MINDIFF

Total minutes between start time and end time ( $((hh*60) + mm)$ ). To express the total difference in minutes, add the HOURLDIFF global (converted to minutes) to the MINDIFF value.

### Examples

This program call:

```
EZLEATDF 06/23/00 06:28 06/24/00 15:50
```

Sets the following task global variables:

```
TIMEDIFF = 09:22
DAYSDIFF = 2
HOURLDIFF = 57:22
MINDIFF = 3442
```



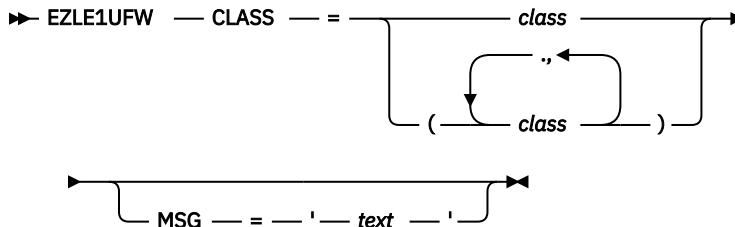
## Forwarding User Messages (EZLE1UFW)

## Purpose

The EZLE1UFW routine forwards messages to an AON focal-point domain.

## Format

## EZLE1UFW



## Parameters

**CLASS=***class*

Specifies message classes used to determine which notification operators receive this message. You can specify up to 10 message classes. There are no default message classes. The AON notification classes are in the range 00–99. The different notification classes are described in *IBM Z NetView Messages and Codes Volume 2 (DUI-IHS)*.

**Note:** The classes must be the same as those defined with the NTFYOP control file entry, as described in Chapter 25, “Using AON Command Processors,” on page 261.

**MSG='text'**

Specifies the text used for this message. If not coded, the text in the message buffer is used. Although this variable is optional, it is required if not called from the automation table.

## Return codes

# O

Program (command list) processed correctly.

**1**

Processing error was encountered.

## Usage

The MSG parameter cannot be used for multiline messages.

## Examples

The following example sends individual messages for each line in the multiline response:

```
IF MSGID='IST075I'
THEN EXEC(CMD('EZLE1UFW CLASS=20')ROUTE(ALL*));
```

## Checking the Timer (EZLEACKT)

## Purpose

The EZLEACKT routine determines whether a timer exists and stores pertinent timer data in task global variables (TGLOBALS).

## Format

### EZLEACKT

➤ EZLEACKT — *timerid* ➤

## Parameters

### *timerid*

Specifies the ID of the timer being searched for. Typically, for AON recovery efforts, this is the resource name.

## Return codes

### 0

Timer does not exist.

### 1

Timer exists.

## Flags

### EZLTIMCM

Command to be issued under timer ID

### EZLTIMDT

Date the timer would pop

### EZLTIMOP

Operator under which the timer would run

### EZLTIMTM

Time of day the timer would pop

## Usage

The EZLEACKT routine is used primarily to determine if automation is already in effect for a resource with an existing timer ID, such as timers from MONIT intervals.

## Recovering Generic Resources (EZLEAGEN)

---

## Purpose

EZLEAGEN is a generic module that can do the following:

- Attempt resource recovery
- Check error thresholds
- Check automation recovery flag
- Check resource status
- Initiate recovery on MONIT intervals

## Format

### EZLEAGEN

➤ EZLEAGEN — *resource* ➤

## Parameters

### *resource*

The resource to be recovered

## Usage

This routine is provided as a model for user-written modules extending AON techniques. The EZLEAGEN module can be started by sending it a valid VTAM resource name.

## Examples

See [“Sample AON extended module” on page 258](#) to see how EZLEAGEN is coded.

## Getting Resource Information (EZLEAGRN)

---

### Purpose

The EZLEAGRN routine retrieves VTAM information about a resource through a VTAM display command and stores the data in task global variables (TGLOBALs).

### Format

#### EZLEAGRN

➤ EZLEAGRN — *res\_name* ➤

### Parameters

#### *res\_name*

The name of the VTAM resource to be displayed

### Return codes

**0**

Resource values assigned

**1**

Module error

### Flags

Table 6 on page 279 lists the EZLEAGRN routine TGLOBALs.

| Table 6. TGLOBALs for EZLEAGRN |                                |                             |
|--------------------------------|--------------------------------|-----------------------------|
| TGLOBAL                        | Description                    | Retrieved from VTAM message |
| RESTYPE                        | Resource type                  | IST075I                     |
| RESSTAT                        | Resource status                | IST486I                     |
| RESMAJ                         | Resource's major node          | IST134I<br>or IST081I       |
| RESLINE                        | Resource's higher node-line    | IST081I                     |
| RESNODE                        | Resource's adjacent major node | IST391I                     |

Table 6. TGLOBALs for EZLEAGRN (continued)

| TGLOBAL | Description                       | Retrieved from VTAM message |
|---------|-----------------------------------|-----------------------------|
| RESPU   | Resource's higher node controller | IST135I                     |
| RESSA   | Resource's subarea                | IST484I                     |
| RESSW   | Resource's switched major node    | IST136I                     |
| RESNET  | Resource's network ID             | IST075I<br>or IST1043I      |

### Usage

VTAM resource types are translated into the following standard automation resource categories:

#### LU

Logical unit

#### PU

Physical unit

#### PU

LCL\*

#### LINE

Line

#### PU

PU\*

#### NCP

PU T4/5

#### NCP

CA Major Node

#### CDRM

CDRMs

#### LINK

Link Station

#### CP

Control Point

All other resource types remain the same. If the resource cannot be displayed in VTAM, a resource type of DEFAULTS is issued and message EZL208I is logged.

The EZLEAGRN routine gathers information about SNA resources. However, your installation might require you to tailor the information gathered by the EZLEAGRN routine. You can code user exits to alter this information. You can update values for the EZLEAGRN routine with the EXIT05 parameter of the ENVIRON EXIT control file entry. Refer to the *IBM Z NetView Administration Reference* for more information about the ENVIRON EXIT entry.

## Updating the Status File and Logging Messages (EZLEASLN)

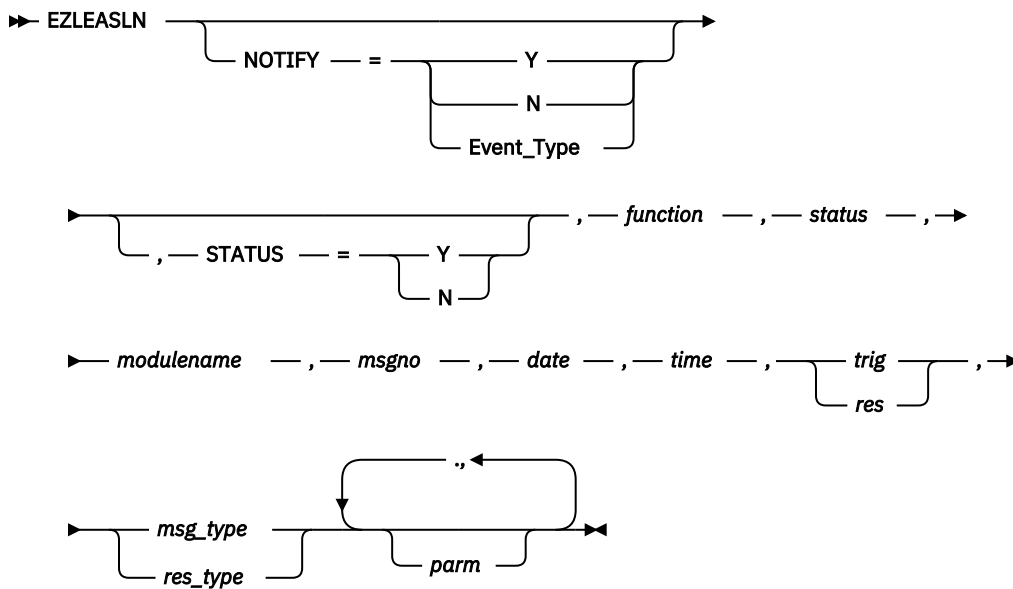
### Purpose

The EZLEASLN routine updates the AON status file, logs associated messages in the AON log file and NetView log, and issues notifications. The content of the log or notification message depends on the message ID and the variable data that is placed in the message.

This routine can be initiated only by another module or by a command processor. Keep the parameters for the EZLEASLN routine in the same order that they are shown in the following syntax diagram. Even though you do not use some of the parameters, include the comma delimiters as if those parameters were in the routine.

### Format

#### EZLEASLN



### Parameters

#### *date*

The date the message was logged. Use the NetView variable, &DATE or REXX function date ('U').

#### *function*

Defines the AON component that wrote the message. The function can be:

- AON
- APPN
- IP390
- SA
- SNA
- SNBU
- TCPIP
- X25

#### *modulename*

Specifies the name of the module or program that is issuing the message. This parameter is used in the AON log, by the tracking and problem determination processes.

***msgno***

Specifies the number of the message to be issued.

***msg\_type/res\_type***

Specifies the resource type or the general cause for the message being issued. For errors and tracking information, the message type is typically ERR or ERROR. The message type can also be INFO, INIT, or similar indicative values.

The *msg\_type* variable can be a maximum of 10 characters in length.

**NOTIFY**

Specifying Y enables notifications to an operator and message forwarding of some event. Specifying Y queries the NOTIFY Policy based on ResName and ResType.

Specifying N does not notify anyone, it only logs the message to DSILOG.

Specifying Event\_Type queries the NOTIFY Policy for the type of event such as:

- CRITTHRS
- REMIND
- NOMOMONS
- NAMESERV

***parms***

User parameters are message-dependent fields that fill in the message content. Up to five user parameters can be coded, but a restriction of 240 characters in the EZLEASLN call can reduce the actual number of parameters that can be coded.

**STATUS**

Specifying Y enables updates to the status file with the following data:

- RESSTAT
- RESTYPE
- OPID

***status***

Specifies the resource status. This value updates the information in the status file and logs it in the AON log. The following is a list of some of the valid status codes generated by automation:

**ACTIVE**

Resource is active.

**ERRORS**

Command list error conditions.

**INACTV**

Resource is inactive.

**INRCVY**

Resource is in recovery.

**IUSER**

Resource deactivated by user.

**MANUAL**

Manual intervention is required.

**NCPDMP**

NCP memory dump has been replied to (AON/SNA only).

**NCPRLD**

NCP reload has been replied to (AON/SNA only).

**RCVAUT**

Resource recovered by AON automation.

**RCVSYs**

Resource recovered by the system.

**REACTV**

Reactivation process enabled.

**RECVRY**

Resource currently being recovered by automation.

**REMIND**

Reminder processing enabled.

**REMTRM**

Reminders halted; reminder intervals exceeded.

**TRACKS**

Command list tracking records.

**TREACT**

Recovery halted; reactivation intervals exceeded.

**TTHRS**

Resource recovery ended by automation thresholds.

**TUSER**

Resource recovery ended by user.

***time***

The time the message was logged. Use the NetView variable, *&TIME* or REXX function *TIME('U')*.

***{trig/res}***

Specifies the ID of the resource or the specific cause for the message being issued. For errors and tracking information, the message trigger typically is the command that caused the error message to be issued. The message trigger can also be the operator ID, a parameter keyword, a control file keyword, or other similar indicative information.

The *trig* variable can be a maximum of 8 characters in length and cannot contain spaces.

**Return codes**

**0**

Message request processed successfully.

**4**

An error has occurred. Browse the log for further details.

**9**

Message request has been routed to an MSGOPER autotask.

**Usage**

The EZLEASLN routine can perform the following functions:

- Message logging
- Status update and message logging
- AON notification processing based on the NOTIFY policy

Use the EZLEASLN routine when logging messages. Messages produced by calling this routine are not displayed at operator stations if NOTIFY=N is specified.

It can also be used to display a message to the operator. Messages produced by calling this routine are displayed at a NetView terminal, if an authorized notify operator is logged on, or at the z/OS system console if the notify operators are not logged on. These messages are forwarded to the focal point before being displayed.

Use the EZLEASLN routine when you change the resource status, and notification of that change is to be displayed to the operator. The status in the AON status file is changed by this routine.

The field values passed to EZLEASLN are evaluated against the DDFGENERIC control file entry definitions that guide DDF updates. The following fields, sent to EZLEASLN, are evaluated for DDF generic values:

## EZLEASLN parameter

### DDF generic field

#### function

*func*

#### status

*status*

#### resource

*res\_name*

#### res\_type

*res\_type*

#### parms1...parms5

*opt1...opt5*

The function value sent is used for the function value for evaluating DDFGENERIC statements.

The parameters passed to the EZLEASLN routine are dependent on the format of the message. Some messages have more parameters than others. AON messages are shipped in object code format for improved performance. To determine the number of inserts in a message, view the online help for the message. The message inserts are passed as user parms on the call to EZLEASLN.

## Examples

This example shows how to issue a simple status update message. The INACTIVE message has been received for LINE01. The message to be issued, EZL531I, indicates the new status value.

The message text for EZL531I follows:

```
EZL531I &4 &3 IS INACTIVE DUE TO OPERATOR &5 INTERVENTION
```

The module to call EZLEASLN is:

```
USERSLN module
  &CGLOBAL ABC XYZ ... ..
  &TGLOBAL
*
*   Perform check automation
*
  EZLECAUT LINE01
:
  &IF &RETCODE EQ 0 &THEN &GOTO -AUTOOK
  &EXIT
-AUTOOK
  EZLEASLN NOTIFY=Y,STATUS=Y,AON,INACTV,USERSLN,EZL531,&DATE,
             &TIME,LINE01,LINE,OPER1
  :
  &EXIT
```

The user-written module calls EZLECAUT to determine if automation is valid. If the return code greater than zero (0), automation is not permitted. If it is 0, the routine continues.

After it is determined that NetView automation is in effect, EZLEASLN is called. The *function* parameter is AON, indicating automation activity triggered the message. *Status* is INACTV, denoting the new status for LINE01. This status value is placed in the AON status file. The *modulename* is USERSLN.

The message number EZL531 is suffixed by an I to build a message identifier of EZL531I. EZL531I is inserted at beginning of the message text.

The date and time are coded as &DATE and &TIME.

The resource is coded as LINE01 (VTAM resource name). The resource type is coded as LINE (VTAM resource type), which is &4.

The first user parameter is OPER1. It replaces &5 in the message text.



## Checking Thresholds (EZLEATHR)

---

### Purpose

The EZLEATHR routine checks the number of errors recorded in the status file against a preset error threshold. It also supports the recording of the error date and time on the status file.

The EZLEATHR routine searches the automation control file for the applicable threshold of a specific resource. The routine then obtains the error status information from the status file and determines whether any of the three definable thresholds have been exceeded. If a threshold is exceeded, AON issues an error message and an appropriate return code is generated.

This routine can be issued only by another module or by a command processor.

Keep the parameters for the EZLEATHR routine in the same order as they are shown in the following syntax diagram. Use commas or spaces as delimiters.

### Format

#### EZLEATHR



### Parameters

#### NEW|CHECK

Specifies whether a new error is added to the error status information. If you specify NEW or use the default, an error is added to the error status information, then the thresholds are checked. If you specify CHECK, the thresholds are checked based on the existing error information and no new information is added. If you specify NEW or CHECK, place the parameter in the fourth position.

#### opt

Specifies the AON automation component issuing this request. The list of options follows

- AON
- APPN
- IP390
- SA
- SNA
- SNBU
- TCPIP
- X25

#### resource

Specifies the name of the resource for which thresholds are to be checked. This resource name can be a generic resource type or any user-defined name up to eight characters long. The resource name is required and must be the first parameter.

#### res\_type

Specifies the type of the resource. The *res\_type* parameter can be any user-defined name up to 10 characters long. The resource type is required and must be the second parameter.

#### serv\_pt

Specifies the name of the TCP/IP service point controlling the specified resource. If specified, *serv\_pt* must be the third parameter. If not specified, the default is the NetView DomainID.

## Return codes

- 0** No threshold has been exceeded.
- 1** Infrequent threshold has been reached.
- 2** Frequent threshold has been reached.
- 3** Critical threshold has been reached.
- 4** Incorrect parameters were used in the call.
- 5** Time-out or other error occurred.

## Usage

The EZLEATHR routine accesses the control file to check the THRESHOLD entries and the status file to check the status of the resource.

You can modify values for the EZLEATHR routine by defining them in the EXIT06 parameter of the ENVIRON EXIT control file entry. Refer to the *IBM Z NetView Administration Reference* for more information. AON uses the values on the ENVIRON EXIT control file entry unless you define a THRESHOLDS control file entry for the specific resource.

The EZLEATHR routine is used primarily to track error conditions that can be recursive. Because EZLEATHR tracks the errors, AON can notify operators of the recursive situation before it causes problems. After a critical threshold exception, AON deactivates resources to stop the recursive error and recovery.

A maximum of 10 errors can be stored in each record in the status file. These errors are stored in order of date and time-of-error. No details about the error are stored.

The VSAM key to the status file is built from the eight-character VTAM resource name. Only one status file record exists for each resource.

Each resource name in the status file must be unique; therefore, select your resource names carefully.

The EZLEATHR routine searches for the threshold in a predefined sequence to find the appropriate threshold. For a description of the threshold search sequence, refer to the *IBM Z NetView Administration Reference*.

AON is designed to fulfill most needs of thresholds automation processing. However, your installation might require you to tailor threshold automation processing. You can code AON user exits for threshold values for a specific resource. These values are specified on the THRESHOLDS statement or on the ENVIRON EXIT control file entry.

## Examples

This example shows the relationship between the EZLEATHR routine and the automation control file. The threshold to be checked is a user-defined name, PU001. The user module checks the thresholds by calling the EZLEATHR routine.

The automation control file entry is:

```
THRESHOLD PU001,CRIT=(8,02:00),FREQ=(4,04:00),INFR=(4,08:00)
```

The module to call the EZLEATHR routine is:

```
USERTHR module
  &CGLOBAL ABC XYZ ... ..
  &TGLOBAL ABC XYZ ... ..
  EZLECAUT ... check whether automation allowed and set
```

```

:
-AUTOOK
  EZLEATHR PU001 USER NEW
  &IF &RETCODE EQ 0 &THEN &GOTO -OKSOFAR
  &IF &RETCODE EQ 1 &THEN &GOTO -EXCEEDINFR
  &IF &RETCODE EQ 2 &THEN &GOTO -EXCEEDFREQ
  &IF &RETCODE EQ 3 &THEN &GOTO -EXCEEDCRIT
  otherwise, an error occurred, RC=4/5, log error msg
  &EXIT
-OKSOFAR
  perform whatever actions may be required if no thresholds
  have been exceeded
  &EXIT
-EXCEEDINFR
  perform whatever actions may be required if infrequent
  thresholds have been exceeded
  &EXIT
-EXCEEDFREQ
  perform whatever actions may be required if frequent
  thresholds have been exceeded
  &EXIT
-EXCEEDCRIT
  perform whatever actions may be required if critical
  thresholds have been exceeded.
  &EXIT

```

The THRESHOLD entries in the automation control file defined a critical threshold as eight errors occurring in 2 hours, frequent threshold as four errors in 4 hours and infrequent threshold as four errors in eight hours.

The user-written module initiates the EZLEATHR routine using resource name, resource type, and NEW. The error information is added to the status file and the thresholds are checked.

Upon return to the user-written module the *&RETCODE* value is checked. If the value indicates that a critical threshold has been exceeded, the module stops recovery, which is consistent with the message issued by the EZLEATHR routine.

## Examples

This example shows a REXX command list that uses the EZLEATHR routine:

```

/* REXX example of EZLEATHR usage */

/* Check thresholds */

'EZLEATHR 'Resource' USER NEW'
Thr_rc = Rc

select
  when Thr_rc = 0 then                                /* None exceeded*/
  do
    /* Actions for no threshold exceeded */
    exit
  end
  when Thr_rc = 1 then                                /* Infrequent */
  do
    /* Infrequent Actions */
    exit
  end
  when Thr_rc = 2 then                                /* Frequent */
  do
    /* Frequent Actions */
    exit
  end
  when Thr_rc = 3 then                                /* Critical */
  do
    /* Critical Exceeded, */
    /* terminate recovery */
    exit
  end
  otherwise
  do
    'MSG LOG, ERROR: RETURN CODE 'Thr_RC' FROM EZLEATHR 'Resource
    exit
  end
end

```

end

## INFORM Action (EZLECALL)

---

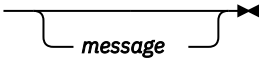
### Purpose

EZLECALL is a REXX routine that generates an immediate INFORM action based on the notification policy. Enter the name of the individual or group policy to contact, and any message text required. EZLECALL does require that an INFORM/CONTACT policy entry exists for this purpose.

The INFORM command is a command synonym (CMDSYN) for EZLECALL.

### Format

#### EZLECALL

➤ EZLECALL — *policy\_name*  *message* ➤

### Parameters

#### *policy\_name*

Specifies the name of the INFORM policy or group name to use, which determines the individuals to contact.

#### *message*

Specifies the message to be sent to the contact. The message must be consistent with the CONNECTION type specified in the INFORM policy. If no message is provided, the default INFORM message or the message specified in the INFORM policy is used.

### Examples

This example shows how the operator can issue an inform for the policy NITEOPS. The text following the policy is sent in an email or to an alpha-pager.

```
INFORM NITEOPS PLEASE CALL THE OFFICE IMMEDIATELY
```

Refer to the *IBM Z NetView Administration Reference* for additional information about how to set up your INFORM policy statements.

## Using Active Monitoring and Recovery (EZLECATV)

---

### Purpose

The EZLECATV routine tries to recover VTAM resources through the VTAM VARY INACT and VARY ACT commands at intervals specified in the MONIT control file entry. Recovery attempts continue until the resource becomes active, interval settings have been exhausted, automation is turned off, or an operator issues a command for that resource (VARY INACT or VARY ACT). If the notify flag for the interval is **Y**, the EZLECATV routine issues a reminder notification.

```
EZL507I resname HAS BEEN UNRECOVERABLE FOR duration
```

At the first interval, message EZL506I issues:

```
EZL506I restype resname ON location  
INACTIVE - RECOVERY MONITORING HAS BEEN INITIATED
```

Keep the parameters for the EZLECATV routine in the same order as they are shown in the following syntax diagram.

### Format

#### EZLECATV

➤ EZLECATV — *resource* — , — *res\_type* — , — *count* — , — *gmtdate* — , — *gmttime* ➤

### Parameters

#### *resource*

Resource name.

#### *res\_type*

Resource type.

#### *count*

MONIT interval count of interval just run. Valid values are 0-99.

#### *gmtdate*

Date of original failure that initiated recovery. The date is given in Greenwich mean time (GMT).

#### *gmttime*

Time of original failure that initiated recovery. The time is given in Greenwich mean time (GMT).

### Usage

Link stations and CDRM are not deactivated during recovery. All other resources are deactivated with a VARY INACT,F command to clear pending states. Activations are done with SCOPE=U for PUs and higher nodes. SCOPE=ONLY is used for LUs and CDRMs. Recovery is stopped if the resource becomes unknown to VTAM in consideration of dynamic CDRSCs. A timer is scheduled at the next MONIT interval with an ID of the resource names.

To tailor reminder interval automation processing code AON user exits for specific MONIT intervals for a resource. Set default exit values for EZLECATV on the MONIT statement or on the ENVIRON EXIT control file entry. Refer to the *IBM Z NetView Administration Reference* for more information about the ENVIRON EXIT entry.

## Checking Automation (EZLECAUT)

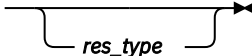
---

### Purpose

The EZLECAUT routine checks recovery automation flags on the RECOVERY control file entry to determine whether automation is in effect for a resource.

### Format

#### EZLECAUT

➤ EZLECAUT — *res\_name* —  ➤

### Parameters

#### *res\_name*

The name of the resource for which to check automation settings

#### *res\_type*

Type of resource being checked

## Return codes

**0**

Automation is in effect.

**1**

Automation is not in effect.

**2**

Automation is defined, but not in effect because of a NOAUTO window.

**4**

Parameters that are not valid.

## Usage

If a NOAUTO parameter is found, calculations determine whether the current time is within the NOAUTO window. The return code is set accordingly.

You can set defaults for the EZLECAUT routine by defining the EXIT07 parameter of the ENVIRON EXIT control file entry. To tailor recovery automation processing, code AON user exits for specific recovery processing.

## Processing Generic Failures (EZLEFAIL)

---

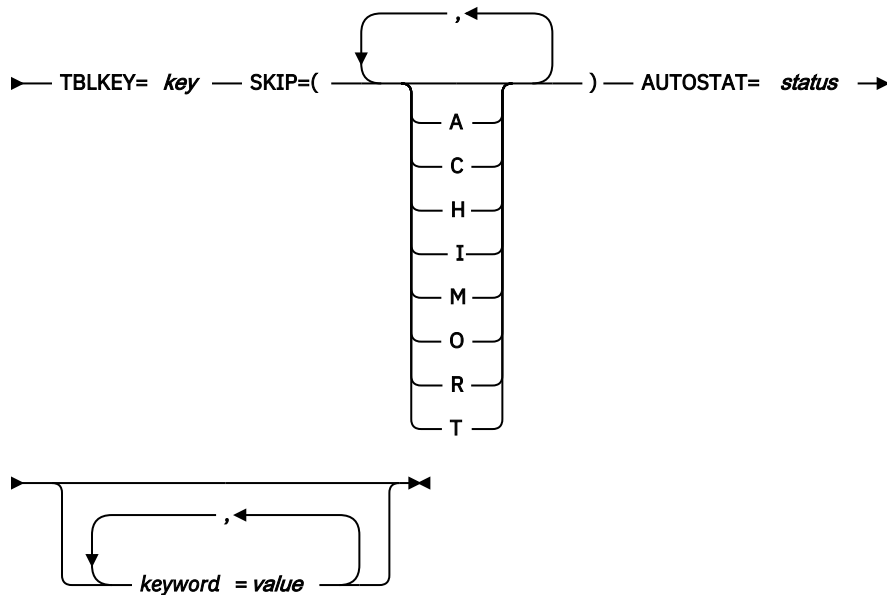
### Purpose

The EZLEFAIL routine processes resource failure events and drives option specific routines from the option tables. The following diagram illustrates the syntax of the EZLEFAIL routine and its parameters.

### Format

#### EZLEFAIL

➤ EZLEFAIL — RESNAME= *resourcename* — RESTYPE= *resourcetype* — OPTION= *optiontype* ➔



### Parameters

#### RESNAME

Specifies the resource name.

**RESTYPE**

Specifies the resource type.

**OPTION**

Specifies the automation component:

- APPN
- IP390
- SA
- SNA
- SNBU
- TCPIP
- X25

**TBLKEY**

Specifies unique handling for the event. The EZLEFAIL routine gets optional processing values from the TBLKEY parameter. If you do not specify the TBLKEY parameter for the EZLEFAIL routine, no optional processing or notification occurs. The values on the TBLKEY parameter specify keywords found in the option definition tables. In the option definition table, the keywords define the actual processing values used for optional processing. AON saves the TBLKEY values in the *outmsgid* and *spec\_function* variables. Message EZL509I is the default *outmsgid*. The value of TBLKEY is in the following format:

```
tblkey_value=(outmsgid,spec_function_call)
```

For example, the EZLEFAIL routine is called with:

```
EZLEFAIL OPTION=SA MSGPRMS=(OPID) TBLKEY=IST105I RESNAME=resname
```

The EZLEFAIL routine gets the values specified on the IST105I keyword in the option definition table. In the option definition table, the values on the IST105I keyword are:

```
IST105I=(EZL531,FKVEAIDA(resname restype))
```

The EZLEFAIL routine issues the EZL531I message and runs FKVEAIDA as a function sending the current value of *resname* (resource name) and *restype* (resource type) for optional processing. An optional processing program performs automation or processing unique to the resource or failure. No optional processing is done and no message is issued if SKIP=(0) is specified on the EZLEFAIL call.

The EZLEFAIL routine issues message EZL509I or EZL510I to all logs and to DDF. EZL509I is issued for resources that continue to be unavailable when EZLEFAIL runs. EZL510I is issued for resources that have been recovered and are in an active state when EZLEFAIL runs. Operators do not receive these messages. These messages are not issued if SKIP=(A) is specified on the EZLEFAIL call.

**SKIP**

Specifies which processing to ignore:

**A**

Availability message is skipped.

**C**

Check that automation processing is bypassed.

**H**

Do not check higher node status.

**I**

Do not gather resource-specific information.

**M**

Option-specific message processing (TBLKEY value) is bypassed.

**O**

Option-specific processing (specified by TBLKEY value) is bypassed.

**R**

Recovery attempt processing is bypassed. Setting the AIP operator status is also bypassed.

**T**

Threshold processing is skipped.

**AUTOSTAT**

Specifies the status.

***keyword=value***

Any valid keyword and value.

**Return codes****00**

EZLEFAIL completed successfully.

**02**

AON initialization has not completed, issue EZL003E.

**03**

Missing parameters, issue EZL203I.

**04**

Incorrect parameters: issue EZL204I.

**05**

Wait time expired: issue EZL205I.

**06**

Command failed: issue EZL206I.

**07**

NOVALUE variable found.

**08**

REXX syntax failure.

**09**

Initialization shipped to automation operator.

**10**

Cannot find resource type/owning option.

**11**

Check recovery, option not enabled.

**12**

Check recovery, higher node down.

**13**

Check recovery, automation flag off.

**14**

Check recovery, resource in available state.

**15**

Check recovery, recovery timer exists.

**16**

Automation is defined, but not in effect because of a NOAUTO window.

**21**

Check thresholding: infrequent routine RC>0.

**22**

Check thresholding: frequent routine RC>0.



**23**

Check thresholding: critical routine RC>0.

**30**

Optional processing routine RC>0.

### Examples

The following example illustrates how you can call EZLEFAIL from the automation table:

```
IF MSGID='IST285I'
& TEXT = . 'DUMP OF ' RESNAME ' FAILED - PERMANENT' .
& HDRMTYPE = 'Q'
THEN
  EXEC(CMD('EZLEFAIL OPTION=SA RESTYPE=NCP'
           ' RESNAME=' RESNAME
           ' SKIP=(C,T,R)'
           ' TBLKEY=IST285IO AUTOSTAT=NCPDUMP'))
  ROUTE(ALL *));
```

The automation flag is not checked, threshold analysis is not performed, and recovery is not initiated. The status sent to DDF is NCPDUMP and IST205I. A keyword in the option definition table specifies any additional message or program to be called.

### Examples

This example shows AON/SNA automation being driven for VTAM message IST105I when a resource fails. EZLEFAIL is called out of the automation table for the AON/SNA subarea (SA) option. Other parameters passed to EZLEFAIL include TBLKEY that identifies an option definition table key to use for recovery of the resource. Additional Information is retrieved from the option definition table during the failure processing.

```
IF TEXT = 'IST105I' resname 'NODE' . & HDRMTYPE = 'Q'
THEN
  EXEC(CMD('EZLEFAIL OPTION=SA MSGPRMS=(OPID) '
           ' TBLKEY=IST105I '
           ' RESNAME=' resname));
```

### Examples

This example shows that AON/SNA automation is driven for VTAM message IST619I when a resource fails. EZLEFAIL is called out of the automation for the AON/SNA subarea option. The option definition table key for additional recovery information and processes is IST619I. Additional information is retrieved from the option definition table during the failure processing.

```
IF MSGID='IST619I'
& TEXT = . 'ID = ' resname ' FAILED' .
& HDRMTYPE = 'Q'
THEN
  EXEC(CMD('EZLEFAIL OPTION=SA TBLKEY=IST619I '
           ' RESNAME=' resname )
           ' SKIP=(R)' ROUTE(ALL *));
```

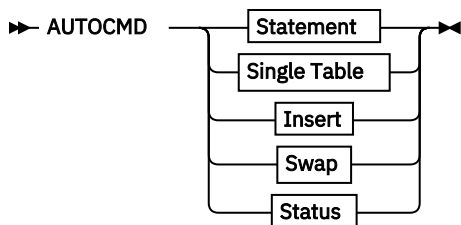
## Managing Automation Tables (AUTOCMD/EZLEF002)

### Purpose

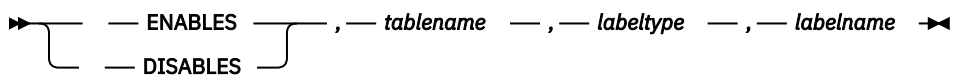
The AUTOCMD routine is used to enable, disable, insert, and get the status of automation tables. The following diagram illustrates the syntax of the AUTOCMD routine and its parameters.

## Format

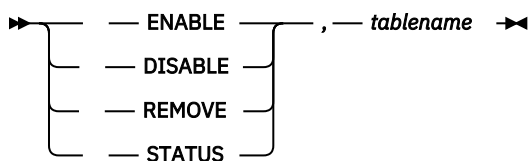
### AUTOCMD



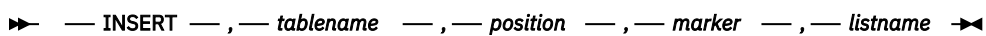
### Statement



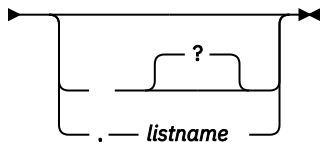
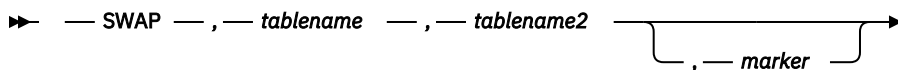
### Single Table



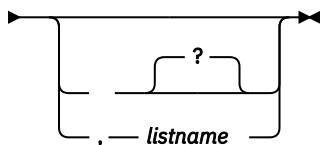
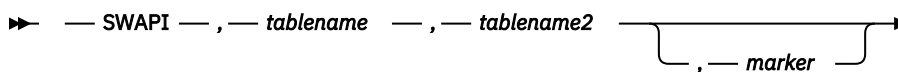
### Insert



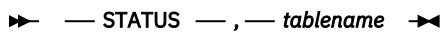
### SWAP



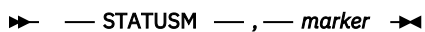
### SWAPI



### Status



### Statusm



## Command

### Synonym

### AUTOCMD

EZLEF002

## Parameters

### **ENABLES**

Activates multiple automation tables.

### **DISABLES**

Deactivates multiple automation tables.

### **ENABLE**

Activates a single automation table.

### **DISABLE**

Deactivates a single automation table.

### **INSERT**

Loads an automation table in the position specified.

### **SWAP**

Replaces an automation table with another automation table.

### **SWAPI**

Replaces an automation table with another automation table and reestablishes all disabled elements.

### **STATUS**

Returns the status of the specified table.

### **STATUSM**

Returns the status of the table with the specified marker.

### ***tablename***

Specifies the name of the automation table to be affected by the chosen keyword.

### ***tablename2***

Specifies the name of the automation table that replaces *tablename*.

### ***labeltype***

Specifies the type of label to be altered as follows:

- LABEL
- ENDLABEL
- BLOCK
- GROUP
- SEQUENCE

### ***labelname***

Specifies a label name, which consists of 1–16 character label or 1–8 character sequence numbers in the automation table.

### ***position***

Specifies the position of the table to be loaded. The position specification follows:

#### **1..n**

The numeric position of the table.

#### **FIRST**

The table loads in the first position.

#### **LAST**

The table loads in the last position.

### ***marker***

Specifies a unique identifier associated with the table being loaded. Markers can be 1–8 characters in length and are optional.

### ***listname***

Specifies a unique name for the listing member associated with the automation table. A list name can be 1–8 characters. If the listing name is omitted, a unique name is created and a listing is generated, automatically. However, if \*NONE\* is provided as a parameter, then no listing is

generated. This is considered an override. No informational messages are generated regarding an override of the listing with \*NONE\*.

### Return codes

For STATUS and STATUSM, the return codes are as follows:

**0**

The search for the automation table or marker was successful.

**4**

The search for the automation table was not successful or an error occurred.

For all other return codes are as follows:

**0**

The request completed successfully.

**n**

The request failed. The error message can be found in the common global variable EZLERRMSG. The common variable stem EZLAUTOR contains messages generated by the request.

### Usage

When loading or swapping tables that replace or become the automation table used by AON, it is necessary to update the EZLMSGTBL and EZLMSGST global variables with the table name and listing name. If AUTOMAN removes the AON table, then these variables are automatically reset.

## Setting Panel Message Color (EZLEMCOL)

---

### Purpose

Use EZLEMCOL to set panel variable attributes for non-AON messages such as VTAM and NetView messages.

### Format

#### EZLEMCOL

➤ EZLEMCOL — ( — *msgnum*, — *msgtext* — ) ➤

### Parameters

#### *msgnum*

Message number

#### *msgtext*

Message text

### Usage

EZLEMCOL can also be called as a command.

Color settings are:

#### Information message

White

#### Warning message

White

#### Error message

Yellow

## Action message

Red

## Examples

Following is a sample of REXX code using EZLEMCOL:

```
$ErrorMsg = EZLEMCOL(MsgId,MsgTxt) /* set panel variable */
'GLOBALV GETT EZLMSGCOLR'          /* get color attribute(s) */
$ERRORMSG = 'EZLMSGCOLR'           /* set panel attribute(s) */
```

## Formatting Panel Messages (EZLEMSG)

---

### Purpose

EZLEMSG is a REXX function that builds messages to be sent to operators while they are viewing an AON panel.

### Format

#### EZLEMSG

➔ *PnlMsg* =EZLEMSG — (*msgnum*, *logupdate*, *clistname*, *parm1*, *parm2*,... *parmn*) ➔

### Parameters

#### *PnlMsg*

The panel variable used to display the message. Upon return from EZLEMSG, this variable is set to the complete message text with inserts included.

#### *msgnum*

Identifies the message to display.

#### *logupdate*

If *Y*, the message is passed back from the function call and is also logged to the NetView log. If *N*, the message is returned from the function call and is not logged.

#### *clistname*

Used to identify the program issuing the message. This is required and issued if an error occurs to properly identify the failing program.

#### *parm1*

First insert for the message. This is optional.

#### *parm2*

Second insert for the message. This is optional.

#### *parmn*

Last insert for the message. This is optional.

### Usage

The default message prefix is EZL. If you generate messages with other prefixes, code the prefix plus the message number. The parameters (*parm1* through *parmn*) are optional and are only required if the message substitutes variable information into the message text as a message insert.

## Examples

The following sample shows REXX code that uses EZLEMSG:

```
PnlMsg = EZLEMSG(901,'N',ident,sel,'1','5')
```

This statement sets the variable *PnlMsg* equal to the text of message EZL901I with inserts of the value of variable *sel*, 1, and 5:

```
PnlMsg = SELECTION 6 IS INVALID. TYPE A NUMBER BETWEEN 1 AND 5
```

## Driving the Inform Policy (EZLENFRM)

---

### Format

#### EZLENFRM

```
➤ EZLENFRM — — policy_name — — resname — — restype — — resdomain — — ➤  
  
➤ — resstat — — aostat ➤
```

### Purpose

EZLENFRM is a REXX routine called by the AON notification policy when an inform action is required. EZLENFRM can also be started by non-AON routines. EZLENFRM checks the inform policy, issues the inform actions, and logs the action, if indicated. All parameters are required and can be substituted in the resulting inform message.

### Parameters

#### *aostat*

Automation status of the failing resource

#### *policy\_name*

Specifies which INFORM policy or group name to use when determining who to contact.

#### *resdomain*

The domain of the failing resource

#### *resname*

The failing resource name

#### *resstat*

The status of the failing resource

#### *restype*

The failing resource type

### Examples

This example shows an invocation of EZLENFRM for policy NITEOPS. A CDRM that is down has reached the end of its monitoring intervals. The NITEOPS policy is consulted and appropriate actions are taken.

```
EZLENFRM NITEOPS NTB6MVS CDRM PACDR NOMOMONS
```

Refer to the *IBM Z NetView Administration Reference* for more information about setting up INFORM policy statements.

## Notify Policy List (EZLENTFY)

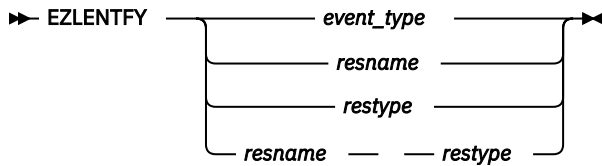
---

### Purpose

EZLENTFY is a REXX routine called to query the AON Notify Policy for a given event type, resource, or resource type.

## Format

### EZLENTFY



## Parameters

### **event\_type**

Query Notify Policy based on the type of event such as CRITTHRS.

### **resname**

Query Notify Policy based on the resource name. Wild cards are supported.

### **restype**

Query Notify Policy based on the resource type.

## Usage

If any Notify Policy keywords are set to *NO*, they are not displayed in the Notify Policy List. Refer to the *IBM Z NetView Administration Reference* for additional information about the NOTIFY statement.

The Notify Policy List can be null. Any caller must be prepared to avoid issuing notifications with a null Notify Policy List.

The following INFORM Policy Names are not valid:

- MSG
- ALERT
- TEC
- DDF

EZLENTFY creates two task global variables:

- EZLNTFYLIST containing the list of notification actions. The format of the list is (Notify1, Notify2, ..., NotifyN)
- EZLNTFYTYPE containing the NOTIFY policy statement used.

User Exit (EXIT10) Functions:

- Override NOTIFY policy based on other criteria, such as your own notify policy database.
- Implement other notifications such as fax.

## Setting the AIP User Status Bit (EZLERAIP)

---

### Purpose

EZLERAIP is a REXX routine that enables the setting and resetting of the Automation in Progress (AIP) user status bit for resources being monitored or viewed in NetView management console (NMC). This routine handles the routing of the request to the domain specified in the control file. On the target domain, EZLERAIP routes the request to the appropriate AUTAIP autotask for processing. This routine can also update the Operator Intervention View (OIV) status, see the request types (*reqtype*) for specific information.

## Format

### EZLERAIP

➤ EZLERAIP — — *resname* — , — *restype* — , — *reqtype* — , — *other\_netid* — , →  
➤ — *nqn\_netid* — , — *prev\_domain* — , — *gw\_flag* — , — *sscp\_name* ➤

## Parameters

### *gw\_flag*

When set to Y causes the AIP operator status to update for both the NCP and Gateway NCP resources. Otherwise, only the NCP resource changes.

### *nqn\_netid*

The NETID of a resource as specified in its network qualified name.

### *other\_netid*

The NETID of the resource if owned by another domain.

### *prev\_domain*

An internal parameter that is filled in by the EZLERAIP call when it reroutes the request to the specified RODMDOM. All other calls must omit this parameter.

### *reqtype*

The operation to be performed. The following values are supported:

#### **RESET**

Resets the AIP status for the resource.

#### **RESET/OIV**

Resets the AIP status and adds the resource to the Operator Intervention View.

#### **SET**

Sets the AIP status for the resource.

#### **SET/OIV**

Sets the AIP status and removes the resource from the Operator Intervention View.

### *resname*

The name of the resource.

### *restype*

The *restype* parameter must be one of the following:

- CDRM
- LINE
- LINKSTA
- NCP
- PU

### *sscp\_name*

The VTAM SNA node name of the reporting VTAM. This is required only when the resource is owned by another VTAM domain.

## Usage

- Commas are required between each of the parameters.



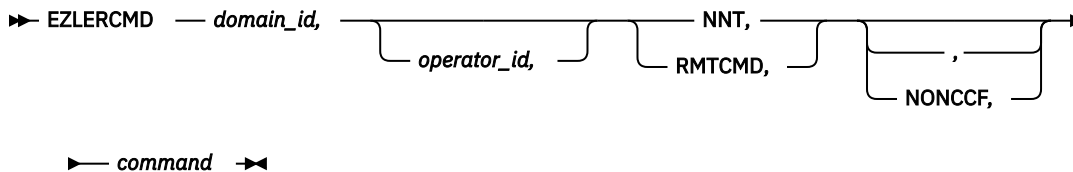
## Routing Commands over Cross-Domain Sessions (EZLERCMD)

### Purpose

EZLERCMD is a REXX routine that routes commands over cross-domain sessions. Both NetView-NetView Task (NNT) and remote command (RMTCMD) session types are supported.

### Format

#### EZLERCMD



### Parameters

#### *domain\_id*

Domain to which the command is sent.

#### *operator\_id*

Operator to issue the command. This parameter is for RMTCMD sessions only.

#### NNT

Specifies that the session the command is routed over is an NNT session.

#### RMTCMD

Specifies that the session the command is routed over is an RMTCMD session.

#### NONCCF

If specified, the command is issued following CMD HIGH. Otherwise, it is issued following CMD HIGH NCCF.

#### *command*

The command and parameters to be routed.

### Usage

The *operator\_id* variable is required for RMTCMD sessions, but not for NNT sessions. No verification is performed on the command. It is routed as is to the target domain. If the specified operator is not logged on to the target domain, the NetView program establishes an RMTCMD session and issues the command upon session initialization. Ensure that the specified operator is logged on to the target domain before calling EZLERCMD.

### Examples

This example routes the LIST STATUS=OPS command over the active NNT session to domain CNM01.

```
EZLERCMD CNM01,,NNT,,LIST STATUS=OPS
```

### Examples

This example shows how to route the LIST STATUS=OPS command over the RMTCMD session to OPER1 on domain CNM01.

```
EZLERCMD CNM01,OPER1,RMTCMD,,LIST STATUS=OPS
```

## Recovering Resources (EZLERECV)

### Purpose

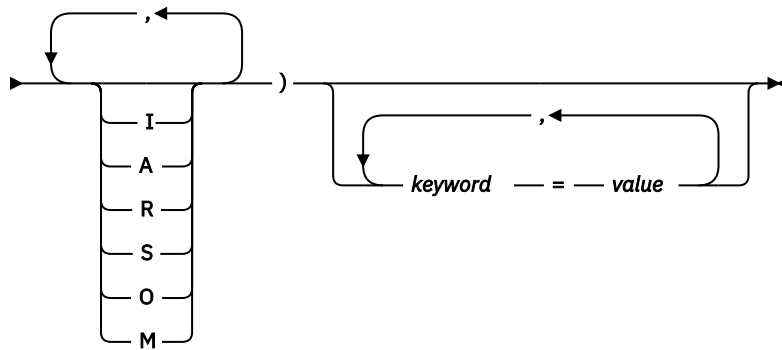
The EZLERECV routine is called for resource recovery and drives option-specific routines from the option tables. The following diagram illustrates the syntax of the EZLERECV routine and its parameters. The AIP operator status is also cleared each time EZLERECV is called for a given resource.

### Format

#### EZLERECV

➤ EZLERECV — RESNAME — = — *resourcename* — RESTYPE — = — *resourcetype* — ➤

➤ OPTION — = — *optiontype* — TBLKEY — = — *key* — SKIP — = — ( ➤



### Parameters

#### RESNAME

Specifies the resource name.

#### RESTYPE

Specifies the resource type.

#### OPTION

Specifies the automation option.

#### TBLKEY

Specifies unique handling for the event.

The EZLERECV routine gets optional processing values from the TBLKEY parameter. If you do not specify the TBLKEY parameter for the EZLERECV routine, no optional processing or notification occurs. The values on the TBLKEY parameter specify keywords found in the option definition tables. In the option definition table, the keywords define the actual processing values used for optional processing. AON saves the TBLKEY values in the *outmsgid* and *spec\_function* variables. Message EZL504I is the default *outmsgid*. The value of TBLKEY is in the following format:

```
tblkey_value=(outmsgid,spec_function_call)
```

For example, if the EZLERECV routine is called with:

```
EZLERECV OPTION=SA MSGPRMS=(OPID) TBLKEY=IST093I RESNAME=resname
```

The EZLERECV routine gets the values specified on the IST093I keyword in the option definition table. In the option definition table, the values on the IST093I keyword are:

```
IST093I=(EZL517,FKVEAIDA(resname restype resstat opid))
```

The EZLRECV routine issues the EZL517I message and runs FKVEAIDA as a function sending the current value of *resname* (resource name) and *restype* (resource type) for optional processing. An optional processing program performs automation or processing unique to the resource or failure. No optional processing is done and no message is issued if SKIP=(0) is specified on the EZLRECV call.

The EZLRECV routine issues message EZL504I to all logs and to DDF. Operators do not receive this message. This message is not issued if SKIP=(A) is specified on the EZLRECV call.

#### **SKIP**

Specifies which processing to ignore:

##### **I**

Do not gather resource-specific information.

##### **A**

Skip availability message.

##### **R**

Bypass recovery stop processing.

##### **S**

Skip active monitoring restart.

##### **O**

Bypass option-specific processing (specified by TBLKEY value).

##### **M**

Bypass option-specific message processing (TBLKEY value).

#### **keyword=value**

Any valid keyword and value.

#### **Return codes**

##### **00**

EZLRECV completed successfully.

##### **03**

Missing parameters, issue EZL203I.

##### **04**

Incorrect parameters, issue EZL204I.

##### **05**

Wait time expired, issue EZL205I.

##### **06**

Command failed, issue EZL206I.

##### **07**

NOVALUE variable found.

##### **08**

REXX syntax failure.

##### **09**

Initialization shipped to automation operator.

##### **10**

Cannot find resource type.

##### **11**

Option not enabled.

##### **30**

Optional processing routine RC>0.

## Examples

This example shows that AON/SNA automation is driven for VTAM message IST093I when a resource is reactivated. EZLERECV is called from the automation table for the AON/SNA SubArea option with optional parameters, such as TBLKEY=IST093I, to identify additional information for the resource recovery actions from the AON/SNA option definition table. Additional information is retrieved from the option definition table during the recovery processing.

```
IF MSGID='IST093I'
  & TEXT = 'IST093I ' resname 'ACTIVE'
  THEN
    EXEC(CMD('EZLERECV OPTION=SA TBLKEY=IST093I '
      'RESNAME=' resname ' MSGPRMS=(OPID) '
      ROUTE(ALL *)));
```

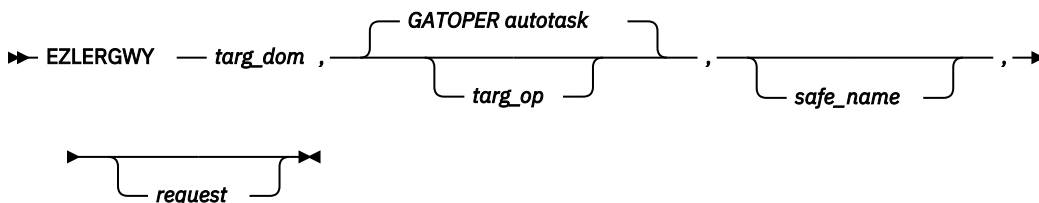
## Routing Commands To Other NetView Domains (EZLERGWY)

### Purpose

EZLERGWY is a REXX routine that uses the RMTCMD command to route commands to other NetView domains.

### Format

#### EZLERGWY



**Note:** If *safe\_name* is coded and contains data, *request* is optional.

### Parameters

#### *targ\_dom*

The target NetView domain to which to route the request.

#### *targ\_op*

The target NetView operator ID to which to route the request. If not specified, the request is routed to the GATOPER autotask as defined in the CDLOG statements in EZLCFG01 in DSIPARM.

#### *safe\_name*

The name of the safe in which to store the response or from which to get the command. If you specify DSILOG, the output is only logged to DSILOG. If not specified, the command is routed and no response is logged or returned to the caller.

#### *request*

The command to run.

### Usage

To enable the remote gateway support, define CDLOG statements for the GATOPER autotask in each NetView domain. These statements are located in member EZLCFG01 in DSIPARM. Using these CDLOG definitions, AON establishes a RMTCMD session between the corresponding NetView domains. For more information, refer to the *IBM Z NetView Administration Reference*.

All parameters are positional and must be delimited by a comma. If you omit a parameter, use a comma to denote its absence.

No syntax checking or security checking is performed for the requested command.

To see the output from your command, specify a *safe\_name* parameter. For debugging, specify DSILOG for the *safe\_name* parameter.

If you are passing a command in the named safe (*safe\_name*) parameter, make sure that your program has placed data (the command) into the safe. If not, you receive message EZL203I. In this case, do not pass the command as a parameter.

### Examples

The following example causes the LIST command to be routed to domain NTV6D. No output is returned to the calling program.

```
EZLERGWY NTV6D,,,LIST
```

Because no *targ\_op* was specified, GATOPER autotask in NTV6D is used.

### Examples

The following example shows how to start EZLERGWY when the command to run is contained in a safe. In this case, the command is LIST DSILOG and is passed to EZLERGWY in a safe called MYSAFE. EZLERGWY builds the command, runs it, and returns the responses back into MYSAFE:

```
...
Command.0=2
Command.1='LIST '
Command.2='DSILOG'
"PIPE STEM Command.|Safe MYSAFE"

"EZLERGWY NTV6D,,MYSAFE,"

"PIPE SAFE Mysafe|Stem Myvar."
Do i=1 to myvar.0
  Say 'Response to command=' Myvar.i
End
```

Using this example, you can send commands longer than 255 bytes (NetView command-line restriction) to remote NetView domains.

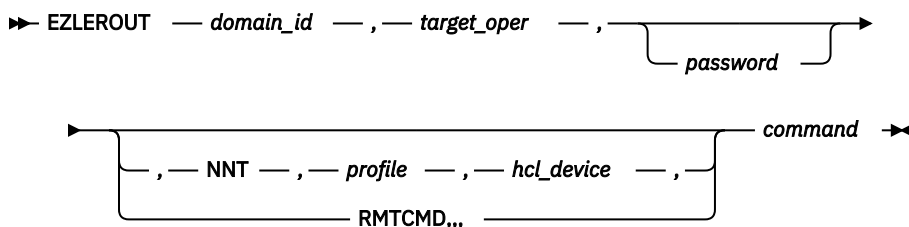
## Routing NNT Cross-Domain Logon Information (EZLEROUT)

### Purpose

EZLEROUT is a REXX routine that routes cross-domain logon information for NNT sessions and automates replies to the NetView DSI809A message. You can also use EZLEROUT to route commands over RMTCMD sessions, similar to the EZLERCMD routine.

### Format

#### EZLEROUT



## Parameters

### ***command***

The command to be issued. Specifying YES for an NNT session runs the initial program defined in the operator logon profile, NNT-Profile. For RMTCMD sessions, this can be any valid NetView command or program (command list).

### ***domain\_id***

The domain to which you are logging on.

### ***hcl\_device***

The hardcopy log device for the NNT session.

### **NNT**

Specifies that you are logging on to an NNT session.

### ***password***

The password to use.

### ***profile***

The NNT session profile.

### **RMTCMD**

Specifies that you are logging on to an RMTCMD session.

### ***target\_oper***

The operator ID to which you are logging on.

## Examples

This example shows how to route the LIST STATUS=OPS command over the active NNT session to domain CNM01. The routine uses the password OPERWORD, and the operator profile OPERPROF. The double-commas (,,) preceding the LIST STATUS=OPS command indicate that the routine ignores the hardcopy log device parameter.

```
EZLEROUT CNM01,,NNT,OPERWORD,OPERPROF,,LIST STATUS=OPS
```

## Examples

This example shows how to route the LIST STATUS=OPS command over the RMTCMD session to OPER1 on domain CNM01.

```
EZLEROUT CNM01,OPER1,,RMTCMD,,LIST STATUS=OPS
```

## Retrieving AON Information (EZLERTVE)

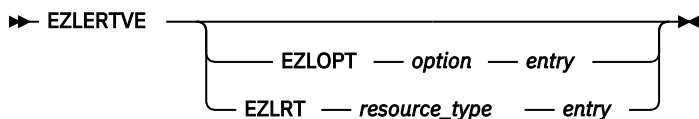
### Purpose

The EZLERTVE routine retrieves data from AON option definition tables to use in automation modules.

### Format

The following diagram illustrates the syntax of the EZLERTVE routine:

### **EZLERTVE**



## Parameters

### **entry**

Option definition table keyword.

### **EZLOPT**

Specifies the option (AON feature) common global variables from the option definition tables.

### **EZLRT**

Specifies the resource type common global variables from the option definition tables.

### **option**

Installed AON feature or suboption. Valid option names are:

#### **AON**

AON base

#### **APPN**

Advanced Peer-to-Peer Networking (APPN) suboption for AON/SNA

#### **IP390**

TCP/IP for z/OS suboption of AON/TCP

#### **SA**

Subarea suboption for AON/SNA

#### **SNA**

AON/SNA automation feature

#### **SNBU**

Switched network backup (SNBU) suboption for AON/SNA

#### **TCPIP**

AON/TCP automation feature

#### **X25**

X.25 suboption for AON/SNA

### **resource\_type**

Any valid resource type supported by AON or its automation features.

## Usage

The EZLERTVE routine must be called from a program, for example, REXX. The results are returned back to the calling program in a local variable, EZLERTVE.

The EZLERTVE routine accesses a variable name of *EZLOPT.option.entry* or *EZLRT.resource\_type.entry* and returns the value of the common global variable.

If no value is found, EZLERTVE returns *N/A*.

## Examples

The following is an example of EZLERTVE in a REXX procedure:

```
opt_id=SNA
Restype=PU

/* Get message class for option and resource type */
'EZLERTVE EZLOPT' opt_id 'MSGCLASS'
opt_class = EZLERTVE
'EZLERTVE EZLRT' Restype 'MSGCLASS'
rt_class = EZLERTVE

if rt_class = "N/A" then
    rt_class = "00"

if opt_class = "N/A" then
    opt_class = "00"
```

The EZLERTVE routine gets the EZLOPT.SNA.MSGCLASS variable and the EZLRT.PU.MSGCLASS variable.

## Issuing Resource State Reminders (EZLESRMD)

---

### Purpose

The EZLESRMD routine checks the status of the resource and issues notifications if the resource is down. The resource is not recovered.

### Format

#### EZLESRMD

➤ EZLESRMD — *resource* — , — *res\_type* — , — *count* — , — *gmtdate* — , — *gmttime* ➤

### Parameters

#### *count*

MONIT interval count of the interval just run. Valid values are 0 through 99.

#### *gmtdate*

Date of the original failure that initiated recovery. The date given is in Greenwich mean time (GMT).

#### *gmttime*

Time of the original failure that initiated recovery. The time given is in Greenwich mean time (GMT).

#### *resource*

Resource name.

#### *res\_type*

Resource type.

### Usage

A timer is scheduled at the next MONIT interval that has the ID of the resource name.

Monitoring is discontinued if the resource becomes unknown to VTAM in consideration of dynamic CDRSCs.

Monitoring continues until the resource becomes active, interval settings have been exhausted, or automation is turned off. If the notify flag for the interval is **Y**, a notification is issued:

```
REMINDER - restype resname HAS BEEN DOWN FOR interval
```

At the first interval, message EZL506I is issued:

```
resname INACTIVE - ATTEMPTING RECOVERY - RECOVERY MAY BE  
TERMINATED BY VARYING THE RESOURCE INACTIVE
```

The EZL540I message is issued when reminders are halted (REMINDERS FOR *restype resname* HALTED - REMINDER THRESHOLDS UNDEFINED).

## Stopping Cross-domain Sessions (EZLESTOP)

---

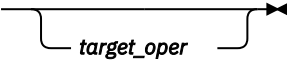
### Purpose

EZLESTOP is a REXX routine supplied with AON to stop cross-domain NetView sessions. Both NNT and RMTCMD sessions are supported.



## Format

### EZLESTOP

➔ EZLESTOP — *domain\_id*, 

## Parameters

### *domain\_id*

Domain with which you have a session.

### *target\_oper*

Operator to be logged off. This is valid only for RMTCMD sessions.

## Usage

EZLESTOP determines the type of session based on the parameters that are provided. If the target operator is specified, then EZLESTOP looks for a RMTCMD session and logs that operator off. If you do not specify a target operator, EZLESTOP stops your NNT session to the specified domain.

## Return codes

**0**

Session was ended.

**4**

Unable to stop cross-domain session. Check task global variable, EXITMSG, for the message that was received.

**5**

Internal AON error. Call IBM Software Support.

**7**

Internal AON error. Call IBM Software Support.

**8**

Internal AON error. Call IBM Software Support.

## Examples

This example logs operator OPER2T off a remote command (RMTCMD) session on domain CNM02.

```
EZLESTOP CNM02,OPER2T
```

## Examples

This example shows how to log off your NetView-NetView task (NNT) session with domain CNM02.

```
EZLESTOP CNM02
```

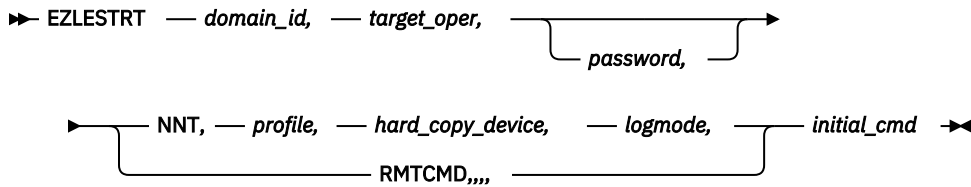
## Starting Cross-domain Sessions (EZLESTRT)

### Purpose

EZLESTRT is a REXX routine that starts cross-domain NetView sessions. The routine supports both NNT and RMTCMD type sessions.

## Format

### EZLESTRT



## Parameters

### *domain\_id*

Target domain to start session.

### *hard\_copy\_device*

NNT hard-copy log device to start.

### *initial\_cmd*

Initial program or command to issue. For an NNT session, YES runs the initial program that is defined in the operator's logon profile. For RMTCMD sessions, this can be any valid NetView command or program.

### *logmode*

Log mode for NNT session.

### *password*

NNT password.

### **NNT**

Start a NetView-to-NetView Task session.

### *profile*

NNT operator profile.

### **RMTCMD**

Start a remote command session.

### *target\_oper*

Operator to be logged on.

## Usage

Delimit all parameters with commas. If you do not specify a parameter, use two commas.

## Return codes

**0**

Cross-domain session started.

**4**

Unable to start cross-domain session. Check the task global variable, EXITMSG, for the message that was received.

**5**

Internal AON error. Call IBM Software Support.

**7**

Internal AON error. Call IBM Software Support.

**8**

Internal AON error. Call IBM Software Support.

## Examples

This example shows how to start a cross-domain RMTCMD session to CNM02 as OPER2T:

```
EZLESTRT CNM02,OPER2T,,RMTCMD,,,OPER2IC
```

The first occurrence of double-commas (,,) means that no password is passed. Other parameters not being passed are the *profile*, *hard\_copy\_device*, and *logmode*.

## Examples

This example shows how to start a cross-domain NNT session to CNM02 as OPER1 using a password of OPERPSWD and a logon profile of OPERPROF:

```
EZLESTRT CNM02,OPER1,OPERPSWD,NNT,OPERPROF,,,YES
```

No parameters are supplied for *hard\_copy\_device* and *logmode*. YES specifies to run the operator's initial program as defined in the profile OPERPROF when the session is established.

## Activating VTAM Resources (EZLEVACT)

---

### Purpose

Use the EZLEVACT REXX routine to reactivate VTAM resources. EZLEVACT issues a VARY NET,ACT command for a specified resource.

### Format

#### EZLEVACT

➡ EZLEVACT — *resource* →

### Parameters

#### *resource*

The name of the VTAM resource to be activated.

### Return codes

**0**

VARY ACT was successful.

**8**

WAIT error or resource is not known.

**9**

Security failure.

**100**

WAIT timeout.

### Usage

EZLEVACT can be called from any NetView program or command processor.

## Examples

The following example shows how to activate resource TA1P523A:

```
EZLEVACT TA1P523A
```

## Deactivating VTAM Resources (EZLEVINA)

---

### Purpose

Use the EZLEVINA REXX routine to deactivate VTAM resources. EZLEVINA issues a VARY NET,INACT,F command for a specified resource.

### Format

#### EZLEVINA

➤ EZLEVINA — *resource* ➤

### Parameters

#### *resource*

The name of the resource to deactivate.

### Return codes

**0**

VARY ACT was successful.

**8**

WAIT error or resource is not known.

**9**

Security failure.

**100**

WAIT timeout.

### Usage

EZLEVINA can be called from any NetView program or command processor.

### Examples

The following example shows how to deactivate resource TA1P523A:

```
EZLEVINA TA1P523A
```

## Moving VTAM Resources (EZLEVMOV)

---

### Purpose

Use the EZLEVMOV REXX routine to move a PU from one line to another. EZLEVMOV issues a MODIFY NET,DR,TYPE=MOVE,FROM=xxxx,TO=yyyy command for a specified resource.

### Format

#### EZLEVMOV

➤ EZLEVMOV — *PU\_name* — , — *from\_line* — , — *to\_line* ➤

### Parameters

#### *from\_line*

The line to which the PU is currently attached

The name of the PU to move

The line to which to move the PU

## Return codes

# O

MOVE was successful.

8

WAIT error, MOVE error, incorrect number of parameters.

9

Security failure.

**100**

WAIT timeout.

## Usage

EZLEVMOV can be called from any NetView program or command processor. If you choose to move a PU, reactivate it after the move is complete. You can use the EZLEVACT for the reactivation.

## Examples

The following example shows how to move PU TA1P523A from line TA1L5023 to line TA1L5024:

EZLEVMOV TA1P523A,TA1L5023,TA1L5024

## Sending MSUs to an MS Transport Application (EZLSMSU)

## Purpose

EZLSMSU can be used to issue the NetView MS Transport send macro. To use EZLSMSU you must build a CP\_MSU major vector and then start EZLSMSU to issue the send macro.

### Format

## EZLSMSU

The diagram illustrates the structure of the EZLSMSU command. It consists of three main fields:

- VARNAME**: A variable name field, represented by a line with a bracket labeled *varname*.
- DESTAPPL**: A destination application field, represented by a line with a bracket labeled *destappl*. It is preceded by a comma and the label **DESTAPPL**.
- DESTLU**: A destination logical unit field, represented by a line with a bracket labeled *destlu*. It is preceded by a comma and the label **DESTLU**.

The fields are connected by hyphens, indicating they are part of a single command structure. The diagram uses a horizontal line with brackets to show the extent of each field and a vertical line to separate the fields.

## Parameters

### ***varname***

Identifies the variable that contains a prebuilt CP\_MSU. The variable is retrieved from your local variable pool. EZLSMSU does not perform data verification; it assumes that the CP\_MSU was built correctly.

### ***destappl***

Identifies the destination MS application for the send. If you do not specify a destination, the default (ALERT) is used and the MSU is sent to the NetView hardware monitor.

### ***destlu***

Identifies the destination MS LU name for the send. For NetView applications, this is the NetView domain ID. If you do not specify a name, the default of the current NetView domain is used.

### ***origappl***

Identifies the application originating the send request. If you do not specify an application, the default (AONALERT) is used.

## Usage

No checking is done on your MSU data. All major vectors, subvectors, and subfields must be built properly. For more information refer to the SNA formats. The data sent to the hardware monitor (DESTAPPL=ALERT) can be an alert, a resolution, or an MSU containing multiple alerts or resolutions. To be notified when a send fails, code an MS Transport application and specify it for the ORIGAPPL. AON registers the AONALERT MS application during initialization. To see an example of how to build a CP\_MSU acceptable for EZLSMSU, browse the EZLEMSU AON program.

## Return codes

EZLSMSU returns RC=0 if the MSU was sent to the NetView MS Transport. If an error occurs, EZLSMSU returns the return code received from NetView program. The return codes are documented in *IBM Z NetView Programming: PL/I and C*.

## Examples

The following example shows how to send the data built in the variable MSUDATA to the ALERT application (hardware monitor) running in your domain from the AONALERT application:

```
EZLSMSU VARNAME=MSUDATA
```

## Examples

The following example shows how to send the MSU data contained in the variable MSUDATA to the MYAPPL application in domain CNM1A from the MYAPPL application in your current NetView domain:

```
EZLSMSU VARNAME=MSUDATA,DESTAPPL=MYAPPL,DESTLU=CNM1A,ORIGAPPL=MYAPPL
```

## Running Entry and Exit Traces (EZLTRACE)

---

### Purpose

Use the EZLTRACE routine in modules and command processors to:

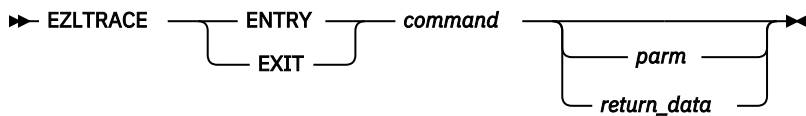
- Provide entry and exit tracing
- Control more detailed levels of tracing

Also, use this routine in your automation modules. Tracing is controlled at the operator panel.

**Note:** This function does not require AON.

## Format

### EZLTRACE



### ENTRY

Indicates start of command

### EXIT

Indicates end of command

### *command*

The module being traced

### *parameters*

Input parameters passed to command (entry)

### *return\_data*

Return values or return code (exit)

## Usage

The *parameters* variables are written to the log when EZLTRACE ENTRY is run. The *return\_data* variable is written to the log when EZLTRACE EXIT is run.

Local variables are set for use by either REXX or NetView command list language for detail tracing:

- EZLTRACEC - NetView command list program
- EZLTRACER - REXX
- Entry message
- Exit message

## Examples

Use the EZLTRACE routine in a REXX procedure to trace entries:

```
parse source . Invoc Ident .
parse upper arg argstring
'EZLTRACE ENTRY 'ident argstring          /* Entry trace */
interpret 'trace' EZLTRACER
```

## Examples

Use the EZLTRACE routine in a REXX procedure to trace exits:

```
ReturnCode = 4
'EZLTRACE EXIT 'ident ReturnCode          /* Exit trace */
exit ReturnCode
```

## Examples

Use the EZLTRACED variable in AON to avoid calls to EZLTRACE when trace is set to NONE:

```
parse source . Invoc Ident .
parse upper arg argstring
'Globalv Getc EZLTRACED'
If Substr(EZLTRACED,1,4)<> 'NONE'
Then Do
    'EZLTRACE ENTRY 'indent argstring
    Trace Value(EZLTRACER)
End
```

## Examples

Use the EZLTRACE routine in a NetView command list to trace entries:

```
MYCODE CLIST
      &CONTROL ERR
      &IDENT = MYCODE
*
      EZLTRACE ENTRY &IDENT &PARMSTR
      &CONTROL &EZLTRACEC
*
```

## Examples

Use the EZLTRACE routine in a NetView command list to trace exits:

```
*
      &RETCODE = 99
      EZLTRACE EXIT &IDENT &RETCODE
      &EXIT &RETCODE
*
```

## SNMP RFC Conversion (FKXECNVT)

---

### Format

Use the FKXECNVT routine to read SNMP MIB RFCs and create entries that can be pasted into */etc/mibs.data*.

### FKXECNVT

►► FKXECNVT — inddname — outddname — mibddname ◄◄

### Parameters

#### *inddname*

Specifies the data set pointer to the input SNMP MIB RFC.

#### *mibddname*

Specifies the data set pointer to the MIB OBJECT file.

#### *outddname*

Specifies the data set pointer of where to put the output *mibs.data* entries.

### Usage

The input file to FKXECNVT must be in textual format. FKXECNVT reads the MIB RFC line by line, and does not handle files stored in string format. Maximum LRECL for all files used by FKXECNVT is 256.

## Examples

```
FKXECNVT USER1.RFCDATA(TN3270E) USER.RFCDATA(TN320UT)
NETVIEW.NETVIEW.V6R3M1.CNMSAMP(FKXMOBJ)
```

This converts the MIB file named TN3270E to a *mibs.data* file called TN320UT and uses FKXMOBJ from the NetView sample library as the base line for the conversion.

## Messages

The following error messages can occur with the FKXECNVT function.

---

| No Files | Explanation                                     |
|----------|-------------------------------------------------|
|          | No parameters were entered for this invocation. |



**Operator response**

Retry the command specifying the Input, Output, and MIB files.

---

**No Output File**
**Explanation**

An output file name was not entered.

**Operator response**

Retry the command specifying the Input, Output, and MIB files.

---

**No MIB Data File name was entered.**
**Explanation**

Parameters were not entered for this invocation.

**Operator response**

Retry the command specifying the Input, Output, and MIB files.

---

**Unable to allocate Input file indsn  
Return Code = RC**
**Explanation**

The ALLOC command failed for the indsn file.

**Operator response**

RC is the return code from the ALLOC command. Use this to determine why the allocate step failed and resolve the problem and then retry the command.

---

**Unable to allocate Output file  
outdsn Return Code = RC**
**Explanation**

The ALLOC command failed for the outdsn file.

**Operator response**

RC is the return code from the ALLOC command. Use this to determine why the allocate command failed, resolve the problem, and then retry the command.

---

**Unable to allocate MIB Data file  
mibdata Return Code = RC**
**Explanation**

The ALLOC command failed for the mibdata file.

**Operator response**

RC is the return code from the ALLOC command. Use this to determine why the allocate command failed, resolve the problem, and then retry the command.

---

**Unable to Read the MIB Data file,  
Received Return Code = RC**
**Explanation**

The REXX EXECIO function failed with return code RC for the MIB data file.

**Operator response**

Use the RC to determine why the EXECIO function failed. Correct the problem and retry the command.

---

**Unable to Write the Output file,  
Received Return Code = RC**
**Explanation**

The REXX EXECIO function failed with return code RC for the Output file.

**Operator response**

Use the RC to determine why the EXECIO failed. Correct the problem and retry the command.

---

**No valid records created for this  
RFC File**
**Explanation**

FKXECNVT was unable to understand any objects in the input RFC file, so no converted records were created.

**Operator response**

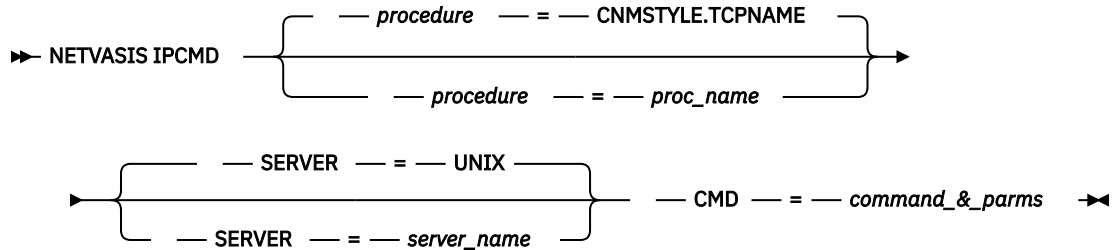
Review the format of the RFC, and ensure that it complies with the format restrictions defined in this book.

## TCP/IP Command Support (IPCMD)

### Format

Use the IPCMD routine to issue any TSO or UNIX command.

### IPCMD



### Parameters

#### *procedure*

Specifies the z/OS procedure. Valid values are PROC and STACK.

#### **CMD**

Specifies the TCP/IP command and parameters.

#### *proc\_name*

Specifies the name of the z/OS procedure. The default is defined in the CNMSTYLE member with the common global variable TCPNAME. The value for TCPNAME is the *proc\_name* of the TCP/IP stack. Refer to *IBM Z NetView Administration Reference* for more information about setting this value.

#### **SERVER**

Specifies the server name, either TSO or UNIX. The default is UNIX.

### Usage

- Specify NETVASIS when entering UNIX commands or commands that are case-sensitive.
- Parameters are not positional, but enter `CMD=command_&_parms` last to enable proper parameter parsing.
- The default stack and server can be changed by modifying the CNMSTYLE member. Refer to the *IBM Z NetView Administration Reference* for more information.

### Examples

To use the UNIX server to ping host name localhost, using the default TCP/IP for z/OS stack name, enter the following command:

```
netvasis IPCMD CMD=oping localhost
```

The response is returned to the caller.

To use the first available TSO server to ping host name localhost, enter the following command:

```
netvasis IPCMD SERVER=TSO CMD=ping localhost
```

The response is returned to the caller.

## Programmatic Interface for IP Trace

---

### Purpose

IP Trace uses an application programming interface and a series of common global variables that must be set before starting the IP Trace program. “Restrictions” on page 319 provides information about these global variables. See the FKXETSMP sample that is shipped with the NetView product for additional information.

### Format

The format of the FKXETRA1 program is as follows:

```
FKXETRA1 sp,type,action,writer,chron_start_date_time,duration,operator
```

### Parameters

#### **sp**

The service point ID. This variable directs FKXETRA1 to use global variables to properly set up the trace.

#### **type**

The type of trace; CTRACE or PKT.

#### **action**

To start (START) or stop (STOP) the trace.

#### **writer**

The name of the source JCL used to create the external writer where trace data is stored. This is used for active and delayed traces. Refer to *MVS Diagnosis: Tools and Service Aids* for more information about creating source JCL for an external writer.

#### **chron\_start\_date\_time**

The date and time to start the trace, in the format YYYY-MM-DD-HH.MM.SS. If date and time are not specified, the trace begins immediately. Refer to the CHRON command in the NetView online help for more information about the date and time format.

#### **duration**

The length of time the trace is to run, in the format HH.MM.SS. This is used for active and delayed traces. Refer to the CHRON command in the NetView online help for more information about the time format.

#### **operator**

The operator task on which to schedule a delayed trace timer for this request. A delayed timer can be set for a time when the requesting operator ID is not logged on to NetView, which causes the timer to fail. This option enables the user to specify the name of the task to be active at the designated time to allow for the successful completion of the task. The owner of the trace is still the original requestor.

**Note:** The task name must be defined in DSIPARM member DSIPOPF and can be specified only for delayed traces.

### Restrictions

To use IP Trace, the following required common global variables must be set:

- **For both Packet Trace and Component Trace**

FT.sp.TARGET

#### **sp**

The service point or stack for which these global variables are set.

**target**

The target system and stack where the trace commands are sent. The target is a stemmed variable, which consists of the NetView domain and TCP/IP procedure name of the stack, in the format DOMAIN.PROC for the stack defined by sp.

- **For Component Trace**

FT.sp.CTRC.OPTS1  
 FT.sp.CTRC.OPTS2  
 FT.sp.CTRC.IPADD1  
 FT.sp.CTRC.IPADD2  
 FT.sp.CTRC.IPADD3  
 FT.sp.CTRC.IPADD4  
 FT.sp.CTRC.PORT  
 FT.sp.CTRC.JOBS  
 FT.sp.CTRC.ASIDS

**sp**

The service point or stack for which these global variables are set.

**OPTS1**

A list of CTRACE options. You can enter up to 20 options, separated by commas.

**OPTS2**

A list of up to 20 additional component trace options.

**IPADD1**

A list of IP addresses. You can enter up to four addresses.

**IPADD2**

A list of up to four additional IP addresses.

**IPADD3**

A list of up to four additional IP addresses.

**IPADD4**

A list of up to four additional IP addresses.

**PORT**

A list of up to 16 IP ports, in the range 0–65535.

**JOBS**

A list of up to 16 z/OS job names, from 1–8 characters.

**ASIDS**

A list of up to 16 z/OS address space identifiers, in the form of four hexadecimal numbers.

- **For Packet Trace**

FT.sp.PKT.LINKS  
 FT.sp.PKT.OPT.n

This global variable contains a total count of optional globals defined for this service point. For example, if there are six FT.sp.PKT.OPT globals defined, the value of FT.sp.PKT.LINKS is 6. Any number of optional globals can be set for all links on a service point. Each of these globals contains the following format:

```
LinkName, Len, Proto, Ipadd, Subnet, DestPort, SourcePort
```

**Note:** Use one global for each service point. When using the 3270 interface, all of these global variables are managed automatically.

**LinkName**

The TCP/IP device name on the service point traced.

**Len**

Specifies that a truncated portion of the IP packet can be traced. You can specify a length in the range 1–65535. The maximum value is FULL, which captures the entire packet.

**Proto**

The protocol collecting data. Valid values are:

- Asterisk (\*), which specifies that packets of any protocol are traced
- ICMP
- RAW
- TCP
- UDP
- *number* (in the range 0–255)

**IPadd**

The IP address that is compared with both the source and destination addresses of inbound and outbound packets. If either the source or destination address of a packet matches the specified IP address, the packet is traced. If *IPadd* is blank or an asterisk (\*) is specified, all IP addresses are traced.

**Subnet**

The subnet mask that applies to the host and network portions of the IP address specified on the corresponding *IPadd*.

**DestPort**

The port number that is compared with the destination port of inbound and outbound packets. The port number is an integer in the range 1–65535. If the destination port of a packet is the same as the specified port number, the packet is traced. This comparison is only performed for packets using either the TCP or UDP protocol. Packets using other protocols are not traced. If *DestPort* is omitted, there is no checking of the destination port of packets. If an asterisk (\*) is specified, packets of any protocol and any destination port are traced.

**SourcePort**

The port number that is compared with the source port of inbound and outbound packets. The port number is an integer in the range 1–65535. If the source port of a packet is the same as the specified port number, the packet is traced. This comparison is only performed for packets using either the TCP or UDP protocol. Packets using other protocols are not traced. If *SourcePort* is omitted, there is no checking of the source port of packets. If an asterisk (\*) is specified, packets of any protocol and any source port are traced.

**Usage**

The IP Trace function uses many z/OS commands to control and manage TCP/IP component and packet traces. The following issues can affect the functionality of the AON IP Trace:

- Each task or operator requires a unique z/OS console to issue z/OS commands and receive command responses.
- The CNMCSSIR task must be active and receiving unsolicited messages.
- The time required to receive responses to IP trace commands can vary greatly. Customize the following statement in the CNMSTYLE member to set the time delay value (in seconds):

```
COMMON.EZLIPTTraceJCLWait = 2 //AON wait time for source JCL errors response.
```

**Examples**

To start Packet Tracing for YOURHOST, you must set up the following common global variables:

```
FT.YOURHOST.PKT.LINKS      = 3
FT.YOURHOST.PKT.OPT.1     = LOOPBACK,FULL,*,*,255.255.255.255,*,*
FT.YOURHOST.PKT.OPT.2     = TCPIPLINK,FULL,*,*,255.255.255.255,*,*
```

```

FT.YOURHOST.PKT.OPT.3    = TCPIPLINKN000000,FULL,*,*,255.255.255.255,*,*
FT.YOURHOST.TARGET      = LOCAL.TCPIP

```

To start a packet trace on this system, now, code the following within a REXX Clist:

```

/*SAMPLE CODE TO START PKT TRACE NOW */
'FKXETRA1 YOURHOST,PKT,START,PKTWITER'
SAY 'PKT TRACE REQUEST ENDED WITH RC' RC
/* DISPLAY FKX MESSAGE, TOO          */
'GLOBALV GETT EXLMSGTXT'
SAY EZLMSGTXT
EXIT

```

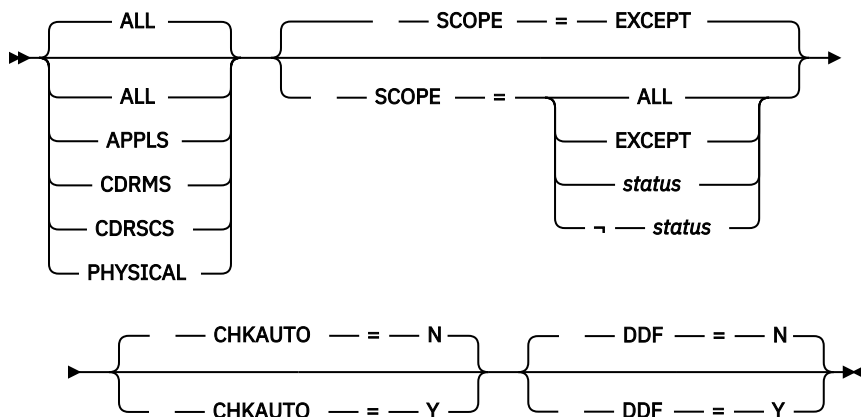
## SNA Resource Automation (FKVESYNC)

### Format

Resources can fail while NetView (or AON/SNA) is down. Because AON was not notified of the failure, AON automation routines do not request to recover the failed resources.

FKVESYNC is provided to assist you with automating your SNA resources. You can use FKVESYNC to determine which resources are down, and then drive AON automation routines to recover the failed resources.

### FKVESYNC



### Parameters

Defines which resource types to automate.

#### **ALL**

Automates all resources regardless of status. This is the default.

#### **APPLS**

Automates only the applications.

#### **CDRMS**

Automates only the cross-domain resource managers.

#### **CDRSCS**

Automates only the cross-domain resources.

#### **PHYSICAL**

Automates only the physical components. This includes NCPs, LUs, PUs, and switched nodes.

**SCOPE**

Defines the extent to which the resource is automated.

**EXCEPT**

Automates all resources that are currently not in an ACTIVE status. This is the default.

**ALL**

Automates all resources regardless of status.

***status***

Automates only resources with this status.

***¬status***

Automate all resources that do not have this status.

**CHKAUTO**

Determines if the resources displayed are to be checked against the recovery statements in the control file when calling the NETSTAT command. The default is N.

**DDF**

Determines if the information is passed to the Dynamic Display Facility (DDF) when calling the NETSTAT command. The default is N.

**Usage**

- FKVESYNC must be called from operator written code that is driven when NetView is initialized.
- FKVESYNC uses the AON NETSTAT command to determine the status of your SNA resources. In large networks with many failed resources, this process can take some time and CPU cycles.

**Examples**

To check the status of all CDRMs, enter the following command:

```
FKVESYNC CDRMS
CNM377I FKVESYNC : INPUT ACCEPTED AND BEING PROCESSED ...
PLEASE WAIT
EZL001I REQUEST NETSTAT WAS SUCCESSFUL FOR OPER4
```

This detects all of your failed CDRM resources and drives AON automation routines to attempt the recovery of each. When you see message EZL001I, then AON recovery has been driven for each failed resource.





---

## Chapter 27. Tailoring gateways and focal points

Gateways are inbound and outbound automated operators. Focal points are domains to which all notifications are forwarded. Gateways and focal points enable you to define the following:

- Focal point host
- Backup focal point host
- Intermediate host or hosts
- Distributed host or hosts
- Adjacent host environment or environments

The message/alert router forwards messages and alerts from multiple hosts to a single host. This enables a network operator to receive all the network alert messages at a single console. The message destination is controlled within control file entries. Routed messages contain host origination identification.

To enable notification forwarding for AON you must complete the following steps:

- Determine the notification forwarding hierarchy.
- Tailor the NetView definitions.
- Define the focal point and backup focal point entries.
- Define the outbound gateway operator entries.
- Add NetView outbound and inbound operator IDs.
- Implement the changes in a host-by-host approach.

Refer to the example following each step for more detail.

### 1. Determine the notification forwarding hierarchy.

You must determine the relationship between the different systems in a notification forwarding network. You must designate a primary focal point where you can send the messages. You can designate a backup for the primary focal point for situations when the primary focal point is unavailable. You must define the connectivity between the hosts in a tree-structured hierarchy, so that all notification messages are forwarded up to the primary focal point.

You should develop a chart depicting the notification forwarding hierarchy. [Figure 201 on page 326](#) shows an example of a notification forwarding chart.

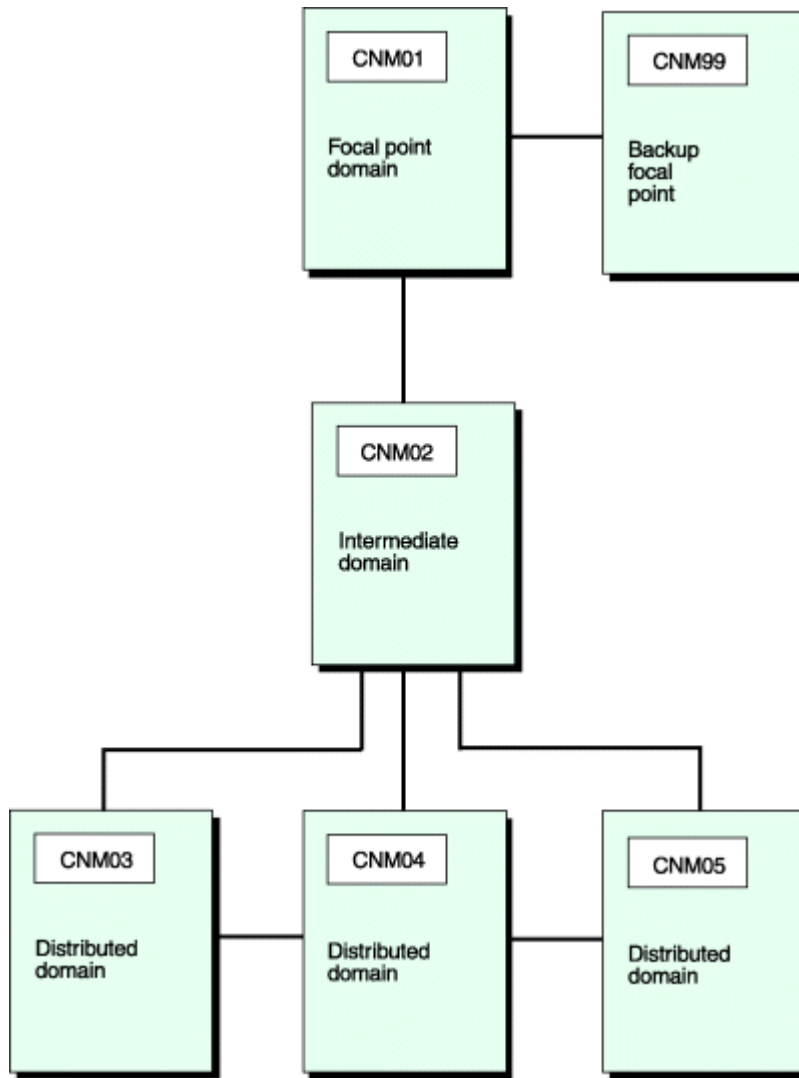


Figure 201. Notification Forwarding Hierarchy Chart Example

In this example, CNM01 is the primary focal point and CNM99 is the backup focal point. The backup focal point does not receive notifications unless the primary focal point is down. As soon as the primary focal point is operational and the gateway connections are reestablished, the backup stops receiving notifications.

2. Tailor the NetView definitions.

**Note:** It is assumed that you have implemented all VTAM definitions required to permit cross-domain logons to remote NetView programs.

Add an RRD statement (if one does not exist) in the CNMSTYLE member for each host with which this host directly communicates. This definition supports notification forwarding.

3. Define the focal point and backup focal point entries.

The focal point control file entry identifies the NetView domain where you forward the notification messages. Each focal point control file entry defines the final destination. Add a focal point definition control file entry to each distributed host.

An example of a focal point definition entry is:

```
FORWARD FOCALPT,PRI=CNM01,BKUP=CNM03
```

4. Define the outbound gateway operator entries.

The outbound gateway operator establishes and maintains all of the outbound connections to the domains defined in the GATEWAY control file entries. The outbound gateway operator is defined in the AUTOOPS entry of the control file as GATOPER.

```
AUTOOPS    GATOPER, ID=name
```

The GATEWAY entries define the other domains with which a session should be established so that notification messages can be forwarded to or from these domains.

```
GATEWAY    CNM02, DESC='NETVIEW CNM02', PASSWORD=password
```

NetView-NetView (NNT) sessions are established to all the domains defined by attempting to log on to the inbound gateway operator of each domain. Each inbound gateway operator has the same operator ID as the sending outbound gateway operator.

5. Define NetView outbound and inbound operator IDs to the DSIOPF profile.

Define the inbound and outbound operator IDs to support notification forwarding. A sample operator ID (GATCNM01) is provided.

Define one outbound operator ID. The outbound operator ID is the ID defined in the AUTOOPS GATOPER control file entry in your domain.

Define one inbound operator ID for each domain defined in a GATEWAY control file entry. For the inbound operator ID, use the ID defined in the AUTOOPS GATOPER control file entry in the other domains.

**Note:** You must also specify the logon ID in the GATEWAY statement of the remote domain.

6. Implement the changes in one NetView host at a time.

The notification forwarding implementation is best approached as a top-down implementation starting with the primary focal point, then the distributed hosts. This approach works best because the focal point is ready to handle the forwarded notifications when notification forwarding is enabled in the remote hosts.

When using a top-down approach, if your installation has not implemented message forwarding, the messages at the remote hosts are displayed to notification operators at the remote hosts. After message forwarding is turned on, the messages are routed to the focal point and appropriately handled. Local notification operators, if any remain, are also notified appropriately.

## AON focal point compatibility

The following restrictions apply to the AON-supported focal point compatibility:

- AON does not support intermediate focal point domains. You can use full-function domains as intermediate routing domains, but AON does not act on data passing through an intermediate routing domain, and does not update DDF in the intermediate domain.
- Downlevel distributed domains must be directly connected to the focal point domain. You cannot use any ADJNETV statements.
- Downlevel domains might not act as an intermediate. You cannot specify them in the DOMAIN or ALTNETV parameters on any ADJNETV statement.
- Downlevel focal point domains must have all distributed domains directly connected. You cannot use any ADJNETV statements.
- Downlevel compatibility is provided to ease migration to current levels of gateway support. Downlevel compatibility is not intended for long-term production operation. Migration of all domains to full-function support should be accomplished as quickly as possible.

Table 7 on page 328 illustrates allowable communications setups. Wherever a downlevel domain is connected to a full-function domain, the full-function domain should be AON. Wherever two full-function domains are connected, the full-function domains can be any supported full-function product.

| Table 7. Communications Setup in Focal Point Domain Environment                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Support type                                                                                                                                                                                                                                                                                                                                                                                                          | Communications setup                                                                                                                                                                                                                                                                                                                                                              |
| Focal point support                                                                                                                                                                                                                                                                                                                                                                                                   | 1. DL(dn) —> AON(fp)<br>2. AON(dn) —> DL(fp) (except AON SolutionPac)<br>3. FF(dn) —> FF(fp)<br>4. FF(dn) —> FF(ir) —> FF(fp)                                                                                                                                                                                                                                                     |
| SEND CMD support                                                                                                                                                                                                                                                                                                                                                                                                      | 1. DL <—> AON <a href="#">“2” on page 328</a><br>2. AON <—> DL <a href="#">“2” on page 328</a><br>3. FF <—> FF<br>4. FF <—> FF <—> FF<br>5. DL <—> AON <—> DL <a href="#">“3” on page 328</a><br>6. DL <—> AON <—> FF <a href="#">“2” on page 328</a> , <a href="#">“3” on page 328</a><br>7. FF <—> AON <—> DL <a href="#">“2” on page 328</a> , <a href="#">“3” on page 328</a> |
| <b>Notes:</b><br>1. The abbreviations in this table have the following meanings:<br><b>AON</b><br>Full-function domain (AON)<br><b>FF</b><br>Full-function domain<br><b>DL</b><br>Any downlevel domain<br><b>(dn)</b><br>Distributed domain<br><b>(fp)</b><br>Focal point domain<br><b>(ir)</b><br>Intermediate routing domain<br>2. SEND CMD support for these is the same.<br>3. Not valid for focal point support. |                                                                                                                                                                                                                                                                                                                                                                                   |

Figure 202 on page 328 illustrates a typical migration environment:

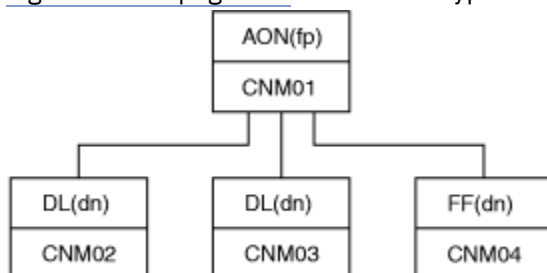


Figure 202. Typical Migration Environment

Table 8 on page 329 illustrates the focal-point and SEND CMD support for a typical migration environment.

| Table 8. Focal-point and SENDCMD Support for Typical Migration Environment |                                |                      |
|----------------------------------------------------------------------------|--------------------------------|----------------------|
| Support type                                                               | Between domains                | Communications setup |
| Focal point                                                                | CNM02 to CNM01                 | DL(dn) —> FF(fp)     |
|                                                                            | CNM03 to CNM01                 | DL(dn) —> FF(fp)     |
|                                                                            | CNM04 to CNM01                 | FF(dn) —> FF(fp)     |
| SENDCMD                                                                    | CNM02 to CNM01                 | DL <—> FF            |
|                                                                            | CNM01 to CNM03                 | FF <—> DL            |
|                                                                            | CNM04 to CNM01                 | FF <—> FF            |
|                                                                            | CNM02 to CNM03 (through CNM01) | DL <—> FF <—> DL     |
|                                                                            | CNM02 to CNM04 (through CNM01) | DL <—> FF <—> FF     |
|                                                                            | CNM04 to CNM03 (through CNM01) | FF <—> FF <—> DL     |

Figure 203 on page 329 shows one possible example of a full-function environment.

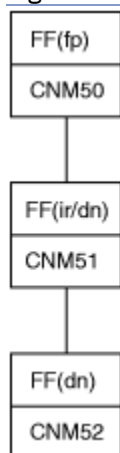


Figure 203. Example Full-function Environment Diagram

Table 9 on page 329 lists focal-point and SENDCMD support for the full-function environment as previously illustrated.

| Table 9. Focal-point and SENDCMD Support for Full-function Environment |                              |                            |
|------------------------------------------------------------------------|------------------------------|----------------------------|
| Support type                                                           | Between domains              | Communications setup       |
| Focal point                                                            | CNM51 to CNM50               | FF(dn) —> FF(fp)           |
|                                                                        | CNM52 to CNM50 through CNM51 | FF(dn) —> FF(ir) —> FF(fp) |
| SENDCMD                                                                | CNM51 to CNM50               | FF <—> FF                  |
|                                                                        | CNM52 to CNM50 through CNM01 | FF <—> FF <—> FF           |

Sample definitions for this full-function environment for each domain follow:

- For domain CNM50:

```
AUTOOPS GATOPER, ID=GATCNM50
GATEWAY CNM51, PASSWORD=password
ADJNETV CNM52, DOMAIN=CNM51
```

- For domain CNM51:

```
AUTOOPS GATOPER, ID=GATCNM51
GATEWAY CNM50, PASSWORD=password
GATEWAY CNM52, PASSWORD=password
FORWARD FOCALPT, PRI=CNM50
```

- For domain CNM52:

```
AUTOOPS GATOPER, ID=GATCNM52
GATEWAY CNM51, PASSWORD=password
ADJNETV CNM50, DOMAIN=CNM51
FORWARD FOCALPT, PRI=CNM50
```

## Notification forwarding example

Assume that the AON operator OUTMAN wants to forward a request (AON notify) to CNM02 from CNM01. The outbound gateway operator for CNM01 is GATCNM01. The outbound gateway operator for CNM02 is GATCNM02. The inbound gateway operator for CNM02 that receives notification messages from CNM01 is GATCNM01.

The program sends the request to the CNM01 outbound gateway operator (GATCNM01). The outbound gateway operator determines if the request is for the CNM01 domain. In this example, it is not. The outbound gateway operator routes the request to the inbound gateway operator in domain CNM02 (GATCNM01) from domain CNM01. When the inbound gateway operator receives the request, it is sent to the outbound gateway operator for CNM02 (GATCNM02). The outbound gateway operator from CNM02 determines that the forwarded request is for this domain and issues the request. Refer to [Figure 204 on page 330](#).

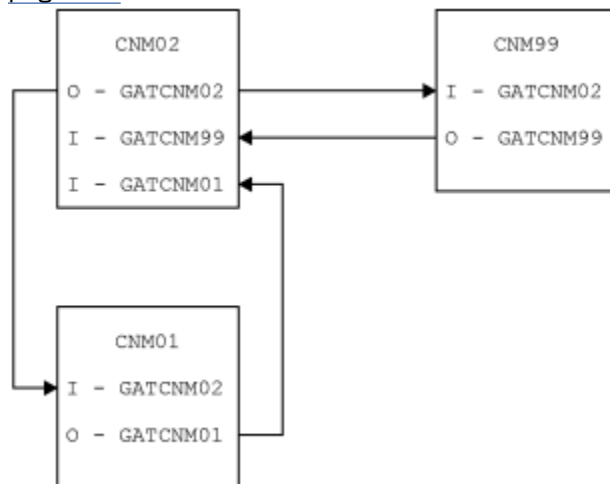


Figure 204. Notification Forwarding Example

[Table 10 on page 331](#) lists the required control file entries for the domains illustrated in [Figure 204 on page 330](#).

Table 10. Required Control File Entries

| Domain | Control file entry                                    | Example                                                                                                                       |
|--------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| CNM01  | AUTOOPS<br>Automation operator                        | AUTOOPS GATOPER, ID=GATCNM01                                                                                                  |
|        | GATEWAY<br>Forwarded domains and operator definitions | GATEWAY CNM02, DESC='NEXT DOMAIN',<br>PASSWORD= <i>pswd</i>                                                                   |
|        | FORWARD FOCALPT<br>Focal point definitions            | FORWARD FOCALPT, PRI=CNM99                                                                                                    |
|        | ADJNETV<br>Focal point definitions                    | ADJNETV CNM99, DOMAIN=CNM02                                                                                                   |
| CNM02  | AUTOOPS<br>Automation operator                        | AUTOOPS GATOPER, ID=GATCNM02                                                                                                  |
|        | GATEWAY<br>Forwarded domains and operator definitions | GATEWAY CNM99, DESC='NEXT DOMAIN',<br>PASSWORD= <i>pswd</i><br><br>GATEWAY CNM01, DESC='DOWNSTREAM',<br>PASSWORD= <i>pswd</i> |
|        | FORWARD FOCALPT<br>Focal point definitions            | FORWARD FOCALPT, PRI=CNM99                                                                                                    |
| CNM99  | AUTOOPS<br>Automation operator                        | AUTOOPS GATOPER, ID=GATCNM99                                                                                                  |
|        | GATEWAY<br>Forwarded domains and operator definitions | GATEWAY CNM02, DESC='DOWNSTREAM',<br>PASSWORD= <i>pswd</i>                                                                    |
|        | FORWARD FOCALPT<br>Focal point definitions            | FORWARD FOCALPT, PRI=CNM99                                                                                                    |
|        | ADJNETV<br>Focal point definitions                    | ADJNETV CNM01, DOMAIN=CNM02                                                                                                   |





## Chapter 28. AON user exits

With AON, you can write user exits to tailor automation for a specific task. An exit can call any command, program, command list, or EXEC that can be issued from a NetView command line. Exits are processed by an automation operator, so processing restrictions that apply to automation operator-run programs also apply to exit programs. See *IBM Z NetView Programming: REXX and the NetView Command List Language* for information about these restrictions.

User exits can both read and alter task global variables or return codes set by some common routines. These common routines guide further automation notification and logging or initiate activity based on failure or other conditions defined to AON. You can write exits that read and alter the task global variables and return codes when the following AON functions are issued:

- SNA resource information gathering (EZLEAGRN)
- Thresholds checking (EZLEATHR)
- Automation flag analysis based on recovery statements (EZLECAUT)
- Resource recovery attempts on MONIT intervals (EZLECATV)
- AON message issued to logs or operators
- NCP recovery

Exits started by common routines can be called with input parameters. Valid input parameters for exit routines include the task global variables available to the user exit (any of the task global variables listed in [Table 12 on page 334](#) can be passed as a parameter by prefixing the task global variable name with an ampersand, thus creating a parameter such as &RESNAME or &RESTYPE, for example), user-defined literals, and the resource name sent to the common routine.

Exits to be called are defined in the control file. Default exits are coded on the ENVIRON EXIT entry. These exits are called every time the common routine is started. [Table 11 on page 333](#) identifies which routine calls which exit.

| Table 11. Common Routines That Call Exits                             |                                |                                            |
|-----------------------------------------------------------------------|--------------------------------|--------------------------------------------|
| Common routine                                                        | Policy entry that sets default | Control file entry that sets specific call |
| EZLEAGRN                                                              | ENVIRON EXIT (EXIT05)*         | n/a                                        |
| EZLEATHR                                                              | ENVIRON EXIT (EXIT06)*         | THRESHOLDS                                 |
| EZLECAUT                                                              | ENVIRON EXIT (EXIT07)*         | RECOVERY                                   |
| AON messaging                                                         | ENVIRON EXIT (EXIT08)*         | n/a                                        |
| EZLECATV                                                              | ENVIRON EXIT (EXIT09)*         | MONIT                                      |
| EZLENTFY                                                              | n/a                            | NOTIFY EXIT (EXIT10)                       |
| EZLENFRM                                                              | SETUP EXIT (EXIT11)+           | n/a                                        |
| EZLENFRM                                                              | SETUP EXIT (EXIT12)+           | n/a                                        |
| <b>Note:</b> Specific entries override the default entries.           |                                |                                            |
| *<br>Indicates that the entry is defined in the control file.         |                                |                                            |
| +<br>Indicates that the entry is defined in the Inform Policy member. |                                |                                            |

You can also define different exits on the control file entry applicable to a particular resource or class of resources for each of the exit points (by using wildcard characters or resource types in definitions). These are defined on the RECOVERY, THRESHOLDS, and MONIT entries for the resource. You can specify up to 10 exits for each common routine. If you specify more than one exit routine, each exit is called until the first nonzero return code is received from the user-written exit routine. When multiple exits are called, AON uses the task global variable values set by the series of exits, up to and including the last exit issued. Table 12 on page 334 identifies the task global variable values set by exits.

| <i>Table 12. Task Global Variables Available to User Exits</i> |                    |                     |                                                                                                                              |                                  |                                                         |
|----------------------------------------------------------------|--------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------|----------------------------------|---------------------------------------------------------|
| <b>Common Routine</b>                                          | <b>Exit Number</b> | <b>Policy Entry</b> | <b>Task Global Variable</b>                                                                                                  | <b>Exit Task Global Variable</b> | <b>Issued</b>                                           |
| EZLEAGRN                                                       | EXIT05             | n/a                 | RESNAME<br>RESTYPE<br>RESEXT<br>RESSTAT<br>RESMAJ<br>RESLINE<br>RESNODE<br>RESPU<br>RESSA<br>RESSW                           | EZLEAGRNRC                       | Before EZLEAGRN returns control to the calling program. |
| EZLEATHR                                                       | EXIT06             | THRESHOLDS*         | RESNAME<br>RESTYPE<br>EZLTRSHLD<br>EZLCRIT<br>EZLFREQ<br>EZLINFR<br>EZLNONE                                                  | EZLEATHRRC                       | After the type of threshold exceeded is determined.     |
| EZLECAUT                                                       | EXIT07             | RECOVERY*           | RESNAME<br>RESTYPE<br>RESEXT<br>RESSAT<br>RESMAJ<br>RESLINE<br>RESNODE<br>RESPU<br>RESSA<br>RESSW<br>EZLTIMERD<br>EZLNWINDOW | EZLECAUTRC                       | Each time the recovery flag is checked for a resource.  |
| AON Messaging                                                  | EXIT08             | n/a                 | EZLCONVERT<br>EZLMSGTXT                                                                                                      | n/a                              | When issuing an AON message.                            |

Table 12. Task Global Variables Available to User Exits (continued)

| Common Routine | Exit Number | Policy Entry | Task Global Variable                                                                                                                                                                                                                         | Exit Task Global Variable | Issued                                       |
|----------------|-------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------------------------------|
| EZLECATV       | EXIT09      | MONIT*       | RESNAME<br>RESTYPE<br>RESSTAT<br>RESMAJ<br>RESLINE<br>RESNODE<br>RESPU<br>RESSA<br>RESSW<br>EZLINTVL<br>EZLMONIT                                                                                                                             | EZLECATVRC                | Before deactivation of a resource.           |
| EZLENTFY       | EXIT10      | NOTIFY*      | RESNAME<br>RESTYPE<br>RESEXT<br>RESSTAT<br>RESMAJ<br>RESLINE<br>RESNODE<br>RESPU<br>RESSA<br>RESSW<br>EZLTIMERD<br>EZLNWINDOW<br>EZLNIFYTYPE<br>EZLINTVL<br>EZLMONIT<br>EZLNIFYLIST<br>EZLTRSHLD<br>EZLCRIT<br>EZLFREQ<br>EZLINFR<br>EZLNONE | n/a                       | Before any notification actions.             |
| EZLENFRM       | EXIT11      | SETUP+       | EZLPOLEX                                                                                                                                                                                                                                     | n/a                       | Before consulting the inform policy.         |
| EZLENFRM       | EXIT12      | SETUP+       |                                                                                                                                                                                                                                              | n/a                       | Before calling the inform interface routine. |

**Attention:** The task global variables contain data that guides AON recovery processing. Indiscriminate changes to the task global variables can cause unpredictable results.

\*

Indicates that the entry is defined in the control file.

+

Indicates that the entry is defined in the Inform Policy member.

During initialization common global variables are set for the default exits for each exit point. If the ENVIRON EXIT control file entry is modified, reload the control file. The new exit common global variable

values are changed. Exit definitions on the THRESHOLDS, RECOVERY, or MONIT entries take effect immediately and do not require the control file to be reloaded when they are updated online.

**RESNAME**

Name of the resource

**RESTYPE**

Type of resource, such as PU or IPHOST

**RESEXT**

For SNA resources; contains data from an IST0751 message (Display NET response)

**RESSTAT**

Current automation status of the resource

**RESMAJ**

For SNA resources; the major node that the resource is attached to

**RESLINE**

For SNA resources; the line that the resource is attached to

**RESNODE**

For SNA resources; the name of the higher resource that the current resource is attached to

**RESPU**

For SNA resources; the physical unit that the resource is attached to

**RESSA**

For SNA resources; the subarea that the resource is in

**RESSW**

For SNA resources; the switched major node that the resource is attached to

**EZLNTIMERD**

The time interval until the end of a NOAUTO window

**EZLNWINDOW**

The name of the RECOVERY policy definition used when in a NOAUTO window

**EZLNTFYTYPE**

The type of notification, such as REMIND

**EZLINTVL**

For SNA resources; the current MONIT interval number

**EZLMONIT**

For SNA resources; the next MONIT interval number.

**EZLNTFYLIST**

The list of Notify Actions, such as MSG or DDF

**EZLTRSHLD**

Set to the type of threshold condition, such as CRIT, FREQ, INFR, or NONE

**EZLCRIT**

Flag that is set to Y when a Critical Threshold condition has occurred (Y, N)

**EZLFREQ**

Flag that is set to Y when a Frequent Threshold condition has occurred (Y, N)

**EZLINFR**

Flag that is set to Y when an Infrequent Threshold condition has occurred. (Y, N)

**EZLNONE**

Flag that is set to Y when no threshold condition has occurred (Y, N)

## **EXIT01 - EXIT04 processing during NCP recovery**

---

With NCP recovery, you can specify user exits to run during different stages. Four separate user exit statements are used for each of the four stages where an exit can be used.

## EXIT01

User-defined command or command list. The program defined in EXIT01 runs after the memory dump message response. Sample exit routines are provided (FKVEX01 in CNMSAMP).

## EXIT02

User-defined command or command list. The program defined in the EXIT02 exit runs after the memory dump is complete and the memory dump time exceeded timer is purged. Sample exit routines are provided (FKVEX02 in CNMSAMP).

## EXIT03

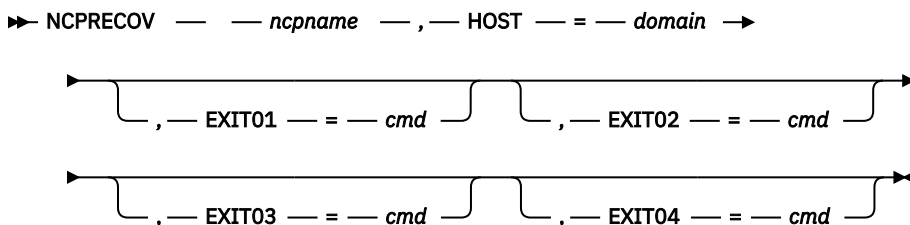
User-defined command or command list. The program defined in the EXIT03 exit runs after the reload message response. Sample exit routines are provided (FKVEX03 in CNMSAMP).

## EXIT04

User-defined command or command list. The program defined in the EXIT04 exit runs after the load is complete and the load time exceeded timer is purged. Sample exit routines are provided (FKVEX04 in CNMSAMP).

## Format

### NCPRECOV



### *command*

Specifies an exit to be run or a command to be issued.

### *ncpname*

Specifies the NCP for which these exits are being used.

## Usage

To use these exits, copy them into your user library from the DSICLD data sets.

If response to the option to dump the NCP is specified as N, then EXIT02 is not run because no message is received, thus indicating that a memory dump is complete.

If response to the option to load the NCP is specified as N, then EXIT04 is not run because no message is received, thus indicating that a load is complete.

Variable parameters cannot be passed for these exits. For example, USERPGM &NCP is not valid.

**Attention:** The task global variable (TGLOBALS) contain data that guides AON/SNA recovery processing. Indiscriminate changes to these task global variables (TGLOBALS) can cause unpredictable results.

## Examples

In the following example, AON runs the FKVEX01 exit when it replies to the request to dump message. AON runs the FKVEX02 exit when the memory dump is complete. AON runs the FKVEX03 exit when it responds to the reloading of the NCP. AON runs the FKVEX04 exit when it completes the reloading of the NCP.

```
NCPRECOV NCPABC,
        HOST=CNM01,DUMP=(Y,N),RELOAD=(Y,N),
        LINKSTA=123-S,DUMPSTA=123-S,
        LOADTIME=00:15,DUMPTIME=00:10,
        EXIT01=FKVEX01,
        EXIT02=FKVEX02,
```

## EXIT05 processing (EZLEAGRN)

### Purpose

Exit processing for EZLEAGRN is issued just before EZLEAGRN returns control to the calling program. AON gathers all information about a resource from a VTAM DISPLAY NET command and sets the following task global variables:

#### RESLINE

The higher node line of the resource

#### RESMAJ

The major node of the resource

#### RESNODE

The adjacent major node of the resource

#### RESPU

The higher node controller of the resource

#### RESSA

The subarea of the resource

#### RESSW

The switched major node of the resource

#### RESSTAT

Resource status

#### RESTYPE

Resource type

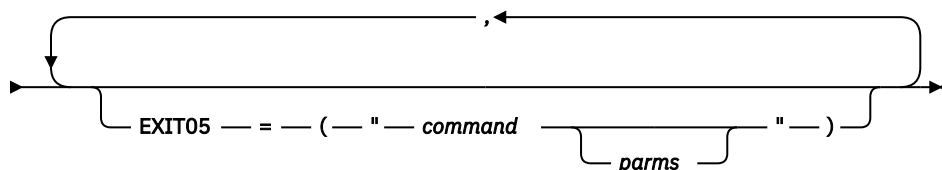
Any of these task global variables can be altered. The return code passed to the calling program by EZLEAGRN can be set from a user exit. The EZLEAGRN return code does not guide automation activity in AON, but the attribute can be adjusted by the exit. The altered return code is set by the user exit and placed in the task global variable, EZLEAGRNC.

EZLEAGRN exits can be defined only in the control file on the ENVIRON EXIT by the EXIT05 parameter. The previously mentioned task global variables can be passed to the exit by specifying the task global variable name preceded by an ampersand (&). The resource name can also be passed to the exit routine by specifying &RESNAME.

### Format

#### ENVIRON EXIT

➡ ENVIRON EXIT ➡



### Parameters

#### command

Specifies the exit or command to be issued.

#### parms

Specifies the parameters to be passed to the exit or the command to be issued.

## Usage

A sample exit is provided in CNMSAMP member FKVEXT05.

Run all exits inline. Do not route exits to another task.

Code user exits to change any of the task global variables set by EZLEAGRN. These task global variables are set from the results of a VTAM DISPLAY command for the resource.

If any user-coded routines are written in REXX and a task global variable is changed, retrieve the new value by using the REXX 'GLOBALV GETT' statement.

Set task global variable EZLEAGRNRC in the user exit to alter the return code for EZLEAGRN.

All defined user exits are called in the order they are coded in the control file. All exits are issued until the user exit returns a nonzero return code. This return code is different from the return code set in task global variable EZLEAGRNRC.

EXIT05 can be used to change resource data used by AON or set additional task global variables from running other commands or message processing. For example, you can set up additional resource types to be used to guide recovery flags and DDF logging by analyzing the resource name and changing the RESTYPE task global variable.



**Attention:** The task global variables contain data that guides AON recovery processing. Indiscriminate changes to the task global variables can cause unpredictable results.

## Examples

For this example, each time EZLEAGRN is issued, FKVEXT05 is issued and passed the value of RESNAME as the first parameter, the literal TEST as the second parameter, and the value of task global variable RESTYPE as the third parameter.

```
ENVIRON EXIT,EXIT05=("FKVEXT05 &RESNAME TEST &RESTYPE")
```

## EXIT06 processing (EZLEATHR)

---

### Purpose

Exit processing for EZLEATHR is issued after EZLEATHR performs threshold analysis and before operator notifications have been issued. The resource name and type must be passed to EZLEATHR. AON sets the following task global variables in EZLEATHR:

#### EZLCRIT

Indicates whether a critical threshold has been exceeded (Y,N)

#### EZLFREQ

Indicates whether a frequent threshold has been exceeded (Y,N)

#### EZLINFR

Indicates whether an infrequent threshold has been exceeded (Y,N)

#### EZLNOTIFY

Specifies whether any operator notifications are to take place (Y,N)

#### EZLRECORD

Specifies whether this error is to be saved in the status file (Y,N)

#### EZLTRSHLD

Threshold exceeded value (NONE,CRIT,FREQ,INFR)

#### RESTYPE

Resource type

The return code passed to the calling program by EZLEATHR can be set from EXIT06. The altered return code is set by EXIT06 and placed in the task global variable EZLEATHRRC. The return codes from EZLEATHR are used by calling programs to determine which threshold has been exceeded and if automation should continue.

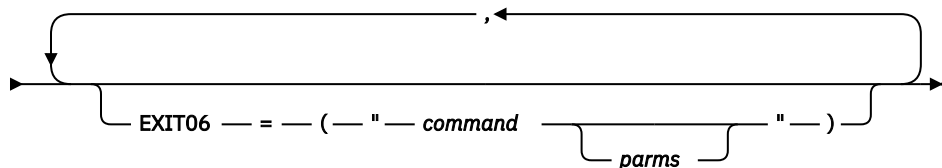
| Table 13. EZLEATHR results                |                                                           |
|-------------------------------------------|-----------------------------------------------------------|
| If EZLEATHR determines...                 | ...then....                                               |
| a critical threshold has been exceeded    | a return code of 3 is passed back to the calling program. |
| a frequent threshold has been exceeded    | a return code of 2 is passed back to the calling program. |
| an infrequent threshold has been exceeded | a return code of 1 is passed back to the calling program. |

EZLEATHR exits can be defined in the control file on the ENVIRON EXIT by the EXIT06 parameter or on the THRESHOLDS control file entry. Task global variables can be passed to the exit by specifying the task global variable name preceded by an ampersand (&). Resource names can also be passed to the exit routine by specifying &RESNAME.

## Format

### ENVIRON EXIT

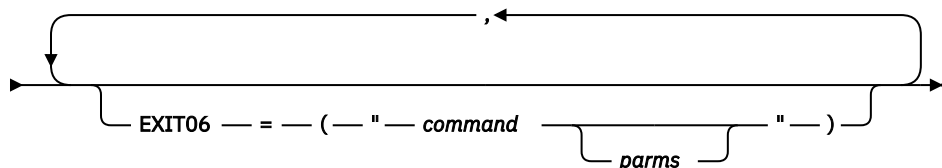
➤ ENVIRON EXIT ➔



THRESHOLDS entry for the specific resource:

### THRESHOLDS

➤ THRESHOLDS — resource ➔



## Parameters

### command

Specifies the exit or command to be issued.

### parms

Specifies the parameters to be passed to the exit or command to be issued.

## Usage

A sample exit is provided in CNMSAMP member EZLEXT06.

Run all exits inline. Do not route exits to another task.



Code user exits to change any of the task global variables set by EZLEATHR.

If any user-coded routines are written in REXX and a task global variable is changed, retrieve the new value by the use of the REXX 'GLOBALV GETT' statement.

Set task global variable EZLEATHRRRC in the user exit to alter the return code for EZLEATHR.

If an exit parameter is coded on the THRESHOLDS entry, the exits coded on the THRESHOLDS entry are issued.

All defined user exits are called in the order they are coded in the control file. All exits are issued until the user exit returns a non-zero return code. This return code is different from the return code set in task global variable EZLEATHRRRC.

If the threshold exceeded type is altered, make the change in the EZLTRSHLD task global variable.

EXIT06 can be used to change the thresholding analysis results used by AON to guide automation and notifications based on thresholds exceeded. This might be desirable if there are no other factors influencing failure rates or resources which are not available to AON. EXIT06 can also perform some function based on thresholding exceptions, but not necessarily change the results of the analysis.



**Attention:** The task global variables contain data that guides AON recovery processing. Indiscriminate changes to the task global variables can cause unpredictable results.

## Examples

For this example, each time the THRESHOLDS FLAG is checked for resource ABPU, EZLEXT06 is issued and passed the value of the task global variable RESNAME as the first parameter, the value of task global variable RESTYPE as the second parameter, and the type of threshold that was exceeded as the third parameter:

```
THRESHOLDS ABPU,
            CRIT=(4,00:12),
            FREQ=(4,01:00),
            INFR=(5,24:00),
            EXIT06=("EZLEXT06 &RESNAME &RESTYPE &EZLTRSHLD")
```

## EXIT07 Processing (EZLECAUT)

---

### Purpose

Exit processing for EZLECAUT is issued when AON checks the automation status of a resource. Automation status is set by analyzing the RECOVERY AUTO and NOAUTO parameter entries. However, if the EZLECAUT higher node of the resource is inactive, AON does not analyze the RECOVERY entries. In this case, EXIT07 is not called. For example, if EZLECAUT is called with ABPU and the line ABLINE1 that is associated with ABPU is not in an active state, exit processing is not issued for resource ABPU.

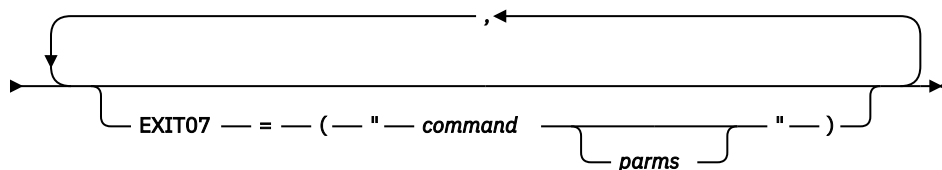
AON uses the EZLECAUT return code to determine whether automation is on or off for the resource. A return code greater than zero (0) indicates that automation should not occur. A return code of zero (0) indicates that automation is to proceed. The return code that is passed to the calling program by EZLECAUT can be set from a user exit. The altered return code is set by the user exit and placed in the task global variable EZLECAUTRC.

EZLECAUT exits can be defined in the control file on the ENVIRON EXIT by the EXIT07 parameter and the RECOVERY entry that applies to the resource. The previously mentioned task global variables can be passed to the exit by specifying the task global variable name preceded by an ampersand (&). The resource name can also be passed to the exit routine by specifying &RESNAME.

## Format

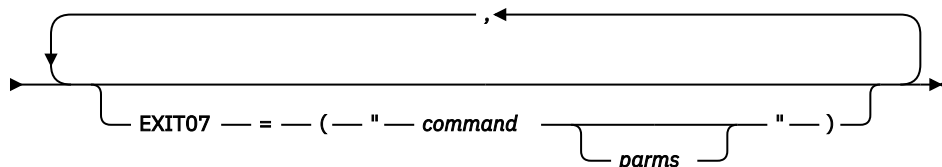
### ENVIRON EXIT

➤ ENVIRON EXIT ➔



### RECOVERY

➤ RECOVERY — *resource* — , — AUTO — = — Y ➔



## Parameters

### command

Specifies the exit or command to be issued.

### parms

Specifies the parameters to be passed to the exit or the command to be issued.

### resource

Specifies the resource name.

## Usage

A sample exit is provided in CNMSAMP member EZLEXT07.

Run all exits inline. Do not route exits to another task.

Code user exits to change task global variables set by EZLECAUT. These task global variables are set from the results of a VTAM DISPLAY command for the resource.

If any user-coded routines are written in REXX and a task global variable is changed, retrieve the new value by the use of the REXX 'GLOBALV GETT' statement.

Set task global variable EZLECAUTRC in the user exit to alter the return code for EZLECAUT.

If an exit parameter is coded on the RECOVERY entry, the exits that are coded on the RECOVERY entry are issued.

All defined user exits are called in the order they are coded in the control file. All exits are issued until the user exit returns a nonzero return code. This return code is different from the return code set in task global variable EZLECAUTRC.

A return code of zero from EZLECAUT indicates that recovery for this resource is to continue. Any other return code from EZLECAUT indicates that automation should stop.

If the parameters sent to EZLECAUT include the resource name, type and higher node name, EZLEAGRN is not called for the resource. If any of those parameters are missing, EZLEAGRN is issued to retrieve the resource type. As a result EXIT05 is issued for the resource. Standard AON processing sends only the resource name; therefore, each call to EZLECAUT and EXIT07 can cause an execution of EZLEAGRN and EXIT05.

You can use EXIT07 to influence the automation flag and AON recovery activity based on data that is not readily available to AON. For example, EXIT07 can check a database or issue commands to an other equipment manufacturer (OEM) device to further determine whether resource recovery is appropriate. EXIT07 can also initiate some activities that do not change the automation flag; however, that should occur only because AON has detected an error condition.



**Attention:** The task global variables contain data that guides AON recovery processing. Indiscriminate changes to the task global variables can cause unpredictable results.

## Examples

For this example, each time the RECOVERY FLAG is checked for resource ABPU, EZLEXT07 is issued and passed the value of the task global variable RESNAME as the first parameter and the value of task global variable RESTYPE as the second parameter:

```
RECOVERY ABPU,
        AUTO=Y,
        EXIT07=("EZLEXT07 &RESNAME &RESTYPE")
```

## EXIT08 processing (AON messaging)

### Purpose

The AON messaging exit enables you to run your own program when an AON message is sent to operators. EXIT08 is defined on the ENVIRON EXIT entry and is specified by the EXIT08 parameter.

The AON messaging exit enables you to analyze the resource name and type and assign a new valid AON message class for this resource when an alert is to be sent to a focal point (either AON or the system operations component of IBM Z System Automation).

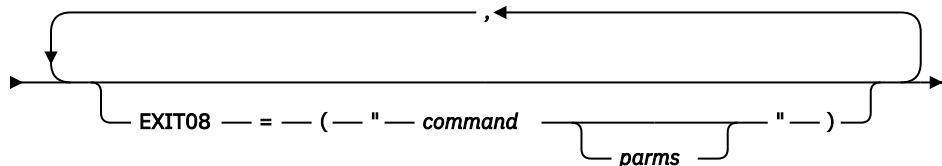
The resource name, type, and any user-defined parameters can be passed to the user exit defined in the EXIT08 parameter. The parameters are received by the exit in the order that they are passed. A sample exit has been provided in CNMSAMP (EZLEXT08). The assigned message class or classes are stored in the task global variable EZLCONVERT. If multiple message classes are assigned, the classes must be separated by blanks. Any return code set by the user exit be ignored. The resource name and resource type can be passed to the exit by specifying the RESNAME and RESTYPE.

You can also pass the message text using variable MSGTEXT,EZLMSGTXT or referring to the task global variable EZLMSGTXT in your program. Your program can then send the message to other operator notification devices like pagers or status boards. It can initiate a note in an electronic mail system. Most external notification devices have workstation interfaces that can be accessed from a service point. Some service points can be driven by NetView or z/OS commands issued from NetView programs.

### Format

#### ENVIRON EXIT

➡ ENVIRON EXIT ➡



### Parameters

#### parms

Specifies parameters that are to be passed to the exit routine

## Usage

A sample exit is provided in CNMSAMP member EZLEXT08.

Run all exits inline. Do not route exits to another task.

All return codes from the user exit be ignored.

Return message classes in task global variable EZLCONVERT. Message classes are a maximum of 2 characters separated by blanks.



**Attention:** The task global variables contain data that guides AON recovery processing. Indiscriminate changes to the task global variables can cause unpredictable results.

## Examples

In this example, EZLEXT08 is issued when a message is issued:

```
ENVIRON EXIT,EXIT08=("EZLEXT08 &RESNAME &RESTYPE")
```

EZLEXT08 is passed the values for resource name and resource type.

## EXIT09 processing (EZLECATV)

---

### Purpose

Exit processing for EZLECATV is issued after the MONIT interval data is returned and before activation or deactivation of the resource occurs. You can set the following TGLOBALs to influence recovery in EXIT09:

#### EZLACT

Change this value to specify whether to attempt to activate the resource. Valid values are yes (Y) and no (N). The default is yes (Y).

#### EZLINACT

Change this value to specify whether to attempt to deactivate the resource. Valid values are yes (Y) and no (N). The default is no (N) for resource types of LINKSTA or CDRSC, otherwise the default is yes (Y).

#### EZLINTVL

This value cannot be changed but can be passed to the exit. This TGLOBAL contains the number of the MONIT interval currently being processed.

#### EZLMONIT

This value cannot be changed but can be passed to the exit. This TGLOBAL contains the MONIT value that is currently being run.

#### EZLNOTIFY

Change this value to specify a different notify flag for the current MONIT interval (can suppress or force operator reminder notification on this interval).

#### EZLNXTINTVL

Change this value to indicate whether MONIT intervals are to continue. The default is yes (Y).

#### EZLSCOPE

Change this value to specify the scope parameter for VTAM VARY ACT command. The default scope value is U for NCPs, lines, and PUs.

#### RESTYPE

Change this value to change the resource type of the resource.

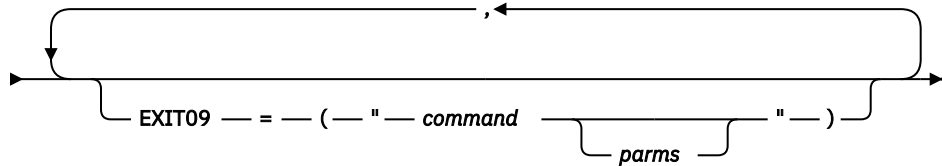
The return code passed to the calling program by EZLECATV can be set from EXIT09. The altered return code is set by the user exit and placed in the TGLOBAL EZLECATVRC. The return code from EZLECATV is not used by AON for automation.

EZLECATV exits can be defined in the control file on the ENVIRON EXIT or the MONIT entries by the EXIT09 parameter. The resource name and resource type can be passed to the exit by specifying the &RESNAME and &RESTYPE.

## Format

### ENVIRON EXIT

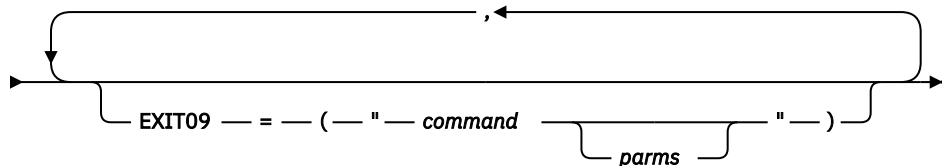
➤ ENVIRON EXIT ➤



Special exit routines can be coded for specific resources on the MONIT entry for the specific resource MONIT.

### MONIT

➤ MONIT — resource ➤



## Parameters

### command

Specifies the exit to be run or the command to be issued.

### parms

Specifies the parameters to be passed to the exit or the command to be issued.

## Usage

A sample exit is provided in CNMSAMP member FKVEXT09.

Run all exits inline. Do not route exits to another task.

Code user exits to change the TGLOBALs set by EZLECATV.

If any user-coded routines are written in REXX and a TGLOBAL is changed, retrieve the new value by using the REXX 'GLOBALV GETT' statement.

Set TGLOBAL EZLECATVRC in the user exit to alter the return code for EZLECATV.

All defined user exits be called in the order they are coded in the control file. All exits be run until the user exit returns a non-zero return code. This return code is different from the return code set in TGLOBAL EZLECATVRC.

EXIT09 can be used to change the way AON recovers a resource, or class of resources by setting the TGLOBALs. EXIT09 can be used to perform some function which is desirable each time a recovery interval is processed, but not necessarily change the AON recovery logic.



**Attention:** The TGLOBALs contain data that guides AON recovery processing. Indiscriminate changes to the TGLOBALs can cause unpredictable results.



## **Parameters**

### **ALERT**

Specifies when to generate MSU notifications. MSUs are required for Event Integration Facility (EIF) events.

#### **YES**

Specifies to generate an MSU.

#### **NO**

Specifies do not generate an MSU. No is also the default.

### **TEC**

Specifies to generate an MSU so that the event server receives the notification.

### ***command***

Specifies the exit or command to be issued.

### **DDF**

Specifies how the DDF component is updated.

#### **YES**

Log the event to DDF.

#### **NO**

Do not log events to DDF. No is the default.

### **DEFAULTS**

Notification policy at a system wide, or defaults, level

### **Event\_Type**

Event\_Type can be one of the following:

#### **ADPCONGEST**

Adapter Congestion Threshold exceeded

#### **BRGCONGEST**

Bridge Congestion Threshold exceeded

#### **CRITTHRS**

Critical automation threshold exceeded

#### **NAMESERV**

NameServer Failure Threshold exceeded

#### **NOMOMONS**

No more monitoring intervals defined

#### **REMIND**

Reminder that a resource is still down

### **EXIT10**

Is a user exit that enables the override of any of the notification actions.

### **INFORM**

Specifies the use of pager/beeper or email that is defined in the CONTACT entry statement of the Inform Policy. You can use different resources and resource types.

### **MSG**

Specifies whether to generate a message or not.

#### **YES**

Specifies to display a message.

#### **NO**

Specifies not to display a message. No is the default.

### ***parms***

Specifies the parameters that are to be passed to the exit routine.

### **ResName**

Notification policy based on a particular resource name

## ResType

Notification policy based on a type of resource

## Res\*

Notification policy based on a range of resource names

## Usage

It is possible for EXIT10 to be driven without any notification policy defined. In that case, EZLNIFYLST be null. This would occur if the defined policy has been customized to not issue any notifications. The user exit can then enable particular notifications.

## Examples

In the following example, CLIST TEXTIT10 be called to override the existing notification policy in EZLNIFYLST for a specific resource:

```
NOTIFY PU0001,MSG=YES,DDF=NO,ALERT=TEC,INFORM=NO,EXIT10=TEXTIT10 &RESNAME
```

## EXIT11 and EXIT12 Inform Policy processing

### Purpose

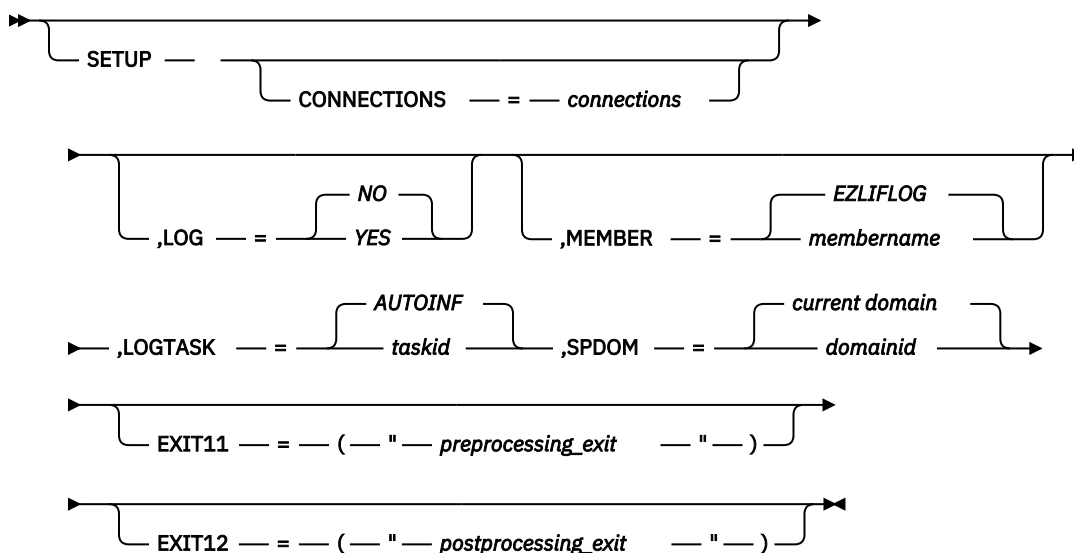
EXIT11 and EXIT12 enable you to alter the inform policy processing.

EXIT11 is called before inform policy processing and can be used to dynamically change the inform policy name, group name, or provide a list of policy or group names as needed to insure that the appropriate contacts are made.

EXIT12 is called after inform policy processing and before starting the interface specified in the policy, enabling the inform action to be changed.

### Format

#### SETUP





## Parameters

### CONNECTIONS

A list of additionally supported connection types. If an interface supports FAX, then FAX must be added to the SETUP CONNECTIONS parameter. It is not necessary to add NUPAGE, ALPHAPAGE, or EMAIL to this list.

### EXIT11

EXIT11 is started before checking the in-storage inform policy information.

### EXIT12

EXIT12 is called before starting the interface specified in the inform policy.

### LOG

Set the LOG keyword to YES if you want the INFORM Log enabled. When enabled, selected INFORM actions are logged, and this list can be displayed using the full screen AON function, ILOG. ILOG enables each INFORM action to be deleted, acknowledged, or reissued as needed. Because I/O is required, the default for SETUP LOG is NO. Actions caused by the use of the INFORM command are not logged.

### LOGTASK

The INFORM log requires an autotask for the sequencing of all updates. If the log is enabled, the autotask indicated is started, or AUTOINF is started by default.

### MEMBER

If LOG=YES is specified, then *membername* is the member that AON INFORM uses to write the log records. The records are written to the first data set name found in the DSILIST data set definition. If no member name is specified, EZLIFLOG is used by default. The write protection key >INFORM or the first line of the member starting in column 1 is written. Therefore, the INFORM log function overwrites only other INFORM logs.

### *postprocessing\_exit*

Specifies the post-processing exit for the inform policy. The parameters passed to EXIT12 are the same as those passed to the INTERFACE routine. For more information about the parameters that are passed, see [“Usage” on page 349](#).

### *preprocessing\_exit*

Specifies the pre-processing exit for the inform policy. For information about the parameters that are passed, see [“Usage” on page 349](#).

### SPDOM

The domain ID of the network NetView that owns the INFORM service point or application. The current domain is the default.

## Usage

When EXIT11 is called from EZLENFRM or a notification policy, it receives the policy name, resource name, resource type, domain ID, resource status, and automation status. When EXIT11 is called from EZLECALL or the INFORM command, it receives the policy name and the domain ID. The return codes that can be set by the exit are:

**0**

Continue normal inform policy processing.

**4**

Continue normal inform policy processing using changed information. This new information is specified using a task global variable EZLPOLEX which contains the value policy name or group name or list of common delimited policy and group names.

**8**

Discontinue processing. This enables linkage to other inform technologies without returning to this inform policy flow.

When EXIT12 is called before starting the inform action routine, it receives information related to the inform action. Refer to the EZLENETF sample in the CNMCLST data set for a description and the format of these parameters. The return codes that can be set by the exit are:

0

Call the interface code as specified in the inform policy contact statement.

n

Discontinue processing. Before returning this value, you can call other interface routines directly.

## EXIT13 Socket Monitoring

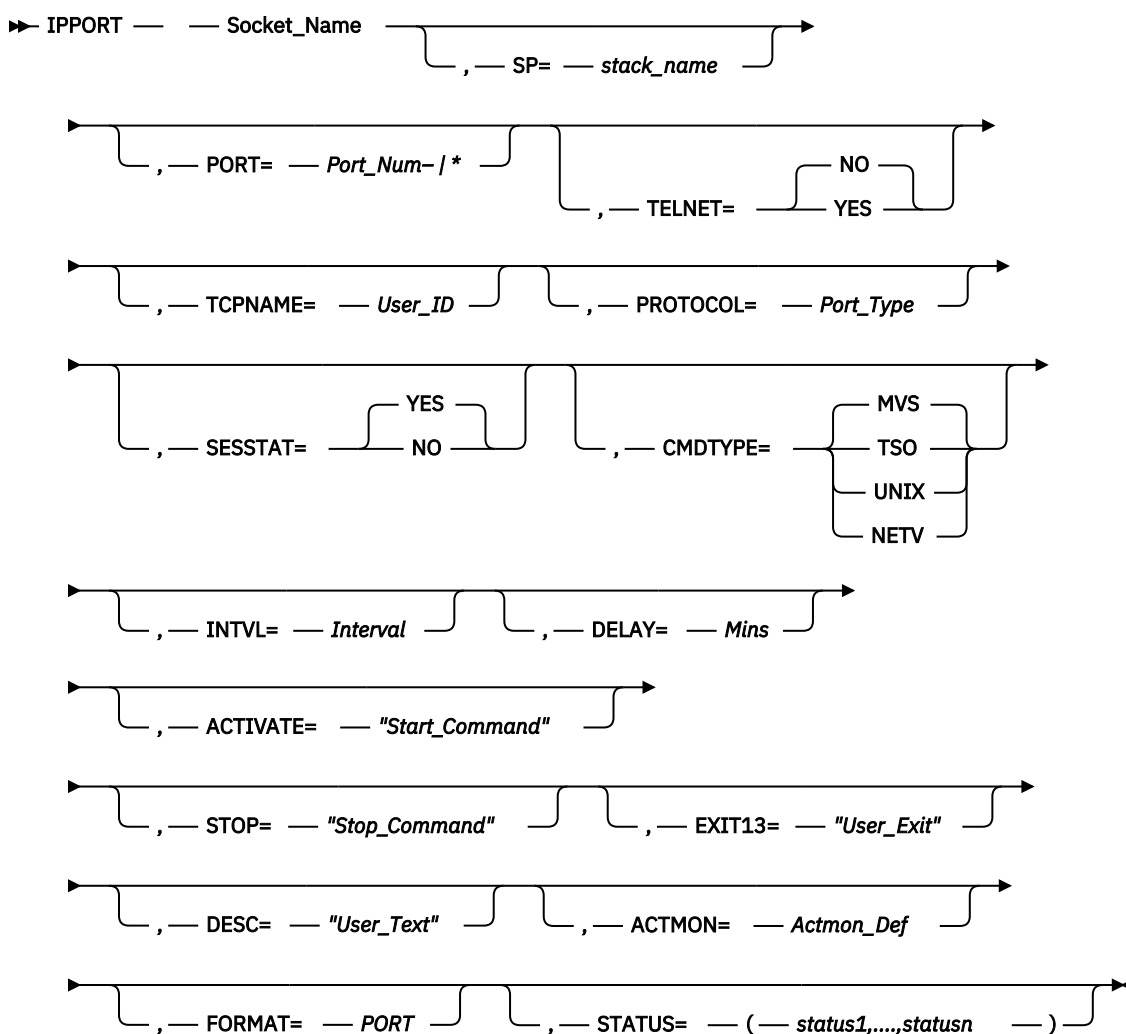
### Purpose

EXIT13 can be used to prevent the STOP and ACTIVATE commands from being issued against a socket. You can also use it to override the STOP or ACTIVATE commands.

EXIT13 is called for both STOP and ACTIVATE processing.

### Format

#### IPPORT



### Usage

You can use Exit13 to prevent the STOP and ACTIVATE commands from being issued against a socket, or to override the STOP or ACTIVATE commands. Exit13 be called for both STOP and ACTIVATE processing. The input that is available to this exit includes the following task global variables:

**FKX\_Command**

STOP or ACTIVATE command as coded in the socket policy

**FKX\_Delay**

Delay time between STOP and ACTIVATE commands

**FKX\_Action**

Identifies type of action, STOP or ACTIVATE

**FKX\_CmdType**

Identifies the type of command z/OS, NETV, TSO, or UNIX

You can choose the following task global variables with this exit:

**FKX\_Command**

Modifies the STOP or ACTIVATE command

**FKX\_Delay**

Modifies the delay time between the STOP and ACTIVATE commands

**FKX\_CmdType**

Modifies the type of command, z/OS , NETV, TSO, or UNIX

When zero (0) is the return code either the STOP or ACTIVATE command runs.

When a nonzero return code is received from the user exit, the STOP command not be issued, a DELAY is not valid, and the user exit attempt to process the ACTIVATE command. If the nonzero return code is for an ACTIVATE command, ACTIVATE is not attempted. AON recovery monitoring still occurs.

## EXIT14 SNMP MIB polling

---

**Purpose**

EXIT14 is driven during SNMP MIB polling processing. AON/TCP has retrieved the status of a resource interface table (IFTable).

Use EXIT14 to perform further processing, such as retrieving the IpAddr Table from the resource to correlate the interfaces of the resource with their IP addresses.

EXIT14 can be coded for any of the IP390 resource definitions in the following list. For complete syntax, refer to the *IBM Z NetView Administration Reference*.

- IPHOST
- IPINFC
- IPNAMESERV
- IPROUTER
- IPTN3270

**Note:** Because EXIT14 is used during SNMP MIB polling, specify FORMAT=SNMP on your resource definitions.

**Format**

## IPROUTER

```

➤➤ IPROUTER — — AON_Name — , — SP= — stack_name — , — OPTION= — IP390 — , ➤

➤ IPADDR= — ipaddr — , — HOSTNAME= — host_name ➤

┌ — , — INTVL= — interval ─┐ ┌ — ACTMON= — Actmon_Def ─┐
└──────────────────────────┘ └──────────────────────────┘

┌ — , — FORMAT= ─┐ ┌ — PING ─┐
└────────────────┘ └──────────┘
┌ — SNMP ─┐
└──────────┘

┌ — , — STATUS= — ( — status1, ..., statusn — ) ─┐
└──────────────────────────────────────────────────┘

┌ — MIBVARn= — ( — mib_var, operator, value — ) — ... ─┐
└──────────────────────────────────────────────────────────┘

┌ — CORRELATE ─┐ ┌ — N ─┐ ┌ — , — EXIT14= — ' — FKXEXT14 ... — ' ─┐
└──────────┘ ┌──┴──┘ └──────────────────────────────────────────────────┘
┌ — Y ─┐
└──────────┘

┌ — , — EXIT15= — ' — FKXEXT15 ... — ' ─┐
└──────────────────────────────────────────┘

```

### Parameters

#### ACTMON\_Def

Defines active monitoring for groups of resources in the network. For additional information refer to the *IBM Z NetView Administration Reference*.

#### AON\_Name

The unique name associated with the router that is used by AON/TCP.

#### CORRELATE Y | N

Set to Y for trap correlation of IPHOSTs/IPROUTERS/IPTN3270s and their respective interface. Set the CORRELATE parameter to Y for resources only when SNMP is available. All known interfaces are used to correlate the status. Use the CORRELATE parameter only for interfaces installed on critical IPHOSTs or IPROUTERS. N is the default.

#### EXIT14

Defines additional processing of SNMP interface table. The default value is NONE. Started for FORMAT=SNMP.

#### EXIT15

Defines additional processing of user-defined thresholds (MIBVAR). The default value is NONE.

#### FORMAT

Specifies one of the following options to be used to determine the resource status:

##### PING

Pings the resource to check its status.

##### SNMP

Uses SNMP MIB polling to check the status of the resource.

**HOSTNAME**

Indicates the fully qualified TCP/IP host name, using up to 30 characters including dots. The *host\_name* variable is case-sensitive. The following is an example of a host name:

```
mipres.whitehouse.capital.usa
```

**INTERVAL**

Defines the monitoring interval in hh:mm format. This is required for proactive monitoring. If the interval is defined for IPSTAT but not specified monitoring does not occur.

**IPADDR**

The IP address. Alphabetic characters are not valid.

**MIBVARn**

Multiple MIBVAR statements can be defined and are needed only for SNMP thresholding (requires FORMAT=SNMP).

**OPTION=IP390**

The only valid entry is IP390.

**Stack\_Name**

The name of the z/OS stack to use for TCP/IP commands. This name is synonymous with the z/OS service point.

**STATUS**

Defines the expected (AON) status of the resource.

**Examples**

The following example defines ROUTER1 as part of the NYROUTERS policy grouping to be monitored every 15 minutes through SNMP MIB polling:

```
ACTMON    IP390,OPTION=IP390,INTVL=01:,STATUS=NORMAL
ACTMON    IPROUTER,OPTION=IP390,INTVL=00:30,STATUS=NORMAL
ACTMON    NYROUTERS,OPTION=IP390,INTVL=00:15,
          FORMAT=SNMP

IPROUTER  ROUTER1,
          SP=NMPIPL10,
          OPTION=IP390,
          IPADDR=1.2.3.4,
          HOSTNAME=yourhost.plant.floor.co,
          ACTMON=NYROUTERS,
          EXIT14=MYEXIT14
```

## EXIT15 SNMP MIB thresholding

---

**Purpose**

EXIT15 is driven during SNMP MIB Thresholding processing. AON/TCP has issued SNMP GET requests for each MIBVARx variable for a given resource.

EXIT15 enables you to perform further processing such as determining which MIB variable and value did not meet the threshold specification.

EXIT15 can be coded for any IP390 resource definition in the following list. For additional information refer to the *IBM Z NetView Administration Reference*.

- IPHOST
- IPINFC
- INPNAMESERV
- IPROUTER
- IPTN3270

## Format

### IPHOST

➤ IPHOST — — AON\_Name — , — SP= — *stack\_name* — , — OPTION= — *IP390* — , ➤

➤ IPADDR= — *ipaddr* — , — HOSTNAME= — *host\_name* ➤

┌ — , — INTVL= — *interval* ┐ ┌ ACTMON= — *Actmon\_Def* ┐

┌ — , — FORMAT= ┐  
└ *PING* ┘  
└ *SNMP* ┘

┌ — , — STATUS= — ( — ? — *status1*, ..., *statusn* — ) ┐

┌ MIBVAR1, ..., MIBVARn= — ( — *mib\_var*, *operator*, *value* — ) ┐

┌ CORRELATE ┐ ┌ N ┐ ┌ — , — EXIT14= — *FKXEXT14* ... ┐  
└ Y ┘ └ Y ┘

┌ — , — EXIT15= — *FKXEXT15* ... ┐ ➤

## Parameters

### ACTMON\_Def

Defines active monitoring for groups of resources in the network. For additional information, refer to the *IBM Z NetView Administration Reference*.

### AON\_Name

The unique name associated with the TCP/IP host that is used by AON/TCP.

### CORRELATE Y|N

Set to Y for trap correlation of IPHOSTs/IPROUTERS/IPTN3270s and their respective interface. The CORRELATE parameter be set to Y only for resources when SNMP is available. All known interfaces are used to correlate the status. Use the CORRELATE parameter only for interfaces installed on critical IPHOSTs or IPROUTERS. N is the default.

### EXIT14

Defines additional processing of SNMP interface table. The default value is NONE. Started for FORMAT=SNMP.

### EXIT15

Defines additional processing of user-defined thresholds (MIBVAR). The default value is NONE.

### FORMAT

Specifies one of the following options to be used to determine the status of the resource.

#### PING

Pings the resource to check its status.

#### SNMP

Uses SNMP MIB polling to check the status of the resource.

**HOSTNAME**

Indicates the fully qualified TCP/IP host name, using up to 30 characters including dots. The *host\_name* variable is case-sensitive. The following is an example of a *host\_name*:

```
mpres.whitehouse.capital.usa
```

**INTERVAL**

Defines the monitoring interval in hh:mm format. This is required for proactive monitoring.

**IPADDR**

Consists of IP addresses up to 17 characters in length. Alphabetic characters are not valid.

**MIBVAR1 – n**

Multiple MIBVAR statements can be defined and is needed only for SNMP thresholding (requires FORMAT=SNMP).

**MVS\_Stack\_Name**

The name of the MVS stack to use for TCP/IP commands. This name is synonymous with the MVS service point.

**OPTION=IP390**

Only valid entry is IP390.

**STATUS**

Defines the expected (AON) status of the resource.

**Examples**

This example defines host *yourhost* for proactive monitoring using SNMP polling, looking for a status of NORMAL, THRESH\*, or DEGR\*:

```
ACTMON    IP390,OPTION=IP390,INTVL=01:00,STATUS=NORMAL

IPHOST    PKOCH,SP=NMPIPL10,
          OPTION=IP390,
          HOSTNAME=yourhost.yourcompany.com,
          INTVL=45
          FORMAT=SNMP,
          STATUS=(NORMAL,THRESH*,DEGR*),
          MIBVAR1=(tcpActiveOpens.0,LT,5000),
          MIBVAR2=(tcpInErs.0,GT,5),
          MIBVAR3=(lpInDiscards.0,EQ,1),
          MIBVAR4=(ipRoutingDiscards.0,GE,3),
          EXIT15=MYEXIT15
```





---

## Chapter 29. AON option definition tables

AON option definition tables specify programs that perform resource-dependent functions. The option definition tables also provide literals used by the AON operator interface to create menus containing only selections available in your installation. Each AON function has an option definition table. Some functions, such as AON/SNA, have several options, such as subarea, APPN, SNBU, and X25, that can be enabled independently. Option-specific definitions in the tables are found on the EZLOPT entries. Definitions that are resource-specific are on the EZLRT entries in the option definition tables.

---

### How AON uses option definition tables

AON uses the EZLERTVE routine to access the option definition tables. For resource-specific definitions, the option definition tables include EZLRT entries. When the EZLERTVE routine is called to look up an EZLRT table value, it first looks for the keyword of the resource type. If the keyword does not exist for that resource type, EZLERTVE obtains the OPTION keyword for the resource type and looks up that keyword. If the keyword is not found there, and the option has an OPTION keyword, EZLERTVE continues looking for an entry in the parent definition of the option.

For example, if the EZLERTVE routine is called to find EZLRT CDRM RESINFO, it searches table EZLRT for resource type CDRM and keyword RESINFO. The EZLERTVE routine looks for RESINFO in the EZLRT CDRM parameter. If it is not found, the EZLERTVE routine obtains the OPTION value of SA from the EZLRT CDRM parameter. The EZLERTVE routine looks for RESINFO on the EZLOPT SA parameter. Again, if it is not found, the EZLERTVE routine gets the OPTION value of SNA from the EZLOPT SA parameter. The EZLERTVE routine looks for RESINFO on the EZLOPT SNA parameter. If it is still not found, the EZLERTVE routine returns N/A to the calling routine.

The first time RESINFO is found on any parameter, the value found there is returned to the calling routine and the EZLERTVE routine exits.

The option definition table uses a tree structure. AON uses the definition found at the lowest level as follows:

```
EZLOPT SNA → EZLOPT SA → EZLRT CDRM
```

For more information about EZLERTVE, see [Chapter 26, “Coding common routines,” on page 273](#).

---

### Displaying or replacing a definition

The values in the option definition tables are placed in NetView common global variables for fast access during automation. The common global variable format for resource type values is `EZLRT.resource_type.keyword`. The common global variable format for option values is `EZLOPT.option.keyword`. You can display these common global variables by selecting one of the options on the AON Common Global Editor panel shown in [Figure 205 on page 358](#). To access this panel, type **CGED** and press **Enter**.

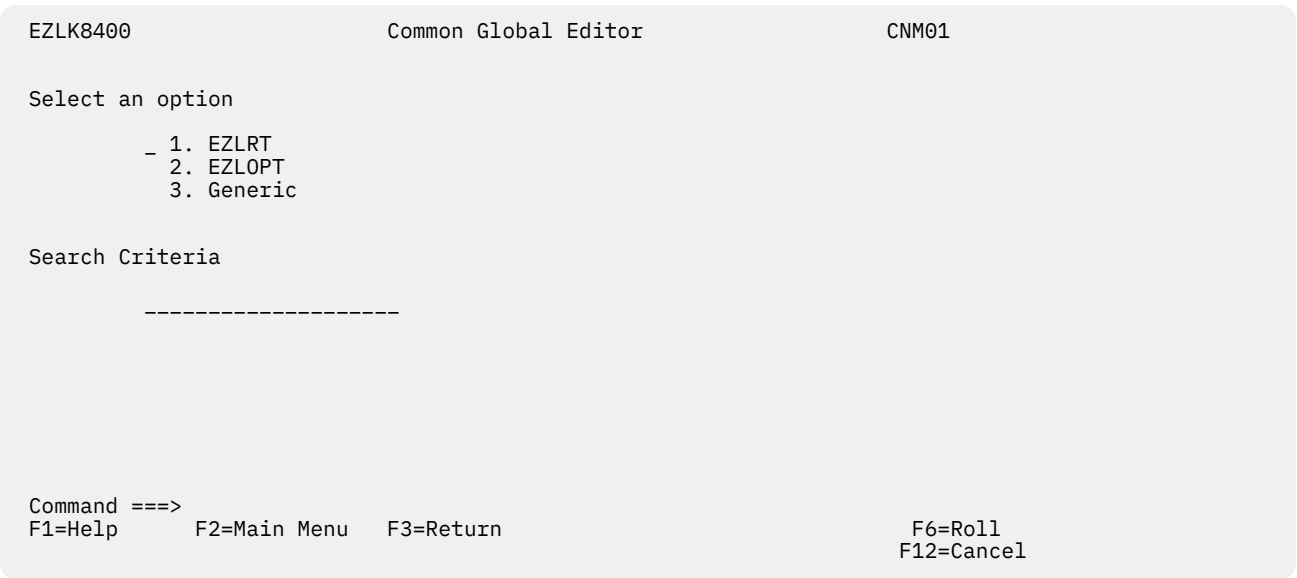


Figure 205. Common Global Editor Panel

The selection options are shown in [Table 14 on page 358](#):

| Table 14. Display selection options |                                                                                                                                                                                                     |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Values to display                   | What you select                                                                                                                                                                                     |
| All the resource type values        | <b>1</b>                                                                                                                                                                                            |
| All the option values               | <b>2</b>                                                                                                                                                                                            |
| User-specified values               | <b>3</b> and type a search criterion. For example, EZL*. *. IDENTIFY lists all of the IDENTIFY routines defined by option or by resource type, as shown in <a href="#">Figure 206 on page 358</a> . |

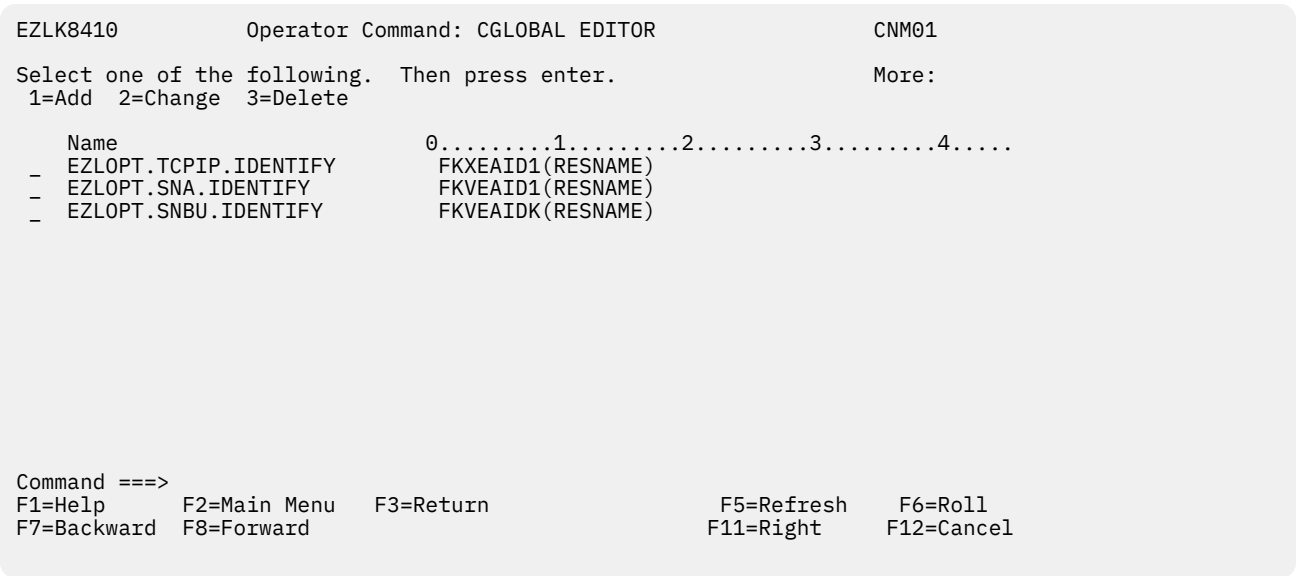


Figure 206. Operator Command: CGLOBAL EDITOR Panel

To change the value of an EZLOPT entry, use the change function on this panel and type over the current value with the new value. The value is changed immediately. It does not change the option definition table. If you reload your option definition table or recycle the NetView program, the definition reverts to what is defined in the option definition table. If you need permanent changes, update the definition and

replace it, as described in the next section, “[Displaying or replacing an option definition table](#)” on page 359.

## Displaying or replacing an option definition table

The AON option definition tables are kept in the NetView DSIPARM data set. You can browse or reload one of these tables into the common global variables used by the AON programs by using the AON Loader Tables panel. To display this panel, type the fast path **AON 1.8.6** and press **Enter**. The panel shown in Figure 207 on page 359 is displayed.

|                                                                       |              |                       |            |            |
|-----------------------------------------------------------------------|--------------|-----------------------|------------|------------|
| EZLK8600                                                              |              | AON: Loader Tables    |            | CNM01      |
| Type one or more action codes. Then press enter.<br>1=Browse 2=Reload |              |                       |            | More:      |
| Type                                                                  | Table        | Description           | Status     |            |
| - AON                                                                 | EZLTABLE     | AON Base              |            |            |
| - SNA                                                                 | FKVTABLE     | AON SNA Automation    |            |            |
| - TCP/IP                                                              | FKXTABLE     | AON TCP/IP Automation |            |            |
| -                                                                     | -----        |                       |            |            |
| Command ==>                                                           |              |                       |            |            |
| F1=Help                                                               | F2=Main Menu | F3=Return             | F5=Refresh | F6=Roll    |
| F7=Backward                                                           | F8=Forward   |                       |            | F12=Cancel |

Figure 207. AON: Loader Tables Panel

The Loader Tables panel lists the tables that are currently loaded. In Figure 207 on page 359, option definition tables are installed for the AON base and the SNA Automation and TCP/IP Automation features. The panel also contains a blank table-name entry field where you can type a table name not already listed on the panel. You can use the browse option to display the table in DSIPARM. If you select option 2, the table is retrieved from DSIPARM and the common global variables generated by the table are refreshed from the current table. Refreshing the table places the last load of the table into effect immediately.

## Guidelines for option definition table entries

Follow the guidelines described in this topic to create or modify option definition table entries.

### Defining literals

Quotations are not valid unless the literal consists of more than one token. Semicolons (;) are not valid in literals. For example, ENABLE=YES has no semicolon.

### Defining text

Place text inside single quotation marks. Semicolons (;) are not valid. The following is an example:

```
CMD=' "RECYCLE" RESNAME '
```

### Defining lists

Enclose lists in parentheses. A list can contain a space or comma delimiter, but cannot contain a semicolon (;) as a delimiter. The following is an example:

```
IST105I=(EZL531,FKVEAIDA(resname restype))
```

In the previous example, the first element in the list is a literal, and the second element is a function call.

## Defining function calls

Code function calls in the following format:

```
keyword=function_name(parm1 parm2 parmn)
```

Spaces are not valid between the function name and the opening parenthesis. Enclose the parameter list after the function name in parentheses. An empty parameter list is indicated by (). Quotation marks are not valid in the program call. Use spaces as delimiters in the parameters in the list. Semicolons are not valid in function calls. If one of the parameters sent to the function is a literal, enclose the literal in double quotation marks, as follows:

```
Keyword=function_name(parm1 "lit2" parmn).
```

The function must return data in *keyword=variable* format or a numeric return code. Use spaces as delimiters in return data. Enclose values containing spaces in double quotation marks or parentheses.

For example, a RESINFO call for PU resolves to that defined for SNA: RESINFO=FKVEAID2(*resname*). If the resource name in the calling program is PU01, the function call resolves to *return\_string* = FKVEAID2(PU01).

In the previous example, the only parameter being sent to function FKVEAID2 is the name of the resource, what the variable *resname* contains in the calling program. The calling program must ensure that *resname* has a value. The returning data from FKVEAID2 is stored in *Return\_string*. The *Return\_string* would contain a data string such as the following:

```
'RESTYPE=PU HIGHNODE=LINE01 STATUS=INACT RESHIER="NCP01 NCP LINE01 LINE PU01 CTRL"'
```

If the returning data is numeric, it should be a return code.

## Defining command calls

Command calls in option definition tables are enclosed entirely in single quotation marks. The input parameters should use spaces as delimiters for variable names and double quotation marks for literals. For example:

```
keyword='program_name "var1 var2 "'lit1" var3'
```

A program should return only numeric return codes to the calling program.

In the next example, the program name is EZLECAUT; *resname* is a variable that takes on whatever value *resname* has in the calling program:

```
CHKAUTO='EZLECAUT " resname " PU''
```

PU is being sent as a literal. If the *resname* had the value of PU01, the call is:

```
EZLECAUT PU01 PU
```

The program returns a return code for analysis by the calling routine. The program can also alter the common global variables or task global variables (TGLOBALS) used by the calling program to influence the processing in the calling program.

## Common global variable format

Common global variables are created in the following format:

```
Table_name.Entry.Kwx
```

## Error checking

The table loader stops if it encounters an error. Any common global variables created at that point are defined. AON checks for the following errors:

- Entry = ''.
- There are no keywords.
- There are no closing parentheses.
- Keyword values are uneven.
- A semicolon is missing after an entry.
- Value *x* exceeds the maximum amount of valid data in a common global variable (currently set at 255).
- The *option\_definition\_table* is not in the DSIPARM data set.
- The member is empty.

## Error return codes

Error message 218 is issued for errors found by EZLEALDR after the syntax check. The following return codes are issued to help you determine the problem with the table being loaded:

|           |                             |
|-----------|-----------------------------|
| <b>5</b>  | Missing entry ID            |
| <b>6</b>  | Missing key values          |
| <b>7</b>  | Missing key parameters      |
| <b>8</b>  | Uneven keyword in key value |
| <b>9</b>  | Missing ending semicolon    |
| <b>10</b> | Missing comma               |
| <b>11</b> | Keyword value entry > 255   |

## Format for option definition table entries

---

This topic illustrates the syntax of an option definition table entry and describes the parameters that can be used in option definition table entries.

### table\_name entry

#### table\_name entry

➤ *table\_name entry* — , — *kw1= val1* — , — *kw2= val2* — , — *kwn= valn* ➤

#### Parameters

##### *table\_name*

The name of the option definition table.

##### **EZLOPT**

The option definition table contains the automation definitions for an entire option or suboption. Define only those keywords that apply. The highest level option does not have an OPTION keyword. The OPTION keyword points to the higher level option of the suboption.

**EZLRT**

The resource type option definition table contains an entry for every valid resource type under any option or suboption. An **OPTION** keyword must be defined to identify the option that owns this resource type. There are no current provisions for resource types owned by multiple options.

**entry**

Option name or resource type under an option.

**Option names (EZLOPT) are:****SNA**

This option provides automation definitions and programs for AON/SNA network operation and automation. This option is specifically for VTAM.

**SA**

This suboption of AON/SNA provides automation definitions and programs for subarea SNA networks. Specifically, NCP, LINE, PU, LU, CDRMS, CDRSCs, and APPLs are included.

**APPN**

This suboption of AON/SNA provides automation definitions and programs for Advanced Peer-to-Peer Networking (APPN) SNA networks. This option is specifically for control points (hosts), end nodes, and network nodes.

**SNBU**

This suboption of AON/SNA provides automatic dial backup for failed PUs between two SNA subarea nodes.

**X25**

This suboption of SNA provides active monitoring of X25 components defined in NCP and NCP LUDRPOOL availability.

**TCPIP**

This option provides drop-in tables for operating and monitoring of Internet Protocol networks using simple network management protocol (SNMP) for network management.

**IP390**

This suboption of TCPIP provides tables for the operation and monitoring of IP resources using z/OS Communications Server IP.

**Resource type names (EZLRT) are:****APPL**

Option SNA; Suboption SA

**STG**

Option SNA; Suboption SA

**CDRM**

Option SNA; Suboption SA

**CDRSC**

Option SNA; Suboption SA

**NCP**

Option SNA; Suboption SA

**LINKSTA**

Option SNA; Suboption SA

**LINE**

Option SNA; Suboption SA

**SESSION**

Option SNA; Suboption SA

**PU**

Option SNA; Suboption SA

**LU**

Option SNA; Suboption SA

**CP**  
Option SNA; Suboption APPN

**CPCPSESS**  
Option SNA; Suboption APPN

**SNBUPU**  
Option SNA; Suboption SNBU

**X25MCH**  
Option SNA; Suboption X25

**X25PU**  
Option SNA; Suboption X25

**IPROUTER**  
Option TCPIP

**SERVER**  
Option TCPIP

**SP**  
Option TCPIP

**IPPORT**  
Option TCPIP; Suboption IP390

**IPTELNET**  
Option TCPIP; Suboption IP390

**IPHOST**  
Option TCPIP; Suboption IP390

**IPINFC**  
Option TCPIP; Suboption IP390

**IPNAMESERV**  
Option TCPIP; Suboption IP390

**IPTN3270**  
Option TCPIP; Suboption IP390

**IPCONN**  
Option TCPIP; Suboption IP390

***kw1=val1***

Keyword associated with the entry in the table.

**Keyword names**  
**Description**

**ACTMON='program\_call'**

Called by the EZLERECV routine and AON initialization to actively monitor a network resource for availability. Do not use this routine to call EZLEFAIL when the monitored resource is not in an acceptable state. After calling EZLEFAIL, ACTMON should not reschedule itself because EZLEFAIL starts recovery monitoring and EZLERECV restarts this process when the resource is available again.

If the routine is in a normal status, the program should reschedule itself using the NetView AT or AFTER command.

This program should be called as a program because the results of the processed call and the return code is not checked. This process is important to AON/SNA, APPN, and AON/TCP options.

**AUTOVIEW='program\_call'**

When AutoView is started, the presence of this definition adds the option on the AutoView option panel. If the option with this definition is selected by the user, the program is called to provide the user a custom information screen about the resource managed by that option.

**CHKAUTO='program\_call'**

This program is called to discover whether automated recovery, tracking, and notification should be performed for a resource. This program determines whether the resource is to be automated.

For most resources, the RECOVERY statement of the control file is processed to determine the value of the AUTO parameter and whether the current time is within an applicable NOAUTO window. It can also involve commands and other program calls to determine whether this is an automated resource. EZLECAUT is currently being used as a generic routine to check the RECOVERY entry in the control file. This program is called by EZLEFAIL and EZLERECV. Its call can be skipped by defining SKIP=C on the call to EZLEFAIL/EZLERECV.

Call CHKAUTO as a command because the return code is checked. A return code of zero (0) means automation is in effect. A return code of 1 means that automation is not in effect. If this call is made as a function, the receiving alphanumeric data back from the function is treated as a return code of zero (0).

**CHKHIGH='program\_call'**

This program is called by EZLEFAIL to check that the higher node for the resource is in a state for which recovery on the resource can occur. The higher node can be in terms of hierarchy, connectivity, network management, or priority. For example, if a Line to a PU is inactive, it is impossible to activate the PU because the physical connection is not present; therefore, the call to CHKHIGH should return a 1 (do not continue processing). This is an example of a connectivity higher node.

**CHKTHR='program\_call'**

This program is called by EZLEFAIL to determine whether threshold has been exceeded. Check the control file THRESHOLDS setting for the resource and analyze the status file error log to determine whether threshold setting has been exceeded. This call in EZLEFAIL can be skipped by coding SKIP=(T) on the EZLEFAIL invocation.

Set return codes to:

- RC=0 No threshold has been exceeded.
- RC=1 Infrequent threshold has been exceeded.
- RC=2 Frequent threshold has been exceeded.
- RC=3 Critical threshold has been exceeded.

**CRITACT='prog/func\_call'**

This program is called when a critical threshold (RC=3 from the CHKTHR command call) has been exceeded in EZLEFAIL. Its purpose is to take action when a critical threshold is exceeded. If processing in the program should not continue, a nonzero return code should be returned. If data is returned from a function call, the return code is treated as if it were zero(0), action that should occur when a critical threshold has been exceeded. In the SA option, a critical threshold exception for a PU causes the PU to be deactivated.

**ENABLE=Y|N**

The enable flag indicates whether an installed function should be enabled. The flag is checked in EZLEFAIL and EZLERECV. Those programs exit if the ENABLE parameter is set to no (ENABLE=N) for the option. This flag is not valid at the EZLRT table level.

**FREQACT='prog/func\_call'**

This program is called when a frequent threshold (RC=3 from the CHKTHR command call) has been exceeded in EZLEFAIL. Its purpose is to take action when a frequent threshold is exceeded. If processing in the program should not continue, a nonzero return code is returned. If data is returned from a function call, the return code is treated as if it were zero (0). Issue actions that should occur when a frequent threshold has been exceeded

**HELPDESK='program\_call'**

The presence of this definition adds the option into the AON help desk menu panel (EZLK1000). If the option is selected, this program is called to provide a help desk problem determination function for the operator.



**IDENTIFY='prog/func\_call'**

This program is called at the beginning of EZLEFAIL and EZLERECV to determine which option a resource belongs to and what its resource type is. The only parameter that the IDENTIFY program should require is the resource name. The IDENTIFY program should be able to determine whether the resource belongs to this option (where the IDENTIFY program is coded) and what the resource type is. If the resource type is to be returned, the IDENTIFY program should be a function call returning RESTYPE=*restype*. The return code is assumed to be zero and processing continues. If the return code is not zero, the resource is not managed by this option, and the calling program must try another option or discontinue processing.

**INFRACT='prog/func\_call'**

This program is called when an infrequent threshold (RC=3 from the CHKTHR command call) has been exceeded in EZLEFAIL. Its purpose is to take action when an infrequent threshold has been exceeded. If processing in the program should not continue, a nonzero return code should be returned. If data is returned from a function call, then the return code is treated as if it were zero (0). Issue the action that should occur when an infrequent threshold has been exceeded from this program.

**MAINPANELPOS=selection\_number\_on\_AON\_panel**

This literal is used to format the main operator interface panel. This literal specifies where, in the list of installed options, this selection is displayed. This keyword is used by the 3270 operator interface only and is only valid in the EZLOPT table.

**MESSAGING='prog/func\_call'**

This program is called when operator messaging, logging, and DDF updates are done. It is called from EZLEASLN. You can set special variables for messaging, resource types can be initialized, special processing for updating a particular option, suboption, or resource type can be done.

**MSGCLASS=(msgclass\_number,...)**

MSGCLASS is a two-digit number used to assign AON notifications to notification operators (as assigned on the NTFYOPS control file entry). Notification operators use this *msgclass\_number* in their CLASS list (on the NTFYOPS control file entry) to receive the message. A message accumulates message classes from the EZLRT MSGCLASS= table entry, EZLOPT MSGCLASS entry for the suboption and option, and the call to search *pppMTxx* (*xx* is the first two digits of the message number and *ppp* is the message prefix, *pppxxns*). If an entry in this DSIPARM member has this message ID starting in column one, the MSGCLASS defined for it is added to the MSGCLASS list.

**OCMDCMD=program\_name**

Use this command call to issue the 3270 operator interface for an option or suboption. This keyword is used by the 3270 operator interface only and is only valid in the EZLOPT table.

**OCMDDESC='literal\_for\_panel\_id'**

This literal is displayed on the main panel of the AON operator interface menu (EZLR0000) to describe the installed options and suboptions. This keyword is used by the 3270 operator interface only and is only valid in the EZLOPT table.

**OPERLIST=(operglob1 operglob2 operglob3 ... operglobn)**

Use OPERLIST when allocating work to automation operators to enable multithreading of the automation work load. If the automation operator is not active when a command is routed to an automation operator, it is routed to the next automation operator in the list. If a command is routed to *operglob2*, it is shipped to the operator ID stored the common global variable, *operglob2*. Operator IDs are defined in the AUTOOPS entries in the control file. Refer to the *IBM Z NetView Administration Reference* for more information. If operator2 (from *oprglob2*) is inactive, the command is routed to *operglob3*. The routine continues until an active automation operator is found to issue the command or *operglobn* is reached.

**OPTION=immediate\_superoption\_owning\_the\_suboption\_or\_resource\_type**

This literal should reflect the immediate owner of this option or resource type. For a suboption, it reflects the option with which it is shipped (as specified in its EZLOPT definition). For a resource type, the literal reflects the option or suboption that is responsible for managing the resource type that is specified in its EZLOPT definition.

For example, the SNA option ships with the following table definitions for its options and suboptions:

```

EZLOPT AON,ENABLE=Y,...
EZLOPT SNA,OPTION=AON,ENABLE=Y,...
...
EZLOPT SA,OPTION=SNA,ENABLE=Y,...
EZLRT NCP,OPTION=SA,...
EZLRT PU,OPTION=SA,...

EZLOPT APPN,OPTION=SNA,ENABLE=Y,...
EZLRT CP,OPTION=APPN,...

EZLOPT SNBU,OPTION=SNA,ENABLE=Y,...
EZLRT MODEM,OPTION=SNBU,...
...

```

The AON option does not have an OPTION parameter because it is the highest level in the EZLOPT table for this grouping.

This keyword is required on every entry except the highest option level.

### **RECOVMON='program\_call'**

This program is called by EZLEFAIL as the last action before notifying operations of the failure. RECOVMON date and time entries are given in GMTDATE and GMTTIME parameters. This command call can be skipped by coding SKIP=(R) on the EZLEFAIL call. These are some steps that you can follow:

1. Check the automation flag to ensure that automation has not been turned off for this resource since the last execution of this program.
2. Check the status of the resource to ensure that it is still not in an acceptable status (it might have recovered since the last execution of this program).
3. Take action to return the resource to an acceptable status.
4. Post an availability message to the logs.
5. If recovery was not successful, notify operators that the resource is still inactive. If recovery was successful, write a message to the log saying that recovery was successful.
6. Reschedule this program to run on a user-defined interval, typically, the control file MONIT intervals, although some options might choose to use a constant interval. Customers can define the interval without modifying the program.

This should be a command call because the return string and return codes are not checked. It is an asynchronous monitoring process.

### **REPORTER=variable\_name**

REPORTER is a variable that completes the Reported By field used by AON and some AON function messages. The default variable used is DOMAINID. AON/TCP uses SP.

### **RESINFO='function\_call'**

The program returns all information required by the calling program about a resource. The only parameters required should be the resource name and perhaps the resource type (if known). A function call passes back data. The return string should be in the "varname=varvalue..." format. The calling program assigns the variables when processing the return string.

### **RESLIST='program\_call'**

The operator interface calls this program to create a select list of resources of a particular type or for an option (depending on where in the tables it is coded). This keyword is used by the 3270 operator interface only and is only valid in the EZLOPT table.

### **STSCMD=program\_name**

This is the name of the command processor used by an option to update the AON VSAM status file.

### **STSPIPE=program\_name**

This is the name of the command processor used by an option to update the AON VSAM status file (PIPE version).

**SUMCMD=program\_name**

This is the program name called by the operator interface to create a resource information summary panel for a resource type or option. This keyword is used by the 3270 operator interface only and is only valid in the EZLOPT table.

**tblkey\_value**

The value on the TBLKEY= keyword specifies optional processing values used by the EZLEFAIL or EZLERECV routine. If you do not specify the TBLKEY parameter for the EZLEFAIL or EZLERECV routine, no optional processing or notification occurs. The values on the TBLKEY parameter specify keywords found in the option definition tables. In the option definition table, the keywords define the actual processing values used for optional processing. AON saves the TBLKEY values in the *outmsgid* and *spec\_function* variables. Message EZL509I is the default *outmsgid* for EZLEFAIL. EZL504I is the default *outmsgid* for EZLERECV. The value of TBLKEY is in the following format:

```
tblkey_value=(outmsgid,spec_function_call)
```

For example, the EZLEFAIL routine can be called with:

```
EZLEFAIL OPTION=SA MSGPRMS=(OPID) TBLKEY=IST105I  
RESNAME=resname
```

The EZLEFAIL routine obtains the values specified on the IST105I keyword in the option definition table, for example:

```
IST105I=(EZL531,FKVEAIDA(resname restype))
```

In the previous example, the EZLEFAIL routine issues the EZL531I message and runs FKVEAIDA as a function sending the current value of *resname* (resource name) and *restype* (resource type) for optional processing. An optional processing program would perform any automation or processing unique to the resource or failure. No optional processing is done and no message is issued if SKIP=(0) is specified on the EZLEFAIL or EZLERECV call.

The EZLEFAIL routine issues message EZL509I or EZL510I to all logs and to DDF. The EZLERECV routine issues EZL504I to all logs and to DDF. Operators do not receive this message. This message is not issued if SKIP=(A) is specified on the EZLEFAIL or EZLERECV call.

**Usage**

Sequence numbers in option definition tables can cause unpredictable results.



---

## Appendix A. Implementing X.25 Monitoring Support

AON/SNA X.25 enhances problem determination for packet networks connected to your SNA network. AON/SNA X.25 provides support for the X.25 interface between data terminal equipment and packet-switching networks.

AON/SNA X.25 provides an exit to trap hardware alerts from X.25 resources and translate them into meaningful alerts. These alerts give the operator a clear interpretation of the error, including the meaning of the error bytes, and more specific suggested action. The AON/SNA Help Desk provides NPSI diagnostic and cause code translation. AON/SNA X.25 monitors LUDRPOOLS in NCPs for threshold availability. Users can view a full-screen display of AON/SNA X.25 virtual circuits being monitored by AON/SNA X.25. Thresholding can be done in the number of sessions on a switched virtual circuit (SVC). Threshold exceptions and status changes are reflected in DDF and the NetView log.

AON/SNA X.25 also inserts a message in the NetView log. This message correlates the INOP message generated by VTAM and the corresponding NPSI alert. Additionally, the operator can display the available number of LU control blocks. An alert is sent to the hardware monitor each time an incorrect XID problem is encountered.

Users can monitor switched connections through full-screen panels. Each connection or disconnection, related to a monitored line, updates the panel. Switched group lines are defined in the control file.

---

### Understanding the NPSI Hardware Monitor Enhancement

This section describes the FKVXITAN exit routine and the BNJ146 message automation.

#### Using the FKVXITAN Exit Routine

The FKVXITAN user exit plays a role in the generic alerts generation from NPSI alerts. This user exit traps alerts from NPSI and translates the diagnostic, clear, and cause codes. The AON/SNA X.25 code then issues a GENALERT with the translated information so that operators can more easily understand X.25 NPSI alerts in the NetView hardware monitor.

#### Understanding the CNM Interface

All unsolicited communication network management (CNM) messages are sent by VTAM to the NetView product as part of the specialized DELIVER RU (X'810812') through the CNM interface. This includes the alerts coming from X.25 NPSI. The DSICRTR NetView subtask is the CNM Router task which gets those unsolicited CNM messages and later dispatches them to other tasks (for example, the hardware monitor).

#### Understanding the XITCI Exit Routine for the DSICRTR DST

The XITCI exit routine for the DSICRTR DST receives control for each incoming unsolicited CNM message. The user exit routine must be defined in the DSICRTTD initialization member for the DSICRTR task.

#### Understanding the FKVXITAN User Exit

A XITCI user exit routine is coded for the DSICRTR subtask, FKVXITAN. The FKVXITAN exit is used for each unsolicited CNM message, including the RECFMS type 00 (alerts) coming from X.25 NCP Packet Switching Interface (NPSI).

#### Understanding FKVXITAN Logic

The following list explains the flow of the FKVXITAN exit:

1. The DELIVER RU is parsed to recognize the embedded RU and the block ID (identifying the originating product). If it is not a RECFMS coming from NPSI, no processing occurs and the exit returns control to normal NetView processing.

2. If it is an alert record, the rest of the input is parsed and the following fields are extracted:

- alert type
- major code
- minor code
- action code
- text (when present),
- three qualifiers
- resource hierarchy

Format conversions take place so you can display these fields in a message.

3. The BNJ146I message is created with the same format as the standard NetView BNJ146I message for RECFMS except that the token 2 and 3 (date and time) are not inserted.
4. The message is sent to the X.25 network management task for further automation.
5. The RECFMS that triggered the FKVXITAN user exit is suppressed by setting the return code to 4. The alert RECFMS coming from NPSI disappears and does not go to the hardware monitor.

## Understanding BNJ146 Message Automation

Figure 208 on page 370 shows a sample entry that automates message BNJ146.

```
IF MSGID = 'BNJ146I'
& TOKEN(2) = 'R'
& TEXT = 'BKID=FEF'
& TEXT = 'TYPE='BKID=FEF'. 'ACT='ACT'QUAL='
QUAL TEXT='VCN 'HIER='HIER
'DOMID='
THEN EXEC (CMD('FKVEOG01 T' TYPE ACT 'Q('QUAL')Q' 'T('VCN')T'
HIER) ROUTE(ONE *)) DISPLAY(N) NETLOG(Y) SYSLOG(N);
*
IF MSGID = 'BNJ146I'
& TOKEN(2) = 'R'
& TEXT = 'BKID=FEF'
& TEXT = 'TYPE='TYPE'BKID=FEF'. 'ACT='ACT'QUAL='
QUAL 'HIER='HIER
'DOMID='
THEN EXEC (CMD('FKVEOG01 Q ' TYPE ACT 'Q('QUAL')Q' HIER)
ROUTE(ONE *)) DISPLAY(N) NETLOG(Y) SYSLOG(N);
```

Figure 208. Automation Member Entries for BNJ146I

The first entry corresponds to alerts sent by NPSI V2/V3 and the second entry to alerts sent by NPSI V1. The parameters from the BNJ146I message that are needed by the program are extracted from the message and passed as variables to the program.

## Understanding Code Point tables

The following code points are part of these tables:

- BNJ92UTB alert description code points
- BNJ93UTB probable cause code points
- BNJ94UTB user cause code points
- BNJ95UTB install cause code points
- BNJ96UTB failure cause code points
- BNJ81UTB possible action code points

NPSI sends basic alerts (RECFMS type 00). Only a few codes are supplied by IBM to cover the range of possible AON/SNA X.25 problems. This package provides user code points to complement the code supplied by IBM. The code point range reserved for the user is between X'E000' and X'FFFF'. The range of

code points used in this package is from X'EE00' to X'EEFF'. Thus, AON/SNA provides more details and more advice on the possible causes and solutions for an alert.

## Describing the correlation between INOP messages and NPSI alerts

When alerts are sent against the virtual circuit line resources, the station (PU) is not owned at the time the alert is sent. This makes using the NPSI alerts difficult.

This difficulty affects the outgoing call refused alerts (either refused by NPSI or cleared by the AON/SNA X.25 PSDN or the DTE). AON/SNA X.25 NPSI sends on the SSCP-PU (the NCP PU) session, first an INOP type 1 on the address of the station, then a RECFMS on the address of the VC line. As a result, on the host side, a message is received:

```
IST259I INOP RECEIVED FOR nodename CODE = 01
```

The *nodename* is the name of the switched PU that attempted to call out. This message identifies neither the path nor the link on which the callout failed. An NPSI alert gives the hierarchy down to the virtual circuit and indicates why the outgoing failure occurred, but it does not give the name of the switched PU that attempted to call out and is affected by the failure.

For installations that frequently make outgoing calls on AON/SNA X.25 switched virtual circuits (SVCs), trying to correlate INOP messages and corresponding NPSI alerts is a challenge. AON/SNA X.25 responds to this need, but because of the asynchronous nature of this process, the tool cannot be considered as reliable.

Message IST259I is automated in the automation table and puts the switched PU name into a task global variable. A queue of 10 such global variables (X25PU0 to X25PU9) is necessary because several IST259I messages can arrive before the corresponding BNJ146I messages arrive and AON/SNA X.25 can pick up the PU names. This occurs when several PATHS statements are active for a switched PU, leading to successive outgoing call failures. If you must store more than 10 PU names in global variables, you can easily modify the FKVEOI00 and FKVEOG02 routines. Replace the number 10 with the new value. One statement is in FKVEOI00 and two are statements in FKVEOG02.

A sample message automation member statement follows:

```
IF MSGID = 'IST259I' & TEXT =. 'FOR X' .  
& TEXT =. 'FOR' PUN 'CODE = 01'  
THEN EXEC (CMD('FKVEOI00 ' PUN) ROUTE(ONE *))  
NETLOG(Y) SYSLOG(N) DISPLAY(Y);
```

AON/SNA X.25 generates the generic alerts from the NPSI alerts on virtual circuits. It retrieves the oldest switched PU name in the queue, if one is present. It inserts the PU name at the end of the hierarchy, behind the line name if no PU name is currently in the hierarchy.

AON/SNA X.25 records the time together with the PU name in a queue. When a PU name has been in the queue for more than 3 minutes, AON/SNA assumes the correspondence between the INOP messages and the alerts has been lost. It then cleans up the whole queue.

A message is written to the NetView log that shows the diagnostic code, the cause code, and all the hierarchies for the problem.

Figure 209 on page 371 shows an example of the message written to the NetView log.

```
NCCF          N E T V I E W   L O G          CNM01          02/06/19 14:05:44  
C CNM01      CLEARCAUSE BYTE = 05, DIAG BYTE = 00 RECEIVED FOR  
             FOLLOWING RESOURCE HIER=RADNCPE,NPSI,MCH01,MCH,000001,VCN,  
             XL01002,V.C.,X.25PUT,RDTE
```

Figure 209. NetView Log

## Monitoring Switched Virtual Circuit (SVC) resource utilization

---

An AON/SNA X.25 subscription supplies, with one physical link, a number of virtual circuits, both *private virtual circuits* (PVCs) and *switched virtual circuits* (SVCs). These switched virtual circuits (SVCs) are often used for frequent and short duration connections. The number of switched virtual circuits (SVCs) in use increase and decrease with starting and ending calls from programs.

The primary purpose of this monitoring facility is to provide one screen that displays the number of switched virtual circuit (SVC) lines as available or busy. The system automatically updates this information, but the operator manually refreshes the screen to receive the current information.

Other features of this facility include the ability to:

- Apply threshold values for the number of free switched virtual circuits (SVCs) to highlight switched virtual circuit (SVC) shortages
- Add, change, or delete a temporary link definition
- Define individual timers for critical links to refresh status information about a fixed interval
- Link to DDF to enable DDF monitoring of link availability and threshold status

## Understanding the X25INIT command

---

To start the switched virtual circuit (SVC) monitoring facility, initialize the environment by running the X25INIT command. Call this command from the configuration file at AON startup. The X25INIT command performs the following:

- reads the X25MONIT control file entries in the configuration file
- checks these entries for validity
- sets up timers for links that you request
- obtains status information for the links

If X25INIT detects errors while validating the parameters for a link, AON/SNA issues a message and bypasses that link.

Switched virtual circuit (SVC) monitoring works only with the naming conventions given by NPSI for the group name and the VC line name. The NPSI naming convention follows:

**pLxxxlcn**

The name of the VC line.

**p25Sxxxxy**

The name of the switched virtual circuit group name.

where:

**p**

Is the prefix coded in the X25BUILD macro.

**xxx**

Is the address specified in the AddRESS operand of the X25MCH macro.

**lcn**

Is the logical channel number.

**y**

Is the group sequence on the physical link.

## Understanding the LUDRPOOL command

---

The LUDRPOOL routine is based on the NCP control blocks structure and works with different versions of NCP. The supported versions are NCP V4R2 through V7.



If you specify a time interval, AON/SNA X.25 issues a NetView EVERY command to run the LUDRPOOL check at regular intervals. It sets a threshold as a global variable so that an alert is sent when the pool of available LUs falls under the threshold.

## Monitoring LUDRPOOL utilization

One of the critical resources in switched SNA is the pool of LU control blocks created within the NCP to be used dynamically for dynamic reconfiguration and switched data links.

The pool of LU blocks is defined by the NCP LUDRPOOL macro:

```
DRP00LPU PUDRPOOL NUMBER=0

DRP00LLU LUDRPOOL NUMTYP1=10, RESERVE 10 LUS ON PU.T1 PUS
              NUMTYP2=90,   RESERVE 90  LUS ON PU.T2 PUS
              NUMILU=20     RESERVE 20  LUS FOR IND.LU
```

The PUDRPOOL statement is necessary to use the LUDRPOOL utilization counters.

AON/SNA X.25 is more generally used for switched virtual circuits (SVCs). LUDRPOOL is one of the resources that major AON/SNA X.25 NPSI users want to monitor, especially if they have hundreds or thousands of available switched virtual circuits (SVCs), controlled by one NPSI.

PU Type 2 represents the vast majority of devices. In the previous example, the pool of LUs for switched PU 2 (operand NUMTYP2) is a common installation. PU Type 1 is relevant for NPSI (non-SNA connections), but there is always one simulated LU per virtual channel.

## Understanding security alerts for an incorrect XID

A frequent cause of failure in switched SNA connections occurs in the XID exchange, when the SNA ID (IDBLK + IDNUM) sent by the device attempting a switched connection is not recognized by the host.

VTAM compares the station ID it receives in the XID with the values coded for the IDBLK and IDNUM parameters of the PU statements in the switched major nodes.

When VTAM does not find a match, the call is refused and a message issued:

```
IST690I CONNECTION REQUEST DENIED - INVALID STATION ID = aaaaxxyyyyyy
```

The message might be an indication of:

- A security violation. Somebody might have tried to break into your network.
- An operations error. A switched major node might not have been activated.
- A system programming error. The good values of IDBLK-IDNUM might not have been specified in a switched major node.

All of these events are likely to happen regularly with AON/SNA X.25 NPSI switched virtual circuits (SVCs), in particular when the number of connections is high.

A specific NPSI difficulty with IDBLK-IDNUM is encountered for non-SNA AON/SNA X.25 connections (PCNE, PAD, GATE, DATE). In this case, the IDBLK and IDNUM do not come from the remote DTE but are generated by NPSI, according to the virtual circuit on which the call is received and according to the order of the virtual circuit definitions in the NCP generation.

## Implementing an alert through GENALERT

In the following example, the IST690I message is automated in the automation table and causes an alert through GENALERT. The security type is (SCUR).

```
IF MSGID = 'IST690I' & TEXT = '.'= ' STATID  
THEN EXEC (CMD('FKVE0SEC ' STATID) ROUTE(ONE *)) DISPLAY(N)  
NETLOG(Y) SYSLOG(N);
```

---

## Appendix B. Using the browse facility

AON uses the NetView browse facility, which enables certain commands and functions that you can use to browse log files. When you display any of the AON logs, AON enables certain commands and function keys to help you to browse the logs more efficiently.

In addition to the function keys used throughout NetView and described in [“Function keys” on page 7](#), the browse facility provides the following additional function keys to help you browse log files:

**F4 Top**

Moves to the beginning of the information being browsed.

**F5 Bottom**

Moves to the end of the information being browsed.



## Appendix C. AON command synonyms

The following tables show how to use fastpaths to reach each AON panel.

| <i>Table 15. Fastpath Commands Reference Table</i> |                 |                                          |
|----------------------------------------------------|-----------------|------------------------------------------|
| <b>Command</b>                                     | <b>FastPath</b> | <b>Panel</b>                             |
| AON                                                | AON             | AON: Operator Commands Main Menu panel   |
| AONINFO                                            | AON 0           | AON tutorial panel                       |
|                                                    | AON 1           | AON: Base Functions panel                |
|                                                    | AON 1.0         | AON Base Functions tutorial panel        |
| AONHD                                              | AON 1.1         | AON: Help Desk panel                     |
| AUTOVIEW                                           | AON 1.2         | AON: AutoView panel                      |
| DDF                                                | AON 1.3         | Data Center Networks panel               |
|                                                    | AON 1.4         | AON: Automation Settings panel           |
| SETAUTO                                            | AON 1.4.1       | Recovery Settings panel                  |
| DISAUTO                                            | AON 1.4.1       | Recovery Settings panel                  |
| DELAUTO                                            | AON 1.4.1       | Recovery Settings panel                  |
| SETNTFY                                            | AON 1.4.2       | Notification Operators panel             |
| DELNTFY                                            | AON 1.4.2       | Notification Operators panel             |
| SETTHRES                                           | AON 1.4.3       | Thresholds panel                         |
| DELTHRES                                           | AON 1.4.3       | Thresholds panel                         |
| SETMONIT                                           | AON 1.4.4       | Monitor Intervals panel                  |
| DELMONIT                                           | AON 1.4.4       | Monitor Intervals panel                  |
| ACTMON                                             | AON 1.4.5       | Active Monitor Settings panel            |
|                                                    | AON 1.5         | AON: Cross Domain Functions panel        |
| CDLOG                                              | AON 1.5.1       | AON: Cross Domain Logon panel            |
| AONGW                                              | AON 1.5.2       | AON: Cross Domain Gateway Display panel  |
| AONTAF                                             | AON 1.5.3       | AON: Terminal Access Facility Menu panel |
| TIMER                                              | AON 1.6         | AON: Timer Management panel              |
| AONMAINT                                           | AON 1.7         | AON: Task and Log Maintenance panel      |
| DSPCFG                                             | AON 1.7.1       | AON: Configuration Data Display panel    |
| DSPSTS                                             | AON 1.7.2       | Display Status Data panel                |
|                                                    | AON 1.7.3       | NetView log panel                        |
| NLOG                                               | AON 1.7.4       | Automation log panel                     |
| DBMAINT                                            | AON 1.7.5       | AON: Database Maintenance panel          |
| STARTEZL                                           | AON 1.7.6       | AON: Start Automation Components panel   |

*Table 15. Fastpath Commands Reference Table (continued)*

| <b>Command</b> | <b>FastPath</b> | <b>Panel</b>                                   |
|----------------|-----------------|------------------------------------------------|
| STOPEZL        | AON 1.7.6       | AON: Start Automation Components panel         |
| AONTASK        | AON 1.7.7       | AON: Task/Operator Display panel               |
|                | AON 1.8         | AON: Support Functions panel                   |
| AONTRACE       | AON 1.8.1       | AON: Set Trace panel                           |
| AONINIT        | AON 1.8.2       | AON: Reinitialize Automation panel             |
| AONENABL       | AON 1.8.3       | AON: Enable/Disable Automation panel           |
| CGED           | AON 1.8.4       | AON: Common Global Editor panel                |
|                | AON 1.8.5       | AON: Automation Table panel                    |
| LOADTBL        | AON 1.8.6       | AON: Loader Tables panel                       |
| ILOG           | AON 1.9         | Inform Log Utility panel                       |
| MARK           |                 | DDF mark                                       |
| UNMARK         |                 | DDF unmark                                     |
| DM             |                 | Clear held message                             |
| AONAIP         |                 | Set or reset the AIP operator status           |
| AONSNA         | AON 2           | SNA Automation: Menu panel                     |
|                | AON 2.0         | Tutorial                                       |
| SNAHD          | AON 2.1         | SNA Help Desk panel                            |
| SNAMAP         | AON 2.2         | SNAMAP panel                                   |
| VTAMOPT        | AON 2.3         | VTAM Options Management panel                  |
| NETSTAT        | AON 2.4         | NetStat panel                                  |
| VTAMCMD        | AON 2.5         | VTAM Commands panel                            |
| APPN           | AON 2.6         | APPN Commands Menu panel                       |
|                | AON 2.6.0       | APPN Tutorial panel                            |
|                | AON 2.6.1       | APPN Issue Checkpoint Commands panel           |
|                | AON 2.6.2       | APPN Display Control Points panel              |
|                | AON 2.6.3       | APPN Display Directory panel                   |
|                | AON 2.6.4       | APPN Display Transmission Group Profiles panel |
| SNBU           | AON 2.7         | Switched Network Backup Menu panel             |
|                | AON 2.7.0       | SNBU Tutorial                                  |
| LISTSNBU       | AON 2.7.1       | Display SNBU Resource List panel               |
| DISSNBU        | AON 2.7.2       | Display Information on SNBU Resources panel    |
| SETSNBU        | AON 2.7.3       | Control SNBU Resources panel                   |
| DISPOOL        | AON 2.7.4       | Display Modem Pools panel                      |
| SETPOOL        | AON 2.7.5       | Add or Delete Modem Pool Resources panel       |

*Table 15. Fastpath Commands Reference Table (continued)*

| <b>Command</b> | <b>FastPath</b> | <b>Panel</b>                                                              |
|----------------|-----------------|---------------------------------------------------------------------------|
| CHGSNBU        | AON 2.7.6       | Control SNBU Connections Manually panel                                   |
| CHGSPEED       | AON 2.7.7       | Change Modem Speed Manually panel                                         |
| QRYSNBU        | AON 2.7.8       | Display SNBU Status panel                                                 |
| X25            | AON 2.8         | X.25 Menu panel                                                           |
|                | AON 2.8.0       | X.25 Tutorial                                                             |
| X25MONIT       | AON 2.8.1       | X.25 SVC Monitoring panel                                                 |
| LUDRPOOL       | AON 2.8.2       | X.25 LUDR Pool Management panel                                           |
| DSPSNBU        |                 | Displays SNBU Status Data from status file                                |
| SNAVIEW        |                 | Displays the AutoView resources for AON/SNA                               |
| X25INIT        |                 | Initializes X.25                                                          |
| AONTCP         | AON 3           | TCP/IP Automation: Commands Menu panel                                    |
|                | AON 3.1         | TCP/IP Automation: Ping a Service Point panel                             |
| NV6KPERF       | AON 3.1.4       | TCP/IP Automation: Performance Thresholds panel                           |
| NV6KLIST       | AON 3.1.5       | TCP/IP Automation: Display the TCP/IP Critical Resource List panel        |
|                | AON 3.2         | TCP/IP Automation: Issue Command to Service Point panel                   |
| MVSPING        | AON 3.2.1       | TCP/IP Automation: Ping a resource through TCP/IP for MVS                 |
| IPSTAT         | AON 3.2.2       | TCP/IP Automation: Manager Telnet and FTP sessions through TCP/IP for MVS |
| NVSNMP         | AON 3.2.5       | TCP/IP Automation: Display the SNMP Menu                                  |
|                | AON 3.2.6       | TCP/IP Automation: Server Management                                      |
| IPMAN          | AON 3.2.7       | TCP/IP Automation: IP Resource Management                                 |
| TCPLIST        | AON 3.2.7       | TCP/IP Automation: IP Resource Management                                 |
| NV6KVIEW       |                 | TCP/IP Automation: AutoView panel                                         |





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## Appendix D. Customizing the SNMP Group Definition File (FKXSNMP)

**Note:** This function no longer requires AON. The information about this function is included in both *IBM Z NetView IP Management* and the *IBM Z NetView User's Guide: Automated Operations Network*.

Use the following rules for creating new entries in the SNMP group definition (FKXSNMP) file:

- The Group name must be from 1 to 15 characters and must start in column 1.
- The Group name cannot be duplicated.
- There must be at least one space between the Group Name, the GROUP, the Group type, and the base MIB for Table type Groups.
- There can be up to three lines of abstract definition for a Group. The abstract lines can be up to 72 positions and must start with a question mark (?) in column 1.
- The Abstract lines for a Group must follow the GROUP statement for the group.
- Valid Group types are:
  - LIST
  - LIST+
  - TABLE
  - WALK

A **LIST** group type must include the EXACT MIB variable names to be collected.

A **LIST+** group works almost the same as a LIST type Group, but enables the definition of variable data.

The LIST+ group enables you to specify a variable field to be appended to the list of MIB objects in the group. This enables a single group definition to be used for various MIB object groups. For example, a group can contain objects that relate to a specific interface number. If you use traditional LIST type groups, you need multiple groups, one to define each interface. A LIST+ group can be defined to ask prompt for an interface number, when selected, enabling only one group definition to be needed. LIST+ adds keywords that are used to set up the variable data. All of these keywords must start in column 1.

### PANELINPUT

Defines this as a LIST+ group

### PANELCONST

A user-customizable field that is displayed in the input panel and must be delineated with double quotation marks ("")

### PANELVAR

An input field where the data is collected from the screen, for example:

```
PANELCONST "PLEASE ENTER AN INTERFACE NUMBER:  
"PANELVAR "___"
```

Displays as:

```
PLEASE ENTER AN INTERFACE NUMBER:
```

```
---
```

VAR keywords in LIST and LIST+ groups indicate the starting of varbind lists. This helps in parsing in UNIX. VAR must start in column 1.

A **TABLE** group type must have a base variable to start the search. This is the index variable for the table. A Table group type does not need to have variables listed. The table is Walked and all variables in the table are collected. MIB variables listed in this group are for documentation only.

A **WALK** group type must be a known group name (defined in an existing RFC). A walk command is issued against the name. MIB variables listed in this group are for documentation only. Walk groups, as defined in some RFCs, might be much larger than the 15-character limit. To work with this limit, the Full Name for the Walk Group might be entered on the Group line after the WALK type (for example: groupname GROUP WALK veryLongGroupName).

Comments must have an asterisk (\*) in column 1.

## Appendix E. VTAM messages

This section explains the purpose of the VTAM messages that AON/SNA uses, shows the operator response, and any special processing.

### General Resource VTAM messages

|                 |                                                                      |
|-----------------|----------------------------------------------------------------------|
| <b>IST093I</b>  | <b><i>resource_name</i> ACTIVE</b>                                   |
| <b>IST1132I</b> | <b><i>resource_name</i> IS ACTIVE,<br/>TYPE=<i>resource_type</i></b> |

#### Explanation

Indicates when a resource becomes active. AON/SNA posts the resource as available.

#### Operator response

- EZL504I : *resource\_type resource\_name* IS AVAILABLE
- EZL517I : *resource\_type resource\_name* HAS BECOME ACTIVE FROM INTERVENTION BY OPERATOR *operator\_id*

If the operator that recovered the resource is an AON/SNA automation operator, suppress the EZL517I operator notification because the IST093I was issued as a result of automation activity.

Clear the threshold settings from the status file in preparation for the next failure.

If recovery is in effect, the status of the resource is not CONCT, and the resource type is a PU, LINE, or LINKSTA, check to see if this is part of a two-line TG with lines of unequal speeds and switch traffic flow to the faster (primary) line and relegate the slower line to a back-up function in case the faster line fails. This is defined by the TGSWITCH control file definitions.

|                 |                                                                            |
|-----------------|----------------------------------------------------------------------------|
| <b>IST105I</b>  | <b><i>resource_name</i> NODE NOW<br/>INACTIVE</b>                          |
| <b>IST1133I</b> | <b><i>resource_name</i> IS NOW INACTIVE,<br/>TYPE=<i>resource_type</i></b> |

#### Explanation

If the VTAM program issues IST105I or IST1133I as a result of an operator (OST) command, AON/SNA posts status and stops automated recovery. This enables the operator to work with a resource without AON intervention. The operator is now responsible for recovery of the resource. If the VTAM program issues the message as unsolicited, AON/SNA initiates standard EZLEFAIL recovery.

#### Operator response

- EZL505I : RECOVERY TERMINATED FOR *resource\_type resource\_name* DUE TO ACTION BY OPERATOR *operator\_id*
- EZL509I : *resource\_type resource\_name* IS UNAVAILABLE
- EZL531I : *resource\_type resource\_name* IS INACTIVE DUE TO OPERATOR *operator\_id* INTERVENTION

If an AON/SNA automation operator issues the INACT command, ignore the message because it is part of a recovery attempt.

If an operator issues the INACT command, stop recovery and issue the EZL531I.

If a timer exists for recovery monitoring, purge the timer and start recovery monitoring again.

If the automation status is INRCVY REMIND or REACTV, issue the EZL505I message to all operators.

|                 |                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------|
| <b>IST129I</b>  | <b>UNRECOVERABLE OR FORCED<br/>ERROR ON NODE <i>resource_name</i> -<br/>VARY INACT SCHED</b> |
| <b>IST1135I</b> | <b>FORCED VARY INACT SCHEDULED<br/>FOR <i>resource_name</i></b>                              |
| <b>IST1136I</b> | <b>VARY INACT <i>resource_name</i><br/>SCHEDULED - UNRECOVERABLE<br/>ERROR</b>               |

#### Explanation

Detects if resource is in a state that AON/SNA cannot recover it from. Stop automation and notify operators that intervention might be required.

#### Operator response

- FKV526I : *resource\_type resource\_name* IS IN AN INVALID STATE: CURRENT STATUS IS *resource\_status*

Thresholding, availability, messaging, and recovery are not performed by the EZLEFAIL program.

Stops recovery attempts and messaging if the resource is in an unrecoverable state: not ACT\* CON\* INA\* or IIN\*.

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| <b>IST383I</b>  | <b>DEACTIVATION OF ID =<br/>resource_name FAILED -<br/>REQUEST: rcmd SENSE : rsense.</b> |
| <b>IST1268I</b> | <b>resource_name DEACTIVATION<br/>rcmd FAILED : rsense</b>                               |

#### Explanation

Detects failed resource deactivations. AON/SNA initiates recovery for the resource with the EZLEFAIL program.

#### Operator response

- EZL509I : *resource\_type resource\_name* IS UNAVAILABLE
- FKV527I : DEACTIVATION OF *resource\_name* CANNOT BE COMPLETED BECAUSE *resource\_name* HAS FAILED WITH SENSE: *rsense*

|                 |                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------|
| <b>IST608I</b>  | <b>VARY ACT FOR ID =<br/>minor_resource_name FAILED-<br/>HIGHER NODE : resource_name</b>    |
| <b>IST1274I</b> | <b>VARY ACT <i>minor_resource_name</i><br/>FAILED = <i>resource_name</i> NOT<br/>ACTIVE</b> |

#### Explanation

Detect when AON/SNA cannot recover a resource because its higher node is not active. AON initiates recovery for the higher node and the lower node with the EZLEFAIL program.

#### Operator response

- FKV529I : ACTIVATION OF *minor\_resource\_name* FAILED DUE TO INACTIVE HIGHER NODE - *resource\_name*. ACTIVATION OF *resource\_name* IS ATTEMPTED
- EZL509I : *resource\_type resource\_name* IS UNAVAILABLE

## CDRM VTAM messages

|                |                                                                                   |
|----------------|-----------------------------------------------------------------------------------|
| <b>IST727I</b> | <b>COMMUNICATION WITH CDRM<br/>resource_name LOST -<br/>REASON=X'cause code'.</b> |
|----------------|-----------------------------------------------------------------------------------|

Thresholding is not performed for the resource.

|                 |                                                                 |
|-----------------|-----------------------------------------------------------------|
| <b>IST619I</b>  | <b>ID=<i>resource_name</i> FAILED-<br/>RECOVERY IN PROGRESS</b> |
| <b>IST1416I</b> | <b>ID=<i>resource_name</i> FAILED-<br/>RECOVERY IN PROGRESS</b> |

#### Explanation

Detects a resource failure that the VTAM program is attempting to recover. AON/SNA enables the VTAM MONIT interval to recover the resource. If the VTAM program is unsuccessful, AON/SNA initiates recovery.

#### Operator response

- EZL509I : *resource\_type resource\_name* IS UNAVAILABLE

Thresholding and recovery are performed after the MONIT interval delay.

|                |                                                                      |
|----------------|----------------------------------------------------------------------|
| <b>IST621I</b> | <b>RECOVERY SUCCESSFUL FOR<br/>NETWORK NODE <i>resource_name</i></b> |
|----------------|----------------------------------------------------------------------|

#### Explanation

Detects successful VTAM recovery of a resource. Posts active resource status and discontinues AON/SNA recovery efforts.

#### Operator response

- EZL504I : *resource\_type resource\_name* IS AVAILABLE

Clear the status file THRSOLD setting if it is CRIT.

If RECOVERY flags are on for the resource (call EZLECAUT), the resource status is not CON\* or REC\*, and the resource type is a PU, LINE, or LINKSTA, check to see if the resource is one of a two line TG with lines of unequal speeds. Manage the line so that the faster line (the primary line) is carrying all the traffic and relegate the slower line to a back-up function in case the faster line fails. This is defined by the control file TGSWITCH definitions.

#### Explanation

Detects and notifies operators of a CDRM failure. AON/SNA tries to recover CDRM failures with the EZLEFAIL program.

### Operator response

- FKV520I : COMMUNICATION WITH CDRM *resource\_name* LOST DUE TO FORCED INACTIVATE OF THE VR (RC = *rc*); AUTOMATIC RECOVERY IN PROGRESS
- FKV521I : COMMUNICATION WITH CDRM *resource\_name* LOST DUE TO VR INOP (RC = *cause\_code*.); AUTOMATIC RECOVER IN PROGRESS
- FKV522I : COMMUNICATION WITH CDRM *resource\_name* LOST DUE TO SSCP - FAILURE (RC = *rc*); AUTOMATIC RECOVERY IN PROGRESS
- FKV525I : COMMUNICATION WITH CDRM *resource\_name* LOST DUE TO SESSION OVERRIDE - ACTIVATE ALREADY IN PROGRESS (RC = *cause\_code*.)
- FKV511I : COMMUNICATION WITH CDRM *resource\_name* LOST DUE TO SSCP FAILURE - (RC=*rc*); AUTOMATIC RECOVERY IN PROGRESS
- FKV517I : COMMUNICATION WITH CDRM *resource\_name* LOST DUE TO CLEANUP - THE SSCP IS RESETTING (RC = *cause\_code*)
- FKV519I : COMMUNICATION WITH CDRM *resource\_name* LOST DUE TO SSCP CONTENTION (RC = *cause\_code*)
- FKV531I : COMMUNICATION WITH CDRM *resource\_name* LOST DUE TO GATEWAY NODE CLEANUP (RC = *cause\_code*)

EZL504I is not issued for the CDRM by the EZLEFAIL program.

### IST742I

**ACTIVATION OF CDRM  
*resource\_name* FAILED GWN PATH  
NOT AVAILABLE**

### Explanation

Detects and notifies operators of a CDRM failure. AON/SNA initiates CDRM recovery with the EZLEFAIL program.

### Operator response

- FKV541I : ACTIVATION FAILED FOR CDRM *resource\_name*; GATEWAY PATH NOT AVAILABLE

### IST742I

**ACTIVATION OF CDRM  
*resource\_name* QUEUED GWN  
PATH NOT AVAILABLE**

### Explanation

Detects and notifies operators of CDRM queuing.

### Operator response

- FKV541I : ACTIVATION queued FOR CDRM *resource\_name*; GATEWAY PATH NOT AVAILABLE

Thresholding and recovery not run by the EZLEFAIL program.

## Application messages

### IST400I

**TERMINATION IN PROGRESS FOR  
APPLID *resource\_name***

### Explanation

Detects and notifies operators that the application stopped.

### Operator response

- FKV552I : APPLICATION *resource\_name* HAS BEEN TERMINATED BY VTAM

Availability messaging thresholding and recovery are not run by the EZLEFAIL program.

### Explanation

Detects a VTAM ACB close has started. AON/SNA monitors the process to be sure that the close is successful. IST805I is issued. This prevents ACBs from being in an endless loop.

### Operator response

- EZL550I : APPLICATION *appl* WAS NOT CLOSED CORRECTLY BY *operator\_id*

EZLEFAIL processing for availability messages, thresholding, messaging, and recovery is not done.

Check the control file recovery flag for JOBNAME. If recovery is off for it then exit.

Start timers for NOTIFY and CHECK intervals. When running on CHECK intervals (defined in the control file RECOVERY flag for APPLs) then the status of the application is checked but if it is down, operators are not notified. If it is up, recovery processing is done.

### IST804I

**VTAM CLOSE IN PROGRESS FOR  
*applname* OPENED BY *jobname***

When running on NOTIFY intervals, operators are notified if the application is still unavailable.

After two minutes, issue the EZL550 message. This is to give the VTAM program a chance to stop the application correctly before notifying operators. If it stops correctly, IST805I processing purges this timer and the operators are not notified.

---

|                |                                                    |
|----------------|----------------------------------------------------|
| <b>IST805I</b> | <b>VTAM CLOSE COMPLETE FOR<br/><i>applname</i></b> |
|----------------|----------------------------------------------------|

---

### Explanation

The VTAM close is complete for the ACB. AON/SNA stops monitoring for successful close.

Only optional processing is performed by the EZLEFAIL program. Purge the timer set from IST804I to issue a message to the operator that the application has not stopped normally.

## Host VTAM messages

---

---

|                |                                                                                                         |
|----------------|---------------------------------------------------------------------------------------------------------|
| <b>IST348I</b> | <b>UNABLE TO PROCESS<br/>DISCONNECTION FOR PU =<br/><i>resource_name</i> DUE TO LACK OF<br/>STORAGE</b> |
|----------------|---------------------------------------------------------------------------------------------------------|

---

### Explanation

Notify operators of storage shortage problems during disconnection of a PU.

### Operator response

- EZL509I : *resource\_type resource\_name* IS UNAVAILABLE
- FKV518I : *resource\_type resource\_name* DISCONNECTION FAILED DUE TO LACK OF STORAGE - STATUS IS *resource\_status*

'D NET,BFRUSE'

Thresholding is not performed by the EZLEFAIL program.

---

|                |                                                         |
|----------------|---------------------------------------------------------|
| <b>IST561I</b> | <b>STORAGE UNAVAILABLE: <i>pool</i><br/>BUFFER POOL</b> |
|----------------|---------------------------------------------------------|

---

### Explanation

Detects and notifies operators of storage shortage problems.

### Operator response

- FKV514I : STORAGE UNAVAILABLE FOR *resource\_name* BUFFER POOL

'D NET,BFRUSE'

The EZLEFAIL program performs only optional processing and messaging.

---

|                |                                                     |
|----------------|-----------------------------------------------------|
| <b>IST562I</b> | <b>STORAGE UNAVAILABLE: <i>pool</i><br/>REACHED</b> |
|----------------|-----------------------------------------------------|

---

### Explanation

Detects and notifies operators of storage shortage problems in CSA.

### Operator response

- FKV515I : STORAGE UNAVAILABLE - *resource\_name* HAS BEEN REACHED

'D NET,BFRUSE'

The EZLEFAIL program performs only optional processing and messaging.

---

|                |                                                                |
|----------------|----------------------------------------------------------------|
| <b>IST564I</b> | <b>STORAGE UNAVAILABLE COMMON<br/>AREA SUBPOOL <i>pool</i></b> |
|----------------|----------------------------------------------------------------|

---

### Explanation

Detects and notifies operators of storage shortage problems in CSA.

### Operator response

- FKV516I : STORAGE UNAVAILABLE FOR SUBPOOL *resource\_name*

'D NET,BFRUSE'

The EZLEFAIL program performs only optional processing and messaging.

---

|                |                                                           |
|----------------|-----------------------------------------------------------|
| <b>IST693I</b> | <b>UNABLE TO DISCONNECT ID =<br/><i>resource_name</i></b> |
|----------------|-----------------------------------------------------------|

---

### Explanation

Detects disconnection failure. AON/SNA initiates recovery with the EZLEFAIL program.

### Operator response

- EZL509I : *resource\_type resource\_name* IS UNAVAILABLE
- FKV528I : DISCONNECT OF *resource\_name* FAILED DUE TO I/O ERROR OR INSUFFICIENT STORAGE

'D NET,BFRUSE'

---

**IST706I**      **ADJSSCP TABLE FOR**  
***resource\_name* IGNORED -**  
**INSUFFICIENT STORAGE**

#### Explanation

Detects and notifies operators of a storage shortage problem.

#### Operator response

- FKV513I : STORAGE UNAVAILABLE FOR ADJSSCP TABLE FOR *resource\_name*

'D NET,BFRUSE'

The EZLEFAIL program performs only optional processing and messaging.

## NCP VTAM messages

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---

**IST095A**      ***replyid*: OPTION TO DUMP**  
***resource\_name* AVAILABLE -**  
**REPLY 'YES' or 'NO' or**  
**'YES,DUMPSTA=LINKSTANAME'**

#### Explanation

Detects NCP option to dump WTOR. AON responds to the WTOR according to the NCPRECOV definitions. EXIT01 is used. A timer is set to ensure that the dump does not take longer than the time specified in the DUMPTIME parameter. The EZLEFAIL program is not used.

#### Operator response

- EZL509I : *resource\_type resource\_name* IS UNAVAILABLE
- FKV538I : REPLY OF *reply* WAS ISSUED BY AUTOMATION FOR *ncpname* FROM *host*: CRITICAL RELOAD REPLY FROM NON-RECOVERY HOST
- FKV535I : REPLY OF *reply* WAS ISSUED BY AUTOMATION FOR *ncpname* FROM *host*: NON-CRITICAL RELOAD REPLY FROM NON-RECOVERY HOST
- EZL227E : *ident* COULD NOT FIND EXPECTED CONFIGURATION DATA FROM NCPRECOV COMMAND : *errmsg*

---

**IST260I**      ***resource\_name* - *sscpname***  
**SESSION LOST SA *saname* CODE**  
***code*.**

#### Explanation

The EZLEFAIL program is called. Notifies operators of a host to NCP session loss.

#### Operator response

- FKV524I : SESSION LOST BETWEEN *resource\_name* AND *sscp\_name* IN SA *sa*

- FKV539I : SESSION LOST BETWEEN *resource\_name* AND *reply* IN SA *sa* DUE TO A FORCED DEACTIVATION OF THE SSCP-PU SESSION
- FKV543I : SESSION LOST BETWEEN *resource\_name* AND *resource\_name2* DUE TO DEACTIVATION OF THE VIRTUAL ROUTE
- FKV545I : SESSION LOST BETWEEN *resource\_name* AND *resource\_name2* IN SA *sa* DUE TO AN SSCP FAILURE

The EZLEFAIL program only runs messaging.

---

**IST270I**      **LOAD OF *resource\_name***  
**COMPLETE-LOAD MODULE**  
***mod\_name***

#### Explanation

Finishes NCP recovery by indicating that the load of the NCP is complete. AON/SNA issues EXIT04. The timer for LOADTIME is purged because the load of the NCP is now complete. The EZLEFAIL program is not used.

#### Operator response

- FKV544I : RELOAD WAS SUCCESSFUL FOR *resource\_name* AND IS AVAILABLE

---

**IST272A**      ***replyid* NO INITIAL TEST FOR**  
***resource\_name* REPLY 'U' TO**  
**BYPASS - OR CANCEL**

#### Explanation

Responds to the bypass initial load WTOR for the NCP. Replies U. AON/SNA uses the EZLEFAIL program to run.

#### Operator response

- EZL509I : *resource\_type resource\_name* IS UNAVAILABLE
- FKV530I : BYPASS THE INITIAL TEST ROUTINE FOR *resource\_name* - REPLY "U" TO BYPASS WAS ISSUED

- FKV551I : REPLY FOR BYPASS INITIAL TEST FOR NCP *resource\_name* NOT ISSUED; AUTOMATION FOR *resource\_name* IS OFF: REPLY 'U' TO BYPASS OR CANCEL FOR REPLY ID reply

AON/SNA runs only special processing in the EZLEFAIL program to respond to the outstanding reply.

---

**IST278A**      *replyid* 'INVALID' REPLY FOR 'ID =' *resource\_name* LOAD - ENTER 'U' - OR CANCEL

---

### Explanation

Responds to a bad reply to IST272A. Replies U.

### Operator response

- EZL509I : *resource\_type resource\_name* IS UNAVAILABLE
- FKV530I : BYPASS THE INITIAL TEST ROUTINE FOR *resource\_name* - REPLY "U" TO BYPASS WAS ISSUED
- FKV551I : REPLY FOR BYPASS INITIAL TEST FOR NCP *resource\_name* NOT ISSUED; AUTOMATION FOR *resource\_name* IS OFF: REPLY 'U' TO BYPASS OR CANCEL FOR REPLY ID *reply*

AON/SNA runs only special processing in the EZLEFAIL program to respond to the outstanding reply.

---

**IST284A**      *replyid* OPTION TO RELOAD *resource\_name* AVAILABLE - REPLY 'YES' OR 'NO' OR 'YES,LOADSTA=LINKSTANAME'

---

### Explanation

Responds to the option to reload NCP WTOR according to the NCPRECOV definitions. AON/SNA uses EXIT03 and sets a timer to ensure that the time taken to reload the NCP does not exceed LOADTIME. The EZLEFAIL program is not used.

### Operator response

- FZL227E : *ident* COULD NOT FIND EXPECTED CONFIGURATION DATA FROM NCPRECOV COMMAND : *errmsg*
- FKV551I : REPLY FOR BYPASS INITIAL TEST FOR NCP *resource\_name* NOT ISSUED; AUTOMATION IS OFF: REPLY 'U' TO BYPASS OR CANCEL FOR REPLYID *replyid*
- FKV537I : REPLY OF reply WAS ISSUED BY AUTOMATION FOR *resource\_name* FROM *host*: NON-CRITICAL RELOAD REPLY FROM RECOVERY HOST

- FKV538I : REPLY OF reply WAS ISSUED BY AUTOMATION FOR *resource\_name* FROM *host*: CRITICAL RELOAD REPLY FROM RECOVERY HOST

---

**IST285I**      *dumptype* DUMP OF *resource\_name* FAILED - PERMANENT

---



---

**IST285I**      *dumptype* DUMP OF *resource\_name* FAILED - ddname CANNOT

---



---

**IST285I**      *dumptype* DUMP OF *resource\_name* FAILED - UNSUPPORTED

---



---

**IST285I**      *dumptype* DUMP OF *resource\_name* FAILED - COMPLETE

---

### Explanation

AON/SNA checks the progress of the NCP dump. EXIT02 is run if the dump is complete. If the dump failed, notify the operators that the dump of the NCP cannot complete. AON/SNA purges the timer set for DUMPTIME. The EZLEFAIL program is not called when the dump is COMPLETE. The other three variations of IST285I call EZLEFAIL.

### Operator response

- EZL504I *resource\_type resource\_name* IS AVAILABLE
- FKV550I DUMP OF NCP *resource\_name* FAILED - PERMANENT IO ERROR ON NCP OR DUMP DATASET
- FKV554I DUMP OF NCP *resource\_name* FAILED - DUMP DATASET ddname CANNOT BE OPENED
- FKV559I DUMP OF NCP *resource\_name* FAILED - DUMP DATASET ON AN UNSUPPORTED DEVICE TYPE
- FKV558I DUMP of *resource\_name* COMPLETE

Thresholding and recovery are not run in the EZLEFAIL program.

---

**IST361A**      *replyid resource\_name* FOUND LOADED WITH *loadmod* REPLY 'YES' TO RELOAD OR 'NO' TO CANCEL ACTIVATION

---

### Explanation

Notify operators when the NCP load module does not match the currently loaded module. AON/SNA replies NO if the NCP is to be automatically recovered.



### Operator response

- EZL509I *resource\_type resource\_name* IS UNAVAILABLE
- FKV510I LOAD OF NCP *resource\_name* STOPPED BECAUSE LOAD MODULE DOES NOT MATCH NEW NCP; AUTOMATION IS OFF, REPLY 'NO' TO CANCEL OR 'YES' TO RELOAD FOR REPLYID *replyid*
- FKV512I : LOAD OF NCP *resource\_name* CANCELLED - LOAD MODULE DOES NOT MATCH NEW NCP

The EZLEFAIL program runs the only optional processing.

|                 |                                                                                                |
|-----------------|------------------------------------------------------------------------------------------------|
| <b>IST380I</b>  | <b>ERROR FOR ID = <i>resource_name</i> FAILED - REQUEST : <i>rcmd</i>SENSE : <i>rsense</i></b> |
| <b>IST1139I</b> | <b><i>rcmd</i> FOR <i>resource_name</i> FAILED - SENSE: <i>rsense</i></b>                      |

### Explanation

If *rcmd* is REQDUMP, AON/SNA purges the memory dump time timer. If *rcmd* is REQLOAD, AON/SNA purges the load time timer. Notification operators are notified that the LOAD/DUMP has failed. The EZLEFAIL program is not used.

### Operator response

- FKV501I DUMP OF NCP FAILED - *action reason*
- FKV502I LOAD OF NCP FAILED - *action reason*

|                |                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------|
| <b>IST464I</b> | <b>LINK STATION <i>resource_name1</i> has CONTACTED <i>resource_name2</i> SA <i>subarea</i></b> |
|----------------|-------------------------------------------------------------------------------------------------|

### Explanation

Detects a link station connection to the NCP from AON/SNA. AON/SNA posts link station and NCP as available if the NCP is an automated NCP with a NCPRECOV statement. DUMP and LOAD timers are purged. The EZLEFAIL program is not used.

### Operator response

- EZL504I *resource\_type resource\_name* IS AVAILABLE
- FKV548I *resource\_name* HAS CONTACTED *resource\_type resource\_name* - *resource\_name* IS AVAILABLE

|                 |                                                                                           |
|-----------------|-------------------------------------------------------------------------------------------|
| <b>IST530I</b>  | <b><i>ru</i> PENDING FROM <i>resource</i> TO <i>resource</i> FOR <i>resource_name</i></b> |
| <b>IST1278I</b> | <b><i>ru</i> PENDING FROM <i>netid</i> to <i>netid</i> FOR <i>resource_name</i></b>       |

### Explanation

Detects an NCP unavailability from a channel attached non-recovery host. If the NCP status indicates that it is being recovered by another host, AON/SNA enacts/ acts the NCP to recover connection to this host. Otherwise, the operator AON/SNA notifies the operator, who manually recovers the NCP. The EZLEFAIL program is not used.

### Operator response

- EZL504I : *resource\_type resource\_name* IS AVAILABLE
- FKV542I : NCP *ncpname* REQUIRES A MANUAL ACTIVATION

|                |                                                            |
|----------------|------------------------------------------------------------|
| <b>IST881I</b> | <b>UNABLE TO CONTACT LINK STATION <i>resource_name</i></b> |
| <b>IST881I</b> | <b>LOST CONTACT TO LINK STATION <i>resource_name</i></b>   |

### Explanation

The NCP name is retrieved from the NCPRECOV control file entry link station *resource\_name*.

This message detects and reminds operators that the link station is not in contact with the NCP. Stops DUMP/LOAD completion monitoring. The EZLEFAIL program is not used.

### Operator response

- EZL509I *restype resource\_name* is UNAVAILABLE - *ncp\_name* MAY NOT BE AVAILABLE
- EZL555I : LINK STATION *linksta* HAS LOST CONTACT WITH NCP *ncp\_name*

|                |                                                                                   |
|----------------|-----------------------------------------------------------------------------------|
| <b>IST897I</b> | <b>NONDISRUPTIVE LOAD OF <i>resource_name</i> WITH <i>module_name</i> STARTED</b> |
|----------------|-----------------------------------------------------------------------------------|

### Explanation

Detects that the load of an NCP has started. The EZLEFAIL program is not used.

### Operator response

- FKV556I LOAD OF *ncpname* BY OPERATOR *operator\_id* STARTED

|                |                                                                            |
|----------------|----------------------------------------------------------------------------|
| <b>IST961I</b> | <b>NONDISRUPTIVE LOAD OF <i>ncpname</i> WITH <i>module_name</i> FAILED</b> |
| <b>IST523I</b> | <b>REASON = <i>reason_text</i></b>                                         |

**Explanation**

Detects when the load of an NCP failed. The EZLEFAIL program is called from another program which parses and passes the second IST523I message of the MLWTO.

**Operator response**

- FKV560I LOAD OF *ncp\_resource\_name* FAILED -  
RECEIVE *.reason\_text*

Thresholding and recovery are not run by the EZLEFAIL program.

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